CRITICAL LITERACY TEACHING SERIES: CHALLENGING AUTHORS AND GENRE

Rachel Carson

Challenging Authors

Karen F. Stein

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Critical Literacy Teaching Series: Challenging Authors and Genre

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Rachel Carson

Challenging Authors

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DEDICATION

For my daughters Arielle and Lisa who are working to build a better world,
and to the future generations who will inherit the world we leave.

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INTRODUCTION

RACHEL CARSON, ENVIRONMENTALISM, AND THE WEB OF LIFE

Rachel Carson is perhaps the most significant environmentalist of the twentieth century. In fact, she is often considered the founder of contemporary environmentalism. Her books about the sea invite readers to share her celebration of its wonders. *Silent Spring,* her graphic and compelling description of the damage caused by the widespread aerial spraying of organochlorine pesticides such as DDT, opened our eyes to the issue of pollution and to the interconnectedness of systems, indeed, of all living beings with each other and with the planet itself.

During World War II DDT was hailed as a hero, a wonder drug that saved the lives of soldiers and civilians threatened by insect-borne diseases. Paul Müller, the man who discovered DDT's potent insecticidal properties, was awarded the Nobel Prize for Physiology and Medicine. After the war DDT became the pesticide of choice to protect agricultural crops from insect damage. But it was widely spread in urban as well as rural areas. Spray trucks traveled through city streets spraying DDT widely in hopes of both curtailing a polio epidemic and protecting trees planted along city streets from insect attack. Only a few birdwatchers and scientists noticed that birds, fish, and other wildlife were dying in areas that had been sprayed.

Carson, a quiet, reticent woman, took on the task of investigating the consequences of such widespread use of DDT. At a time when science and technology were hailed as the tools that had won the war and now would lead America forward to even greater heights of power and well-being, her work challenged the fundamental assumptions of science and technology. She questioned the practices and belief systems of economic entomologists, pesticide manufacturers, agribusiness, government regulatory agencies and common citizens. Indeed, she challenged the very vision of American scientific, technological, moral, and political supremacy in the Cold War period.

Silent Spring immediately became a best–seller. The book was effective because it assembled scientific evidence and interpreted complex technical information so that it was accessible to a wide reading public. It resonated with mounting fears of the dangers of pollution, sparked by the knowledge of radioactive fallout, pesticide residues in food, and the tragedy of birth defects caused by thalidomide, a drug prescribed to pregnant women. The book provoked denunciations by chemical companies, but it led to public outrage, and to Congressional hearings that ultimately resulted in a ban of DDT in the U.S.

Although Carson is perhaps now best known for *Silent Spring,* she had previously achieved recognition and won awards and honors for three lyrical natural history studies of the ocean: *Under the Sea Wind* 1941, *The Sea Around Us* 1951, and *The Edge of the Sea,* 1955.

My introduction to Rachel Carson came one summer many years ago when I vacationed with friends on an island off the coast of Massachusetts. My hosts had a copy of Carson's *The Edge of the Sea*, a book that describes different shoreline environments. We spent

several days visiting the beaches on the island, taking along Carson's book as a guide to what we saw. Since that summer, much of what Carson wrote about has resonated throughout my life.

I've been fortunate to live on the East Coast and to enjoy the seashore. I have eaten fresh-caught fish at local restaurants, canoed and kayaked down a river to the ocean. I walk along the shore throughout the year, and swim in the ocean during the summer at various beaches. The beach and ocean present a different spectacle every day, as human and animal visitors come and go; the colors, textures, and the atmosphere are shaped by the sky, the wind, the temperature, and the tides. Like Carson, I have seen the ocean in many moods. I have watched sunrise paint the sand and the ocean gold, rose, and violet. I have experienced nature's awesome power in the hurricanes that batter the shore, and I have enjoyed the calm of sunset at low tide. But I have also seen the damage that humans can inflict.

On our local beaches in mid-March the small sand-colored piping plovers return from their southern sojourns and scrape shallow depressions in the sand to lay their eggs. When they hatch, the tiny plover chicks hop along the sand, looking like cotton balls on sticks. Human encroachment on plover habitat dramatically reduced their numbers to levels where they were declared an endangered species. Recent government policies, such as closing some beach areas and monitoring plover nesting areas are helping to build up the population, although the birds remain vulnerable to attacks by raccoons, foxes, rats, gulls, and dogs.

One of the many threats to the ocean is pollution from a variety of sources, including sewage. When I arrived at the university where I currently teach, several years after the publication of *Silent Spring,* the town was planning to install sewers in some areas to replace some of the septic systems then in use. I attended a talk by a civil engineer about sewage. At that stage the sewage plant was only going to include primary treatment (rather than the more thorough treatment which is now in place) of the waste. When I inquired if that minimal treatment was going to cause ocean pollution, he replied that "The solution to pollution is dilution," in other words, spread the waste around in the water and dilute it. I was appalled at the response, but I came to learn that it was the standard thinking about waste disposal at the time. Curious about the current state of our wastewater treatment, I was happy to note that the town recently received from the Environmental Protection Agency a Regional "Operations and Maintenance Wastewater Excellence Award" as reported on March 5, 2008 (http://yosemite.epa.gov/opa/admpress).

Carson, along with many of us and with most others who studied or fished in the ocean during her lifetime, believed that the ocean's resources of fish stocks were copious and inexhaustible; for a time she even thought that we would turn increasingly to fish for food as we damaged the land. However, as large fishing fleets trawl the ocean bottom the fish stocks are declining. In her non-fiction work *Payback,* Canadian author Margaret Atwood points to the harm that trawling inflicts on the ocean as she depicts a society that has lost respect for nature. Trawling "is like taking a front-end loader and scraping up your entire front garden and shredding it, keeping a few pebbles, and dumping the rest of it down the drain" (Payback 191). Due to declining fish stock many of the fishermen in my state have sold their boats, the local fish processing cooperative has closed, and an increasing number of expensive pleasure yachts are replacing the fishing boats that used to ply the waters.

People come to the tidal flats to dig up quahogs (edible hard shell clams), but every day the local newspaper prints a map of the areas closed to shell-fishing because of pollution. Our beaches are wide expanses of white sand, and tourists as well as residents enjoy the

pleasures of sand and surf. However, there are a number of days each summer when swimming is prohibited due to pollution when bacterial counts are high. Management of coastal resources has become a significant and controversial issue.

Sadly, I saw firsthand the havoc and destruction that an oil spill can bring, when on January 19, 1996, a barge transporting heating oil ran aground at Moonstone Beach, one of Rhode Island's most undeveloped beaches, adjacent to a national wildlife refuge. The barge spilled over 2700 metric tons of No. 2 fuel oil that were spread by the high winds and rough seas. Stinging, acrid fumes assailed our nostrils and brought tears to our eyes as far as the town six miles away. Community members gathered to attempt to clean the blackened, oil-covered shorebirds—usually unsuccessfully. It was estimated that approximately 9,000,000 lobsters were destroyed as well as loons and other shorebirds. Fisheries in the area were closed for a time, which brought economic hardship to the local fishing industry, an important component of the community's economic structure. Legal suits followed as damage estimates were set and appealed. It took four years for the court cases to be resolved, and 10 years for a lobster restoration program to be implemented. According to oceanographers C. M. Reddy and J. G. Quinn "this spill was the most damaging man-made accident in the history of the state of Rhode Island" (Reddy and Quinn 446).

Another of the issues that Carson addressed, that of toxic pesticides quite literally came home to me. (Of course, no doubt I am unknowingly eating food that contains all manner of pesticide residues, and wearing clothing, walking on rugs, and drinking from bottles treated or produced with various toxic chemicals.) One of the government spraying programs that Carson castigated in *Silent Spring* is the gypsy moth spray program. The gypsy moth is an invasive species that was brought to Medford, Massachusetts by a researcher who hoped to crossbreed the moths with silkworms and create an American silk industry. Some of the moths escaped and reproduced, spreading widely through the Northeast and now as far west as Wisconsin. Years after the publication of *Silent Spring* spraying programs to control gypsy moths continued although the insecticide had changed. Planes flew overhead in my rural neighborhood in Rhode Island spraying a chemical somewhat less toxic to humans, Sevin (carbaryl). Nevertheless, there are precautionary statements for people to heed when using this chemical, and it is toxic to aquatic wildlife and especially to bees. At the time of the spraying I was keeping a beehive, and requested that they not spray my property. The town posted a helium balloon on our lot boundary to signal not to spray that area. I doubted that the spray would respect the marked property lines on a half-acre lot, and I suspected that the bees would not be especially careful to stay on their side of the boundary. In any event I moved my hive elsewhere. Rhode Island has long ago stopped the spraying program. There are natural predators of the gypsy moth in New England that keep its population in check most of the time. Gypsy moth populations have a cycle of proliferating for a few years and then crashing from virus disease when the population gets too large. (Please see chapter 5 for further discussion of the gypsy moth spray program.) Rachel Carson would have recommended that authorities rely on such natural cycles and other forms of biological controls. Although she did not state that chemical means should never be used, she urged that research into all the effects of pesticides be considered, and that caution and moderation prevail if and when spraying was deemed necessary.

Carson's vision of the interconnected web of life informed her understanding of ecology and inspired new ways of thinking about natural systems and about the human impact on the world in which we live. Her words invite us to learn about the environment we inhabit and

challenge us to become better informed about the manifold implications of environmental issues such as global warming and pollution. Her concept of the interconnectedness of natural systems is increasingly relevant to our time as globalization produces a more intricate and tightly woven fabric of connections across the planet.

This book participates in the *Critical Literacy Series: Challenging Authors and Genres*, edited by Paul Thomas. The book is intended for the general reader, and for teachers and students. This project takes a new approach to Rachel Carson, using the perspectives of ecofeminist theory and other theories of environmentalism, and demonstrating Carson's relevance to contemporary issues. Carson is a challenging author. She challenges our belief that science and technology can control the natural world. She asks us to recognize our place in the world around us and to treat the earth respectfully. Her work calls us to rekindle our sense of wonder at nature's power and beauty, and to tread lightly on the earth so that it will continue to sustain us and our descendants.

Carson teaches us that the earth and the oceans are living entities, shaping and shaped by their interactions with the living creatures that inhabit them. Among these creatures, humans are having the greatest impact, and our alteration of the environment is turning out to be greater than Carson imagined possible.

Carson's short life was richer and fuller than we had realized before Linda Lear's biography was published in 1977. She visited and corresponded with a host of friends and colleagues. Her life and writings integrate a broad spectrum of ideas and issues. All her work melds scientific information with literary grace, and draws upon reason and emotion, fact and imagination. She infused her science and nature writing with poetry, explaining that she did not put the poetry into her writing about the sea; the poetry was already there. She argued for government transparency, for people's civic rights, for public engagement and activism.

Writing this book led me in many more directions than I anticipated when the project began. I learned about Rachel Carson and about the position of women scientists in the U.S. I learned about the oceans and their creatures, about ecology, environmental ethics, environmental justice, the U.S. governmental regulatory process, urban environments, and about the ecological challenges we are currently facing, such as factory farms, toxic pollution, and climate change. Study of Carson's work may lead in a whole range of directions, to the study of oceanography, of the natural environment, or of the ethics, economics, and politics of the environment. This book aims to set Rachel Carson's work into the context of her life and the times in which she lived; to summarize and review her contributions to the literature of the environment; to indicate her continuing influence; and to point to her many legacies across a spectrum of environmental research and activism.

I will address Carson's relationship to ecofeminism and other theories of environmentalism and demonstrate Carson's continuing importance to contemporary issues. Her vision of the interconnectedness of natural systems is increasingly relevant to our time. Her understanding of nature's web of life challenges us to address the manifold implications of global warming and environmental pollution. To enrich and expand our understanding of Carson's work, this book will:

—Analyze Rachel Carson's struggles and achievements as a woman scientist in a male-dominated field

—Discuss briefly several important American environmentalists and nature writers, both those who paved the way for Carson's work and those who learned from and are extending her work

—Discuss the history, economics, and ethics of pesticide use in the U.S.

—Analyze the ethical implications of using DDT to combat mosquito-borne diseases such as malaria

—Analyze the arguments Carson put forth in *Silent Spring* and her other books

—Relate Carson's ideas to current thinking about global warming, ocean pollution and the impact of pesticides and other contaminants on the environment.

Foremost, my aim is to convey the passion for life and the world around us that animated and sustained Carson, and to awaken the sense of wonder that may enrich all our lives.

The book is organized in eight chapters.

Chapter one reviews Carson's biography. In this chapter I draw mainly from the work of Linda Lear who wrote the definitive biography of Carson, *Rachel Carson: Witness for Nature*. Lear consulted archival materials extensively, worked painstakingly with Carson's papers at the Beinecke Library at Yale University, and with collections of papers of people who were important in Carson's life, including George Crile, the physician who supervised her treatment for cancer at the end of her life. Lear also interviewed many of Carson's friends, relatives, colleagues, editors, and others. As a result of her wide-ranging and thorough research she was able to demonstrate Carson's wide network of friends and her passionate engagement with life, and thus to rectify the misimpression left by some earlier biographies that Carson was a reclusive and isolated spinster.

Chapter two looks briefly at Carson's forerunners in the conservation and preservation movements. It looks briefly at the work of such nature writers as Henry David Thoreau, John Muir, John Burroughs, Henry Beston, and especially at several authors that Carson listed as important influences on her own work: William Beebe, Gilbert Klingel, and Thor Heyerdahl.

Chapter three describes the three best-selling books that Carson wrote about the sea, *Under the Sea Wind, The Sea Around Us*, and *The Edge of the Sea*. I discuss the process of writing the books, summarize key points of each, and consider the responses to and reviews of the books. Each of the books combines lucid poetic language with careful scientific explanations to present compelling narratives of the ocean, its inhabitants, and its importance in the ecosystem of planet Earth.

Chapter four considers Carson's other writings, consisting of government publications about fish and oceans, articles in newspapers and magazines on topics of nature and the environment, a TV script about clouds, speeches Carson delivered when receiving numerous awards, her posthumously published book *The Sense of Wonder,* and her published letters to her dear friend Dorothy Freeman. Reading these works provides us with a fuller picture of Carson's public and private life.

Chapter five, the largest chapter, is devoted to *Silent Spring*, Carson's impassioned exposé of the dangers posed by widespread pesticide spraying. To set Carson's work in the appropriate context, the chapter first explores the development and history of pesticide use in the U.S. and its surprising relevance to American foreign policy. I then proceed to summarize *Silent Spring,* and to review the reactions to the book. Chemical companies that manufactured pesticides hurried to counteract Carson's message by asserting that pesticides

are safe, by critiquing or parodying the book, and even by trying to prevent the book's publication.

Because her previous books were each best sellers, Carson's name was widely known when she published *Silent Spring*. Carson faced many personal, professional, and political challenges when writing this book. She had to contend with serious illness of family members, and she suffered from a series of ailments, including the metastatic breast cancer that led to her death. Because her work challenged popular thinking as well as powerful chemical companies and government agencies, she had to be able to substantiate all her facts, and she had to tread lightly when describing particular products and incidents. Because her work was seen as subversive at a time when the political climate was a conservative one, it was often difficult for her to gain access to government materials necessary for her work. Yet she persevered despite many difficulties and wrote a book that brought awareness of the damage that widespread pesticide spraying was wreaking on the environment and on living creatures. The book, a major achievement, led directly or indirectly to government hearings, to the banning of DDT in the U.S., and to the birth of the environmental movement in the United States. It generated controversies that continue to the present time.

Chapter six, "After *Silent Spring*: Rachel Carson's Legacies," looks briefly at some of the implications of her work and points to the careers of some of the women scientists and environmental activists who continue to write about, study, and protect the environment. It explores some of the ongoing environmental issues and controversies related to the book, especially questions about the uses of pesticides in food production and disease control. Related to the concerns about food production is Carson's Foreword to Ruth Harrison's *Animal Machines,* a book about raising veal, chickens, and beef on factory farms. Carson deplored the inhumane practices that Harrison revealed. The chapter will look at the case of malaria, an insect-borne disease, and examine the allegation that Carson is responsible for the deaths of Africans because DDT fell out of favor. It will close with some considerations of the issue of sustainability.

Chapter seven suggests ways to teach Carson's legacies. When people heard I was writing a book about Carson, I was surprised to find that many did not recognize her name, or knew only one facet of her work. I am convinced that we need to teach her work in several different contexts. I will suggest briefly some of the ways we may learn, read, and teach about Rachel Carson.

Chapter eight is a compilation of resources in print, online, films, and DVDs related to the many issues that Carson's life and work address, and to the organizations that her work inspired.

Three appendices provide brief chronologies of Carson's life, the use of DDT, and U.S. environmental legislation.

As I wrote this book I came to feel that I personally knew Rachel Carson, and my admiration for her grew. I was saddened by her untimely death, and felt that I had lost a friend. Please follow me on the journey to learn more about Rachel Carson, the oceans, and the world of nature we inhabit.

CHAPTER 1

BIOGRAPHY[1]

Rachel Carson: "The Poet Laureate of the Sea" (Ballard)

In 1886, twenty-one years before Rachel Louise Carson was born, Sarah Orne Jewett published one of her best known stories, “A White Heron.” The story tells about Sylvie, a young girl attuned to nature. She knows where the heron roosts, but keeps the location secret in order to protect the bird from a hunter who wished to add it to his collection of stuffed specimens. Like the fictional Sylvie, the real Rachel Carson wandered in the woods in her neighborhood and knew where to find birds and their hidden nests.

Sylvie’s name, of course, comes from the Latin word for forest, a symbol of her connection to the natural world. (The name of the bird has changed since Jewett wrote her story: the white heron would now be called an egret.) The hunter took Sylvie out with him for a day of hunting, and she “could not understand why he killed the very birds he seemed to like so much” (Jewett 5). He promised to pay for the information about the heron’s location. While the hunter wanted to collect a specimen, Sylvie preferred the living bird. Sylvie’s family was poor, and Sylvie was lonely, but she valued the bird’s life above the friendship of the hunter and the monetary reward. Jewett ends the story: “Woodlands and summer-time remember! Bring your gifts and graces and tell your secrets to this lonely country child!” (9).

In any event, the young Rachel must have been somewhat like Sylvie, for both loved to roam in the fields and woods, and delighted in seeing and hearing the birds. In fact, one of Carson’s early essays, “My Favorite Recreation,” published in *St. Nicholas Magazine* in 1922 when she was 15, described wandering with her dog in the woods and finding birds’ nests (Lear Woods 13). During one day she found “bobwhites, orioles, cuckoos, hummingbirds,” and an ovenbird (Lytle 24). Like Sylvie, Carson did not like the idea of killing the songbirds or other living creatures. According to biographer Philip Sterling, her brother Robert remembered that young Rachel asked him to stop hunting, reminding him that although the sport was fun for him “it can’t be much fun for the rabbits” (Sterling 20). Carson wrote that she had been “happiest with wild birds and creatures as companions” (Lytle 18). This love of nature remained with Carson throughout her life, bringing her joy, sustaining her in good and bad times, and animating her books. Indeed, in her essay “Help Your Child to Wonder,” posthumously published in book form as *The Sense of Wonder*, Carson writes that “those who contemplate the beauty of the earth find reserves of strength that will endure as long as life lasts” (Wonder 100). (For further discussion of *The Sense of Wonder* please read chapter 5). And in her great work *Silent Spring* Carson chose the vivid metaphor of silenced songbirds to convey her warning about the threats to life on earth unleashed by widespread pesticide spraying.

It is possible that Rachel Carson read or heard “A White Heron” at some point. Whether or not Carson knew Jewett’s story, it incorporates ideas that were important to the young Rachel, and that would resonate in Carson’s life and works.

Above all else, Carson was a writer and a scientist. Vivian Gornick has described the excitement of being one of these people: "a scientist or a writer is one who ruminates continuously on the nature of physical or imaginative life, experiences repeated relief and excitement when the insight comes, and is endlessly attracted to working out the idea. . . .

These are people whose most absorbed, alive, and focused hours are spent either writing or doing science" (30). In a college essay, Carson demonstrates that she belongs to that group, asking “Is there any mental joy comparable to having conceived and written a story? . . . If there is, I haven’t found it.” (CT COLL Box 2 F 11 dated 11/15/27). Carson combined the vocations of science and writing to convey important scientific ideas in lucid prose. Robert Ballard has called her "the poet laureate of the sea" (xviii).

Carson's earliest biographers promulgated a popular image of Carson as a reclusive, isolated spinster. McCay points out that “several of Carson’s earlier biographers compared her life unfavorably with [Thoreau’s]. His solitary existence at Walden Pond is extolled, while her reclusive life, her attempts to find a quiet spot away from the press of the modern world, are viewed as spinsterish and narrow” (McCay 88). McCay is trying to defend Carson’s quest for privacy here, but in her use of the word “reclusive” she is perpetuating the image of a hermit-like woman, which, as more recent biographer Linda Lear has demonstrated, is far from the case. Even Carson’s friend and editor Paul Brooks helped to perpetuate that stereotype when he claimed that her life lacked "a broad margin" (xx). As her more recent biographer Linda Lear explains, Brooks was borrowing a phrase from Henry David Thoreau. But whereas Thoreau was claiming that he needed this margin for his privacy and solitude, Brooks was implying that Carson's life was narrow because she did not marry. Lear's biography, based on meticulous and extensive research, presents a fuller, more rounded picture of Carson. Lear conducted research in archives, including the Carson papers at the Beinecke Library at Yale University; she interviewed numerous friends, colleagues, correspondents, and others important to Carson; and she was able to gain access to the then unpublished voluminous correspondence between Carson and her dear friend Dorothy Freeman (now published as *Always, Rachel*). Basing her biography on these materials, Lear corrects the stereotyped misimpression, revealing Carson's extensive network of friends, correspondents, and colleagues, and asserts "I can replace the icon [of a reclusive,“spinsterly” Carson] with a more truly heroic, far richer, and more passionate woman than the world has thus far embraced" (Waddell 214). In this chapter and throughout this book I have relied heavily on Lear’s definitive biography of Carson.

Carson studied marine biology and attempted to build a career as a scientist. However, entering the workplace during the Depression at a time when women had few opportunities to do research in scientific fields, she became a science writer, and consequently was able to remain a generalist rather than a specialist. Thus, Carson’s view of life remained a large one. Whereas many researchers focus on a small specialized area, Carson’s vision was more global. She understood the living world as a web of interconnections and transmitted her understanding in poetic essays and books that were accessible to the general public as well as informative to scientists. She wrote lyrically about the ocean and its creatures in her books and essays about the ocean. Concerned by the damaging impact humans were having on the earth and its creatures, Carson wrote *Silent Spring* to warn us about the network of contamination radiating out from government programs of aerial pesticide spraying.

CHILDHOOD

Rachel Carson was born on May 27, 1907 in Springdale, Pennsylvania to Maria McLean and Robert Warden Carson. She was named for her maternal grandmother, a strong woman who raised her two daughters herself after her husband died. Maria McLean had been a

teacher prior to her marriage, and she retained a lifelong interest in reading and education. Rachel Carson's sister Marian was ten years older, and her brother Robert was eight years older. Because of this age difference, combined with the distance of their home from both the town and the schools Rachel attended, and her reticent temperament, young Rachel spent much time alone or in her mother's company, and the pair developed a close relationship that continued until Maria's death. Rachel grew up on a small farm on a hill near the Allegheny River. The town of Springdale itself was growing more industrialized and ugly: Rachel was embarrassed by its smelly glue factory and two power plants. But the 64-acre family homestead contained a small apple and pear orchard, and offered spaces for woodland walks that Rachel and her mother enjoyed. Rachel and her mother were close throughout their lives, and their delight in nature was one of the bonds they shared. Starting when Rachel was one year old, they walked in the woods together and shared experiences of nature. Carson's biographer Linda Lear notes that Carson's "acuity of observation and her eye for detail were shaped on these childhood outings" (Witness 16).

In her childhood Carson, guided by her mother, developed the two passions that would shape her future life and career: love of nature and of writing. Her mother was an enthusiastic follower of the nature-study movement that originated at Cornell University in the period from 1884-1890. According to the noted botanist Liberty Hyde Bailey, one of the proponents of that movement, children should be encouraged to discover nature through direct experience, rather than through books. Bailey explains that nature-study aims "to open the pupil's mind by direct observation to a knowledge and love of the common things in the child's environment" (4). Its goal is "to put the pupil in a sympathetic attitude toward nature for the purpose of increasing the joy of living . . . to enable every person to live a richer life, whatever his business or profession may be. Nature-study is a revolt from the teaching of mere science in the elementary grades. . . .Nature-study . . . is spirit" (4-5). Moreover, Bailey extols the experience of encountering living nature above the "dry-as-dust" study of dead specimens. Nature-study avoids mechanical memorization and cataloging of specimens, and tries to make knowledge a "comprehensive whole" (8). Bailey advocates following the seasons to observe the cycles of nature. Instead of "close and specialized study of inert or dead form, [one should] place the children in the fields and woods that they might study all nature at work" (10), for "mere facts are dead, but the meaning of the facts is life" (14). Therefore, although we may "begin with the fact . . . the lesson is the significance of the fact" (17).The nature-study philosophy also encouraged reading stories and poetry about nature.

Bailey's colleague Anna Botsford Comstock wrote many widely-used "books and leaflets on nature study for students and teachers" (Kass- Simon 256). The books contained exercises and experiments that introduced children to the exploration of nature through the changing seasons. Maria Carson used the books that her children brought home from school to engage the family in outdoor nature-study activities.

From her earliest childhood throughout her life Carson lived according to the nature-study principles. As an adult she went birding with friends and with the Audubon Society; she took field trips related to her studies of marine biology and to her work with the Bureau of Fisheries; she spent many happy hours exploring tide pools and walking in the woods; and in preparation for her book *The Sea Around Us* she went for a dive.

The values embraced by nature-study resonate in Carson's writings as well. (See Lisa Sideris for a fuller discussion of the influence of nature-study on Carson.) Her philosophy is

most directly expressed in her short essay "Help Your Child to Wonder" (later published in book form as *A Sense of Wonder),* when she advises parents to awaken the child's emotions and interest, rather than naming the animals and plants they find. Observation of nature would lead to wonder, and wonder would be the motivating force that would lead the child to learn the names and the life stories of the observed creatures and phenomena.

The nature-study values of sensory experience, direct observation, and engagement with living nature appear in all of her texts as well. Her application for a Guggenheim Foundation grant explains that she is interested not so much in facts per se but in the interpretation of facts. In *Under the Sea Wind* Carson attempted to portray the world from the point of view of the birds and fish, instead of the usual human perspective. In line with the ideology of nature-study, Carson was fundamentally a generalist who saw nature as a holistic whole, rather than a specialist who focused on a small part of that whole. This vision enabled her to analyze and understand the human impact on the environment and see the consequences of human interventions.

Carson began writing as a young child. When she was nine years old she made a book for her father containing illustrated poems about animals. By the time she was ten she had published one of her stories in the children's magazine, *St. Nicholas Magazine.* Other notable writers who published stories as children in that magazine include William Faulkner, F Scott Fitzgerald, e.e. cummings, Edna St. Vincent Millay, and E. B. White. Carson's story, "A Battle in the Clouds," tells about a Canadian air force pilot, based on a story her brother wrote in a letter home while he was serving in the Army Air Service. Carson published several more stories in that magazine. Even as a young child, she approached her writing in an organized and business-line manner, keeping careful records of her submissions, and listing in a ledger the places she sent her stories, the cost of postage, and the responses of the magazines (Lear Witness 18-20). The seriousness with which she conducted research and kept careful files of materials is reflected in the fact that she saved many of her high school and college papers, which are now preserved in the Carson archives at the Beinecke Rare Book and Manuscript Library at Yale University.

EDUCATION

In her senior year of high school Carson wrote an essay, "Intellectual Dissipation" about the independent search for truth, and the wise use of one's intellectual power. She argued that mental development derives from three factors, from "contact with the great minds of all ages" by reading great books, from true friendship, and from clear thought and "sound reasoning" The essay indicates Carson's seriousness and sense of purpose. She explains, "Friendship should be one of the most sacred relationships we are permitted to sustain on earth. It should result in a mutual improvement of character and raising of intellectual standards" (CT Coll Box 1 F 12). Carson was to find such a friendship at least once in her close relationship with Dorothy Freeman. According to Carson the third pillar of intellectual development requires that we follow our own truths, whether or not others agree with us. Through talent, determination, and perseverance Carson was able to fulfill the lofty standards she set for herself.

A bright and conscientious student, she graduated first in her high school class and went to Pennsylvania College for Women (which later became Chatham College) in Pittsburgh.

At that time Pittsburgh was a prosperous, but grimy and soot-covered city, as a result of its iron and steel industries.

Carson maintained a close relationship with her mother throughout her life. When she was in college, Maria came on weekends to visit, often bringing cookies. Later, when Carson got a job and began to write essays and books about nature, she would read her work aloud to her mother. Maria would retype the works in progress and leave clean manuscript copy for Carson to find when she came home from work.

Carson was a serious student. In a college essay she set forth her beliefs about the process of education: “Scholarship . . . must be intensely individualistic. Each student is essentially a law unto himself. He must analyze his personal needs, take stock of his individual assets, and determine for himself the best means of reaching his individual goal. . . . Education is a spiritual adventure . . . [that comes] not by academic spoonfeeding, but through the undaunted efforts of an adventurous mind” (Ct Coll Box 2 F 11).

During her years at college Carson excelled in writing, her intended career. Stories and essays she wrote for her classes and for the school newspaper and its literary supplement reveal hints of the direction her future would take. In an essay for her composition class she wrote "I love all the beautiful things of nature, and the wild creatures are my friends" (Lear Witness 32). According to Mary McCay, one of her stories, “The Golden Apple,” raises the issue of women’s subservience to men. Carson's version is about Paris who chooses the most beautiful woman to award the golden fruit. McCay discusses Carson’s story and speculates that it asks why women must submit to men who make the choices for them. McCay suggests that perhaps Carson was thinking of her own family where her mother was in a subservient position to a husband who was unsuccessful in his business ventures (McCay 4-5).

Miss Grace Croff, Carson’s English teacher, commented about one of her essays, "The Master of the Ship's Light," "your style is so good because you have made what might be a relatively technical subject very intelligible to the reader. The use of incident and narrative is particularly good" (Lear Witness 33- 34). This was a predictive indicator of Carson’s future direction as a writer with the gift of translating technical scientific material into lively and readable prose.

While at college Carson pursued her career aspirations by submitting some of her compositions for publication in magazines such as *Colliers, Saturday Evening Post, Good Housekeeping*, *Harper’s,* and *Ladies Home Journal*. Although her essays were good college papers, they were not yet at a level suitable for publication in in these popular mass market magazines.

During Carson’s sophomore year she experienced an epiphany about her love of the ocean (even though she had not yet seen an ocean), inspired by a line from a poem of Alfred, Lord Tennyson, “Locksley Hall”: “For the mighty wind arises, roaring seaward, and I go.” She interpreted the line as a call to her, linking her destiny with the sea. As it turned out, the sea did indeed become a lifelong fascination and shaped her life’s work.

However, although she began her studies in college as an English major, Carson was inspired by her biology courses, and especially by her charismatic teacher Mary Scott Skinker. Carson agonized over whether to switch her major field of study from English to biology. She did, however, decide to make that switch. With some friends she formed a science club, named Mu Sigma based on Mary Skinker’s initials. To become a scientist was an unusual and difficult path for a woman at that time: teaching seemed the only available

career for a woman scientist. Indeed, the college president, Cora Coolidge, shared the common belief that science was an inappropriate career for women. Consequently there were tensions between Skinker, the demanding and inspiring science teacher, and Coolidge, who favored the humanities and arts as more fitting subjects than science for young women to study. Although Skinker left Pennsylvania College for Women she remained Carson's friend and mentor.

When she switched her major in college from English to biology Carson at first believed that she gave up the possibility of a writing career. Serendipitously, she eventually came to realize that she could combine her two passions, and to utilize both her love and knowledge of biology and her writing talent. Years later, in 1954, she reflected on this fortuitous career development in a talk she gave when she was inducted into the national organization of women in journalism, Theta Sigma Phi. In her talk to the gathering, called "The Real World Around Us," she noted that through her turn to science "I was merely getting something to write about" (Lear Witness 80).

Reserved and often serious, Rachel focused on her academic work and sports. She formed close friendships with several fellow students who were also interested in science, and with the two teachers, Grace Croff and Mary Scott Skinker, who had most inspired and encouraged her. She graduated magna cum laude, with high honors.

After graduating from college in 1929, Carson spent six weeks in the summer at the Woods Hole Marine Biological Laboratory on Cape Cod. Now that she could actually experience the ocean, she was enchanted. She swam, walked along the beaches, and went on dredging boats, observing the sea creatures that the nets captured (Lytle 35). She enjoyed walks by the shore, and spent much time investigating tide pools.

Later, remembering that time, Carson wrote:

> I began to get my first real understanding of the real sea world--that is, the world as it is known by shore birds and fishes and beach crabs and . . . other creatures that live in the sea and along its edges. . . . That was when I first began to let my imagination go down through the water and piece together bits of scientific fact until I could see the whole of life of those creatures as they lived them in that strange sea world (Carson, undated memo to Mrs. Eales, qtd in Lear Lost 54).

The Marine Biological Laboratory at woods Hole was a congenial place where men and women worked together and there was little hierarchical formality. Carson began research on the cranial nerves of reptiles which she hoped would become her master's project. However, she later changed her research focus.

In the fall she entered Johns Hopkins University to study for the Master's degree in zoology. She enjoyed living in Baltimore, but missed her mother. There was little money for visits home or even for long distance phone calls. To bring her family closer, she found a house for rent in a suburb, Stemmers Run. Her extended family—her parents, her now divorced sister, Marian, and Marian's two daughters, Marjorie and Virginia--moved there to join her in the spring of 1930. Her brother joined them shortly after. Maria Carson kept house, freeing Rachel Carson from house-keeping duties.

While at Johns Hopkins Rachel Carson served as a lab and a teaching assistant in a summer course in biology. She wrote her master's thesis on the pronephros, an organ that performed the function of a kidney in the early stages of catfish development. Her thesis

was titled "The Development of the Pronephros during the Embryonic and Early Larval Life of the Catfish." She received her Masters degree in zoology in May 1932.

After graduation she spent another six weeks at Woods Hole and then went on to work towards a PhD at Johns Hopkins University, where she studied genetics under Herbert Spencer Jennings and general physiology under S. O. Mast. She worked part time in the genetics laboratory of Raymond Pearl at the Institute for Biological Research in the Johns Hopkins School of Hygiene and Public Health.

Unfortunately, she had to give up her doctoral studies in 1934 because her father's health was poor. When her father died suddenly in July 1935, Carson, now age 28, took on the heavy responsibility of supporting her extended family: herself, her mother, her sister Marian, and Marian's two daughters. Entering the job market during the Great Depression, she became the main supporter of her family, a role she was to retain throughout her life. Marian suffered from diabetes and was often too ill to work. Their brother Robert Carson worked sporadically, but did not contribute much to the family income.

When her sister Marian died in 1937, leaving behind two daughters, Carson and her mother assumed the responsibility of raising the two girls, Virginia and Marjorie. In 1937 the family moved to Silver Spring, Maryland, so Carson could be closer to her office.

CAREER AT THE FISH AND WILDLIFE SERVICE, AND EARLY PUBLICATIONS

Rachel Carson sought a job teaching biology in college or junior college, but such jobs were scarce, especially for women, during the Depression. At the advice of Mary Skinker she took and scored well on three federal Civil Service exams, junior parasitologist, junior wildlife biologist, and junior aquatic biologist. However, there were few jobs for women in scientific fields, and, indeed, in the midst of the U.S. Great Depression, jobs were scarce in all fields. Carson did not obtain a job as a researcher for which she was well-qualified, because those jobs went to men. Instead, she was lucky to find a temporary job writing radio scripts for the U.S. Bureau of Fisheries.

Serendipitously, in this position Rachel Carson found a way to combine her interests in marine biology and writing. In 1935 Elmer Higgins, Division Chief of the U. S. Bureau of Fisheries (which became the U. S. Fish and Wildlife Service in 1939), hired her for a temporary assignment to write a series of fifty-two short scripts for radio programs on marine life that the Bureau was preparing. Before Carson began her work, they had found no one who could write interestingly about the material: the scientists could not make the information interesting to the public, and the writers lacked the scientific knowledge. The series was called "Romance Under the Waters," but to the Fisheries personnel it was known as "seven-minute fish tales." Her work received high praise.

While the job was not as prestigious or well paid as a research job, it did help Carson to support her family (although her continuing struggle over finances motivated her to write newspaper and magazine articles for pay), and it led to a new career direction. In fact, the job enabled her to remain a generalist and, therefore, it allowed her to retain a broader view than a specialist's viewpoint. As a writer and editor who reviewed and edited research reports on a range of topics she gained information about many related fields. Her position primed her to write her lyrical books about the ocean and to maintain a holistic picture of the environment, including the impacts of pesticides. Had she become a researcher she might

have needed to develop a more narrow focus, and she might have written academic rather than popular books and articles.

As part of her temporary job in the Bureau of Fisheries Carson wrote an introduction to a brochure. But her supervisor Elmer Higgins rejected it and asked for a re-write: her introduction was far too lyrical and more appropriate for a general interest magazine than a government brochure. Instead, Higgins suggested, Carson should submit the text to the *Atlantic* magazine.

She sent the revised essay first to *The Reader's Digest* but never got a reply from them. She then submitted it to the *Atlantic* which published it in September 1937 as "Undersea." Her name was listed as R. L. Carson, because, as she wrote to the magazine, people felt that such scientific essays "would be more effective . . . if they were presumably written by a man". However, her full name appeared in the Contributors' Column. This was her first appearance in an adult mass market magazine, and it opened the door to a new direction for Carson as a writer for the public. Quincy Howe, the senior editor at the publishing house Simon and Schuster, wrote to Carson, telling her how much he enjoyed the essay and asking if she planned to write a book about the sea. She had not considered such an idea, but now she began to think about it, and to gather material. The book, *Under the Sea Wind*, appeared in 1941. Unfortunately, when the book was published, the public's attention was focused on U.S. entry into World War II, and sales of Carson's book were disappointingly limited. (For a discussion of the book please see chapter 3, "The Sea Books.")

In August of 1936 Carson was hired into a continuing position as a junior aquatic biologist in the Bureau of Fisheries Division of Scientific Inquiry in the Baltimore field office. Her job was to work for the assistant bureau chief who was conducting a study of the fish of the Chesapeake Bay. She was one of only two women employed at a professional level in the Bureau. The laboratory and field research jobs went to men; Carson's job involved editing reports and analyzing the findings of these researchers. She had a gift of translating scientific and technical writing into accessible, lively prose that spoke to common readers, thus conveying the results of research into the popular domain. Through her job she learned the most up-to-date facts that researchers were discovering and the theories they were developing (Lear Witness 82). Carson sought opportunities for field work, and proposed trips to national Wildlife Refuges and National Parks in order to research material for informative conservation booklets.

Carson developed a network of colleagues in the Fish and Wildlife Service and in the Audubon Society who shared similar ecological values. As an active member and sometime board member of the Audubon Society she participated in many birding trips and bird walks. She also began to build up a network of colleagues and scientific advisors who respected her precision and clarity; many of them became lifelong friends. She was able to draw upon this network for factual data and emotional support as she conducted her investigations into the lives of the ocean's creatures and, later, into the ramifications of insecticidal spray programs.

During World War II Carson wished she could be more involved in contributing to the war effort, but did not see opportunities. However, she did write pamphlets intended to introduce people to food fish as a source of protein when meat was in short supply.

She was involved in the policy-making staff of the Division of Information. As she wrote speeches on conservation matters she observed how groups such as "timber, mining, grazing

and even recreation" sought to advance their particular interests and to modify legislation that affected them. Even within her own agency differences arose between divisions regarding the ways to protect wildlife (Lear Witness 180).

Given the political nature of her position, Carson began to have reservations about the work of the agency. She found the constraints of writing within the government increasingly frustrating. Hoping to get a wider scope for her writing, she applied to the Reader's Digest and the New York Zoological Society, identifying herself as a scientist as well as a writer. On October 26, 1945 she wrote to Dr. William Beebe, an oceanographer and ornithologist at the New York Zoological Society: "I don't want my own thinking in regard to 'living natural history' to become set in the molds which hard necessity sometimes imposes upon Government conservationists" (YCAL MSS 46 box 4 f67). She hoped to find a "broader field" that would give her more freedom to write in accord with her beliefs. But jobs were going to returning servicemen in the post-war period, and Carson stayed with her job at Fish and Wildlife. Nevertheless, she determined to devote as much time as she could to her own writing. At a friend's advice, Carson secured a literary agent, Marie Rodell, who became a close friend as well as a helpful agent.

While she was involved in the pre-publication business surrounding her second book *The Sea Around Us,* in September 1950 Carson had a small breast tumor removed. The surgeon told her it was not malignant and no further treatment was indicated. After the surgery, Carson vacationed at Virginia Beach, Virginia, and Nags Head, North Carolina, where she walked along the shore and in the dunes, making notes and preparing to write the guide to the seashore that eventually became her third book, *The Edge of the Sea.* The original idea was a kind of handbook to identifying the seashore creatures. But Carson "was more interested in understanding why creatures live where they do and how they adapt to their natural environment than she was in a mere catalog" (Lear Witness 186). She applied for a Guggenheim Fellowship "to prepare a guide to seashore life on the Atlantic coast of the U.S." It would be an original, creative book, not only a "handbook for identification, but [it would also] provide an understanding of the biological principles that control life in this zone. An ecological concept will dominate the book. It will be so organized and written as to be practically useful in the field" (YCAL MSS 46 box 108 f 2077). It would contribute "to a better understanding of an interesting and important region of our world" (Lear Witness 187). When the prestigious award was granted in March 1951 Carson applied for and received a year's leave from the Fish and Wildlife Service to work on her shore guide.

CAREER AFTER LEAVING THE FISH AND WILDLIFE SERVICE

Although she was now freed of her work responsibilities, family ones took up much of her time, and caused her pain. Maria Carson was now 83 and growing frail and dependent on Carson. Her niece Marjorie was pregnant out of wedlock, a scandal and stigma in the 1950s. Because she was diabetic an abortion was not possible. Their brother Robert Carson was scandalized by the pregnancy. Always an intensely private person, Carson struggled to keep these difficult family matters private, just at the time when publication of her new book *The Sea Around Us* thrust her into the public spotlight. When Marjorie's son Roger Christie was born February 18, 1952, Rachel and Maria Carson added him to their growing family that still required Rachel's support.

Buoyed by the success of her second book, *The Sea Around Us* (please see a discussion of this in chapter three), Carson was now financially secure. She resigned from her position at Fish and Wildlife in May 1952 to devote her energies full time to writing.

Ever since a vacation in 1946 spent in a cottage on the Sheepscot River, west of Boothbay Harbor in Maine, Carson dreamed of owning a small vacation home in that area. She was now, in 1952, able to realize that dream. She had a house built for her, that she named Silverledges. When there she spent time walking along the bluffs, exploring tide pools or wandering in the nearby woods.

It was here on the Maine coast that Carson realized another long-held dream, the dream of an intimate, mutually-enriching, friendship that she had written about in her high school senior essay: "Friendship should be one of the most sacred relationships we are permitted to sustain on earth. It should result in a mutual improvement of character and raising of intellectual standards" (CT Coll Box 1 F 12). For in that neighborhood was the vacation home of neighbors Dorothy and Stanley Freeman. Their son had given them a copy of *The Sea Around Us* which they read aloud to each other. When they learned that Carson was also planning to spend summers in their neighborhood they wrote to welcome her. Almost immediately Dorothy and Rachel became close friends, spending the summers exploring nature together. They watched the birds and butterflies migrating, watched the sky and its constellations at night, and investigated the tide pools on the rocky coast at low tide. They often brought the small creatures they found in these pools back to Carson's cottage to view them in a small aquarium or under a microscope. After their observations, Carson carefully replaced the creatures exactly where she had found them. During the remainder of the year, back at their other homes (Carson's in Maryland; Freeman's in Maine), Carson and Freeman kept in touch though letters, phone calls, and occasional visits. (Please see chapter 4 for further discussion of their correspondence.) The friendship was an important one for both women. Each felt that she had found someone who fully understood her. Such a friendship was particularly important for Carson, for she felt that her life as a writer was a lonely one.

Another new friend for Carson was Paul Brooks, her editor at Houghton Mifflin. Brooks encouraged Carson in her writing, and was understanding about delays and revisions as she worked on the book first named *A Guide to Seashore Life on the Atlantic Coast.* Carson originally planned the book to be similar to the series of field guides to the birds authored by Roger Tory Peterson and edited by Brooks. However, as she worked on the book, Carson found that writing thumbnail sketches of different marine creatures was tedious. She reconceived her plan for the book as a description of four types of shores: the rocky coast of Maine, the sandy coast of the mid-Atlantic seaboard, and the coral coast and mangrove coast of southern Florida. The book was published in 1955 as *The Edge of the Sea* (see chapter 3 for a discussion of this book).

Following the publication of *The Edge of the Sea* Carson received many offers to write for a variety of venues. One such offer came from CBS TV asking her to write "something about the sky" that became a TV script about clouds (see chapter 5).

Family problems again slowed down Carson's writing. Her mother Maria's arthritis made walking difficult, and her niece Marjorie's diabetes was worsening. Carson resisted her agent Marie Rodell's suggestions of putting her mother in a nursing home or of getting household help (Lytle 116). Instead of hiring helpers, Carson fantasized about having a twin "who could do everything I do except write, and let me do it!" (Letter to DF 2/3/56 in *Always* p 150). Marjorie contracted pneumonia and needed hospitalization. When she

returned home she seemed to be recovering, but her condition worsened rapidly and she died in 1957. Carson had been close to her niece and was devastated by her unexpected death at the young age of thirty-one. Carson was then 50, and her mother was 88 years old. They adopted Marjorie's son Roger Chrisite, now a lively but bewildered orphan aged five.

In order to have a quiet place to write, Carson planned to buy a bigger house. To raise the money she wrote a child's version of *The Sea Around Us*. She planned, but never completed, an anthology for the World of Nature that would include "the whole story of Life on our earth" (Lytle p 118, fn 34). Carson was now able to have a larger house built for the family, and she planned to add a room for Roger in her Maine vacation house.

POLITICAL ACTIVITY

Carson was politically active to a small degree in the usual sense of direct involvement in a governmental political process. But her major political achievement was the book *Silent Spring* that awakened the public and the government to the dangers of indiscriminate pesticide spraying, and initiated the environmental movement in the United States. Her vision of the connectedness of all life is an ideology that leads those who embrace it to respect living beings and the earth we all inhabit. The book turned the emphasis of the young U.S. environmental movement in a more activist, political direction.

While she was employed as a researcher, editor and information specialist with the Fish and Wildlife Service, Carson felt that she could not speak out publicly against programs with which she disagreed. She looked for other writing jobs because she was eager to be able to speak out more freely. When she earned enough money from her books to leave the Service she became more politically involved. When the election of President Dwight David Eisenhower in 1952 initiated a change of environmental policy favoring private business interests over conservation interests Carson was dismayed. The new Secretary of the Interior, Oregon businessman Douglas McKay, dismissed Albert M. Day, the director of the Fish and Wildlife Service, and put in less experienced political appointees in important positions within the Department of the Interior. Carson, who was at this time a respected and well-known author of two best-selling books, wrote a letter to the editor protesting Day's dismissal as "an ominous threat to the cause of conservation" (Lear Witness 257). The letter was published in the *Washington Post* and widely reprinted in other venues, including in the *Reader's Digest*.

When a change in governmental policies seemed possible, in 1960 Carson served on the Natural Resources Committee of the Democratic Advisory Council helping to draft the platform for the election that brought John F. Kennedy to power. In this capacity she wrote about pollution, radioactive waste dumping in the sea, and chemical poisoning for the platform. And, when *Silent Spring* stirred up protests from chemical companies, Carson testified in two Congressional hearings about the dangers of pesticide spraying.

Carson's critique put forward in her monumental *Silent Spring* proposes an important approach to politics. She wrote this book to inform the public about the environmental contamination that indiscriminate pesticide spraying inflicts on communities. In the book Carson challenged federal and state programs that sprayed pesticides in residential areas without informing the residents about the toxicity of the sprays. Setting this book in the context of civil rights she argued "if the Bill of Rights contains no guarantee that a citizen shall be secure against lethal poisons distributed either by private individuals or by public

officials, it is surely only because our forefathers, despite their considerable wisdom and foresight, could conceive of no such problem" (Spring12-13). This concern for protecting citizens "against lethal poisons distributed either by private individuals or by public officials" is an important aspect of the environmental justice movement. Frequently, waste treatment plants are built, and toxic materials are stored and processed in urban inner cities, leading to environmental degradation and high rates of asthma and other respiratory diseases in these neighborhoods. (For more discussion and information about this issue please see chapter 6, "After *Silent Spring* " and chapter 8, "Resources'.)

Writing *Silent Spring* was a courageous act for Carson, for she was challenging both the government and the powerful chemical companies. She was the sole supporter of an extended family, and had no institutional base. If she were sued by a chemical company, legal fees would have drained her resources. Her risk-taking was rewarded: the book succeeded beyond the expectations of Carson and her friends. The book galvanized public opinion and led directly to investigation of the spraying by a Congressional Committee, and indirectly to the banning of DDT in the U.S., and to the formation of the Environmental Protection Agency.

Carson urged people to take action, to make their voices heard, asserting "The public must decide whether it wishes to continue on the present road, and it can do so only when in full possession of the facts" (Spring 13). Her goal was to inform citizens and motivate them to engage in the political process. Her book reached a wide audience because she was already widely known as a best-selling author of three beautifully crafted and prize-winning books about the ocean. Because Carson was a private person the platform she chose to spread her message was not the podium but the written word. However she did express a political philosophy in some of the talks she gave. When she spoke to the Federation of Homemakers in 1963 after the publication of *Silent Spring* she remarked "I personally have no patience with the . . . view that we dare not tell the American people the truth lest it scare them. . . . I have more respect for the public than that." She found in the many letters written to her about *Silent Spring* "not panic . . . perhaps shock and dismay, but . . . also a firm determination to do something to correct the situation." She encouraged her audience "You do have something to say about what is done. Democracy still works if we would have it so, and the individual citizen does have a voice to which our legislators and our administrators listen" (YCAL MSS 46 box 99 f1856).

HONORS, AWARDS AND PUBLIC SPEAKING

Carson's fame after publication of *The Sea around Us* led to many speaking invitations, most of which she rejected. However, when she won The National Book Award for nonfiction in 1952 she attended the reception and award ceremony on January 27. In her acceptance speech she asserted the linkage between literature, life, and science:

> the materials of science are the materials of life itself. Science is part of the reality of living; it is the what, the how, and the why of everything in our experience . . . [rather than] something that belongs in a separate compartment . . . apart from everyday life. . . . It is impossible to understand man without understanding his environment and the forces that have molded him physically and mentally (Lear Witness 219; YCAL MSS 46 box 108 f 2065).

She explained that the goals of both science and literature are "to discover and illuminate truth" (Lear Witness 219). She was especially happy to receive another prestigious award that year, the John Burroughs Medal for excellence in nature writing. In her acceptance speech Carson alluded to the destructive power of the atomic bomb: “intoxicated with the sense of his own power, [mankind] seems to be going farther and farther into more experiments for the destruction of himself and his world" (Lear Witness 221). Carson also urged naturalists to seek a wider popular audience rather than to write just for other naturalists.

Other awards followed, including the Henry Grier Bryant Gold Medal for distinguished services to Geography from the Geographical Society of Philadelphia, a Gold Medal from the New York Zoological Society (where she had once applied for a job), and honorary degrees from several colleges and universities. Carson declined the honorary degrees except for the Drexel Institute of Technology (doctorate of letters), the Pennsylvania College for Women, her alma mater (doctorate in literature), Oberlin College (doctorate in science), and Smith College (doctorate in literature). The awards celebrate the power of Carson’s work, her ability to link the separate disciplines of literature and science, and to convey scientific information in clear and luminous prose.

Speaking at the Zoological Society award dinner Carson spoke about "the relatively new science of oceanography and the importance of creative imagination as a critical adjunct of technology and scientific discovery" (Lear Witness 224).

At this time Carson was invited to join a Scripps Institute of Oceanography research expedition to study “the biological effects of radiation from the 1946 atomic tests in the Bikini Islands.” But the pressures of writing her next book and arranging for her mother's care led her to decline the invitation (Lear Witness 237).

In December 1953 Carson participated in a scientific seminar about the ocean sponsored by the American Association for the Advancement of Science (AAAS) and held in Boston, Massachusetts. She spoke about the research she was doing, and described how metabolic products of marine organisms excreted into the water might influence the development and/or reproduction of other organisms. She explained

> we are brought back to the fundamental truth that nothing lives to itself. The water is altered, in its chemical nature and in its capacity for inducing metabolic change, by the fact that certain organisms have lived within it and by doing so have transmitted to it new properties with powerful and far-reaching effects.

She invited other scientists to participate in creative and imaginative research in this new area. She reiterated a recurring theme of her research: "we are brought face to face with one of the great mysteries of the sea" (YCAL MSS 46 box 99 f 1856).

In 1954 Carson was elected to the Royal Society of Literature in England, and made an honorary member of the honor society of women journalists, Theta Sigma Phi. In her talk to Theta Sigma Phi Carson spoke about the importance of the natural world which was being threatened by “this trend toward a perilously artificial world” because of a mounting tendency toward replacing “natural feature[s] of the earth” with “man-made-ugliness.” Carson affirmed "I believe natural beauty has a necessary place in the spiritual development of any individual or society. . . . The more clearly we can focus our attention on the wonders and realities of the universe about us, the less taste we shall have for destruction" (Lear Witness 259-60).

A new series of awards followed publication of *Silent Spring.* Carson received several prestigious awards and honors in 1963: the Albert Schweitzer Medal of the Animal Welfare Institute, the Paul Bartsch Award from the Audubon Naturalist Society for distinguished contributions to natural history (she accepted this in absentia); an Audubon Society Medal for distinguished service to conservation; the Cullum Medal of the American Geographic Society; and election to the American Academy of Arts and Letters. The National Wildlife Foundation named her Conservationist of the Year. In some cases she was the first or one of the few women to receive the award: she was the first woman to receive the Audubon society medal, and there were only three other women members of the American Academy of Arts and Letters. Unfortunately declining health made it difficult or impossible for her to accept all of them in person. However Carson did manage to spend six days in New York in December attending dinners and luncheons at which she received these awards and spoke. Lear writes "Carson's six days in New York were triumphant and happy ones surrounded by her dearest friends and many admirers who delighted in the recognition given to her.… She managed all these public occasions beautifully," but needed more treatments when she returned home (Witness 471 – 73).

CARSON'S LAST YEARS

While working on *Silent Spring* Carson suffered from a variety of ills. In January of 1960 she began to experience a string of ailments that impeded her research and writing, and that would ultimately lead to her death. Biographer Linda Lear writes: "At times during these months only her indefatigable determination and inextinguishable spirit kept her from giving up" (Witness 364). Her first health problem that year was a duodenal ulcer which compelled her to eat baby food and take antacids until it cleared up. When she recovered from the ulcer she became ill with viral pneumonia and then a sinus infection, both of which led to more delays in her writing.

Ironically, just as she sent out to professional colleagues for review two chapters about possible links between pesticides and cancer, she found two small tumors in her left breast and went for surgery at Doctor's Hospital in Washington, D. C. on April 4. One appeared benign, but the other was suspicious, and Dr. Fred Sanderson performed a radical mastectomy. Although Carson wished to know the truth of her condition, her physician did not divulge the fact that her tumor was indeed malignant. At the time when Carson's cancer was diagnosed physicians were often less open and less communicative with their female patients than is the practice at the present time. Lear notes that a physician might not have told a married woman the diagnosis, but would have told her husband the diagnosis if he asked directly (Witness 368). Lisa Sideris explains that the "concepts of informed consent and patients' rights were not well developed. At the time of Carson's diagnosis, it was not unusual for biopsy and mastectomy to be performed as a single procedure, with no inquiry into the patient's wishes regarding her own body" (Sideris Body 140).

The radical mastectomy, an operation defined as "the removal of the affected breast, the underarm lymph nodes, and both chest wall muscles," and first performed by Dr. William Stewart Halsted in 1882, was the standard treatment for breast cancer at that time (Lerner 15-40, 303). However, after the surgery Dr. Sanderson recommended no additional treatment although, in fact, he was aware that the tumor was malignant and had metastasized. When Carson inquired if the tumor was malignant, the surgeon told her it was

"a condition bordering on malignancy" (Lear Witness 367). Lear believes that physicians tried "to hide the truth of her condition for as long as possible" (Witness 368). And, at that time, in the U.S. the word cancer was so terrifying, it was seldom mentioned. Barron H. Lerner writes that women often delayed consulting physicians about suspicious lumps in their breasts because the prognosis was “grave” and the standard treatment, the radical mastectomy, was a difficult and disfiguring operation (Lerner 29). In contrast, Carson wished to know the details of her condition, for she found relief in knowing the facts, however unpleasant. However, although she sought information for herself, she tried to keep her surgery and other health issues quiet, speaking of them only to a few close friends. She shunned publicity and did not want her illness to dominate discussions of the important book she was writing. She asked Dorothy Freeman to speak of her health only if asked, and then to say that Carson had never looked better. The only additional treatment for breast cancer available at that time was a debilitating course of radiation, and the cure rate was low. So, perhaps delaying radiation had the result of providing Carson the time she needed to complete the book to which she was so passionately dedicated.

In late November Carson found a hard swelling on her left side, and doctors recommended radiation therapy, claiming to be puzzled by the mass. Just as Carson believed the public has a right to know the truth about the toxic substances that were being sprayed and rejected the government “sugar coating of unpalatable facts” and the “little tranquilizing pills of half truth,” she was determined to discover the full truth of her health and rejected the sugar coating of the information her doctor gave her (Spring 13). She grew concerned that her doctors were not being honest with her, and in late 1960 she wrote to Dr. George Crile, a friend, and a cancer specialist at the Cleveland Clinic. Crile was an advocate of the modified radical or partial mastectomy, rather than the full radical, and he believed in talking frankly with his patients and giving choices of treatments, whenever possible. He later wrote a book, *What Women Should Know about the Breast Cancer Controversy.* Dr. Crile asked for her medical records, and learned that Dr. Sanderson knew her original tumor was malignant. Crile recommended a course of ten radiation treatments started in January 1961. Carson thanked Crile and wrote “I appreciate, too, your having enough respect for my mentality and emotional stability to discuss all this frankly.” Knowing the facts, “even though I might wish they were different [provided] more peace of mind” (YCAL MSS 46 box 102 f 1938). On December 22, 1960 Crile answered: "Dear Rachel, I have always believed that intelligent people in responsible positions not only wish to know as much as possible about any ailment they have, but also that such people are entitled to know everything that is known about such ailments." He thanked her for sending him a "lovely blonde poinsettia" and extended his best wishes for the New Year (YCAL MSS 46 box 102 f 1938).

Soon after, Carson developed a staphylococcus infection that required hospitalization. Other maladies followed. The cancer metastasized, and radiation treatments weakened her and drained her energy. Carson remained realistic but hopeful in January 1961 as she started her course of 10 radiation treatments. A series of health problems escalated: Carson suffered from a bladder infection, and then phlebitis in both legs that made it too painful to walk or stand.

As the radiation left her nauseated and weak and walking remained painful Carson was beginning to experience despair. A brief passage in one of the small notebooks that she kept by her bed for the purpose of entering scientific information and observations of nature testifies poignantly to her suffering: "I moan inside—and I wake in the night and cry out

silently for Maine. I prayed very graciously to God that he would make it a nice day." After this outburst of despair, the next page continues with her usual scientific observations. But a later entry provides a "handwritten ledger of her financial holdings, including her investments, real estate, retirement, insurance, and royalties" indicating her awareness of her declining health and her wish to settle her estate (Lear Witness 384).

By late March 1961 Carson was feeling more energetic and was able to resume work on *Silent Spring*, and to anticipate returning to her summer cottage in Southport, Maine. Once again, she was well enough to work seriously on the manuscript and to take walks with her dear friend Dorothy Freeman. Then in November 1961 iritis, an infection of the iris of her eye, left her unable to read for several months. She continued to work with the help of her assistant, Jeanne Davis who read portions of the book to her. Nevertheless Carson told Dorothy Freeman that she was primarily “visual minded,” and needed to read and re-read the manuscript herself.

In a letter to Freeman Carson wrote about how important finishing the book was to her. She hoped that the ailments that prevented her from writing more quickly might have actually benefited the book’s reception, because the public opinion was now more receptive for the book (given the public’s concern with radiation and the recent thalidomide scare). Nevertheless, she mused, "if one were superstitious it would be easy to believe in some malevolent influence at work, determined . . . to keep the book from being finished” (Always 6 January, 1962 pp. 390–391). Later, putting a positive spin on her difficulties, she came to feel that the time away from her usual intense concentration on the manuscript helped to give her perspective and to shape the material better.

Carson's health continued to deteriorate. She began to suffer from angina. The cancer had spread to her bones, making it difficult for her to walk. She spent one last summer in her beloved cottage in Southport, Maine and tried to spend as much time as she could enjoying the woods and tide pools, and sharing walks, talks, and experiences with her friend Dorothy Freeman. One morning in particular she felt was especially memorable. She wrote to Dorothy, expressing things she could write better than speak.

The letter was intended as a farewell to Dorothy. In it Carson implicitly acknowledged that her death was imminent. The letter spoke of watching the migrating monarch butterflies earlier that day, on an especially lovely morning they had spent together, a morning she would remember. “I shall remember the monarchs. . . . We felt no sadness” knowing that this migration marked the end of their lives and they would not return.

> When any living thing has come to the end of its lifecycle we accept that end as natural. . . . For ourselves . . . when that . . . cycle has run its course it is a natural and not unhappy thing that a life comes to its end.
>
> That is what those brightly fluttering bits of life taught me this morning. I found a deep happiness in it–so, I hope, may you” (Always, September 10, 1963 p. 468).

Freeman wrote back “I am so grateful to the Butterflies for bringing to me the precious thoughts you captured tangibly. Please know this message before me now is one of the loveliest possessions I shall cherish always” (Always September 12, 1963 p. 468).

Carson continued to hide the reality of her deteriorating health from the public and from all but her closest friends and family. She claimed that walking was difficult because of her arthritis and she pretended that her hospitalizations and treatments were due to arthritis. She traveled to San Francisco to present a talk at a Kaiser Foundation symposium on “Man

Against Himself." She used a cane to go on the stage, and sat while delivering an hour-long talk, "The Pollution of the Environment" on October 18. Lear comments that "it was one of her finest speeches, beautifully crafted and expertly delivered before a hushed and appreciative audience" (Witness 464).

The next day, in a wheelchair, Carson toured Muir Woods and a beach with David Brower, then the executive director of the Sierra Club. Brower pushed her chair on the beach at Rodeo Lagoon. He later wrote that they were surprised and elated to see "a whole gaggle of Brown pelicans . . . [who could now thrive] thanks to Miss Carson and her book" (Lear Witness 465).

Meanwhile, on her return home, Carson had to make difficult decisions about settling her finances, putting her estate in order, and providing for the care of her grandnephew Roger Christie. Bequeathing her papers and her financial resources was the simpler part. Carson appointed her literary agent Marie Rodell as her literary executor. Rodell learned that Yale University in New Haven, Connecticut had recently opened the Beinecke Rare Book and Manuscript Library. Carson agreed that this would be the best depository for her manuscripts and papers, and arranged the bequest. She asked Dorothy Freeman to keep the letters she had sent her, and these were edited and published after Freeman's death by her granddaughter Martha Freeman.

She left cash bequests to family members, the Nature Conservancy and the Sierra Club. She set up a trust for Roger's support, but determining his guardianship was more difficult, because it was emotionally painful. She apparently hoped that her niece Virginia and her husband Lee would offer to adopt him, but they did not. Carson considered several possibilities but never asked anyone directly, perhaps because she feared refusal. She added a codicil to her will nominating Mr. and Mrs. Paul Brooks, and Professor and Mrs. Stanley L. Freeman, Jr. (son of Dorothy and Stanley Freeman) as her nominees for guardians, believing that his "best interests will be served" if he were raised by friends who had children close to his age and who would "rear him with affectionate care." After Carson's death Paul and Susie Brooks agreed to become Roger's guardians (Lear Witness 466- 67, 477, 481).

Carson asked Reverend Duncan Howlett to hold a simple funeral service for her at All Souls Unitarian Church. She suggested that he might read a passage from the final section of her book *The Edge of the Sea* at the service. She died from cancer and from a heart attack on April 14, 1964, at age 56.

Among the obituaries for Carson are the following:

Washington Post 4/16/64

> Her competence as a scientist was demonstrated by the failure of her critics to cite a major factual error in her powerful polemic against the indiscriminate use of chemical pesticides. . . . It was a tribute to her skill as a writer that she made her case so forcefully that President Kennedy appointed a panel of scientists to investigate her charges -- and the scientists upheld Miss Carson. (YCAL MSS 46 box 112 f 2174)

Marjorie Spock wrote in "Bio-Dynamics," summer 1964

> Seldom has the death of a public figure aroused as sharp a sense of loss and grief as that of Rachel Carson . . . [She was a] champion of scientific sanity. . .Here was a woman who risked everything to tell the truth about pesticides, that all of mankind and

> the whole of nature might be benefited." Spock claims that Rachel Carson "understated the case [of pesticides]-to withhold---for the sake of credibility---much of the true horror of the spraying picture. Future developments will show that *Silent Spring* is a model of conservatism." (YCAL MSS 46 box 112 F 2173)

Counter to Carson's wishes, her brother Robert planned an elaborate state funeral at the Washington National Cathedral on Friday, April 17. Carson had told friends she wished to be cremated, and Robert wanted the ashes to be buried at Parklawn Cemetery alongside their mother. He consented to let Dorothy Freeman have half of them, which she scattered along the coast of the Sheepscot River in Newagen, a place where the two friends had spent much time together. Lisa Sideris considers the implications of the conflict between Robert's and Rachel's desires:

> This struggle over Carson's body, and perhaps her soul, may appear sadly symbolic of her tireless efforts to unite those things – knowledge and wonder, fact and poetry, reason and emotion, even evolutionism and Presbyterianism – that humans seem determined to drive apart. But Carson might have taken a more reverential, and hopeful, view of this final partitioning of her remains between land and water. After all, as she stressed repeatedly, the dividing line between land and sea is itself an illusion that falls away when we expand our vision. The sea encircles us always; no separation exists in the world as it really is. Such was her view as she expressed it in the final passage of *The Sea around Us* (Sideris Secular 246).

The final passage of that book that she wished Reverend Howlett to read at her funeral reads: "in its mysterious past [the ocean] encompasses all the dim origins of life and receives in the end, after, it may be, many transmutations, the dead husks of that same life. For all at last returned to the sea – to Oceanus, the ocean river, like the ever-flowing stream of time, the beginning and the end" (The Sea Around Us 196).

Because he could not speak at the funeral, Reverend Howlett held a memorial service more in accordance with Carson's wishes at All Souls on April 19. He said "last week one of the true prophets of our time, Rachel Carson died. . . . In her memory [I] shall read a passage from her own hand which expresses in a remarkable way the strength, the simplicity, and the serenity that marked her character." Dorothy had given him Carson's farewell letter and he read "I shall remember the monarchs," a fitting and beautiful tribute to its author, Rachel Carson (Lear Witness 477-83).

NOTES

[1] Much of the information in this chapter is derived from the definitive biography of Rachel Carson by Linda Lear, *Rachel Carson, Witness for Nature*.

CHAPTER 2

NATURE WRITING: A WHIRLWIND TOUR

Nature writing reflects the attitudes and belief systems of the cultures that produce it; consequently it has appeared in great variety. When writing about nature exposes conflicts, or, as Bill McKibben terms them “collisions between people and the rest of the world” it becomes environmental writing (McKibben xxii).

In agrarian cultures stories about nature and people's relationship to it have been passed on orally. People in such societies feel connected to a community grounded in and directly related to nature and its recurring cycle of the year. Mircea Eliade theorizes: "Sacred time appears under the paradoxical aspect of a circular time, reversible and recoverable, a sort of eternal mythical present that is periodically reintegrated by means of rites" (Eliade 1987, 70).

Russian theorist Mikhail Bakhtin explicates the evolution of linear temporal consciousness in tandem with estrangement from nature and community. According to Bakhtin, in agrarian communities personal time is not yet differentiated; people believe themselves part of the collective, the unity, the social body. Death is part of a cycle of life, connected to regeneration and new birth. This time is not separated from the earth or from nature. The literature of the agrarian community is grounded in myth, in the stories that explain the culture’s origins and its relationship to the divine beings that pervade and control nature and human life. This unified agrarian relationship to nature becomes fragmented as culture evolves from a classless society to one based in property ownership and capitalism. Individuals become isolated. Bakhtin tells us that nature itself becomes a backdrop: “it was turned into landscape, it was fragmented into metaphors and comparisons serving to sublimate individual and private affairs and adventures not connected in any real or intrinsic way with nature itself” (217).

Along with the separation from nature came the impulse to dominate and control it. Notions of dominion appear in the Old Testament book of Genesis: "God said to them: bear fruit and be many and fill the earth and subdue it! Have dominion over the fish of the sea, the fowl of the heavens, and all living things that crawl about upon the earth!" (Genesis 1:28 Everett Fox translation). Greek and Roman culture shared the belief that humans had dominion over nature. Indeed, this is the belief that undergirds the rise of modern science. In the 17th and 18th centuries the Enlightenment philosophers divided reason from emotion, intellect from feeling, and humans from nature. Francis Bacon writes "nature being known, it may be mastered, managed, and used in the services of human life" (quoted in McDonough 84).

During the Romantic period of the early 19th century the concept of domination changed to appreciation of nature. Ramachandra Guha argues that the first wave of the environmental movement arose with the Industrial Revolution, when people began to seek the calm, peace, and quiet of nature as a refuge from the chaotic noise and pollution of the cities. The primary focus of this phase of environmentalism became preservation or conservation, the practice of setting aside places that would be preserved as natural spaces. Examples of this movement were the establishment of national forests, wildlife refuges, and national parks.

People began to value nature and to experience feelings of awe and the sublime in its presence. Poets such as Samuel Coleridge, John Keats, William Wordsworth, and Percy Shelley found meaning, insight, inspiration, and delight in nature. When Wordsworth wrote

his sonnet "The World Is Too Much with Us" (1806) the “world” in the title is the world of commerce and of cities. He sought respite from the dirt and dreary confinement of cities in a connection to nature:

> The world is too much with us; late and soon,
>
> Getting and spending, we lay waste our powers:
> Little we see in Nature that is ours

He wished to experience the power and sacredness that primitive people had, to

> Have sight of Proteus rising from the sea;
> Or hear old Triton blow his wreathed horn.

The Romantic period "was the turning point in the long Western tradition of human transcendence and domination over nature. The central [Romantic] view… was that the root of the modern human malaise is its separation, or 'alienation' from its original unity with nature, and that the cure of this disease of civilization lies in a reunion between humanity and nature that will restore concreteness and values to a natural world in which we can once more feel thoroughly at home, in a joyous consonance and reciprocity with all living things" (Abrams and Harpham 75).

When we consider American nature writing, we find a robust tradition of literature spanning a broad spectrum. Thomas J. Lyon lists a continuum of different types of “Writing About Nature” that contains categories such as Field Guides (for example Roger Tory Peterson’s field guides to the birds), Natural History Essays (in which he includes Carson’s books about the sea), Rambles (for example the more recent writers Annie Dillard, *Pilgrim at Tinker Creek*, and Terry Tempest Williams, *Refuge*), Solitude and Back-Country Living (Henry David Thoreau, *Walden),* Travel and Adventure (Barry Lopez, *Arctic Dreams*), Farm Life (Wendell Berry, *A Continuous Harmony,* Michael Pollan, *Second Nature*), and Man’s Role in Nature (Joseph Wood Krutch, *The Great Chain of Life*, Bill McKibben, *The End of Nature*). Lyon explains that the writing grows more personal as it moves from the field guides to his last category, “Man’s Role in Nature.” Lyon’s book, *This Incomparable Land: A Guide to American Nature Writing*, starts with a chronology beginning in 1492 and ending in 2000. The book contains four chapters discussing the American environment and the tradition of nature writing, followed by a substantial annotated bibliography of one hundred thirty-four pages ranging from Edward Abbey’s books about the desert to Donald Worster’s “fascinating intellectual history [of how] ecological thought over the past three centuries has been shaped by large cultural currents” (Lyon 266).

In the Romantic tradition, the American naturalist Henry David Thoreau (1817-1862) enjoyed walking and observing the natural world. He kept a journal of his observations while he lived for over a year in a cabin he built on the shores of Walden Pond in Massachusetts, claiming "I went to the woods because I wished to live deliberately, to front only the essential facts of life" (Walden 88). In his essay "Walking" published posthumously in 1862 he stated his credo, a reverence for unspoiled nature. "In wildness is the preservation of the world . . . When I would recreate myself, I seek the darkest wood, the

thickest and most interminable . . . swamp. I enter a swamp as a sacred place. . . . There is the strength, the marrow of nature" (Thoreau Walking 30, 35).

Here are some of the nature writers Carson read. This very short list hints at the richness of American nature writing in the nineteenth and early twentieth century.

— Henry Beston (1888 – 1968), emulating Thoreau, lived in a small cabin on Nauset Beach, Cape Cod for a year and wrote lyrically about his experiences there in *Outermost House* (1928).

— John Burroughs (1837 – 1921) lived and wrote about the farm country of the Catskills in New York State.

— Ada Govan wrote *Wings at My Window* and contributed magazine articles about birds. Carson wrote to her regarding an article about bird banding, and the two corresponded. Linda Lear believes that Govan's writings were one of the influences on Carson's "Help Your Child to Wonder" (Lear 512 n 57).

— Olive Thorne Miller (1831-1918), author of books for children and adults on bird watching and preservation.

— John Muir (1838 – 1914) explored the Sierra Mountains and kept extensive journals. He founded the Sierra Club and was instrumental in preserving Yosemite as a national park.

— Dallas Lore Sharp (1870-1929) wrote magazine articles about wildlife

— Edwin Way Teale (1899-1980) was a friend of Carson's. Among his books are a series about the changing seasons including *North with the Spring* (1951), *Autumn Across America* (1956), and *Journey into Summer* (1960).

— Henry Williamson (1895-1977) English naturalist wrote novels with animals as the subjects including *Salar the Salmon* (1935) and *Tarka the Otter* (1927)

— Mabel Osgood Wright, (1859-1934) edited the magazine *Bird Lore* that became the *Audubon* magazine. Wrote popular children's books about nature, especially birds, in the 1890s

There are many excellent anthologies that include writing by these and other authors. (Chapter 8, Resources, lists anthologies of nature writing and works by individual authors.)

All these authors' works and more were known to Carson and probably influenced her in some ways. From her earliest years her mother guided the Carson family to experience nature, inspired by the nature-study movement promulgated by writers such as Olive Thorne Miller, Mabel Osgood Wright and Liberty Hyde Bailey. Her early love for nature was an important influence on Carson throughout her life, resonating in her writing, and perhaps most clearly evident in "Help Your Child to Wonder." (For more discussion of this topic see Lisa H. Sideris "The Secular and Religious Sources of Rachel Carson's Sense of Wonder" in Sideris and Moore.)

Notably missing from this list is the naturalist Aldo Leopold. Carson's biographer Linda Lear believes that Carson may not have read Leopold's *A Sand County Almanac* (1949), attributing Carson's lack of interest to her criticism of some of the Fish and Wildlife refuge management policies (Lear 521 n.6). Possibly also Carson rejected his view of hunting. Leopold had been in the forestry service, and he remained a hunter throughout his life. However, according to Marybeth Lorbiecki, author of the biography *Aldo Leopold: A Fierce Green Fire* "it was through hunting that he came to fully understand wildlife's needs

for sufficient habitat. It was also through hunting that he saw his biggest mistake played out -- the unfortunate results of his youthful hubris regarding game and predators" (Lorbiecki web). For, in his essay "Thinking Like a Mountain," Leopold describes his realization that predators are an essential part of nature, and help to maintain a balance. When the wolves were extirpated, the deer roamed free and stripped the vegetation, leaving the mountain bare and subject to erosion. "While a buck pulled down by wolves can be replaced in two or three years, a range pulled down by too many deer may fail of replacement in as many decades" (Leopold 276). Leopold's essay "The Land Ethic" has become a guiding principle of environmentalism. Much as Carson asserted that humans participate in an all-inclusive web of life, Leopold saw that humans must recognize that we are part of a "community of interdependent parts. . . . The land ethic simply enlarges the boundaries of the community to include soils, waters, plants, and animals, or collectively: the land" (Leopold 278). Leopold continues: "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise" (293).

I would like to focus next on several books that Carson herself selected as her favorite sea books, and to discuss in particular the less-well known books among them. *The Washington Post* asked Carson to list her favorite sea books. Here is her list on 2 December, 1951, headlined "Rachel Carson's Treasure Chest of Sea Books."

— Beebe, William. *Half Mile Down*, 1943.

— ---. *Beneath Tropic Seas: A Record of Diving Among the Coral Reefs of Haiti*, 1928.

— Beston, Henry. *Outermost House*: *A Year of Life on the Great Beach of Cape Cod,* 1928.

— Conrad, Joseph. *The Mirror of the Sea.*

— Heyerdahl, Thor. *Kon-Tiki: Across the Pacific in a Raft* 1950.

— Klingel, Gilbert. *The Bay,* 1951.

— Melville, Herman. *Moby-Dick or the Whale,* 1851.

— Tomlinson, Henry Major. *The Sea & the Jungle*, 1912.

— *Coast Pilots and Sailing Directions* especially for Norway, the Shetlands and Orkneys. (YCAL MSS 46 box 98 f 1802).

These texts are all about adventurers and explorers: sailors, divers, and travelers.

William Beebe (1877-1962) was an ornithologist, explorer, and the author of numerous natural history books and articles written both for scholarly and for popular audiences. He was the curator of ornithology for the New York Zoological Park (the Bronx Zoo) and became the Director of its Department of Tropical Research. He wrote several books about jungles, and about the sea. The two that Carson listed for the newspaper article describe Beebe's adventures diving in tropical waters.

Half Mile Down is written for a general audience but contains appendices that categorize and describe according to scientific nomenclature the organisms he saw underwater. Beebe reviews a history of diving, starting with a legend that Alexander the Great descended in a glass bubble. Beebe's early dives were made using a diving helmet (a glass front helmet with an air hose attached). He explains that this is an inexpensive and simple way to experience the underwater beauties and mysteries. It was the type of diving possible before the development of the aqualung and scuba (self-contained breathing apparatus) gear.

He sees this as a vision of possible family entertainment in the future. All one need wear is sneakers, a bathing suit and the helmet. He sets forty feet as the limit for this kind of diving.

He describes various dives in different places, and explains his wonder at the world opened to him undersea.

> In this Kingdom most of the plants are animals, the fish are friends, colors are unearthly in their shift and delicacy; here miracles become marvels, and marvels recurring wonders. . . In hundreds of dives we have never encountered [dangers]. One thing we cannot escape-- forever afterward . . . the memory of the magic of water and its life, of the home which was once our own-- this will never leave us (10).

In *Beneath Tropic Seas* Beebe details the dives he made in Haiti for the Tenth Expedition of the New York Zoological Society. "All I ask of each reader is this, --Don't die without having borrowed, stolen, purchased or made a helmet of sorts, to glimpse for yourself this new world. Books, aquaria and glass-bottomed boats are, to such an experience, only what a time-table is to an actual tour" (6). He laments the difficulty of describing what he has seen on his dives, yet manages to portray the magic of the scene:

> We need a whole new vocabulary, new adjectives, adequately to describe the designs and colors of under sea. . . . If one asks for modernist or futuristic designs, no opium dream can compare with a batfish or an angry octopus. The night overhead glories in a single moon; here, whole schools of silvery moonfish rise, pass and set before us, while at our feet rest constellations of star-fish—crimson, sepia and mauve (36).

For these deeper descents Beebe used a sphere designed and constructed by Otis Barton, called the bathysphere, with windows of fused quartz three inches thick. The two broke records for diving depth when they descended 3028 feet in 1934. The book details 26 bathysphere descents in the Atlantic Ocean off Nonsuch Island. Appendices list the fish and other organisms seen at different depths and provide technical details of different dives.

The *Coast Pilots* are annual compendiums of useful information for navigating coastal waters. The ones for the United States each cover a small region. They describe the typical weather at different seasons including the times when fog is most likely. They enumerate obvious landmarks such as water towers, and indicate coastal features such as average and extreme wave heights, shoals, islands, rocks, and depths.

Gilbert Klingel (1908-1983) was a boat builder and naturalist. On an exploratory trip to study natural history Klingel was shipwrecked on Inagua, one of the islands of the Bahamas. He remained on the island for some time and published *Inagua,* a natural history of the site. His second book *The Bay* is compiled from articles he wrote for *The Baltimore Sun* recounting his experiences diving in the Chesapeake Bay and other areas in Maryland. His first chapter describes a descent wearing a diving helmet to the Chesapeake Bay on a night in May. The water is cold and magically illuminated with phosphorescence from glowing creatures.

> Like so many dwellers in the undersea, which are pale and dull when viewed in the upper air, these fishes seen in their own element, and lighted with the rays of the lamp were creatures of surprising beauty. . . . Their scales flashed delicate iridescent pinks and lavenders tinged with overtones of shimmering blues and with glaucous greens (6).

Carson reviewed the book in *The New York Times*, October 14, 1951. She wrote: "Gilbert Klingel shares with Fabre and a few other naturalists a rare ability to describe the life of a restricted area in terms that invest it not only with fascination but with rich meaning."Klingel received the prestigious Burroughs Medal of the American Museum of Natural History for the book in 1953.

When Thor Heyerdahl (1914-2002) lived on the Polynesian island of Fatu Hiva doing research on the island's natural history, he became intrigued with the question of the origin of the Polynesians. Where had they come from? Theories abounded, but all seemed to have flaws. Heyerdahl pondered the question and propounded a unique theory based on commonalities in language and legends he saw between Polynesia and South America. Like Carson, Heyerdahl was a generalist. He built his theories on research from different disciplines, drawing from biology, geography, linguistics, and oral legends. Both the early South America Indians and the Polynesians cultivated sweet potatoes, both the Incas of Peru and the Polynesians had a system of using knots on twisted strings as memory aids in recounting genealogies of a long line of ancestors, both peoples had built similar pyramids and large stone statues such as the ones on Rapa Nui or Easter Island (18-26). Based on these correspondences Heyerdahl theorized that the original settlers of Polynesia migrated from Peru. But how could primitive people have made such a journey, about 4000 miles across the ocean?

To determine if such a journey was possible Heyerdahl and a crew of 5 other men built a raft in the traditional pae pae style out of 9 large balsa logs tied together with hemp rope (83-86). The raft had a sail, and a cabin built of bamboo reeds with banana leaves for a roof. People warned them that hemp ropes would rot and disintegrate before they arrived at their destination, but they came to realize that hemp was superior to metal cables; the cables would have cut through the logs during the course of the trip.

Heyerdahl's book, *Kon Tiki* records the adventurers' 101 day journey across the Pacific from Peru to the Raroia reef pulled by the Humboldt current and pushed by a mostly Westerly wind. The raft's design proved adequate to the task, and as they traveled they learned more about steering. They ate supplies they had stored on board supplemented abundantly by fish. *Kon Tiki* explains that every morning the person assigned cooking duty would gather the fish that had landed on the deck overnight and fry them up for breakfast. The book, translated from the Norwegian by F. H. Lyon, is lively and very readable.

Carson includes a summary of the journey in *The Sea Around Us* and writes that she asked Heyerdahl for "his impressions, especially of the sea at night." He wrote about seeing phosphorescence, and about his impression that some fish from the depths swam up closer to the surface at night (in *Kon Tiki* he mentions seeing a snake mackerel, *Gempylus*). Carson also includes Heyerdahl's description of flying fish and squids which leapt out of the water and sometimes landed on the raft (32-33, 53).

Heyerdahl shares Carson's love of the sea and appreciation of nature. His book resonates with his admiration for the world of the sea and its creatures that he came to know on the journey. He writes: "Coal-black seas towered up on all sides, and a glittering myriad of tropical stars drew a faint reflection from plankton in the water. The world was simple—stars in the darkness" (173). And he reveals a sensibility akin to Carson's "sense of wonder":

> The sea contains many surprises for him who has his floor on a level with the surface and drifts along slowly and noiselessly. . . . We usually plow across it with roaring

> engines . . . with the water foaming round our bow. Then we come back and say that there is nothing to see far out on the ocean (117).

On a subsequent ocean adventure Heyerdahl collected samples of the pollution he observed and warned that we are treating it like a sewer.

The *Kon Tiki* expedition proved that it was possible for people to make such a journey before the advent of printed maps, metal tools, and steam engines. But the question of the origin of the Polynesians remains controversial. Recent excavations of the huge stone statues at Easter Island reveals that they are partially buried underground, and marked with petroglyphs; DNA samples of the various island groups are being analyzed. Perhaps new evidence will emerge to settle the question.

Moby Dick, Herman Melville's great narrative of an ocean voyage with its Captain Ahab obsessed by his passion for revenge on the white whale responsible for losing one of his legs also appears on Carson's list. This major novel has been discussed and interpreted at great length in numerous publications, books and articles. Movies have been made of the book. *Moby Dick* is an encyclopedic book, part novel, part natural history, part psychological study, part philosophical meditation, part narrative of the nineteenth century New England whaling industry.

A recent interpretation of the novel by Toni Morrison reads the story of the white whale as Melville's rumination on the emerging American ideology of whiteness that he perceived as dangerous. The book may also be read as a story of a community of workers formed among men from different nations who bond on the ship.

I'd like to explicate the novel here from a sort of Carsonian perspective. Let's think of Ahab's ship as the ship of state, the white whale as Nature, and Ahab as representing humans (scientists perhaps). So Ahab (science, technology, hubristic humanity) sets out—headstrong and full of confidence in his own power but with little regard for the consequences and for the dangers he unleashes—to overcome the forces of nature (the white whale). He seeks to avenge an injury that is painfully real, but his maniacal quest for revenge is out of proportion to the offense. The ship is small and weak when confronting the size and power of the whale. Ahab's misguided project fails, destroying the ship and wrecking lives, including his own. Only Ishmael, the narrator, returns alive to bring back the story, a warning to his fellows of the danger of hubris, the folly of trying to overcome the power of nature.

Carson listed this book among her favorites years before she wrote *Silent Spring,* but perhaps she already saw herself as the one who bears witness, akin in some way to Ishmael.

Henry Beston's *Outermost House* documents the cycle of a year in a small cottage which he named The F'castle on Nauset Beach in Cape Cod. He starts with a description of the beach itself, then tells the story of the turning year with its changes of weather and of the animals that pass through or remain. He marvels at the mystery of flocking birds as they feed individually but suddenly form "shorebird constellations" and fly off together according to the will of the group (23). In chapter two he observes "the strangest and most beautiful of the migrations over the dunes," a migration of orange and black monarch butterflies, reminding us of Carson's farewell letter to Dorothy Freeman. Beston's book closes with an August night when he sleeps on the beach but wakes up while it is still dark to see the constellation Orion, the hunter.

> In the luminous east, two great stars aslant were rising clear of the . . . darkness gathered at the rim of night and ocean—Betelgeuse and Bellatrix, the shoulders of Orion. Autumn had come, and the Giant stood again at the horizon of day and the ebbing year, his belt still hidden in the bank of cloud, his feet in the deeps of space and the far surges of the sea. My year upon the beach had come full circle; it was time to close the door (215).

When Beston left his cottage he had several notebooks. He proposed to his fiancée, author Elizabeth Coatsworth, that they set a wedding date. Her response was "No book, no marriage." He completed the book and they married the following June (Finch Introduction xvi). Beston wrote nine more books about nature. In 1960 he donated the cottage to the Massachusetts Audubon Society. It had to be moved back from the ocean several times due to erosion. The Audubon Society rented the cottage out to society members in summers until a winter storm swept it out to sea in February 1978 (Finch Introduction xxxii).

Joseph Conrad's *The Mirror of the Sea* is a collection of essays first published in various magazines from 1904–6 and published in book form in 1906. As Zdzislaw Najder, editor of the Oxford edition of the book, points out, the title works two ways: Conrad's book is a mirror or reflection of the ocean, and the ocean itself is a mirror of nature and of life (viii). But there is another meaning to the idea of mirror, captured well in Conrad's book. A mirror reflects what is in front of it at a particular time. Unlike a painting or photograph, a mirror's images are evanescent, constantly changing, as are the moods and images held fleetingly by the sea.

In fact, the book is about sailing and seamanship, a nostalgic celebration of the art of sailing wooden ships before iron steamships replaced them.

> The genuine masters of their craft [Conrad means craft here as art or skill as well as ship] . . . have thought of nothing but of doing their very best by the vessel under their charge. To forget one's self, to surrender all personal feeling in the service of that fine art" is the art of sailing (29).

Conrad asserts that "the taking of a modern steamship about the world . . . has not the same quality of intimacy with nature" (30). The book reverberates with an appreciation of ships and the men who sail them. "Ships do want humouring. . . .. Your ship is a tender creature , whose idiosyncrasies must be attended to" (51-52).

Conrad describes his "initiation" into an understanding of the sea on a clear and calm day when his ship rescues nine sailors whose ship was damaged in a previous hurricane and is about to sink. He recognizes "the cynical indifference of the sea to the merits of human suffering and courage" and feels his romantic illusions about the sea slip away, "but its fascination remained. I had become a seaman at last" (141-2).

Henry Major Tomlinson (1873- 1958) was a British journalist and war correspondent and novelist. His editor sent him to the Amazon to report on Sir Roger Casement's investigation of abuses of native people by the Peruvian Amazon company that made huge profits in the rubber trade but treated the workers badly.

The Sea and the Jungle is a fictionalized travelogue of Tomlinson's experience of sailing on the Amazon River. The book starts in November when the narrator's friend entices him to leave a boring job in a rainy and cold London to sail on the Amazon River in Brazil as the ship's purser. At that point the story is told as dated journal entries of the trip.

The boat's mission is to bring supplies to workers engaged in the difficult job of building a railroad in the jungle. Their task is a perilous enterprise. Previous attempts have ended in failure because of the treacherous conditions: malaria, yellow fever, poisonous snake bites, beriberi, uncertain transportation and supply lines dependent on the unpredictable weather. (In fact the railroad when finally completed was short-lived, and the Peruvian rubber boom collapsed when East Indian and African plantations undercut the price of Peruvian wild rubber, rendering the railroad uneconomical (http://www.answers.com/topic/1912).

En route the narrator walks in the jungle, or explores by canoe and mule. He describes the flora and fauna, the brilliant flowers and parrots, the monkeys, the natives and their leaf-roofed houses. "The Amazon was an immensity of water, a plain of burnished silver, where headlands, islands, and lines of cliff were all cut in one level mass of emerald veined with white" (93). He describes the jungle: "Individual sprays and fronds project from the mass [of leaves] in parabolas with flamboyant abandon. . . . I could believe the forest afloat, an archipelago of opaque green vapours" (84). He finds the scenery of the Madeira River more impressive than the Amazon and the land warmer and more welcoming than the dreary London November, and remarks,

> Here on the Madeira I had a vision . . . of the earth as a great and shining sphere. There were no fences and private bounds. . . . Our earth had celestial magnitude. It was warm, a living body. The abundant rain was vital, and the forest I saw, nobler in stature [than the Amazon forests], . . . rose like a sign of life triumphant (113).

Insects of all kinds are plentiful. Many are dangerous, such as the malarial and other mosquitoes compared to which "the dragons of mythology were lambs" (186). Some are beautiful, especially the butterflies. A blue morpho butterfly resembles a bit of the sky. The narrator is so entranced following it that he trips over a log.

The narrator has a dry wit. He reflects on the British poor laws, and how he donates money to charity for the poor natives who seem to be happily doing very well without ever hearing about the poor laws (98). At one of the stops on the voyage the narrator meets another Englishman who had been similarly entrapped in a routine office job and opted for adventure working for a company in the Amazon. But the man found he was then a prisoner / victim of the company. He was a clerk, but shipments of materials did not match the invoices; for example a steam shovel came without its shovel, and none of his urgent queries were answered. Then shipments of food and medicine stopped: unbeknown to the workers the company had gone out of business and been sold to another company (146-161). As far as rubber is concerned, Tomlinson devotes three sentences to it:

> But away with their rubber! I am tired of it, and will keep it out of this book if I can. For it is blasphemous that in such a potentially opulent land the juice of one of its wild trees should be dwelt upon—as it is in the states of Amazonas and Para—as though it were the sole act of Providence" (128).

In March the narrator returns home. The boat sails to Tampa, Florida, and he makes his way home to London. Carson shared her pleasure in reading Tomlinson with her friend Dorothy Freeman.

Late 20th century nature and environmental writing has turned more frequently toward warning of an environmental crisis. According to M. H. Abrams and Geoffrey Galt Harpham: "It was in this climate of crisis, or even imminent catastrophe, that ecocriticism

was inaugurated" (72). While there are many strands in ecocritical thought, most agree that the human relationship to nature must be one that includes "the moral responsibility of human beings to maintain and transmit a livable, diverse, and enjoyable world to their immediate posterity" (Abrams, and Harpham 75).

> Despite . . . disagreements, all ecocritics concur that science– based knowledge of looming ecological disaster is not enough, because knowledge can lead to effective political and social action only when informed and impelled, as it is in literature, by imagination and feeling (Abrams and Harpham 75).

Contemporary Western attitudes toward nature range across a spectrum running from the ideas of dominion to stewardship or partnership. In contrast to the concept of dominion, recent ecological thinking recognizes the magnitude of humanity's impact on the earth, and emphasizes stewardship. Ideas of stewardship are the focus of the writings of contemporary "deep ecologist" Arne Naess, who argues for a more eco-centric, less anthropocentric value system.

Growing out of these opposite positions of dominion or stewardship, two strands of thought about the environment recur throughout America history: a wish to exploit natural resources contrasted with the appreciation of a pristine nature and the wish to protect and preserve it. Our challenge today is to find a sustainable balance between these approaches. William McDonough and Michael Braungart, co-authors of the book about sustainable design, *Cradle to Cradle,* argue that both views, dominion and stewardship, are two sides of the same coin, because in order to dominate something we must protect it so that it remains there to be dominated. On the other hand, stewardship implies control and mastery: if one is a steward, one is superior to that over which one has stewardship. McDonough and Braungart prefer the idea of partnership. Carson's warnings of the dangerous hubris of human attempts to control nature, and her vision that all life is interconnected in a living web would align her with their position and affirm her role as an early practitioner of ecocriticism.

CHAPTER 3

BOOKS ABOUT THE SEA

"Who has known the Ocean?"

> Who has known the ocean? Neither you nor I, with our Earth-bound senses know the foam and surge of the tide… Or the lilt of the long, slow swells of mid-ocean, where shoals of wandering fish prey and are preyed upon.… Nor can we know the vicissitudes of life on the ocean floor, where the sunlight, filtering through a hundred feet of water, makes but a fleeting, bluish twilight, where swarms of diminutive fish twinkle through the dusk like a silver rain of meteors (Rachel Carson "Undersea").

With these words, the start of her essay "Undersea" published in *The Atlantic Monthly* in September 1937, Rachel Carson announced her arrival on the American literary scene. The essay invites us to read aloud, to hear its poetic rhythms, and to imagine the scenes it paints. "Undersea," first intended as the introduction to a government brochure about fish, brought Carson to the attention of the author Willem Van Loon and to Quincy Howe, an editor at Simon and Schuster, both of whom encouraged her to develop the piece into the book that became *Under the Sea Wind*. The article and the book launched the literary career she had planned for herself, and Van Loon became a mentor and friend. (Please see chapter 4 for a discussion of "Undersea.")

Although Carson is perhaps best remembered for *Silent Spring*, her three previous best-selling books about different aspects of the ocean—*Under the Sea Wind: A Naturalist's Picture of Ocean Life, The Sea Around Us,* and *The Edge of the Sea*— first garnered her recognition. The resulting book sales and awards enabled her to resign from a full-time job, and gave her the time, the income, and the freedom to write about what most interested her, the natural world with its non-human and human inhabitants. The three sea books are beautifully written and informative. *The Sea around Us* won several awards and was made into an award-winning documentary film.

The lively and engaging sea books provide an introduction to what was at the time of writing the most current knowledge of oceanography and marine biology. They "lead readers to some of the wildest and hardest to imagine places on earth: Arctic tundra in the grip of winter; the weird dark depths of the ocean; microscopic planktonic worlds" (Cafaro in Sideris & Moore 67). But perhaps Carson's greatest contributions are her syntheses of science/ reason and poetry/emotion. Her recognition of the interconnectedness of all life, her respect for the natural world, and her stance of non-anthropocentrism are the principles that became the basis of a new sub-field of philosophy, environmental ethics.

I will provide here a summary of each of Carson's sea books, and point to the highlights of each section. *Under the Sea Wind* is a story in narrative form about the life cycles of some of the birds and fish of the coastal regions. *The Sea Around Us* reviews the then-current scientific information about the ocean in readable, clear prose. *The Edge of the Sea* describes four different coastal environments, from the rocky coast of Maine, to the sand beaches of the mid-Atlantic, to the coral reefs and mangrove swamps of the Florida Keys. It is remarkable how many creatures eke out their lives in the difficult

environment at the edge of the sea. Carson presents this web of life in vivid prose, from the tides to the sand fleas to the crabs, the birds and the fishes.

Carson was a careful observer and took notes on what she saw. Field notes she jotted down in 1938 on a trip to a research station at Beaufort along the Outer Banks of North Carolina indicate her remarkable facility of observation and her poetic language: "The crests of the waves just before they toppled caught the gold of the setting sun then dissolved in a mist of silver. The sand in the path of each receding wave was amethyst, topaz, and blue-black" (YCAL MSS 46 box 1 F1 quoted in Lytle 45 – 46).

She determined from the first to use the perspective of the ocean's creatures, rather than humans in her writing. She named the animals she described using their Latin classification (for example, the black skimmer is called Rynchops). Strikingly, other women natural scientists have developed similar perspectives that recognize the subjectivity of the plant or animal they study. Evelyn Fox Keller describes a similar phenomenon in her biography of Nobel-Prize-winning botanist Barbara McClintock. Keller explains that McClintock experienced a connection with the corn plants she studied, an attitude that Keller used as the title for her biography, *A Feeling for the Organism*. Similarly, primatologists Dian Fossey gave names to the mountain gorillas she studied, Birute Galdikas named the orangutans, and Jane Goodall named the chimpanzees. In each case this refocuses the observer's relationship to the subject observed and moves toward non-anthropocentrism. Sue V. Rosser, a scholar of science, gender, and technology, analyzes a variety of feminist scientific methodologies and finds that according to psychoanalytic feminism, "women scientists might be more likely to use approaches that shorten the distance between them as observer and their object of study, might develop a relationship with their object of study, and might appear to be less objective" (Methodologies 135). According to Rosser, a related feminist methodology, radical feminism, holds that "the world [is] an organic whole, [and rejects] dualistic and hierarchical approaches. Dichotomies such as rational/feeling, objective/ subjective, mind/ body, culture/ nature, and theory / practice are viewed as patriarchal conceptions which fragment the organic whole of reality. . . . Radical feminists view all human beings . . . as connected to the living and non-living world" (Methodologies 137). Although Carson died before the flowering of the second wave of feminism, her values and her approach to the study of nature incorporate these feminist values, and it is probable that her values inspired the theorists of ecofeminism. A relatively recent development within feminism, ecofeminism includes the tenets that "life on earth is an interconnected web, not a hierarchy; . . . a healthy, balanced ecosystem, including human and nonhuman inhabitants, must maintain diversity; . . . [and] the survival of the species necessitates a renewed understanding of our relationship to nature, of our own bodily nature, and of non-human nature around us" (King 408). Carson espoused these principles, and spoke often of the interconnections among organisms in a web of life. In her sea books and in "Teach Your Child to Wonder," she invited us to appreciate and value our relationship to nature; in *Silent Spring* she urged us to understand and respect that relationship.

UNDER THE SEA WIND

Writing the Book

Carson told a college friend that she aimed in her writing to make the animals of the natural world "as alive to others as they are to me" (Seif qtd in McCay 6). This was her intent when writing *Under the Sea Wind*.

In a letter to Edward Weeks, Editor of *The Atlantic Monthly*, Carson describes her progress on *Under the Sea Wind*. She hopes that her first chapter may be publishable in the magazine. “If the chapter does what I want it to do, it will make the reader, for a time, become one with those whose timepiece consists of the tidal rhythms of the sea and whose whole world is sand and sky and water. I want to express the stillness and peacefulness of a wide, salty, marshy place and to make it clear that beneath the calm exterior of Nature the bitter struggle for life goes on” (YCAL MSS 46 box 3 F 45). She wrote to her friend and mentor Henrik van Loon on 5 February 1937 her plan to make the book a narrative, with the “fish and other creatures [as] . . . the central characters . . . [whose] world must be portrayed as it looks and feels to them.” From the fishes’ point of view humans would appear only “as predator and destroyer. . . . It seems to me that the principal thing the book must accomplish is the creation of undersea atmosphere.” (YCAL MSS 46 box 3 f 57). And indeed the animals are the central characters, with names (often scientific names such as Rynchops, the black skimmer, or Scomber the mackerel, or descriptive names such as Blackfoot and Silverbar the sanderlings) and personalities.

In giving the animals names Carson was departing from traditions of scientific objectivity and distance. Yet she was careful to avoid the danger of the anthropomorphism sometimes present in the works of previous nature writers such as Henry Williamson, whose books *Salar the Salmon* and *Takar the Otter* she had enjoyed. She made the distinction clear in the foreword to *Under the Sea Wind.* "I have spoken of a fish fearing his enemies… Not because I suppose a fish experiences fear in the same way that we do, but because I think he *behaves as though he were frightened*. . . . If the behavior of the fish is to be understandable to us, we must describe it in the words that most properly belong to human psychological states" (quoted in Lear Witness 91). In fact, when RKO made a film of her next book, *The Sea around Us,* she objected to the anthropomorphism used in telling the story of the ocean creatures. She wrote to her film agent "the practice of attributing human vices and virtues to the lower animals went out of fashion many years ago" (Lear Witness 239).

Reviewers praised the book for its accuracy and its graceful writing. The edge of the sea teems with life and motion, and Carson evokes it all in luminous detail, from the tides to the sand fleas to the crabs, to the birds and the fish, all taking their place in a great food chain, an interlinked web of life.

Summary of the Book

Under the Sea Wind is organized around the life cycle of various creatures, focusing on migrations and mating. The book’s movement is from the shore to the ocean’s mid depths, to the rivers and deeper ocean waters. The first section explores the lives of coastal birds, the black skimmer and the sanderlings; the second deals with the life of a mackerel; and the third section follows the migration of an eel. Each of these three sections depicts a cycle of life, death, and new birth as the animal—bird, fish, eel—is born, matures, bears new life, and dies, so that its body may be reclaimed by other creatures or by the ocean itself, and the cycle continues. Carson’s vision of this cycle is vast: it encompasses geological ages and the earth itself, for the mountains will erode, and in the distant future “all the coast would be water again, and . . . [the] cities and towns would belong to the sea” (132).

Book I “Edge of the Sea" takes the perspective of birds, chiefly the black skimmer, Rynchops, and two sanderlings, Blackfoot and his mate Silverbar. The sanderlings migrate

to the Arctic to mate and brood their young. As she sits on the eggs, “an abiding fear entered the heart of Silverbar—the fear of all wild things for the safety of their helpless young. With quickened senses she perceived the life of the tundra—with ears sharpened to hear the screams of the jaegers harrying the shore birds on the tide flats—with eyes quickened to note the white flicker of a gyrfalcon’s wing” (37). After the four chicks have hatched Silverbar carefully carries all the pieces of eggshell away from the nest and buries them in gravel to keep predators away from her young. She feigns an injury to herself to lure a fox away from the chicks. She guides her young away from the nesting area through the tundra, teaching them to hide from predators and to find insects and larvae to eat. In August she joins a gathering flock of sanderlings to fly south; the young will migrate in another flock after they have grown strong enough for the journey.

Book II “The Gull’s Way” recounts the life of the mackerel, Scomber. He is born in the inshore waters as “a tiny globule no larger than a poppy seed, drifting in the surface layers of pale green water” (60). He spends his first summer in the harbor, where he is lucky and skillful enough to avoid predators, and he soon learns to become a predator himself. When autumn comes he joins the large schools of mackerel to migrate to his winter habitat, “the deep quiet waters along the edge of the continental shelf, off the Capes of Virginia” (102).

Book III “River and Sea” focuses on the journey of a female eel, Anguilla, who migrates in autumn from a freshwater pond hundreds of miles through streams and rivers to the abyssal depths off the continental shelf to spawn. Carson describes the other creatures that live in those cold, dark depths, the voracious anglerfish, the eight-armed octopus, and the small fish that swim in and around the corners and crevices of a wrecked fishing boat, now covered in seaweed and barnacles. The newly hatched larval eels drift in the water and grow to become elvers. The young male elvers remain in the brackish river estuaries, while the females swim up the rivers in long columns, “each elver pressing close to the tail of the next before it, the whole like a serpent of monstrous length” so that the cycle of life will repeat (132).

Under the Sea Wind is a sensuous book, appealing to all the senses. Visually, Carson paints a canvas with a rich palette of colors. The first paragraph describes the colors of the sea and the beach as night arrives at an island: "both water and sand were the color of steel overlaid with the sheen of silver, so that it was hard to say where water ended and land began” (9). Carson finds beauty where others might not see it. In her eyes a clam worm is transformed into a “bronze water sprite with a scarlet girdle about his middle” (74). The menhaden “flash bronze and silver in the sun” (66). The moon jellies “shimmer in opalescent splendor” (78). The small fish in a cove swim in “shining green and silver caravans, . . . swerving, diverging, and merging again, or at a sudden fright darting away like a shower of silver meteors” (72).

When Carson describes the sounds the language itself sings. Repeated consonance and assonance echo the sibilant sounds of the waves lapping the shore. We hear "the whisper song of the water turning over the shells on the wet sand" (9) and we hear "the water . . . moving with soft tinkling sounds among the windrows of jingle shells and young scallop shells" (11).

Before long Carson even gives us the smells and tastes. A rat comes to feast, "snuffling and squeaking in excitement," lured by "the scent of terrapin and terrapin eggs, fresh laid" (13).

The night is alive with movement. In this nocturnal world the animals are active while humans are asleep indoors. Carson draws us fully into the lives of the creatures she describes so that they become like characters in a novel. We read about the animals, their

feeding behaviors, and the way they move. The first character who arrives at the island that night is a bird, the black skimmer, Rynchops. He flies low with his long lower bill cutting through the water to make vibrations that attract blennies and killifish for him to scoop up.

The tide comes in, bringing with it spawning shad, many of whom are caught in fishermen's gill nets that tear their gills. The eels taste the fish blood in the water and feast on the shad before the fishermen arrive in the morning to find their catch reduced to skeletal fragments. The gulls arrive to scavenge the remnants of shad that the fishermen toss on shore. There is drama overhead as an osprey and an eagle swoop and dive in an aerial ballet, sparring over the catfish the osprey clutches in his talons (47-48).

Carson develops a holistic view of ecology through her beautifully written story of the life in the ocean and on its shores. She views the natural world as a harmonious whole with each creature occupying its place in the pattern. All living beings participate in the ongoing cycle of life and death; each has its place in the scheme of "material immortality." For example, when fishermen litter the beach with fish too small to sell or eat, scavengers arrive to reclaim the dead bodies and turn them into other living creatures. First the gulls and the fish crows, then the crabs come to scavenge the remains. Finally "the sand hoppers [start] . . . their work of reclaiming to life in their own beings the materials of the fishes' bodies. For in the sea, nothing is lost. One dies, another lives, as the precious elements of life are passed on and on in endless chains" (54). Similarly, speaking of the life cycle of jellyfish Carson writes

> their battered bodies became once more a part of the sea, but not until the larvae . . . had been liberated into the shallow waters. Thus the cycle came to the full, for even as the substance of the moon jellies was reclaimed for other uses by the sea, the young larvae were settling down for the winter . . . so that in the spring a new swarm of tiny bells might rise and float away (79).

As we can see from her poetic prose, Carson was a careful writer. She read her words aloud, and revised repeatedly to perfect her sentences. Her mother re-typed the revisions and had them ready when Carson returned from work. I am reminded of the words of another great literary stylist, Gustave Flaubert, explaining a similar concern for perfecting the language of his texts. While he was writing *Madame Bovary* he wrote to Louise Colet on April 7, 1854 lamenting that he had written only thirteen pages in seven weeks. He had burned what he wrote in January, and had been revising the manuscript since February. Now, he writes, the thirteen pages “are in shape, I think, and as perfect as I can make them. There are only two or three repetitions of the same word which must be removed, and two turns of phrase that are still too much alike" (Gioia and Gwynn 286). If Carson had written about people rather than animals and the sea, she would likely be studied in American literature classes along with other classic stylists such as Herman Melville, Nathaniel Hawthorne, and Katherine Anne Porter.

Responses to the Book

Carson rejoiced at the publication of her first book. Unfortunately, *Under the Sea Wind* came out in November, 1941, just before the Japanese bombed American warships based in Pearl Harbor, Hawaii, prompting the U.S. to enter World War II. Public attention focused on the war. Consequently as Carson explained in “The Real World Around Us,” her talk to

Theta Sigma Phi, the Society of Women Journalists in 1954 “the rush to the bookstore that is every author’s dream, never materialized” (Lear Lost 151). The original printing consisted of 3600 copies. Paper shortages prevented the publication of a British edition, and the book did not sell well; as of June 1942 slightly more than 1200 copies had been sold. Carson received royalties on sales of 1639 copies. Simon and Schuster remaindered the book, selling it below cost to jobbers in 1944 and 1945 without notifying Carson or offering her the opportunity to purchase copies. In 1948 when Carson inquired about how many copies were in stock, she learned of the remaindering. She asked to buy some copies, and the publisher, embarrassed at its failure to keep her informed, sent her 20 copies at no cost (YCAL MSS 46 box 3 F 55). Carson distributed them to friends, and sent them to introduce herself to prospective publishers and others.

When *The Sea Around Us* was published in 1951 and quickly became a best seller its publisher, Oxford University Press, reissued *Under the Sea Wind.*

Reviews of the Book

Although sales were slim, the reviews were full of praise for the book’s accuracy and graceful writing. Among the reviews were one written by William Beebe, noted American naturalist, ornithologist, marine biologist, entomologist, explorer, and author, and one by ornithologist George Miksch Sutton. Beebe wrote in *The Saturday Review of Literature* "the purpose of this book is to make the sea and its life a vivid reality, says Ms. Carson, and she has succeeded.… The author is at her best in the complete life histories.… The account of the surface plankton, and the gauntlet drifting of the egg and larval mackerel is full of excitement" (12/27/41 YCAL MSS 46 box3 f 37). Beebe became a mentor for Carson, and the two corresponded with each other.

George Miksch Sutton, who later became Professor of ornithology at the University of Oklahoma, wrote "the author of *Under the Sea Wind* may be ever so expert an ichthyologist, mammalogist, limnologist, or ornithologist – the glossary and certain factual sentences reveal her as such – but she is primarily a lover of the sea. … We feel… that we have explored dark, violet-blue depths; . . . lived in a world among the beings that are grotesque yet beautiful; glimpsed life from a wholly new viewpoint” (*New York Herald Tribune* 12/14/41 YCAL MSS 46 box 3 f 38).

When the book was reprinted in 1951 naturalist and author Henry Beston wrote an adulatory review that Carson felt was the best review the book received. He wrote that “the poetic sense is the justification of man’s humanity. . . . It is Miss Carson’s particular gift to be able to blend scientific knowledge with the spirit of poetic awareness, thus restoring to us a true sense of the world.” Although she hesitated to write to him at first, Carson eventually did, and the two became friends (Lear Witness 238, 261).

THE SEA AROUND US

Writing the Book

After completing *Under the Sea Wind* Carson began to make plans for a book to be called “Return to the Sea,” for she believed that “we will become more dependent on the ocean as we destroy the land” (Lear Witness 161). Along with most lay people and experts at the

time, she believed (as it turns out mistakenly) that the oceans are "too big and vast . . . to be much affected by human activity" (Lear Witness 90). In fact, Carson began to write a chapter "Ocean and a hungry world" suggesting that because erosion is depleting the land's fertility we would turn to the "inexhaustible ocean resources" for food. She explained that the world harvest of sea foods was then about 35 billion pounds per year. While phytoplankton is the most prolific food source in the ocean, the taste might be objectionable, and therefore perhaps the best approach for harvesting food would be fish farming. However, Carson labeled this an "unpublishable chapter," possibly because she feared the damage fish farming might inflict on the ocean. (YCAL MSS 46 Box 7 f 134 Unpublishable chapter) And, indeed, there are now serious concerns that fish farming might be causing ocean pollution and that genetically engineered fish developed for farming may be problematic for humans and for the populations of wild fish.

Oxford University Press agreed to publish Carson's second book. Carson had long felt that she would describe the ocean better if she had diving experience that would help her to develop a better perspective on her topic (Lear Witness 91). In July, 1948, Carson was invited by Dr. F. G. Walton-Smith, a biologist at the Miami Marine Laboratory, to join his summer class in Miami. She seized that opportunity, and was able to combine several activities—research for her book, her dive, an excursion to the Everglades, and a trip to the Florida Keys in connection with a Fish and Wildlife publication. With Walton-Smith's help, Carson arranged a dive. Bad weather for several days in Miami spoiled their plans, but she was finally able to descend the ship's ladder wearing a diving mask, a method then commonly used by divers before the marketing of Jacques Cousteau's aqua lung in the U.S. Despite the imperfect visibility and the limited time of her undersea adventure she believed that this was one of the "milestones of [her] life." She later wrote about "how exquisitely delicate and varied [were] the colors displayed by the animals of the reef, …and the misty green vistas of a strange, nonhuman world" (Lear Witness 165-69).

During her years as an editor Carson had developed a strong sense of the aesthetic considerations of published texts. She wanted her book to avoid the appearance of a textbook, and she voiced her concerns to her editor. She wanted the book "to be read and enjoyed by everyone who has ever seen [the ocean] or who has felt its fascination even before standing on its shores," as Carson herself had (Lear Witness 162 – 63). She sought a title that would not sound overly academic, and tried out several, such as *The Story of the Ocean,* and *Empire of the Ocean* (Lytle 75). The title that she found at last was *The Sea Around Us*. She was a slow and painstaking writer, revising and re-writing. As she finished each chapter she sent it out for review to scientists who could offer feedback. The acknowledgements list numerous names of librarians, scientists, and researchers who helped her check facts and obtain research materials. These experts enjoyed her work and complimented her on her accuracy and her ability to make complex scientific ideas understandable.

Whereas the previous book, *Under the Sea Wind,* told the story from the animals' points of view, this book takes the human observer's point of view. The narratives in *Under the Sea Wind* tell adventure stories of the animals and recount the life cycles of fish and birds. *The Sea Around Us* takes a longer view: it begins more than two billion years ago with the start of planet earth and concludes with the most current information available to Carson when she wrote the book.

Oceanography is a truly interdisciplinary field, and Carson's book covers many areas of knowledge. *The Sea Around Us* conveys the excitement of ongoing exploration and discovery. It asks questions many of us must have wondered about: How deep is the ocean? How deep have humans descended into the depths? What kinds of fish inhabit the abyssal depths? How did the underwater mountains form? Why is the sea salty? Was there ever a sunken continent such as Atlantis? And how did the legend of Atlantis begin? Carson explains clearly what was then known, tells readers how scientists learned this information, and informs us what questions remained about the ocean, ranging over topics such as the ocean's depths, its age, its geology, its tides, currents, and waves, and the multitudes of living creatures that make their home in its depths. The reader follows the stories of scientific exploration and research. When a second edition of *The Sea Around Us* was published in 1960, Carson included an appendix that updates some of the information where relevant.

Of her three ocean books, this is the one that has become somewhat dated. The information in both *Under the Sea Wind* and *The Edge of the Sea* remains valid. However, research subsequent to *The Sea Around Us* has generated deeper ocean probes, more explorations, and new technologies that have yielded more knowledge of the ocean and its role in our climate.

The first chapter deals with the origins of the earth, its moon, and its life forms, topics that are still the subjects of unfolding theories as our research horizons and exploratory opportunities evolve. Some of the theories that Carson describes here have been superseded by more current ones. For example, one of the current competing theories argues that the earth was formed, not as Carson explains, by ejection of a mass from the sun, but by the condensation of a nebula which formed our Sun and its planets. One of the present theories about the moon's origins holds that it was formed when "a planetary body somewhat larger than Mars smashed into the young earth about 4.4 billion years ago" ejecting material that formed the moon. One of the main theories developed after Carson wrote her book is the now generally accepted concept of plate tectonics propounded in the mid-1960s. After Carson wrote her updated edition, evidence accumulated to support a previously weak theory known as continental drift or sea-floor spreading. This theory argues that the continents move apart due to slow flowing currents of material within the earth's mantle (Garrison 43 – 47, 68 – 72).

Although several of the theories that Carson discusses have been superseded as scientists have learned more information, her book still has value. She is careful to explain uncertainties, and to label scientific explanations as theories rather than as facts when appropriate. Further, she introduces her book by describing the excitement of gathering increasing information. Thus her book allows the reader or teacher to measure the shifts in scientific thought as knowledge accumulates.

Summary of the Book

Part I "Mother Sea"

"The Gray Beginnings" The first chapter starts the narrative more than two billion years ago when the earth began as a fiery ball of gas torn from the sun (her update in 1960 extends the

time to 4 1/2 billion years). Her language here is again lyrical and sensuous; its cadences and image patterns echo the Old Testament book of Genesis. Many of her sentences can be rewritten as poetry, for example:

> Imagine a land of stone
>
> a silent land
>
> except for the sound of the rain
>
> and winds that swept across it.
>
> For there was no living voice,
>
> and no living thing
>
> moved over the surface of the rocks (24).

Carson includes a "Chart of the History of the Earth and its Life," indicating the development of mountains, volcanoes, glaciers, the sea, and living creatures through the geologic eras. As the planet cooled, a chunk of matter was flung off to become the moon, leaving a deep scar that became the basin of the Pacific Ocean. Dense clouds covered the earth and heavy rains fell, forming the oceans. The land was bare and rocky. Life began—in ways that remain mysterious to us—in the warm waters of the primeval ocean. But there is an irony here. Carson warns us: "It is a curious situation that the sea, from which life first arose, should now be threatened by the activities of one form of that life. But the sea, though changed in a sinister way, will continue to exist; the threat is rather to life itself" (xii).

In this passage Carson was referring to the practice of storing radioactive waste materials in containers in the ocean. She was concerned that the containers would degrade and leak their contents. At the time she wrote the book, she did not realize the extent of the damage (from such activities as garbage dumping, oil spills from tankers and from ocean drilling, and pollution from runoff) that humans could inflict on the ocean.

"The Pattern of the Surface" describes "the bewildering abundance" of life in the surface waters, and speaks of the intricate interrelationships and the delicate adjustments of living creatures in all the parts of the ocean to variations in light, temperature, color, pressure, and salinity, as they range from its sunlit surface to its cold, dark depths: "what happens to a diatom in the upper sunlit strata of the sea may well determine what happens to a cod lying on a ledge of some rocky canyon a hundred fathoms below" (33). Carson depicts the food chain that starts with microscopic plants such as the diatoms, and continues through the plankton, to "the small carnivores," through the plankton-eating fish to the squid and whales. Plankton is defined as "The collection of small or microscopic organisms, including algae and protozoans, that float or drift in great numbers in fresh or salt water, especially at or near the surface, and serve as food for fish and other larger organisms" (Dictionary.com). The word plankton comes from the Greek word "to wander," for these

creatures wander about here and there as the water carries them for they lack any other means of locomotion.

In "The Changing Year" Carson informs us that "the face of the sea is always changing" (41), for the sea has its seasons as does the land. She reminds us again of her concept of "material immortality": "Nothing is ever wasted in the sea; every particle of material is used over and over again, first by one creature, then by another" (42). The cycle of the year that we are familiar with on land takes another form in the sea. In spring in the temperate zone the mineral rich warm water rises, diatoms bloom, plankton multiply, and fish migrate (as we have seen in *Under the Sea Wind*). Summer brings "a hard, brilliant, coruscating phosphorescence" where certain protozoa abound (44). The sea in autumn is also ablaze with a phosphorescence that now signals the approach of winter when much of the life of the sea settles into "the torpor of semi-hibernation" (46).

"The Sunless Sea" reveals what science has learned about the dark, cold depths of the ocean and the remarkable creatures that dwell there. At the time of the first edition Otis Barton had recently made, in 1949, the deepest known human descent to a depth of 4500 feet in "a steel sphere known as the benthosphere" (48). In the second edition (1961), a superscript guides the reader to the appendix where Carson tells about a later descent, made by Jacques Piccard and Don Walsh in a submersible vessel called the bathyscaphe. Their amazing descent to a depth of 35,800 feet, almost seven miles, took place on January 23, 1960. They descended to the deepest known point on earth, in the Mariana Trench near Guam. The dive took nearly five hours to descend and slightly more than three hours to re-surface. The U.S. Office of Naval Research bought the craft, called the *Trieste*, and it is now in the Navy Museum in Washington, D. C. To this day no human has descended deeper. (In 1961 Piccard published an account of the descent in his book *Seven Miles Down*). Currently, "the deepest-diving human-carrying research submersible" is the Shinkai 6500 which "safely descended to a depth of 21,409 feet on 11 August, 1989" (Garrison 123).

As a result of similar descents and by means of hydrophones, coring tubes, sampling nets, underwater photography, submarines, and other devices, we have learned about the horizontal zones of the sea and the creatures that inhabit each. At the lightless lower depths there are fish that make their own luminescence, and blind fish that use feelers to find their way. No plants can live in these depths where the sunlight does not penetrate, and food is scarce. Consequently, because the fish of the depths are carnivores and must rely on catching prey, many of them must alternate between feast and famine. Some of them have evolved to have "immense mouths and elastic and distensible bodies" that allow them to swallow fish larger than they are in order to obtain the food that will sustain them until their next meal (59).

"Hidden Lands" tells about the undersea topography, and the explorers and scientists who have mapped it. It is only since the mid-twentieth century that we have learned that the ocean floor is not flat, but full of mountains, valleys, and canyons as well as level plains. Carson speculates that the legend of Atlantis, the tale first told by Plato of a continent of fierce warriors that disappeared in one night into the sea, may indeed have an element of truth. The Dogger Bank, a fishing area of the North Sea, was dry land during the Pleistocene era, but submerged as glaciers melted. This relatively shallow plateau was discovered when fishers raised bones and crude stone implements in their nets. Scientists identified the bones

as those of large mammals: wolves, hyenas, bison, and mammoths, and the tools as the work of Pleistocene people.

"The Long Snowfall" refers to the deposition of sediment in the ocean from the erosion of the land. Various methods have been used to measure the depth of the sediment, including core samples, seismic refraction, and measuring the echoes when depth charges were exploded.

"The Birth of an Island" describes how volcanic islands rise from the sea. The chapter devotes an equal amount of space to the death of islands, for, ironically, they are often destroyed by the same volcanic processes that created them, as in the case of Krakatoa which surfaced through volcanic action in "some remote time," and exploded on August 27, 1883, creating a hundred-foot tsunami, and leaving a residue of ash in the atmosphere that generated spectacular sunsets around the globe for almost a year (87).

Carson explains how vegetation and animal life arrives at islands distant from mainlands. Insects and seeds may be carried by winds, or hitch rides on the feathers of migrating birds that stop at the islands. Interestingly, spiders appear to be among the earliest settlers. Floating rafts of timber or vegetation form another means of transportation, carrying wood-boring insects and other small animals that could withstand such journeys, chiefly insects, mollusks and reptiles. Once they arrive, the birds and animals develop in unique ways, as Charles Darwin discovered from his voyage to the Galapagos archipelago. Carson believes that these islands ought to be treated as museums, and their unique flora and fauna left undisturbed. Unfortunately, no such pristine islands remain, for, as Carson laments, "man, unhappily, has written one of his blackest records as a destroyer on the oceanic islands" (93). No mammals colonized these remote islands by natural means, but human activities have brought them. Rats may invade the islands when ships sink nearby. Rats that swam ashore from a wreck off the coast of Lord Howe Island nearly exterminated all the native birds. When goats were brought to St. Helena they ate all the young seedlings of the gumwood, ebony and brazilwood forest trees, eventually leaving the island desolate (95). Sometimes people come to live on these remote islands and bring plants or animals that wreak havoc among the island denizens. Such is the case of plants such as lantana and pamakani brought by settlers to Maui for their private gardens. These plants found favorable conditions and spread into the wild, outcompeting native plants. (Please chapter 8, "Resources" for more information about the Galapagos archipelago.)

"The Shape of Ancient Seas" discusses the alternation of rising and falling sea levels as glaciers form and melt. This chapter starts with Carson's comment that "we live in an age of rising seas" and explains that sea level along the east coast of the U.S. rose about four inches from 1930-1948 (97).

Carson was aware of the rising sea level, but when she wrote this we had not yet become aware of the magnitude of global warming and the large role humans have in accelerating climate change. (For more information about this topic please see chapter 8.)

Part II The Restless Sea

World War II brought advances in the study of waves and surf, as this information was important for the military in planning coast landings, and Carson draws on this material.

"Wind and Water" answers questions about the origin, height, strength and speed of waves. With the exception of tsunamis that are generated by undersea earthquakes, waves are produced by wind. Recorders on the ocean near Cornwall, England, off the coast of California, and along the east coast of the U.S. provide information about the origin and speed of the waves that roll over them. Carson explains that observers can guess the distance that the waves travel by their shapes. A young wave has a peaked shape, while one coming from a great distance will "rear high as though gathering strength for the final act of its life," form a crest along its advancing line, and "plunge suddenly with a booming roar into its trough" (111).

Carson explains that an earthquake off an Aleutian island produced a tsunami on April 1, 1946. The resulting waves traveled at about 470 miles per hour, reaching Hawaii, 2300 miles away, in less than 5 hours. After this a system of tsunami warnings was put in place that has saved many lives. The drawback of this system is that it cannot predict the height of the waves at the shore (119-21).

Wave heights of 25 feet or more are rare, although a wave height of 112 feet has been documented. Waves carry great force. In 1877, a pier weighing about 2600 tons in Wick, Scotland, was carried away in a storm (116). Carson closes this chapter with the acknowledgement that "the deep and turbulent recesses of the sea are hidden mysteries" (124). (For information and resources about hurricanes, earthquakes and tsunamis, please see chapter 8 "Resources")

"Wind, Sun and the Spinning of the Earth" explains how the earth's rotation, the wind, and the sun combine to produce the planetary currents. To interpret the effects of the Gulf Stream, Carson relates a trip she took to the Georges Bank fishing area on the *Albatross III*. The air that was warmed as it blew over the Gulf Stream carried moisture that turned into fog when it crossed the colder Atlantic waters.

Benjamin Franklin directed the first charting of the Gulf Stream when he was Deputy Postmaster General of the American colonies. Boston Customs authorities complained that the mail boats coming from England took two weeks longer than the Rhode Island packet boats traveling to England. When Franklin learned from a Nantucket sea captain that the current would work against boats sailing against it, he had the Gulf Stream marked on a nautical chart. But the English sea captains ignored this useful information (127-28).

Another interesting current featured here is the Humboldt Current that flows along the west coast of South America, bringing cold temperatures to the equatorial coast. The Humboldt is cold because of the upwelling of deep ocean waters, and it is these low temperatures that make it a habitat for penguins near the equator.

"The Moving Tides" are the strongest forces of the sea, responding to the gravitational pull of the sun and moon. The moon, because it is closer, exerts a stronger pull than the sun. Yet the tides may rise to different heights in places near each other, because the tides are affected by the oscillation—the rocking up and down of the water—within different tidal basins. The water at the center of the basin moves less than that at its perimeter (145). Thus, at the edge of the Minas Basin in the Bay of Fundy, which grows narrower and shallower at its mouth, there are spring tides that may rise 50 feet, the world's highest tides.

Carson reveals that the tides have been steadily decreasing over the life of the earth. When the moon was first formed it was much closer to the earth, so its gravitational pull was much stronger. The tidal friction itself is slowing down the earth's rotation, and this

retarding is causing the moon to accelerate its rotation, thus pushing the moon away through centrifugal force (148).

The last part of this chapter explicates the life cycles of some sea creatures, such as the grunion of California, small fish whose lives are governed by the tidal rhythms, even after they are taken from the ocean and placed in aquariums (152-155).

Part III Man and the Sea about Him

"The Global Thermostat" clarifies how the ocean moderates the temperatures of planet earth. If we had no oceans, temperature extremes would be "harsh" (158). Carson explains how heat and cold are transported by the circulation of air and ocean currents. The Humboldt, which Carson discussed in a previous chapter, sometimes changes course, and the resulting warm temperature of the El Nino brings heavy rains and landslides to the Peruvian coast, while the fish that live in the usually colder waters of the Humboldt die from the heat.

Carson spends the last half of this chapter explaining the theory of Otto Pettersson who theorized that cycles of warming and cooling are linked to submarine tidal flows resulting from differing alignments of the sun, moon, and earth. She concludes: "the long trend is toward a warmer earth; the pendulum is swinging" (172).

Carson does not address the issue of the human impact on climate change, for the phenomenon had not been studied at the time she wrote this book This issue, of course, is one that concerns us greatly now. (For resources about climate change, please see chapter 8.)

"Wealth from the Salt Seas" addresses the abundant minerals to be found in the oceans. The steady process of erosion over the centuries has brought minerals from dissolved rocks into the ocean. Carson observes that there are treasures of gold and silver dissolved in the ocean but it is impractical with present technology to extract these precious metals. Instead, salt is extracted by evaporation. The other major resource is petroleum. At the time that Carson wrote the book, large companies were exploring the ocean and digging deep wells to extract petroleum.

When Carson wrote this book there had not yet been major oil spill disasters from ocean drilling (such as the April 2010 explosion of the Deep Horizon oil rig leased to British Petroleum for exploration) or from tankers that crash (such as the March 23, 1989 grounding of the Exxon Valdez in Prince William Sound, Alaska). (Please see chapter 8 for more resources about the impacts of ocean drilling.)

"The Encircling Sea" retells some of the early legends of a dangerous sea full of monsters and whirlpools encircling the land and ending in darkness. Early voyagers such as the Phoenicians kept their trade routes secret to forestall competition. Sailors in the Pacific followed the stars and bird migration routes, while sailors in the Northern seas were hampered by fog and by ice. In fact, sailors often carried birds with them that acted like pilots leading them to land. Carson quotes from the Norse Sagas a report reminiscent of Noah's release of the three doves to find land after the biblical flood. In the Norse account one seafarer released three ravens successively; the first two returned and the third "flew forward over the prow, where they found land" (192).

Matthew Fontaine Maury, a Lieutenant in the U. S. Navy, is considered the founder of oceanography. In the late nineteenth century he organized a world-wide cooperative system

of gathering information regarding winds and currents from ships' officers. From the ships' logs he compiled data in charts that helped ocean-going vessels to reduce sailing times. The contemporary Pilot Charts of the Hydrographic Office are the descendants of Maury's charts.

Carson ends this book with an overview of the long cycles of geography:

> The continents . . . dissolve and pass to the sea, in grain after grain of eroded land. So the rains that rose from it return again in rivers. In its mysterious past it encompasses all the dim origins of life and receives in the end, after . . . many transmutations, the dead husks of that same life. For all at last returns to the sea—to Oceanus, the ocean river, like the ever-flowing stream of time, the beginning and the end (196).

This last passage is the piece that Carson wished to have read at her funeral. She was cremated; half of her ashes were buried next to her mother's grave, and the other half were broadcast in the Sheepscot River in Maine that she loved dearly. So Carson herself returned to the sea.

Carson includes an annotated bibliography of books suggested for further reading.

Responses to the Book

An extract of the book appeared in *The New Yorker* immediately prior to publication. William Shawn, editor of *The New Yorker*, condensed nine chapters into three long profiles for publication. Carson worked on the page proofs with him and wrote her agent, Marie Rodell, that "it has been a great experience to work with him" (Lear Witness 199). The *New Yorker* profiles were usually about prominent people; this one was about the sea. Shawn devoted great care to the project, and Carson appreciated it. The *New Yorker* articles provided much advance publicity for the book and for Carson, its author. Alice Roosevelt Longworth, President Theodore Roosevelt's daughter, called Carson to praise the book. At a book-signing party Carson explained that her reasons for writing the book were to provide interesting information and to promote conservation because "what has taken centuries to develop is being destroyed in a few years" (Lear Witness 202).

A worry surfaced as the publication date neared: would the Korean War limit publicity and sales, as World War II had limited her first book? Fortunately, in this event her worries were not fulfilled. Her book was a great success. It won many accolades and sold well. The "rush to the bookstore" that failed to occur for *Under the Sea Wind* took place now for the new book, keeping it on the *New York Times* bestseller list for eighty-six weeks, and at the top of the list for thirty-two. The first edition sold over 1.3 million copies, and the book was translated into thirty-one languages. It was a Book of the Month Club alternate selection, and a Reader's Digest condensed book. The popularity surprised the publishers, and the book was sold out the day after publication.

The book garnered many prizes. It won the National Book Award for non-fiction. Carson was featured on the cover of The *Saturday Review of Literature*. In December 1950 the

article "Birth of an Island," one of the chapters in *The Sea Around Us*, which had previously been published in *The Yale Review,* won the prestigious Westinghouse Prize for science writing that included a $1000 award. On July 7, 1951 Another welcome award was a Guggenheim Fellowship that enabled Carson to take a year's leave from the Fish and Wildlife Agency.

What, apart from its well-written explanations of natural phenomena, propelled the book to such extraordinary heights of popularity for a book about science? There were a number of books about the sea that were also on the best-seller lists in 1951, including James Jones's novel *From Here to Eternity*, and Thor Heyerdahl's non-fiction story of his sea journey aboard a raft, *Kon-Tiki*. Lear hypothesizes that the post-World War II years were times of uncertainty, as the U.S. felt its vulnerability, and faced the anxieties of the Cold War. In such unsettled times the American public was hungry for "a longer perspective on their problems and a larger dimension by which to measure human achievement . . . [Indeed, a] long view of 'the stream of time' became the leitmotif of all Carson's writing" (Witness 205).

RKO made a movie of *The Sea around Us*, although Carson was disappointed and felt that the film had departed from her text to include some scientific inaccuracies and poor writing. Nevertheless the film won an Academy Award for the best documentary film of 1953.

However, despite the book's popular success, and its literary and scientific merit, many of the male reviewers wondered about who the woman that wrote it could be. One reader wrote "I assume from the author's knowledge that he must be a man" (Lear Witness 206). After the publication of *The Sea Around Us* on July 2, 1951, Carson left for Woods Hole, leaving her 82-year-old mother to handle the fan mail, and to forward any letters from publishers to her. One interviewer who came to meet Carson at Woods Hole, Cyrus Durgin of the *Boston Globe* focused most of his article on Carson's appearance and femininity, and mistakenly called her first book *Under the Seaweed* (Lear Witness 210).

James Bennet wrote a critical letter charging that Carson did not acknowledge the role of God in creating the ocean and the world. She replied "there is absolutely no conflict between a belief in evolution and a belief in God as the creator. Believing as I do in evolution, I merely believe that is the method by which God created and is still creating life on earth" (Lear Witness 227). Indeed, Carson affirmed that the study of evolution would increase "one's reverence and awe both for the Creator and the process" (Lear Witness 227). When the young adult version of the book was issued in 1958 a review published in the Catholic Publishing Company's weekly newsletter, *Our Sunday Visitor,* on November 2, 1958, issued a word of caution because Carson presents the theory of evolution "as indisputed fact" although it is a hypothesis.

Carson's view of the world was a biocentric one in which humans were one species among many. Her view paralleled that of the Ecology Group, which flourished in the 1920s and included Alfred North Whitehead (Lytle 88-89 and Worster *Nature's Economy* 316-332). The Ecology Group propounded an ethic of cooperation that included humans. But Carson's biocentric view focused on nature rather than humans. The moral imperative of her work was the preservation of nature and its varied ecosystems; she did not see nature as providing a political moral for humans (Lytle 90).

In 1958 a special illustrated edition of the *Sea Around Us* was adapted by Anne Terry White for young readers. The book has maps, photos, charts, diagrams and other illustrations to accompany the original text. Reviewers of the book suggested that it is appropriate for readers aged 10 or older, in grades 5-9 (YCAL MSS 46 box 15 f 278).

Most of the reviews of the book praised it. Gustaf Arrhenius, a geologist on the Swedish Deep Sea Expedition that returned in 1948, wrote to Carson that he was asked by a Swedish publishing company to review her book for possible publication in Swedish translation. He told her that he found her book fascinating although it's a field he knows well. "This brilliant synthesis didn't let my interest down for a moment. So I felt obliged to write to you to express my admiration to you for having been able to compress such an enormous material in such a small volume and in such a charming form without losing anything of correctness and depth."

The Atlantic called it "a first-rate scientific tract with the charm of an elegant novelist and the lyric persuasiveness of a poet" ("The Multifaced Ocean" *Atlantic* 8/51). R. Tucker Abbott , a prominent conchologist, who was at the time the Associate Curator of the Department of Mollusks at the National Museum of Natural History, Smithsonian Institution, wrote that "rarely has the soul of the sea been so adequately fathomed or its thousand faces so delightfully pictured as Rachel Carson has done in her latest book. The many technical divisions of oceanography have been blended in a manner that gives the reader a full appreciation of the history, nature, and cyclic functions of the ocean" ("History and Nature of the Ocean" *Scientific Monthly* 8/51).

In 2003 The Oxford University Press reprinted *The Sea Around Us* with updates by oceanographers Robert D. Ballard (introduction) and Brian J. Skinner (afterword) explaining succinctly some of the more recent discoveries such as black smokers and thermal vents, and new theories such as plate tectonics. The book is beautifully illustrated with photographs. Ballard suggests that as the earth's growing population overtaxes our land resources, people may come to find habitation in specialized stable structures anchored in the coastal waters (xliii-xlv). Skinner closes the afterword by reminding us that human society changes more rapidly than the environment. He cautions us that the imbalance between the two rates may cause suffering to society and or the environment. Finding a balance is "one of the greatest challenges facing the human race in the 21st century. Rachel Carson recognized the imbalance long ago, and the situation has become worse since she wrote *The Sea Around Us.* Exploitation of mineral resources on the sea floor [such as Carson described in chapter 13 "Wealth from the Salt Seas"] will surely start in the 21st century, and with sufficient care the needed tasks can probably be carried out with a minimum of environmental disruption. But do we have the will and vision to do so?" (Skinner 273).

THE EDGE OF THE SEA

Writing the Book

Carson's next book project was one she developed at the suggestion of editor Paul Brooks. He proposed that she write a field guide to the life of the seashore, along the lines of the famous field guides to the birds written and illustrated by their mutual friend, Roger Tory Peterson. Her plan for this book was to describe the environment as well as the creatures that inhabited it (see Lytle 102) rather than just snippets of information about individual animals and plants. For her research, Carson continued her investigations of the shore near her Maine cottage, and took trips to Woods Hole, the Carolinas and the Florida Keys. She invited Bob Hines, an associate from Fish and Wildlife, to illustrate the book.

In the early stages Carson struggled to find a structure to accommodate the new book. She wrote to Brooks that describing all of the creatures in “thumbnail biographies . . . was driving me mad” (Brooks House of Life 160 qtd in Lytle 104). Eventually the structure the book required became clear to her; she would discuss the coastal habitats and describe the individual animals in relation to these habitats. Consequently, she divided the book into four ecosystems: the rocky shore north of Cape Cod, the sand beaches from the Cape south to Florida, and the coral and mangrove coasts of Florida. In its earliest versions, the book would have value as a guide to a limited and local audience, and Carson assumed that there would be no magazine serializations or foreign sales. But as she came to re-configure it, she saw it as a companion or sequel to *The Sea Around Us.* The new name would be *The Edge of the Sea*. It would have relevance to similar ecological systems in other parts of the world.

The illustrator, Bob Hines, had to accept small jobs to support himself and was falling behind in his illustrations. To rectify the situation, Carson asked Brooks to give Hines an advance, so as to give him the time to work on the material for the book. Brooks complied.

Summary of the book

The copy of *The Edge of the Sea* I bought in a secondhand store has hand written notes on the end paper as follows "spend an hour on low of the spring tide; October through November lowest tide... look inside the rock weeds." I think Carson would be happy to know that her book led this reader—like many others--to explore the seacoast.

In this book Carson introduces four different shoreline worlds through her own experience of places she loved. She draws an overview of the kinds of life that live in each of these places. She describes the rocky shores of Northern New England, the sandy beaches of the central Atlantic states, and the coral coast and mangrove swamps of the Florida Keys. In depicting each of these four ecosystems Carson follows a similar format. First she provides the geological history of how this particular environment was shaped, then she moves vertically –in the case of the New England shores where life is arranged according to the distance above sea level at neap tide—or horizontally—in the case of the tidal flats—through the area, describing the flora and fauna of each part. It is indeed a magic kingdom that Carson presents to us here, a world of strange creatures who live in borrowed shells or in the body cavities of other living creatures, who wave feathery tentacles in water or extrude siphons through sand to gather their food. Here, too, Carson finds beauty where others might not: the brittle star is like a ballet dancer (224), a hydroid holds up fragile crystalline cups (116).

I Marginal World Here Carson sets out her overview of the sea as the source of the life and mentions visits she makes to three of her favorite places in three of the different types of seashore environments that this book will be about: the rocky shore of Maine and Cape Cod, the sandy beach that stretches along the Atlantic coast from Cape Hatteras to Cape Cod, and the coral coast of the southern Florida Keys. Carson expresses her sense of the links between the sea and land and the way that the sea evokes "the continuing flow of time" (6). In these places she finds the drama of life from its origins in the sea to the present time.

II Patterns of Shore Life At the edge of the sea life is abundant with each creature evolving its own unique adjustment to the exigencies of coastal living. On the rocky shore strong surf requires particular adaptations. Barnacles have a conical shape that deflects the waves, and they cement themselves to the rocks. Yet Carson is awed by the fact that it is not these crusty barnacles in the adult form that attach to the rocks, it is the larvae buffeted by the surf that attach before they have developed hard shells (15-16). Other living beings hold themselves to the rocks with suction cups (for example, sea urchins) or anchor lines (mussels). To resist the force of the waves in the surf zone, the sponges and algae flatten themselves out into the shape of thin mats, whereas in the less turbulent tide pools they may build thicker layers, such as the “cone-and-crater structure that is one of the marks of the” crumb-of-bread sponge (17).

Carson discusses the range of temperature that allows life to flourish, from 32°F to 210°F. The tropical animals are more sensitive to temperature change, especially rising temperatures. (As we now know, global warming is threatening coral colonies in the warmer regions). In contrast to the more fragile coral, the Arctic jellyfish, Cyanea, may actually revive after being frozen for several hours. Some creatures of the shore dig holes in the sand and hibernate when the temperatures become too cold.

The Gulf Stream follows the continental shelf, bringing with it changes in temperature. Where it is closer to the shore as in Florida the ocean temperature is warmer; as it approaches Cape Cod the water becomes cold. Labrador, Greenland, and parts of Newfoundland are sub-Arctic. Carson notes that there has been a general warming up since the start of the 20th century. In *The Sea Around Us* she pointed to changes in the steadily rising sea temperature. Here she explains how some of the sea creatures have extended their ranges. The green crab has moved from Cape Cod at the start of the 20th century as far north as Nova Scotia by the 1950s. The menhaden, in the herring family, disappeared from the waters of Maine around 1900, and returned in the 1950s (18-24).

The rhythms of the inter-tidal zone are controlled by the tides, in their daily cycles of high and low and in their monthly cycles of neap and spring. The neap tides, whose name comes from an old Scandinavian word meaning "hardly enough," are the low tides that occur in the first and third quarters of the lunar cycle. The spring tides, whose name derives from the Saxon word meaning to leap or to spring, occur immediately after the full and the new moon, when the earth, moon, and sun are aligned. The creatures that live between high and low water of the neap tides are in daily contact with the sea, but can live under dry conditions for limited times. Higher zones are visited by the sea for only a short period each month during the spring tides. Barnacles in these higher zones come to life and feed during the periods when they are wet; then they close the plates of their shells and live in a state resembling hibernation until the return of the next spring tide. Spawning is often synchronized with the tides, and the eggs are released either at the neap or the spring tides according to the species. Some of these creatures retain their tidal rhythms even when taken into laboratory aquariums.

Carson was ahead of her time in seeing that ocean water is not inert, but an active participant in the life processes of the creatures that inhabit it. Here again she stresses the interconnectedness of all living beings. She writes "in the sea nothing lives to itself. The very water is altered, in its chemical nature and in its capacity for influencing life processes, by the fact that certain forms have lived within it and have passed on to it new substances

capable of inducing far-reaching effects. So the present is linked with past and future, and each living thing with all that surrounds it" (37).

III The Rocky Shores The rocky shores are those of Northern New England. Carson begins this section with a brief description of the seashore and of her favorite approach to it, from a rough path through an evergreen forest near her house in Maine. She then explains the geological formation of the region through glaciations, as eastern Maine and Nova Scotia were depressed by the weight of glaciers, forming a coastal plain that was flooded as the glaciers melted (39-44).

How did life come to occupy this rocky shore? The sea brought the earliest living creatures, "but only those able to find food could survive on the new shore" (45). The plankton that are carried in by the tide provide food for the earliest inhabitants, the “plankton-strainers such as the barnacles and mussels” (45). Carson takes us on a journey through the intertidal zone, from the snails and the lichen of the uppermost level to the coralline algae and the sea urchins of the lowest level. Here again Carson finds abundant beauty in the tiny strange plants and animals, the feathery tendrils of the barnacles, the Sertularian hydroid whose branching cups look like a cut glass chandelier (116), or the petal-like tentacles of the hydroid Clava, the lacy sponges (84). She describes a palette of pale green (algae in the crumb-of-bread sponge), rose (coralline algae), apricot (bryozoans), amber (snail), black (mussels), white (barnacles) and the transparent comb jellies.

Carson begins our exploration of the intertidal zone with a group of snails, the periwinkles. By explaining their habits and habitats, she shows how these water creatures moved into the intertidal zone and are evolving to become land creatures. The smooth periwinkle is still very much a sea creature. It hides in wet seaweed at low tide, and deposits its eggs on the rockweed (84). The common periwinkle also casts its eggs in the sea, but it often lives in places where it is underwater only at high tide. In contrast, the rough periwinkle lives in rock cavities above the neap tide line, and can live up to 31 days out of the water. In fact it is viviparous, carrying its eggs in cocoons inside the mother’s body. The newly hatched rough periwinkles emerge as tiny fully formed shelled creatures, each about the size of a grain of coffee. All of these periwinkles feed on the algae that cover the rocks, scraping them with a tongue-like organ, the radula (49-51).

Similar to the rough periwinkle, the barnacle eggs are hatched inside the parent’s shell; however they emerge in an immature state as larvae which undergo several transformations. Eventually the young barnacles choose sites on a rough rock, and extrude cement to anchor themselves securely. Once attached, the barnacles enclose themselves in their shells when the tide is out. But they feed during the incoming tides, sending out feathery appendages that sweep the water and strain out diatoms (51-56).

The varieties of seaweeds are like undersea forests. The laminaria attach to substrates with many-branching holdfasts, and in the shelter of these holdfasts many small creatures make their homes. In the more sheltered areas rockweeds form a “fantastic jungle,” for the seaweeds rise with the rising tide and flatten out when the tides retreat. This sea wrack gets both its support and its nutrients from the surrounding water; therefore, it does not need roots, stems, or leaves (78). When the tide is high the sea creatures awaken and start to eat what the tide has brought (71-75).

Carson describes many interesting creatures of this region. One of them is the hydroid that undergoes “a curious alternation of generations.” The parent generation produces buds

asexually, and these buds release into the sea the eggs and sperm that combine to form the next generation (85-89).

Next Carson takes us on a tour of the tide pools that may be as small as a teacup yet contain the beauty of the sea in miniature (110). Tide pool inhabitants vary as the pools are higher or lower, and thus bathed more or less frequently by the ocean. The highest pools have great variations in temperature and salinity as the rocks heat up and cool down, and rains dilute the water.

IV. The Rim of Sand This section describes the ecosystem of the Atlantic coastal plain. Carson explains that sand grains result from the erosion of rocks; the sand on different beaches derives from minerals in varying proportions, resulting in different shadings of color. The bulk of beach sand on the Atlantic coast is quartz, but in some areas there is also beryl, garnet, tourmaline, or glauconite. Further south white sands result from the breakdown of shell fragments. According to Carson "a grain of sand is almost indestructible" (130). Each grain holds a film of water that cushions and protects the sand grains from rubbing together. Amazingly, in this world between grains of sand, tiny flora and fauna swim, just as larger creatures swim in the ocean. For these creatures, "the micro-droplet of water separating one grain of sand from another is like a vast, dark sea" (131).

On the quiet beaches where the surf is gentle because the shore is protected by outlying islands, the creatures of the sand survive by burying themselves. Carson remarks that when she walks along the beach she has the sensation that she is "treading on thin rooftops of an underground city" (140). These sand creatures have developed unique and fascinating ways of "feeding, breathing, and reproducing while lying below reach of the surf" (133). Some of them extend tubes or siphons up during high tide and strain their food from water that is passed through their bodies. The ghost crabs, lugworms, sand anemones, trumpet worms, sand dollars, and razor clams live under the surface of the sand in tunnels and burrows which they may close with sand “doors” when the tide has receded.

Where the coastline is more exposed and there is more surf there are fewer inhabitants and those are very specialized. These animals tend to be small and fast. Some of them feed along the line of the surf where "each wave breaking on the beach is at once their friend and enemy; though it brings food, it threatens to carry them out to sea in its swirling backwash" (153). For example the mole crab backs into wet sand and extends its antennae into the receding waves to filter out its microscopic bits of food. The ghost crab, denizen of the high tide line, burrows into sand tunnels, and feeds at night.

Carson next turns her attention to the wrack line containing flotsam such as shells, egg cases, remains of flora and fauna, beach glass, and other debris carried by the ocean storms. Interestingly some of the creatures who inhabit the open ocean have been discovered first when they appeared as flotsam in such wrack lines, for example the ramshorn shell. The creature that builds this shell is a squid-like animal, the Spirula, whose shell is actually internal. How does this shell of the deep ocean creature arrive on the shoreline? Apparently when the Spirula dies and decomposes, gases lift the shell to the surface where it is carried ashore by waves.

Another component of sea wrack is driftwood, filled with tunnels bored by shipworms. Carson notes that in the 18th century shipworms bored into the dikes protecting Holland from the ocean, and "by doing so [they] threatened the very life of Holland" (186).

Carson asks how animals such as shipworms or barnacles find and colonize the surfaces of dikes, piers, and bridges. Her answer is "the ocean currents are . . . a stream of life, carrying always the eggs and young of countless sea creatures. . . . As long as the currents move . . . there is the possibility, the probability, even the certainty, that some particular form of life . . . will come to occupy new territory. . . . Most of the participants in this cosmic migration are doomed to failure. . . . [But] for all the billions lost, a few succeed" (189).

V The Coral Coast This section describes the Coral Coast of Florida Keys. Coral requires water at a temperature that stays above 70°F. Only the eastern shore of the Keys has coral; the western shore has cold currents that prevent the coral from developing. The Keys themselves are composed of limestone excreted by coral colonies. Offshore the coral reefs are being continually built by the coral polyps, while at the same time they are being attacked by burrowing animals such as mollusks and by sponges that dissolve the lime (194-9).

In the coastal waters there is a lot of white sediment from eroded coral. Some of the creatures that live here are the Vermetid snail that filters water through its gills. These snails form colonies of intertwined shells; they are perfectly adapted to their location. Other creatures are the brittle star, whose movements Carson compares to a ballet dancer. The sea cucumber has a strange method of defense: it thrusts out its internal organs when disturbed. These organs give off a toxin that kills nearby fish. Such a drastic defense often proves suicidal for the sea cucumber, but some do survive (228-9).

In the shallows grasses, rather than algae, grow. In the shelter of these grasses range a number of creatures, many of whom have taken on a bright green camouflage coloring that acts as protection. Among these creatures are the conch, octopus, Seahorse, pipe fish (a species where the male carries the eggs in a pouch).

The corals are on the Atlantic seaward warmer side of the Keys; the mangroves are on the bay side where cooler currents well up. Mangrove seedlings drift in the ocean, perhaps even for months. At first they are horizontal, and then they become vertical. When a seedling reaches the shore it may become embedded and root. Once rooted, the mangrove sets out numerous arching roots around which debris collects. "From such simple beginnings, an island is born" (242). The fauna of the mangrove forests include oysters, raccoons, fiddler crabs, and conchs.

VI The Enduring Sea This two-page chapter celebrates the flow of the oceans, and the flow of life and of time itself. The sea erodes the rocks and grinds them to sand “in a shifting, kaleidoscopic pattern in which there is no finality, no ultimate and fixed reality—earth becoming fluid as the sea itself. . . . The meaning [of each of the ocean’s creatures] haunts and eludes us, and in its very pursuit we approach the ultimate mystery of Life itself” (250).

Appendix: Classification The appendix lists the flora and fauna that Carson discusses in the book, providing the Latin nomenclature, and a thumbnail description of each phylum: Protozoa: one celled plants and animals, Thallophyta: higher algae, Porifera: sponges, and so on. Each of these features an illustration by Bob Hines.

Responses to the Book

When the book was published William Shawn of *The New Yorker* who had worked with Carson to condense the book for magazine publication wrote Marie Rodell, Carson's agent enthusiastically, "She's done it again." Although nominated for a National Book Award, this book lost out to *An American in Italy.* Nevertheless, there were numerous awards. The American Association of University Women honored Carson with its Achievement Award; the National Council of Women of the United States called *The Edge of the Sea* "the outstanding book of the year" (Lytle 107). Carson dedicated the book "To Dorothy and Stanley Freeman who have gone down with me into the low-tide world and have felt its beauty and its mystery." They wrote a note thanking her for unlocking the "beauty and mystery" of the sea for them, and gave her "a diamond pin in the shape of a seashell" (Lear Witness 272).

The New York Times declared the new book "equally wise and wonderful" (Lytle 109). Speaking about her book to The American Association for the Advancement of Science, Carson stressed the theme of the interdependence of life, and the "intricately woven design of the whole" (Lytle 111-112). H. Patricia Hynes writes that Carson's books "stirred people to love the sea because of its beauty for which she was their eyes, for its mystery of which she was the oracle, and for its cadence and sound for which she was its voice" (35).

Because these books were so successful Carson received invitations for speaking and for other writing projects. Her next project turned from the ocean of water to clouds "in the ocean of air," a TV script for the show *Omnibus.* Chapter 4 will look at Carson's other writings. In part, because people knew Carson's three books and responded to a new book by such a well-known author, *Silent Spring* quickly became a bestseller. Chapter 5 will discuss the writing and the impact of *Silent Spring.* In chapter 8, Resources, I suggest some bird guides, oceanography texts, and other resources for those who wish to learn more about the topics Carson broaches.

CHAPTER 4

"WORDS TO LIVE BY:" CARSON'S OTHER WRITING

Field Notes, Essays, Reviews, and Government Brochures

> There is symbolic as well as actual beauty in the migration of birds; in the ebb and flow of the tides. . . . There is something infinitely healing in these repeated refrains of nature, and assurance that after night, dawn comes, and spring after winter (Carson Words to Live By).

Carson repeatedly said that writing about nature was her avocation as well as her vocation. Although her major writings are her three books about the ocean and *Silent Spring*, she wrote many shorter texts for the government agency at which she worked as a fisheries expert and editor, as well as for a variety of newspaper and magazine publications. Additionally, she delivered speeches as the recipient of numerous awards granted her by various environmental and professional organizations. For her all of these—government brochures and reports, essays, and speeches—just as for her books, Carson compiled meticulous research and revised carefully, usually reading her words aloud as she wrote and re-wrote in order to achieve the results she desired. Until arthritis and other ailments prevented her, Maria Carson served as an editorial assistant for her daughter: Maria would type her daughter's manuscripts during the day, and Rachel Carson would read and revise the typescripts in the evenings.

In addition to her public writing, Carson carried on lively private exchanges with numerous friends and colleagues. She had a large network of correspondents in fields related to her work, and exchanged visits and ideas with many of them. Her letters to her dearest friend, Dorothy Freeman, are published in a collection of their voluminous correspondence edited by Freeman's grand- daughter Martha Freeman.

All of these writings offer further insight into Carson's ideas and values, while her personal correspondence provides us a more intimate look at the whole person.

GOVERNMENT PUBLICATIONS

Before Carson wrote the three books about the sea that won her scientific and popular acclaim, she wrote government publications and essays on themes related to fish, the ocean, and ecology. Some of these have been collected in a book edited by Linda Lear, *Lost Woods*. The title refers to a place that Carson loved near her Maine vacation home. She had hoped to be able to acquire the site for a nature reserve, but was not successful. However, after Carson's death, the department of the Interior set aside land along the Maine Coast as the Rachel Carson National Wildlife Refuge

Carson's first publications in her government job in the U.S. Bureau of Fisheries (which later became the U.S. Fish and Wildlife Service) were the scripts for a series of seven minute radio broadcasts about fish called "Romance Under the Water." When she wrote the introduction to these scripts for a booklet, her supervisor, Elmer Higgins, judged that it was too literary for a government publication, and advised her to send it for publication to the *Atlantic Monthly* magazine, where it was published in September, 1937. (see below)

During World War II, in 1943, when meat was rationed, Carson wrote a pamphlet “Food from the Sea” urging people to eat “underexploited” fish such as the wolffish. She wrote several pamphlets about fish and shellfish of various regions of the US. In 1945 she wrote *Fish and Shellfish of the Middle Atlantic Coast* (Conservation Bulletin Number 38) for The Department of the Interior (home of the Bureau of Fisheries). While describing the different kinds of fish Carson noted the total catch and its monetary value. From New York to Virginia in 1940 fishers harvested 388,000,000 pounds of menhaden, more than 50 million pounds of oysters, about 47,000,000 pounds of croakers, approximately 40,000,000 pounds of crabs, and at least 5,000,000 pounds each of sea trout, river herring, porgy, shad, whiting, clams and haddock (2). The total annual yield in the mid-Atlantic states was 676,000,000 pounds, and in the U.S. 4,000,000,000 pounds (YCAL MSS 46 box 96 F 1711). Although it is not stated in this pamphlet, Carson believed, along with the majority of fishers and researchers who studied fisheries, that the resources of fish that could be harvested from the ocean were inexhaustible. Since this pamphlet was published, fish stocks have been vastly depleted, and some fisheries have collapsed. (For more information and resources on declining fisheries and on fish farming, please see chapter 8 .)

Carson was able to initiate projects at the Fish and Wildlife Service, and some of these led her to interesting field work opportunities. She proposed and wrote a series of brochures about national wildlife refuges Bear River (with Vanez T. Wilson) Chincoteague, Mattamuskeet, and Parker river. This project enabled her to visit Chincoteague in Maryland, and Parker River on the northeastern coast of Massachusetts.

ESSAYS IN NEWSPAPERS AND MAGAZINES

“Undersea”

The revised introductory essay that did not suit the “fish tales” pamphlet appeared in *The Atlantic Monthly* as "Undersea" in September, 1937. This short piece opened new doors for Carson and set her career on a new track. It is richly lyrical and descriptive, replete with information, yet in flowing, readable language. It evokes the reverence for nature and the sense of wonder and mystery that are constant throughout Carson’s work. It therefore provides an excellent introduction to her writing. The essay begins "Who has known the ocean? Neither you nor I, with our earth -bound senses, know the foam and surge of the tide that beats over the crab hiding under the sea weed of his tide-pool home” (Lost 4). The essay continues to describe the paradoxes of the ocean that is home to "the great white shark, two-thousand-pound killer of the seas . . . [and] also the home of living things so small that your two hands might scoop up as many of them as there are stars in the Milky Way" (Lost 6).

Carson shows us the magic and beauty of the sea’s inhabitants. For example, she proclaims that the shell of the marine protozoan, the radiolarian, is "a miracle of ephemeral beauty that might be the work of a fairy glass-blower with a snowflake as his pattern" (Lost 10).

In this essay Carson developed a key concept of her worldview, the concept of “material immortality." This idea signified the cycles of life in which animals die and furnish food for others, so that the molecules that comprise life would recycle continually through an ecosystem. Plants provide food for the plankton that in turn provide food for fish,

> "all, in the end, to be redissolved into their component substances . . . . Individual elements are lost to view, only to reappear again and again in different incarnations in a kind of material immortality. . . . The life span of a particular plant or animal appears, not as a drama complete in itself, but only as a brief interlude in a panorama of endless change" (in Lear, Lost 11).

In "Undersea," she also used the term "food chain" which had first been used by Charles Elton in his 1927 book *Animal Ecology*. (Lytle 43 and see Worster *Nature's Economy*).

This essay garnered Carson fan mail from various readers, one of whom was influential in moving Carson's career in a new direction. R.W. Stone of the Pennsylvania Academy of Science wondered "Did you ever try writing poetry? There is a swing and a lilt to your phrases that made me glow with delight and read them to my wife. . . .Your 'Undersea' appeals to me as a gem" (12/12/37 YCAL MSS 46 box 3 f 59). Ernest G. Draper, Assistant Secretary of Commerce, wrote in praise of the essay and expressed his pleasure that Carson was an employee of his department (this was before the move of her department to the Department of the Interior) (9/3/37 YCAL MSS 46 box 3 f 59).

Hendrik Willem Van Loon, historian, journalist and children's book author, was so impressed with the article that he contacted Quincy Howe, editor at Simon and Schuster, to suggest that he encourage Carson to write a book about the sea. Moreover, on February 5, 1938, Van Loon wrote a letter to Carson, mailed in an envelope decorated with his pen and ink sketch of spouting whales. He invited her to visit his home to discuss such a book (YCAL MSS 46 box 3 f57). In response to their suggestions, Carson wrote *Under the Sea Wind*. Van Loon's invitation was the start of a long and fruitful friendship and mentoring relationship between Carson and the Van Loons. Years later, when Van Loon's son, Willem Van Loon, was writing his father's biography, he wrote to Carson for information about their correspondence and friendship, which she remembered with great pleasure.

In order to earn more money to support her family, Carson began to submit short articles about nature to local newspapers, especially *The Baltimore Sun,* and to various general magazines. She wrote about the shad fishery, the mysteries of the red tide, oysters in Chesapeake Bay, starlings, and other subjects that attracted her interest.

One of these articles, "Fight for Wildlife Pushes Ahead," published in 1938 and reprinted in Lear, *Lost Woods*, commemorates National Wildlife Restoration Week. The article stresses the recreational and economic value of wildlife and natural reserves, citing the numbers of fishing and hunting parties that visited these areas. Yet even in this early essay Carson warned that humans were "upsetting the balance of nature by draining marshland, cutting timber, plowing under the grasses that carpeted the prairies" and causing degradation of the natural environment. Carson explained that the dust storms that were then devastating much of the wheat-growing regions of the Great Plains (Oklahoma, Kansas, New Mexico, Colorado, and Texas) resulted from human over-exploitation of the land (Lost 18. For more information about dust storms see the list of resources in chapter 8). Her concern for the degradation of the environment by irresponsible and thoughtless human actions remained a constant in her life and work, culminating in *Silent Spring*.

The war generated much oceanographic research, as submarines needed information about currents, and amphibious landings required information about water temperature, waves, and other features. Working for a Government research agency gave Carson access to this kind of information as soon as it was declassified. She then used this material in

feature articles for magazines. She wrote about the bat's radar system in an essay "The Bat Knew it First" published in *Collier's* 18 November, 1944. The Navy Recruiting Office praised the piece for its clear explanation, and "distributed it to all its recruiting stations and made it required reading for anyone interested in radar technology" (Lear Witness 114).

Carson also wrote book reviews. She reviewed Gilbert Klingel's natural history of Chesapeake Bay, titled simply *The Bay*, in October 1951 in *The New York Times*. She praised the book for Klingel's "charming examples of good nature writing." (Please see chapter 2 for a discussion of Klingel's book.)

Among her book reviews, introductions, and forewords Carson wrote a Foreword to *Animal Machines* by Ruth Harrison. In this short essay, Carson aligns herself with the animal justice movement. (Please see chapter 6 After *Silent Spring* for a discussion of Carson's foreword and Harrison's book).

As Carson's fame grew, many editors asked her to contribute short essays on different themes. In 1952 William R. Nichols of *This Week Magazine* invited Carson to write a short piece for a series called "Words to Live By." She wrote "The Exceeding Beauty of the Earth" setting out the credo that she herself lived by, and that she would reiterate years later when she was suffering from metastatic breast cancer, and especially in need of the consolations that nature could bring. "In these troubled times it is a wholesome and necessary thing for us to turn again to the earth and in contemplation of her beauties to know the sense of wonder and humility. . . . There is symbolic as well as actual beauty in the migration of birds; in the ebb and flow of the tides. . . . There is something infinitely healing in these repeated refrains of nature, and assurance that after night, dawn comes, and spring after winter" (YCAL MSS 46 25 May, 1952, box 96 F1708). These were words that guided Carson's life; indeed, she was to repeat this sentiment when she developed the ideas put forward here more fully in her longer essay "A Sense of Wonder."

SPEECHES

When Carson achieved fame after the publication of *The Sea Around Us,* she received many invitations to give talks, most of which she declined. However, she was persuaded to speak at the *New York Herald Tribune* Book and Author Lunch on October 16, 1951. For her talk, Carson explained that people were surprised to find that a woman wrote a book about the sea, and that she is not a "tall, oversize, Amazon-type" woman. She described her lifelong fascination with the ocean and shared some of her research by telling interesting anecdotes about oceanography. She recalled that she had always been fascinated with the ocean, and that there is a "particular magic," for the sea provides "a sense of the great antiquity of the earth. It seems changeless but it is always changing" (Lear Lost 78). She told the story of the evolution of periwinkles from marine to land creatures, a story she would repeat in greater detail in *The Edge of the Sea*, chapter III "The Rocky Shores." She informed her audience that scientists were beginning to find living creatures in the depths and middle layers of the sea, where previously they had believed these spaces to be void of life. Moreover, whereas scientists previously believed that the seas were silent, they were finding them full of noise. Carson played a hydrophone recording of sounds made by shrimp, whales and other denizens of the sea's middle regions. According to Carson the sea "is a place of mystery" and she "cherish[ed] a very unscientific hope" that some of its mysteries will remain unsolved (Lear Lost 76-82).

In 1952 Carson received the National Book Award for Nonfiction for *The Sea Around Us*. In her acceptance speech on January 29 she sought to remove the aura of mystery surrounding science, the assumption that "science is the prerogative of . . . [an] isolated and priestlike" elite. She asserted that "the materials of science are the materials of life itself," and she stressed the links between science and poetry, arguing "no one could write truthfully about the sea and leave out the poetry" (Lear Lost 90-92 YCAL MSS 46 box 101 f 1883).

After publication of *Silent Spring* there were many requests for speeches and many honors and awards. Carson, then suffering from terminal cancer, denied most of the requests, but she accepted the following ones.

Honorary Degree Scripps College June 1962 "Of Man and the Stream of Time"

> Man has long talked somewhat arrogantly about the conquest of nature; now he has the power to achieve his boast. (5) It is our misfortune—it may well be our final tragedy—that this power has not been tempered with wisdom, but has been marked by irresponsibility; that there is all too little awareness that man is *part* of nature, and that the price of conquest may well be the destruction of man himself (6). *Scripps College Bulletin* vol xxxvi July 1962 #4 (YCAL MSS 46 box 17 f 15)

National Council of Women of the U.S. Oct 11, 1962

"I prefer that we use foresight rather than hindsight. I think we should not wait for the clear and obvious evidence that wholesale tragedy has struck before taking some measures to avert it" (8) She asked the FDA about tests of chemicals for genetic effect and learned "they rely on toxicity tests to give a clue to possible effects on chromosomes and genes. To anyone trained in genetics this is a very unsatisfactory answer" (9). Carson was dismayed to learn that there was no geneticist on the FDA staff. (YCAL MSS 46 box 17 f 21)

National Women's Press Club speech Dec 5, 1962

Carson quoted the "Globe Times of Bethlehem, PA, Oct 12, 1962 item—'No one in either county farm office who was talked to today had read the book, but all disapproved of it heartily'. Carson countered the attacks—discrediting her "I am a 'bird lover – a cat lover—a fish lover' a priestess of nature --a devotee of a mystical cult having to do with the laws of the universe which my critics consider themselves immune to."

Cullum medal of the American Geographic Society December 5, 1963

In her acceptance speech she noted that the there is a widespread "blindness to the interdependence of all life – including the human species – upon the environment." She continued to explain that many people mistakenly believe humans can master and control nature, and have lost sight of our proper relationship to the environment. Therefore, "we who have the understanding and the vision, therefore, have also the responsibility to

promote public understanding of the problems [of pollution, overpopulation, and exploitation of resources]. This is at once our challenge and our opportunity" (YCAL MSS 46 box 99 F1835).

Because of her writings about the ocean, Carson was also invited to write jacket notes for a recording of Debussy's "La Mer" (see Lear Lost Woods).

TV SCRIPT "CLOUDS"

In 1956, Carson accepted an invitation to compose a TV script about clouds for a series, *Omnibus*. According to Tim Brooks and Earle Marsh *Omnibus* was "the most outstanding and longest-running cultural series in the history of commercial network television" (*The Complete Directory to Prime time Network and Cable TV Shows* Tim Brooks and Earle Marsh, 6th edition (767-68). It was a 90 minute show that ran from Oct 4, 1953—March 31, 1957 sponsored by the Ford Foundation. The show aired originally on CBS from 5:00-6:30 PM on Sundays and then on ABC from Oct 1956—Mar 1957 on Sundays from 9:00-10:30 PM hosted by Alistair Cooke. Carson's program was broadcast on Sunday March 11, 1956, the 100th broadcast of the show (CT Coll box 12 f 9).

The topic had been suggested by an eight year old viewer who requested a program featuring "something about the sky." Curious about the medium she would be writing for, Carson bought her first TV set. This especially pleased her mother Maria, who was now pretty much confined to her home. Carson wrote Dorothy Freeman that her mother was happily watching a popular children's program "The Lone Ranger" (Lear Witness 280-82).

In the *Omnibus* program, Carson talks about an "ocean of air." She starts with our memories of looking up at clouds, speaks of the water cycle, and describes different kinds of clouds: stratus are the lowest and they include fog; cumulus are the middle level, including thunderheads and other violent storms; and cirrus are the highest, made of ice crystals.She includes a discussion of the jet stream, which is analogous to the ocean's Gulf Stream. The jet stream is an important influence on the weather. It was discovered by airplane pilots in WWII who found strong headwinds when flying west. The jet stream can shear off the tops of clouds, push storms out to sea and prevent hurricanes from making landfall.

After the "Clouds" script Carson turned her attention to a book on evolution she had agreed to write for Harper and Brothers. While she was happily beginning her research, Julian Huxley published *Evolution in Action*, and she decided that another book on this topic would not succeed at this time. She switched her focus to ecology, and proposed a new title, *Remembrance of the Earth*. She struggled to discover the central thread that would unify the book, and she found it difficult to work with the editor, Ruth Nanda Anshen, writing to Dorothy Freeman that she would be on her way to Antarctica whenever Anshen was coming to Maryland (Lear Witness 287). Eventually, Carson gave up this project and turned instead to the book that became *Silent Spring*.

THE SENSE OF WONDER

While working on the new book, Carson wrote an article for *Woman's Home Companion* called "Help Your Child to Wonder." The book starts at the edge of the sea, where Carson carried her toddler grand-nephew Roger Christie to the beach on a "stormy autumn night." They laughed together at the sound of the crashing waves, and Carson notes "I think we felt

the same spine-tingling response to the vast, roaring ocean and the wild night around us" (15). When Roger's mother, Marjorie, read the article she cried.

Carson hoped to add more material to the piece, and drafted an outline for developing the essay into a book. She wrote to Dorothy Freeman "I want very much to do the Wonder book. That would be Heaven to achieve" (Always November 1963 p. 490). Although she did not live to complete the revisions the article was published after Carson's death in book form as *The Sense of Wonder*. Subsequently it has been republished in various editions with illustrations or photographs.

Lisa Sideris writes

> I think it's significant that the 'wonder' essay is positioned between *The Edge of the Sea*, a close observation of the intricate balance of life, and *Silent Spring*, a plea for its protection. A sense of wonder closes the distance between 'this is wonderful' and 'this must remain,' between the 'is' and the 'ought.' It is a bridge of moral resolve that links the physical world and the moral world. And so I believe that a sense of wonder may well be a moral virtue, perhaps the keystone virtue of an environmental ethic (Truth 269).

Wonder is linked to a host of emotions: amazement, surprise, openness or receptivity, empathy, and passion. And these emotions lead to humility and responsibility for the world of nature. "Wonder is the missing premise that can transform what-is into a moral conviction about how one ought to act in that world" (Sideris Truth 275).

Carson asks "is the exploration of the natural world just a pleasant way to [spend time] . . . or is there something deeper?" (100). She finds that a close relationship to nature holds healing power because it provides "something lasting and significant. Those who dwell . . . among the beauties and mysteries of the earth are never alone or weary of life. . . .Those who contemplate the beauty of the earth find reserves of strength that will endure as long as life lasts. There is symbolic as well as actual beauty in the migration of the birds, the ebb and flow of the tides, the folded bud ready for the spring. There is something infinitely healing in the repeated refrains of nature – the assurance that dawn comes after night, and spring after the winter" (100 – 101). Here again are Carson's "words to live by," given another iteration.

Carson asserts "It is not so important to *know* as to *feel*" (emphasis in original), explaining that emotion would lead the child to seek information and to learn more (Wonder 56). This attitude, so integral to Carson's life, aligns with the approach of the nature-study movement that urged parents and teachers to provide their students and children real encounters with nature.

Similarly, for Carson, writing required a combination of knowledge and emotion. She wrote to Dorothy Freeman about the joy of attaining "unity with one's subject matter." Carson explains that when the writer achieves that unity 'the subject itself takes over and the writer becomes merely the instrument through which the real act of creation is accomplished" (2/1/56 in *Always* 148). She experienced this connection "sometimes in writing Under the Sea Wind," and more frequently in *The Sea Around Us.* She wished to recapture that sense of unity with her subject: "that is the problem I must solve if I am ever again to be the writer I could be" (2/1/56 in *Always* 148). Artists, writers, scientists and mystics sometimes experience this kind of connection. Evelyn Fox Keller, who wrote a biography of Nobel Prize winner Barbara McClintock, a corn geneticist, writes that

women scientists sometimes tend to feel an emotional connection to their subjects. Her title for the biography, *A Feeling for the Organism*, points to this sense of connection. The Vietnamese monk Thich Nhat Hanh explains this connection between the observer and the observed: "when we want to understand something, we cannot just stand outside and observe it. We have to enter deeply into it and be one with it in order to really understand. . . . The word 'comprehend' is made up of the Latin roots *cum,* which means 'with,' and *prehendere*, which means 'to grasp it or pick it up.' To comprehend something means to pick it up and be one with it. There is no other way to understand something" (*Peace is Every Step* 100*).*

It seems appropriate to me that this posthumous book contains the last published words we have of Carson, her celebration of the mystery and beauty of the natural world.

LETTERS TO DOROTHY FREEMAN

In 1952 Rachel Carson met Dorothy Freeman and the two became fast friends almost immediately. The Freemans' son had given his parents a copy of Carson's *The Sea Around Us* that they read together. When they learned that Carson would be building a vacation house near their vacation home in Southport, Maine, Dorothy wrote to welcome her, and to express her pleasure in reading the book. Carson responded, and their friendship blossomed quickly. During the summers they spent much time together, walking through the woods, exploring tide pools, and watching the moon rise in the evening. When they were apart during the rest of the year they corresponded and visited. The published collection of their letters from their first meeting until Carson's death, 1952-1964, edited by Freeman's granddaughter Martha Freeman, runs to 543 pages. Even so, this is not the sum total of their exchanged letters: Carson and Freeman burned some of them, and Martha Freeman eliminated some because of book length limitations. We are fortunate to have these letters, for they provide a window into Carson's thinking and reveal another side of her personality. Few people write personal letters these days, for they correspond on the internet, an ephemeral medium. Consequently, future biographers may have less material for their research, and will lack the insights that personal correspondence offers.

Carson explains her literary influences to Freeman. When she was in college a line of poetry—"For the mighty wind arises, roaring seaward, and I go"—the conclusion of "Locksley Hall" by Alfred, Lord Tennyson, "burned itself into my mind" (8 November 1954, p 59). The line made a deep impression, and seemed "to tell me that my own path led to the sea—which then I had never seen—and that my own destiny was somehow linked with the sea. . . . When finally I became its biographer, the sea brought me recognition and what the world calls success" (8 November 1954, p 59). She rejoices that the sea led her to Southport and to Dorothy Freeman, and that the title of *The Sea Around Us* now has a new meaning for her: "the sea around Us" (8 November 1954, p 59). Writers whose works inspire her include Henry Williamson who wrote *Takar the Otter* and *Salar the Salmon*. Carson notes a lineage of influence, for Williamson claims he was influenced by Richard Jefferies, another writer Carson admires, although she labels him "a prolific and an uneven writer" (Thursday night, November 19-20, 1953 p.11).

The letters are filled with Carson's descriptions of places and the creatures she finds there. For example, she writes of her delight at hearing veeries, small tawny birds in the thrush family, sing (June 1, 1954 p. 42–43).

Carson shared her writing with Freeman, and invited her to read and comment on drafts of *The Edge of the Sea*. When Freeman felt that she was unqualified to give feedback, Carson wrote "You . . . could very well represent my 'ideal reader' . . . And all I want to know is whether it tells you things you're glad to know, in a way you enjoy" (12 March, 1954 p. 33).

Many of the letters describe Carson's delight in their friendship, and her love of Freeman. In the early days of their acquaintance she worries that her status as famous author may intimidate Freeman, and is relieved that this does not happen. Freeman reassures her that, just as she had hoped, there are "no abysses, no barriers, no pedestals—just a warm glowing understanding" (24 November 1955, p 139). Carson jokes about the volume of their correspondence. When Freeman suggests compiling their letters in twenty years, Carson replies: "who is going to provide the horde of slaves necessary to carry the staggering load of letters that will have accumulated by then?" (9 March, 1954 p. 31).

Carson often laments her slow pace in writing her book, and thinks: "Maybe the easiest way to write a chapter of my book would be to type 'Dear Dorothy' on the first page" (6 November, 1953 p. 10). Thinking of the constraints of distance, Carson muses: "It may well be that the enforced separation, and the necessity of writing instead of speaking, have contributed to the depth of love and understanding that have developed" (17 February 1954 p. 25).

Carson dedicated *The Edge of the Sea* "to Dorothy and Stanley Freeman who have gone down with me into the low-tide world and have felt its beauty and its mystery." They gave Carson a diamond pin in the shape of a seashell and wrote to express their pleasure: "we thought we knew the sea, but through you we discovered how slight was our knowledge. It was you who really unlocked its 'beauty and mystery' for us. . . . May we always be worthy of this tribute to our friendship" (26 October 1955, p. 123).

Carson explains that as a writer she faces "heartache, great weariness of mind and body" (February 6, 1954 p. 20). Moreover there are family problems that she confronts as she tries to support her extended family and maintain a comfortable home: "My great problem was how to be a writer and at the same time a member of my family" (25 February 1955, p 98). She shares her anxieties and struggles about finding the time and energy to write while caring for her family and: "what is needed is a near-twin of me who can do everything I do except write, and let me do that!" She thinks of "all the things that seem worth doing in the years that are left" and worries that she must be "a nurse and housemaid" for her ailing niece Marjorie (3 February 1956, p. 151).

To help her surmount these obstacles, and her "occasional black despair" she requires "nourishment" for her spirit, and is glad to have found someone "who is deeply devoted to me as a person, and who also has the capacity and the depth of understanding to share, vicariously, the sometimes crushing burden of creative effort" (6 February 1954 p. 20). Carson found in Freeman someone who could nourish both her creative / intellectual self and her emotive self: "I feel such a joyous surge of wonder every time I . . . think how in such a dark time and when I least expected it, something so lovely and richly satisfying came into my life" (February 6, 1954 p. 20). She counts the days until their occasional visits, remembers those times with pleasure, and wishes they could see each other more frequently during the seasons they are apart: "If we could only be near enough to talk often and be together even once each month" (30 January 1954 p. 19). "The lovely companionship of your letters has become a necessity to me" (6 February 1954 p. 20). They

exchanged Valentine cards, and Carson wrote "I hope you know how wonderfully sustaining is the assurance of your constant, day-and-night devotion and concern" (13 February 1954 p. 22).

After she completed *Edge of the Sea* Carson began work on her next project, a proposed book about evolution to be published by Harpers. She wrote to Dorothy "I am taking to this research like an old alcoholic to his bottle" (2 December 1955, p. 145). She found the research "stimulating," and hoped to alleviate the anxiety and pressure of deadlines by thinking of the book as two magazine articles rather than as a book. The next book became the classic *Silent Spring*.

CHAPTER 5

SILENT SPRING (1962)

> If the Bill of Rights contains no guarantee that a citizen shall be secure against lethal poisons distributed either by private individuals or by public officials, it is surely only because our forefathers, despite their considerable wisdom and foresight, could conceive of no such problem (Carson, *Silent Spring* 12 - 13).

When Rachel Carson wrote *Silent Spring* she challenged DDT, a "war hero," a "magic" insecticide that saved the lives of both soldiers and civilians from insect-borne diseases in World War II and promised to solve mankind's insect problems (Maguire 196). At a time when science and technology were hailed as the tools that had won World War II (the war that many believed would end all wars) and now would lead America forward to even greater heights of power and well-being, her work challenged the beliefs about science and technology. She questioned the practices and belief systems of economic entomologists, pesticide manufacturers, agribusiness, government regulatory agencies and common citizens. Indeed, she challenged the very vision of American scientific, technological, moral, and political supremacy in the Cold War period. The post-World War II period in the U.S. was a time of rising prosperity, rapid growth in productivity, development of consumer goods, and a general feeling of well-being. Yet it was also a time when U. S. and the Soviet Union were rivals for world domination. Many Americans feared Communism, and people who challenged current popular beliefs might be labeled Communists.

Many people today remember Carson more for this book than for her books about the sea. Because of *Silent Spring*, Carson has been praised as a heroine, a prophet, a crusader, an eco-saint, the founder of the environmental movement, or vilified as an hysterical old maid, and a mass murderer. Mark Hamilton Lytle calls her a "gentle subversive." Steve Maguire contends that "both Carson and DDT are powerful symbols capable of rallying supporters and detractors alike and, simply put, neither would be the contested icon it is today without the other" (195). According to historian Kinkela many people felt that "Carson undermined the foundation of American postwar supremacy with her audacity to question the scientific and technological innovations that sustained the most robust economic and military power in the world. Upon its publication, the ruckus over *Silent Spring* was anything but silent" (4). President Kennedy's Science Advisory Committee researched Carson's work and corroborated her conclusions. Partly in response to her book, in 1972 Congress banned the use of DDT in the U.S. Yet thirty-five years later, on the hundredth anniversary of Carson's birth, the controversy over her book arose again. How did the reserved and quiet Rachel Carson come to challenge this wonder chemical and to stir up such a continuing controversy over its appropriate uses?

Rachel Carson did not set out to be a crusader. She was an ecologist who saw all aspects of life on earth as irrevocably interconnected, and all her writing stems from this fundamental belief. While researching a book she planned to write on ecology she grew

increasingly worried about the dangers posed by the extensive spraying of pesticides on food crops and on trees in public places. A letter from a friend, Olga Huckins, a former literary editor of the *Boston Post,* resonated with Carson's thinking and helped to change the book's focus. Huckins wrote indignantly to the *Boston Herald* in January 1957 about the convulsions and deaths of birds at her birdfeeder after the area had been sprayed with DDT for mosquito control. She sent Carson a copy of her letter. Although she shared her friend's concerns, Carson was then occupied with her ecology book project, so at first she tried to find another author who might assume the task of investigating the problem and writing an exposé. She invited E. B. White, a nature writer on the staff of *The New Yorker,* to write about the pesticide issue. White had written feature stories about "the rape of America's natural resources" for that magazine. He declined the invitation and encouraged Carson to write the article herself, commenting that "the whole vast subject of pollution . . . is of the utmost interest and concern to everybody. It starts in the kitchen and extends to Jupiter and Mars. Always some special group or interest is represented, never the Earth itself" (Graham 18-19). But White did not wish to take on the task. Because Carson feared that the harmful impact humans were having on nature was worse than she had previously thought possible, she had already decided to include a warning about the problems of pollution and habitat destruction in her proposed book on ecology. Olga Huckins's letter about the death of birds after DDT spraying gave Carson the theme she needed to focus her book; it would center on the dangers of indiscriminate pesticide use (Lytle 121).

Carson determined to expose the dangers of pesticide spraying herself. In so doing, she became (as White had remarked) the representative not of a "special group or interest" but of "the Earth itself." She researched and wrote *Silent Spring* to warn the public about the environmental contamination that indiscriminate pesticide spraying inflicts on communities. Carson later wrote to Huckins "You deserve credit (or blame, according to the point of view) for having brought my attention back to this problem. . . . You begged me to find someone in Washington who could help. . . I realized I must write the book" (Graham 17).

Carson's previous books about the ocean explained, described, and celebrated nature. *Silent Spring* exposes the human disruption and disordering of nature. In writing this book Carson became an environmental writer, and one of the founders of the contemporary environmental movement. Bill McKibben defines environmental writing as distinguished from nature writing. According to him, environmental writing "takes as its subject the collision between people and the rest of the world, and asks searching questions about that collision: is it necessary? What are its effects? Might there be a better way? . . . [It seeks] answers as well as consolation [and embraces] controversy, sometimes sounding an alarm" (McKibben xxii). Carson's three ocean books and her published articles fit the category of nature writing; *Silent Spring* is environmental writing, describing a collision between government pesticide spraying policy and the natural world with its plant, animal, and human inhabitants.

At the time that Carson composed the book there was a general respect for authority, especially governmental and scientific authority in the U. S. Scientists were held in high esteem as innovators who contributed to winning World War II and who were now leading the nation forward into a future of prosperity. Government money supported research and the development of science curricula in public schools. The U.S. was seen as competing with Soviet Russia for scientific and technological dominance during the Cold War era. In 1957, responding to the Soviet launch of Sputnik, the first earth-orbiting satellite, Congress

had passed the National Defense Education Act, created NASA (the National Aeronautics and Space Administration), and prepared to ramp up science and technology education.

At the time that *Silent Spring* appeared,

> a post-World War II consensus dominated American society. At its core lay a profound anti-Communism. . . . The consensus encouraged social and political conformity, respect for governmental and community authority, uncritical patriotism, religious faith, and a commitment to a vague notion of an American way of life defined by prosperity, material comfort, and a secure home. A person did not have to be a communist to come under suspicion as a subversive. One had only to dissent against commonly accepted values, as Carson intended to do, to be considered disloyal (Lytle 134).

In the conservative climate that prevailed during the Cold War era of the 1950s people who challenged accepted beliefs and government policies were often viewed as subversive or communist. In questioning and challenging scientific authority, public complacency, and government policy Carson acted with courage and prescience. Her friends and colleagues believed the book would be important, but feared that few would read it, that Carson would be vilified, and that her professionalism and personal values would be held up for scorn. Indeed, they were partly correct: the book did stir up a storm of criticism and controversy, including threats of lawsuits to the publisher, and hostile criticism of Carson. But *Silent Spring* was popular beyond all expectations, and its impact was far-reaching; it prompted a government commission on the use of pesticides, it led to several Congressional hearings and to the subsequent banning of DDT in the U.S. It inspired a new generation of environmental organizations and activists.

The book has been compared to other important belief-changing and controversial books such as Charles Darwin's *Origin of the Species*--which changed our views about the origins of living creatures and propounded the theory of evolution--and Harriet Beecher Stowe's *Uncle Tom's Cabin*-- which painted a horrific picture of the American plantation slave system and gave greater impetus to the abolition movement. All three books challenged dominant beliefs and caused scandals in the process. H. Patricia Hynes reports that Stowe's novel was also challenged, and parodied, and she, too, was vilified and demonized for her book. In fact, Darwin's and Carson's books remain controversial in some quarters to this day.

The critique Carson puts forward in this monumental book proposes an important approach to civic political responsibility. It challenges federal and state agencies that sprayed pesticides in residential areas without informing the residents about the toxicity of the sprays. Setting this book in the context of civil rights she argues that people should "be secure against lethal poisons distributed either by private individuals or by public officials" (Spring 12-13).

Contamination caused by pesticide spraying was not a new concern for Carson. She had been collecting information about the dangers of DDT for many years. Indeed, in 1945 she had proposed an article on this subject to *The Reader's Digest.* During that time she was still working for the Fish and Wildlife Service, and some of her colleagues there, including Elmer Higgins and Clarence Cottam, were investigating the impact of DDT on wildlife at the Patuxent Research Refuge in Laurel, Maryland. Carson edited their research reports and shared their worries. She wrote to *The Reader's Digest* about their experiments to determine the effects of DDT on wildlife.

> We have all heard . . . about what DDT will soon do for us by wiping out insect pests. The experiments at Patuxent have been planned to show what other effects DDT may have when it is applied to wide areas: what it will do to insects that are beneficial or even essential, how it may affect waterfowl, or birds that depend on insect food; whether it may upset the whole delicate balance of nature if unwisely used (Lear Witness 118 – 19).

The Reader's Digest was not interested in the article. In fact, some years later, while Carson was writing *Silent Spring, The Reader's Digest* published an article in favor of spraying. Carson tried unsuccessfully to dissuade them from publishing that article, writing to them about the "enormous danger—both to wildlife and, more frighteningly, to public health—in these rapidly growing projects for insect control by poisons, especially as widely and randomly distributed by airplanes" (Lear Witness 315).

Although she did not publish her information in the early stages of her thinking about it in 1945, Carson, a meticulous researcher, continued to maintain files of information she collected on the topic. When she took on the project in 1957, there was much more research and practical experience available to draw from. Carson suggested articles on the dangers of pesticide spraying to other magazines. *Good Housekeeping* doubted that Carson's assertions "could be substantiated" (Lytle 127, qtd Brooks *House of Life* 236-8). William Shawn, editor of *The New Yorker* was the only one to accept. In fact, Shawn remarked "we don't usually think of *The New Yorker* as changing the world, but this is one time it might" (Lear Witness 327). By now Carson had begun to collect information about a legal trial challenging pesticide spraying that was taking place on Long Island, New York. She also uncovered material about government investigations of chemicals in foods and drugs (the congressional Delaney Committee). Her project began to look as though it would be bigger than the article she had at first planned to write on the subject of pesticides. In fact, over the course of five years of research and writing it grew into a book. In June 1962 *The New Yorker* published a three-part series of articles condensed from *Silent Spring* prior to its publication by Houghton Mifflin as a book (Lear Witness 313- 22).

As she worked on the book, Carson came to see it as a "crusade." But she wanted "the beauty of the world" rather than "the ugly facts" to dominate the book (Lytle 133). Toward that end, she included material on less toxic alternatives for insect control, such as biological methods, so as to be in favor of something, rather than just against. Thus her message was both apocalyptic as she revealed the dangers, and hopeful as she pointed to safer solutions, which she called "the other road."

In order to understand the importance of Carson's book, we must first investigate the intersections between humans and insects, a longstanding linkage, which has often been problematic for humans, and probably for the insects as well!

BACKGROUND ON PESTICIDES BEFORE DDT

Pest Control in Agriculture

The recorded history of human attempts to control insects is a long and fascinating one, dating back at least to ancient Greek and Roman times, when biological, physical, and cultural controls were employed. For example, the Romans wrote a law requiring people to

pick locusts off crops three times a year, while farmers welcomed birds that ate insect pests. Greek and Roman authors advised farmers to interplant vetch with cabbage for protection against cabbageworm. On the other hand, in the Middle Ages, people often used symbolic methods rather than practical ones. According to Alex MacGillivray insects were excommunicated during this period "in surprisingly frequent religious show trials" (MGillivray 15-16).

To appreciate the context in which pesticide spraying was developed and practiced in the U.S., we must look at the complex and interwoven history of insecticides in agriculture, and the evolving governmental regulations established to control their use.[1]

In the region that would become New England the native people cleared and cultivated land for their own use, but never more than 0.5 per cent of the available land. Thus the relatively undisturbed "ecosystem enjoyed a high degree of genetic diversity and natural ecological dynamism" (McWilliams 11). The environmental impact of the white European settlers was far greater, because their commercial and agricultural projects were larger, including the growing of crops for export. The more extensive kinds of farming and land clearing practiced by these settlers provided the conditions that enabled insect infestations. In the first place, although inadvertently, they brought with them insects that were not native to the regions where they settled, and thus had no natural predators or parasites. Amazed by the bountiful and seemingly inexhaustible forests they found, settlers cut trees for export as timber, for building houses, barns, and boats, and for firewood to cook food and warm their homes . They cleared land to plant crops not only for their own use but also for export. Rapid deforestation and the simultaneous appearance of new crops in large tracts of similar plants drove insects from the woods into the farms where they multiplied and became pests (McWilliams 13).

The agrarian South in the United States also cut down forests as it developed an economy based on the plantation system and focused on the labor-intensive farming of cotton, rice, tobacco, and indigo. Agricultural practices in the region unintentionally favored the spread of insect pests. Monoculture, the growth of one crop exclusively, provided inviting habitat for insects which fed on that crop. Thus, the boll weevil, for example, a Central American invader that infested cotton crops in Texas at the end of the nineteenth century, found a plentiful supply of food and no native predators or parasites. Farmers asked the government for help in controlling this pest.

Agriculture was an important segment of the economy, and most of the population was engaged in farming. The prevailing Jeffersonian agrarian ideals valorized the "yeoman farmer," who lived close to nature and practiced self-sufficiency, as the bedrock of democracy. Thus, in response to such requests for help, the U.S. government had begun to create agencies to regulate agriculture and assist farmers.

During President Abraham Lincoln's administration the Morrill Act, passed in 1862, gave the federal government the power to sell public lands and to establish land grant universities whose mission was to educate farmers. The U.S. Department of Agriculture established a Division of Entomology in 1878. Outbreaks of grasshoppers and other insects that devastated crops in the 1870s led to the passage of the Hatch Act of 1887 which set up agricultural experiment stations to test new methods, to conduct practical research on soils and plant growth, and to disseminate the results. The agricultural schools trained economic entomologists who staffed the new agencies. Farmers increasingly turned to these agencies for solutions to insect infestations and similar problems.

There are four main ways of controlling insect pests: physical, cultural, biological, and chemical. Physical quarantine is often the first line of defense. To mitigate the boll weevil problem entomologists advocated quarantine and suggested leaving a 50-mile wide strip of land with no cotton between the US and Mexico. Not surprisingly, farmers rejected this plan (Dunlap 26-27).

Suggested cultural controls—including crop rotation, diversification of crops, and burning cotton plants or plowing them under after the fall harvest to destroy the weevil's winter habitat—met with similar rejections. These practices might have worked if they were conscientiously followed by most farmers. However, many resisted this advice and did not follow the guidelines. For example, burning the cotton plants in the fall meant that there was no possibility of a smaller late harvest (called a "top crop") although burning might result in a larger crop free of the weevils the following year. Many of these farmers were tenants who did not want to risk losing the later crop, and were not always sure they would be cultivating the same plot the following year, or that weather conditions would yield a better crop the following year. Many farmers felt that their extra work would not benefit them if their neighbors did not cooperate. And tradition is a powerful force; new ideas are not often widely accepted (Dunlap 26-28).

Biological control systems include introducing diseases, parasites, or predators that would attack the unwanted insects. But biological controls require specific conditions that allow predators to reproduce and to attack the target pests without becoming pests themselves. Moreover predators are not always available, and this was the case with the weevil. Because the physical, cultural, and biological controls did not succeed, farmers and economic entomologists turned to the last resort, chemical controls (Dunlap 29-30).

In contrast to physical, cultural and biological controls, chemical methods often appear to be "magic bullets" that solve insect problems easily and quickly. Farmers accepted these methods readily. Manufacturers who produced the pesticides developed advertising campaigns and networks of salespeople to market their products. The new generation of agricultural agents trained in land grant colleges was more interested in achieving practical, rapid results than in long-term laboratory or field study of destructive insects and their life cycles (Dunlap 35-36). For these reasons, chemical pesticides became the method of choice for insect control.

Early U.S. Government Regulation of Pesticides

Early insecticides often contained arsenic, and both farmers and the public were concerned about the safety of these compounds when used to spray food products. In 1906 Congress passed the Pure Food Act, and set up a system of inspections to monitor residues of pesticides remaining on food products. However, enforcement of the Pure Food Act was in the hands of the U.S. Department of Agriculture which had been established as an advocate of farmers' interests. Thus regulators' attempts to control chemical residues on crops were driven more by the convenience of growers than by research to establish scientific bases for tolerance levels of chemicals on foods. Consequently, due to lobbying by farmers and growers, tolerance levels for residues of chemicals such as lead and arsenic on foods were often set "by what could be achieved, not by what was desirable" (Dunlap 46). Congress attempted to correct this conflict of interests in 1940 by moving the Food and Drug Administration out of the Department of Agriculture.

DEVELOPMENT AND USE OF DDT (DICHLORO-DIPHENYL-TRICHLOROETHANE)

Because people were wary of the toxic chemicals (such as arsenic) used to control agricultural pests, research chemists sought more effective, less dangerous pesticides. At first, DDT appeared to be the solution to their quest. DDT had first been synthesized in 1874 by a PhD student, Othmar Zeidler, as a course project in the laboratory of Adolph von Bayer at the University of Strasbourg. However, Zeidler did not attempt to find a use for this chemical. This was one of the many chemicals scientific researchers produced using chlorine, an element that "particularly fascinated chemists because it can combine to make thousands of long-lasting molecules" (MacGillivray 14).

In the late 1930s, Paul Müller of the pharmaceutical company J. R.Geigy in Basel, Switzerland, sought a pesticide for plant protection that would be wide-ranging in its application, toxic to a variety of insects, safe for mammals, and cheap. In 1939 he believed he had found the answer to his quest in DDT, a powerful and persistent contact insecticide.

Use of DDT in World War II

Wartime shortages of plant-derived insecticides rotenone and pyrethrum led to a search for replacements that would protect civilians and soldiers from the insect-borne diseases endemic in war and in tropical climates, especially typhus and malaria. Once DDT's insecticidal properties were determined, it was put to use to save lives. Soldiers and civilians in World War II were exposed to infestations of lice that spread typhus and to mosquitoes that spread malaria; DDT helped to protect them. According to Professor G. Fischer, a member of the Staff of Professors of the Swedish Royal Caroline Institute:

> The war situation demanded speedy action. DDT was manufactured on a vast scale whilst a series of experiments determined methods of application. In October of 1943 a heavy outbreak of typhus occurred in Naples . . . . Thereupon . . . 1,300,000 people were treated [with DDT] in January 1944 and in a period of three weeks the typhus epidemic was completely mastered. . . . DDT had passed its ordeal by fire with flying colours. (Nobel Prize Web)

People in Naples accepted the DDT dusting, sprayed their clothing, and even threw the powder at each other at weddings instead of rice (Whorton Before Silent Spring 248). Apparently other insecticides had begun to control typhus, but DDT received the credit (Dunlap 62). News of the new chemical's "magical" properties spread. "Such reports have fired the popular imagination, and the symbol DDT is acquiring a mysterious, romantic aura" wrote Brigadier General Simmons (Simmons, 32).

Professor Fischer enthused:

> DDT has been used in large quantities in the evacuation of concentration camps, of prisoners and deportees. Without any doubt, the material has already preserved the life and health of hundreds of thousands. Currently DDT treatment is the sovereign remedy the world over for the prophylaxis of typhus. (Nobel Prize Web)

The researcher who discovered its insecticidal properties, Paul Müller, won the Nobel Prize in the category Physiology and Medicine. Extolling the magnitude of Müller's work, Fischer explained that DDT

> is practically non-toxic to humans, and acts in very small dosages on a large number of various species of insect. Furthermore, it is cheap, easily manufactured and exceedingly stable. A surface treated with DDT maintains its insecticidal properties for a long time, up to several months.

Presenting Müller a Nobel Prize in 1948 Fischer addressed him:

> Your discovery of the strong contact insecticidal action of . . . [DDT] is of the greatest importance in the field of medicine. Thanks to you, preventive medicine is now able to fight many diseases carried by insects in a way totally different from that employed heretofore. Your discovery furthermore has, throughout the world, stimulated successful research into newer insecticides. (Nobel Prize Web)

DDT was called a war hero; it saved thousands of lives of soldiers and civilians in World War II.

> Sprayed liberally by the Allies in all theaters, it killed the mosquitoes that carry malaria, filiariasis and yellow fever, the lice that carried typhus, and the fleas that carried … bubonic plague. This is no small accomplishment. Wars have always been races between the generals and the insects, and the generals have rarely won (Newton and Dillingham 101).

DDT in the Cold War

After the war DDT was welcomed in the U.S. as a new kind of hero, a new weapon in an ongoing war on the home front, the war against insect pests that were devastating commercial farms. DDT was put to the uses in agriculture that Müller had first envisioned. The War Production Board released DDT for general civilian use on 1 August 1945 (Dunlap 63). The pesticide almost immediately entered the public realm: on September 9 Gimbels, a New York Department store, advertised its first shipment of DDT for sale, extolling its "wonders" and "enticing housewives to buy the new 'Aer-A-Sol bomb'" (Kinkela 7-8). Indeed, "the first aerosol products to reach the test marketing stage resemble[d] . . . the aerosol bombs used by the army and navy" ("Aerosol Insecticides" in Dunlap ed, 39). People sprayed or dusted DDT on clothes to prevent moth damage, on shelf paper lining kitchen cabinets, and throughout their homes to prevent or wipe out all types of insect infestations.

Many believed that DDT would eradicate all insect-borne diseases, and eliminate—not just control, but actually eliminate—all noxious insect pests, perhaps even all insects. For example, economic entomologist Clay Lyle listed several types of insects that he expected would be eliminated, and wrote "the time has now arrived for the eradication of the house fly" (Lyle in Dunlap ed 47). DDT was sprayed on the elm trees that lined suburban streets to control Dutch elm disease that was spread by bark beetles. It was praised by farmers and by government agencies; it was a profitable product for the rapidly growing chemical industry.

Whereas cultural and biological controls are based on the theory that humans and insects of necessity co-exist, the chemical controls seemed to promise complete eradication of harmful insects. Military metaphors were used, as in the popular idea of waging war against harmful insects (Dunlap 35-38). Sandra Steingraber writes “returning GIs were urged to grab a bottle of poison and go after dandelions, mosquitoes and grubs. In demonizing the home front's new enemy, one ad even went so far as to place Adolf Hitler's head onto the body of a beetle" (Steingraber in Matthiessen 54).

Carson focused her critique of pesticide use on domestic uses within the U. S. However, American government and private agencies promoted these chemicals widely to developing countries. Moreover, DDT and related pesticides actually came to be linked with American global aspirations. Therefore, in challenging the extensive use of pesticides, Carson was challenging American scientific, technological, and global political supremacy.

After World War II America emerged as an important world power with expansionist ambitions. The U.S. and the USSR were two super-powers poised in conflict. In complex ways DDT came to be inextricably linked to concepts of U.S. scientific and political supremacy, and to a global mission of promulgating health, prosperity, and the American way of life. The U. S. promoted the idea that it shared with its allies common interests including "insect eradication, health, and freedom from communism" (Kinkela 47). Agencies such as the Rockefeller Foundation and the Institute for International Development participated in U.S. government- sponsored programs under President Harry Truman’s Point Four program to promote development.

> Truman's Point Four program was a U.S. foreign aid project aimed at providing technological skills, knowledge, and equipment to poor nations throughout the world. The program also encouraged the flow of private investment capital to these nations. The project received its name from the fourth point of a program set forth in President Truman's 1949 inaugural address. In the cold war the U.S. government used Point Four to win support from uncommitted nations. From 1950 until 1953, Point Four aid was administered by the Technical Cooperation Administration, a separate unit within the Department of State. During the administration of President Eisenhower it was integrated into the overall foreign aid program (*The Columbia Electronic Encyclopedia,* 6th ed., Columbia University Press, 2007).

Truman believed that U.S. aid would provide "improvements to agricultural production and health [that] would win the hearts and minds of civilian populations worldwide" (Kinkela 80).

The Rockefeller Foundation carried out extensive malaria control projects in Italy and Mexico using DDT. Then the U.S. turned its attention to agricultural production in Mexico, introducing synthetic fertilizers and pesticides. Part of the American plan to win allies and strengthen the western alliance in the post-World War II period was to provide developmental aid, which included the malaria programs and agricultural improvement. Policymakers believed that adequate "food production was necessary to ensure political stability abroad" (Kinkela 78). Various governmental agencies such as the Institute for International Development worked with the Rockefeller Foundation in Mexico to offer aid to farmers. Such aid included farm machinery, hybrid seeds, advice on modern methods, and the chemical fertilizers and pesticides. Among the challenges these agencies faced were

resistance by the local populations to the new methods, distribution problems, and poverty. According to Kinkela,

> Pesticides posed one of the greatest challenges because of their high risk [to the health and safety of the users and to the ecosystem]. Unlike other technological inputs that were safer, although not necessarily ecologically sound . . . . pesticides had the potential to undermine the basic logic of [the Rockefeller Foundation plan for] agricultural development (Kinkela 75).

The increases in agricultural production led to a "Green Revolution" in developing countries such as Mexico, the Philippines, and India. According to *The Columbia Electronic Encyclopedia*

> Beginning in the mid-1940s researchers in Mexico developed broadly adapted, short-stemmed, disease-resistant wheats that excelled at converting fertilizer and water into high yields. The improved seeds were instrumental in boosting Mexican wheat production and averting famine in India and Pakistan, earning the 1970 Nobel Peace Prize for American plant breeder Norman E. Borlaug, leader of the Mexican wheat team. Significant though less dramatic improvements followed in corn. The Mexican program inspired a similarly successful rice-research effort in the Philippines and a network of research centers dedicated to the important food crops and environments of the developing world. By 1992 the system included 18 centers, mostly in developing countries, staffed by scientists from around the world, supported by a consortium of foundations, national governments, and international agencies. (*The Columbia Electronic Encyclopedia,* 6th ed., Columbia University Press, 2007).

But there is a downside to the "Green Revolution." *The Columbia Electronic Encyclopedia* continues by noting some areas of criticism. "The Green Revolution depends on fertilizers, irrigation, and other factors that poor farmers cannot afford and that may be ecologically harmful; and . . . it promotes monocultures and loss of genetic diversity." The "other factors" include chemical pesticides such as DDT and other persistent organic pesticides. Unfortunately, increased production has been capital-intensive, and damaging to the ecosystem. Although the yields were higher, the nutritional value may have decreased (Kinkela 79). Carson, of course, warned about extensive reliance on pesticides. Among the most published contemporary critics of the Green Revolution technology is Vandana Shiva, Indian philosopher, environmental activist, and ecofeminist.

Early Challenges to DDT Use

From the early days of DDT use in agriculture there were some who objected to the widespread use of chemical methods, urging biological controls instead. One of these was an ornithologist, William Vogt, who was born in the U. S. and served at one time as head of the Pan American Union. In 1944 Vogt wrote a book *El Hombre y la Tierra* (*Man and the Land)* advocating natural controls. He especially favored encouraging birds that would eat the insects. He wrote to counter the practice of killing birds under the assumption that birds were destroying crops (Kinkela 69-70).

As early as 1944 evidence about DDT toxicity to bees and other beneficial insects and to birds began to appear (Kinkela 31).There were indications that DDT had caused problems

in experimental rats and in people exposed to high doses, and that it showed up in cows' milk; nevertheless, it was widely used in agriculture, and in mosquito control programs for malaria in the southern U.S., Italy, Mexico, and elsewhere. In the U.S. there was extensive spraying for Dutch elm disease in towns and cities, and for gypsy moths and other insects in urban areas and forests. "Farmers tried DDT against almost anything that flew, crawled, or walked" (Dunlap 65).

Clarence Cottam and Elmer Higgins, Carson's colleagues at the Fish and Wildlife Service, conducted wildlife censuses before and after pesticide spraying in 1946, and documented a small amount of bird kills within forty-eight hours, especially at DDT concentrations of five pounds to an acre. They found that fish were even more sensitive to DDT. However, "most groups [of insects] apparently recovered within a month." They recommended using caution when spraying and, if spraying is deemed necessary, leaving untreated strips in order to disturb wildlife as little as possible. They also explained that timing is critical, and advised spraying either before birds arrive in spring migration or after the birds have nested and raised their young (Cottam and Higgins 58-62 in Dunlap ed).

Like many other tools, DDT is a double-edged sword: it is a powerful and effective insecticide, but it remains in the environment and in the bodies of animals and humans. Eventually the Food and Drug Administration set provisional levels for residues of DDT and other pesticides in or on foods.

In 1950 and 1951 the (James J.) Delaney Committee (its full name is the House Select Committee to Investigate the Use of Chemicals in Food Products) held hearings on food additive safety, including chemicals used in cosmetics and fluorides in drinking water. A small part of the Delaney Committee's proceedings included testimony about DDT. The Beech-Nut Packing Company, producers of baby foods, testified that they found it difficult to obtain residue-free vegetables for their products, and were forced to allow some insecticide residue in the baby food they manufactured. Some physicians and scientists asked for further research. FDA Commissioner Paul Dunbar said that using DDT during the war was "a reasonably calculated military risk," but peacetime uses required standards that took into account longer periods of exposure, and the absence of major diseases (Dunlap 66-69).

At the Delaney Committee hearings Dr. Wayland J. Hayes, Jr., a US Public Health Service toxicologist, who had done some research on DDT in humans, testified to its safety. He agreed that DDT had caused liver damage in rats, but claimed that it was reversible. Hayes testified at several important trials, and repeated his assessment that DDT posed no health risk to people (Dunlap 69-70). He conducted his experiments to determine if relatively large amounts of DDT produce acute toxicity. He did not test for long-term effects of small amounts of the chemical. But, since DDT persists in human fatty tissue, and has been documented in the bodies of all people on planet earth, the long-term and smaller amount case is a far more common situation (Steingraber Faith, 136-40). And more recently researchers are seeking to determine if DDT and similar persistent organic pollutants may cause endocrine disruption, and may interfere with embryonic development (Colborn, Dumanoski and Myers).

By the time of the Delaney hearings in 1950-51 agricultural chemicals already comprised a large and increasingly profitable business, and therefore manufacturers of insecticides now entered the regulatory arena, lobbying for less stringent rules. They made exaggerated claims, arguing that research could take a long time, and meanwhile, starvation and epidemic disease would occur without effective insecticides. An outcome of the Delaney

Committee hearings was the Miller Amendment (1954)--named for representative Arthur L Miller of Nebraska—which provided for the registration of pesticides and required manufacturers to provide evidence of the safety of agricultural chemicals. Manufacturers complained that this placed a burden on them; however, it also gave them the opportunity to conduct their own safety trials and, in effect, to control the ways they would be regulated (Dunlap 66 -75).

In addition to its use in large scale agriculture, DDT was also brought into more public spaces, and even urban streets, schools, playgrounds, and swimming pools during the 1950s. There were widespread spraying programs to control the spread of insects such as fire ants in the Southeast, and gypsy moth caterpillars that consumed the leaves of oak trees in the Northeast.

Gypsy moths are a non-native species that had been brought into the U.S. by a scientist in Medford Massachusetts who hoped to cross-breed them with silkworms to develop an American silk industry. His plan failed, and some of the moths escaped and reproduced, first in the Medford area, and then spread outward through the northeastern U.S. and eventually farther west. At lower population levels, mice and birds keep the population in check, and predator insects were imported to help control the moths. When the moth population increased dramatically, foresters feared that the local oak trees would be decimated and believed that spraying was necessary to control the moths. (Please see the discussion of *Silent Spring* chapter 10 "Indiscriminately from the Skies" below for more about the gypsy moth spraying program in New England.)

Most extensive was the program in the East and Midwest of spraying the elm trees lining the streets of towns and cities to eradicate the bark beetles that carry Dutch elm disease. Carson was troubled by these widespread spraying programs because there was insufficient research about the dangers of the pesticides being used, and she believed that the dangers of fire ants and gypsy moths had been exaggerated.

The government programs of spraying city street and suburban neighborhood trees brought the pesticides close to people’s homes, and many began to see dead and dying songbirds. Some people (such as Olga Huckins and others) who saw the damage done to wildlife began to object, and to ask for more research or for a halt to the spraying. In 1957 a group of concerned residents, including ornithologist Robert Cushman Murphy, curator emeritus of the Museum of Natural History, filed a suit in federal court seeking an injunction against aerial spraying of DDT on their property in Long Island as it would prevent them from growing organic produce. One of the plaintiffs in the suit seeking the injunction was Marjorie Spock, younger sister of the pediatrician Benjamin Spock. Marjorie became a friend and colleague of Carson's, providing her with transcripts of the trial and other material (Lear Witness 306, 313).

One of the witnesses for the defense, Dr. Wayland J. Hayes, Jr., the U.S. Public Health Service toxicologist (who had testified about the safety of DDT at the Delaney Committee hearings), argued that DDT was not a danger to humans. He “cited studies on convict volunteers who had ingested massive doses of the chemical” and he explained that research was continuing. There were elevated levels of DDT in their blood, but no evidence of illness or other damage (Dunlap 88). In fact, Carson noted that many of the volunteers in Hayes's experiment dropped out, and Hayes did not keep track of the health of these volunteers (Lear Witness 334). Furthermore, Hayes conducted the experiment and kept records for only eighteen months, so there was no research about the long-term effects of ingesting the

chemical. In fact, since DDT accumulates and persists in the human body, the effects of such chemicals may be passed through the placental barrier to developing embryos that may be more sensitive to such chemicals than their adult parents. Nevertheless, the judge dismissed the suit, ruling that "mass spraying has a reasonable relation to the public objective of combating the evil of the gypsy moth and thus is within the proper exercise of the police power by designated public officials" (Lytle 140). Therefore, aerial spraying of DDT for insect control continued.

The judge castigated the gypsy moth as “evil” and invoked the police power as appropriate for controlling this miscreant. The assumptions that insect pests are “evil,” or that nature is hostile and malevolent, requiring human control, recur in some critiques of Carson’s book. According to such critics, people who believe that nature is benign or neutral are naïve and woefully misguided.

Public concern about the Fire Ant Program prompted Secretary of Agriculture Ezra Taft Benson to ask for an investigation of the program by the National Academy of Sciences (NAS) in 1960. The NAS is a private, nonprofit organization which acts as an advisor to the United States government on scientific matters. Most of its research is conducted by the National Research Council, which draws upon scientists who volunteer their time. The committee set up three subcommittees, each of which would compile one part of the report.

Part I: Evaluation of Pesticide – Wildlife Problems
Part II: Policy and Procedures for Pest Control
Part III: Research Needs

Among other things, the report revealed the differences in outlook between the Agriculture Department which conducted the spraying and the Fish and Wildlife Service which asked for caution and requested more research about the impacts of DDT on wildlife . According to Frank Graham, author of *Since Silent Spring,*

> fully 25 per cent of the booklet is simply a litany of praise for chemical pesticides. There is no clear statement of what the problem is, nor of the specific problems that prompted this study. . . . In contrast to this lengthy promotion of pesticide wonders, only two pages plus five lines are devoted to "Wildlife Values" (43).

Clarence Cottam called the fire ant program “Scalping the patient to cure the dandruff" (“In Memoriam: Clarence Cottam” Eric G. Bolen). Ecologist and prolific author Frank Egler wrote a "scathing" review of the report in the *Atlantic Naturalist* (a publication of the Audubon Society of the District of Columbia) in 1962. He complained "wildlife here seems to be something that annoyingly gets in the way of pest control programs" (qtd in Graham 43). Additionally, Egler wrote a letter to *Science Magazine* regarding the report:

> These two bulletins beautifully exemplify much of what Ms. Carson is writing about. . . . Of the 43 "supporting agencies" of this committee, 19 are chemical corporations, others are trade councils and associations, together representing the massed might of this industry. . . . No pesticide is mentioned by name, as though there were no differences in toxicity and effects on ecological balances. . . . All undue wildlife kills and undesirable effects are brushed aside as "accidents" without discussion or recommendation. . . . These bulletins could serve as promotional literature for the chemical industry. They

> show a remarkable avoidance of ecological sophistication (YCAL MSS 46 box 84 F 1462).

When Carson began to write *Silent Spring* there was widespread respect for the government and scientists by most citizens. Among governmental agencies there were conflicts over jurisdictions and policies regarding wildlife and pesticides. But as Carson researched and wrote the book, information began to emerge about the hazards of such toxic chemical substances as pesticides, radioactive fallout, and the drug thalidomide. *Silent Spring* would speak to and amplify the growing concerns about health and safety, and it would challenge the public confidence about governmental handling of such issues.

CARSON TAKES ON THE CHALLENGE

According to Lear, "the political atmosphere in which Carson pursued her quest for information on the effects of pesticides . . . was hostile" (Lear CT Coll box 30 f5 Smithsonian Colloquium).

> In every area of inquiry, she encountered experts whose careers had been compromised because they dared to challenge the benefits of the new technology. … Pressure had been or would be applied to [Clarence] Cottam, [Frank] Egler, [Robert] Rudd, [Joseph] Hickey, [George] Wallace, [Morton] Biskind and [Malcolm] Hargraves and [W.C.] Hueper (Lear, CT Coll box 30 f 1, 11).

Carson had been keeping extensive files about pesticide issues since she first proposed an article on DDT spraying to the *Reader's Digest* in 1945. But when she decided to make that subject the focus of her book, she realized that much more research would now be needed to make a convincing and incontrovertible case. Since this project would involve a great deal of time and research Carson first thought of working with someone who could attend the Long Island trial and gather transcripts and other material. Edwin Diamond, science editor of *Newsweek*, was proposed as a co-author of the book, but the collaboration did not work out, chiefly because they had different expectations about the nature of the collaboration. In fact, Carson had more need of a research assistant than a co-author. Consequently, the contract was canceled after a few months. Perhaps because of his unhappiness with this situation, Diamond wrote "one of the most devastating critiques of *Silent Spring* in the *Saturday Evening Post*" in 1963 (Lear Witness 323-5). (Dennis T. Avery would later claim incorrectly, that Diamond had resigned, although Carson had canceled the agreement.)

Carson subsequently enlisted a research assistant, Bette Haney, and an administrative assistant, Jeanne Davis, for help on the book project. Davis typed the manuscript and organized files. Haney visited libraries to read, summarize, and photocopy articles and reports that Carson assigned. She also interviewed government officials about various topics such as the registration of certain pesticides. She was surprised that one official cut the interview short when she explained that she was doing research for Rachel Carson. Some agencies would not willingly have shared public information with her, knowing that Carson was writing a book critical of pesticide use. To gain the essential information, Carson was able to draw on her wide network of colleagues, acquaintances and friends in the Audubon Society, and the Fish and Wildlife Service, as well as other scientists and ecologists that she knew from her previous work. A noted entomologist, Reece Sailer, provided much of the

sensitive information. Carson kept his identity confidential. In fact, her agent and literary executor, Marie Rodell, left the correspondence between Carson and Sailer out of the collection of letters donated to the Beinecke Library (Lear Witness 344, 554 fn 17).

Carson was a careful researcher. She kept extensive files of materials, and she consulted with a wide network of other researchers. Her finished book contained fifty-five pages of documented sources, indicative of her extensive investigation. As she completed the chapters in the book, Carson sent them out for review and fact-checking by experts in the fields she was discussing. The reviewers usually found few errors, and found much to praise. Frank Egler and Clarence Cottam gave the chapters especially thorough reviews because they were well aware of the controversial nature of Carson's material and wanted to help her strengthen her case. Cottam read chapters seven through ten, and wrote that he would be more critical than he ordinarily would be, because "if they can find even a minor error it will be magnified out of all proportion in an attempt to discredit the rest of your work" (Feb 26, 1962 Cottam to Rachel Carson YCAL MSS 46 box 102 f 1936). In fact, Cottam's assessment has proved accurate. Cottam himself had personal experience with the Department of Agriculture's approval of pesticide spraying. He had been one of the scientists working on the National Academy of Sciences pesticide report in 1960, and he objected to the finished report because it did not address the wildlife issues adequately.

At the advice of editor Paul Brooks, Rachel Carson agreed to his choice of Lois and Louis Darling to illustrate *Silent Spring*. Their drawings appear at the start of each chapter of the original edition. The Darlings were pleased to work on Carson's book, but wondered about its reception. Louis asked his wife "who would want to read it?" She replied "anyone who ever held a spray can, and we all have" (Lear Witness 392).

Carson admitted that chapter and book titles often posed difficulty, and the title of this book was a particularly vexing problem. Her first idea was *How to Balance Nature*. Her editor, Paul Brooks, proposed *The Control of Nature*. Based on a dissenting opinion by Justice William O. Douglas in the Long Island gypsy moth spraying case, Carson proposed *Dissent in Favor of Man*. Carson suggested increasingly acrimonious titles: *Man Against the Earth,* later *The War Against Nature* and then *At War with Nature*. Paul Brooks thought that "Silent Spring" would be a good title for the bird chapter, and Marie Rodell suggested that it might serve as the title for the entire book (Lear Witness 322, 347-8, 357, 386). Some on the staff at Houghton Mifflin complained that the title was not a good indicator of the subject. Nevertheless, the title is simple, short, and memorable. It creates a striking image of loss and absence without conveying the notion of conflict or war.

In late January 1962 Carson sent the bulk of the manuscript to her agent Marie Rodell and to William Shawn, editor of *The New Yorker* who was going to publish parts of the book in three issues. She wrote to her friend Dorothy Freeman how pleased she was when Shawn called to tell her it was "a brilliant achievement . . . full of beauty and loveliness and depth of feeling." She sat and cuddled her cat and "let the tears come. . . . Suddenly the tensions of four years were broken. . . . I could never again listen happily to a thrush song if I had not done all I could [to protect them]. And last night the thoughts of all the birds and other creatures and all the loveliness that is in nature came to me with such a surge of deep happiness that now I had done what I could. . . . now it [the book] had its own life" (Always 394 Jan 23, 1962).

In the next section of this book I will summarize *Silent Spring*. Comments and relevant updated information about the topics will appear in the discussion section below the summaries.

SILENT SPRING: SUMMARY AND DISCUSSION OF THE BOOK

The book begins by discussing the impact of pesticides on animals, and then turns to their impact on humans.

1. A Fable for Tomorrow

> What has already silenced the voices of spring in countless towns in America? This book is an attempt to explain (3).

Summary This parable is probably the best known section of *Silent Spring*. It tells the story of an imagined American town, a fertile land with prosperous farms "where all life seemed to live in harmony with its surroundings," until a strange blight arrived stilling the birds, preventing farm animals from maturing, and withering the vegetation (1-3). Carson describes a "white granular powder" that had fallen "like snow upon the roofs and the lawns, the fields and the streams" (3). The damage was not caused by an enemy, "the people had done it themselves" (3).

Discussion With this description Carson sets up the analogy between the pesticide sprays and radioactive fallout, a substance causing great alarm and anxiety at the time the book appeared. It is an evocative use of the metaphor of apocalypse, which Lawrence Buell calls "the single most powerful master metaphor that the contemporary environmental imagination has" (Buell 285). Buell argues that apocalyptic narratives "create images of doom to avert doom" (295). And this, of course, was Carson's intent.

2. The Obligation to endure

> The history of life on earth has been a history of interaction between living things and their surroundings (5).

Summary This chapter sets out the main ideas of the book—the dangers of widespread use of insufficiently tested chemicals—and issues a call for citizen action. Carson begins by referring to the threats to humans and animals posed by radioactive fallout, specifically Strontium 90, a grave public concern at the time she wrote the book. Then she points out that we are producing new chemicals whose effects may also be deleterious. Over eons, living beings adjust to their environment, and reach a balance. But many new chemicals are being added at a more rapid pace than would happen in nature. Next, she explains that using chemicals to control insects is problematic for many reasons. Some of the insects that are not killed by the chemicals will reproduce, and new generations of resistant insects will arise.

Discussion It is important to remember that Carson does not claim that pesticides should never be used; she acknowledges that they are sometimes necessary to control disease and to increase agricultural production. She wrote: "It is not my contention that chemical insecticides must never be used" (12). This fact is often overlooked when critics attack her work. Rather, she argues that

> we have put poisonous and biologically potent chemicals indiscriminately into the hands of persons largely or wholly ignorant of their potentials for harm. We have subjected enormous numbers of people to contact with these poisons, without their consent and often without their knowledge (12).

She spells out preferred methods of control that might be as effective and less dangerous than pesticides. For example, monoculture, or the concentration of one crop, makes that crop a target of insect pests. When many elm trees are planted along city streets, the beetles that spread Dutch elm disease have abundant sources of food and multiply. If the plantings had been diversified, damage might occur to only one species, and the others would thrive. A similar situation prevails for food crops.

In this chapter's conclusion Carson poses the pesticide issue as a problem of civic rights. She calls for citizens to become more involved in political action, she asserts the need for more information, and she calls for public input into decisions that affect citizens. She theorizes that America's founding fathers would have put a "guarantee that a citizen shall be secure against lethal poisons distributed either by private individuals or by public officials" in the Bill of Rights, had they imagined such a possibility. The public "must decide whether it wishes to continue on the present road," but it must have the necessary facts to make this decision (12-13).

3. Elixirs of Death

> For the first time in the history of the world, every human being is now subjected to contact with dangerous chemicals, from the moment of conception until death (15).

Summary For many reasons this was the most difficult chapter for Carson to write. She had to be careful not to name specific manufacturers of these chemicals in order to protect herself from lawsuits. She had to explain complex chemical interactions accurately in language clear and simple enough for the general public to understand.

Carson starts this chapter with what were then surprising facts: pesticides have traveled long distances from where they were sprayed; residues remain in the soil, and in the bodies of animals and people, even in mothers' milk.

The next topic is a discussion of organic compounds, especially chlorinated hydrocarbons. It is important to realize that "organic" to a chemist means based on carbon atoms, rather than the popular meaning of crops grown without artificial fertilizers and pesticides. These compounds, including DDT, are stored in fatty tissues in the bodies of living beings. Carson discusses several of these chemicals and explains their actions: DDT, and the even more poisonous chlordane, heptachlor, dieldrin, aldrin, and endrin. Another

group of pesticides are the organic phosphates including parathion and malathion. When residues of two or more different pesticides combine in our bodies, the effect may be multiplied through synergistic interactions. Many herbicides, or weed killers, are similarly toxic.

Discussion Carson writes "In the course of developing agents of chemical warfare, some of the chemicals created in the laboratory were found to be lethal to insects. The discovery did not come by chance: insects were widely used to test chemicals as agents of death for man" (16). One of Carson's critics, J. Gordon Edwards, takes issue with this statement, asserting rather harshly "Carson thus seeks to tie insecticides to chemical warfare. However, DDT was never tested as an 'agent of death for man.' It was always known to be nonhazardous to humans! Her implication is despicable" (Edwards web). In fact, Carson does not name DDT in this case, and whether or not DDT is hazardous to humans may still be in question. Therefore, perhaps Edwards's own implication may be considered improper and excessive.

Jonathan B. Tucker elucidates the relationship between pesticides and chemical warfare. While conducting insecticidal research, a German scientist produced a chemical (later called tabun) that was unusable as a pesticide, for it was found to be toxic to mammals. Tucker writes

> German chemists discovered nerve agents accidentally while doing industrial pesticide research in the mid-1930s. These compounds were then developed into weapons by the Nazi regime. . . . Zyklon B was brought to the main Auschwitz concentration camp (Auschwitz I) in the summer of 1941 for the delousing of prisoners. In September, however, the SS conducted experiments to test the suitability of the poison for the mass killing of inmates in gas chambers (Tucker 3, 45).

And so Zyklon B, originally an insecticide, came to be an agent of mass destruction.

Physicians and scientists are still investigating these pesticide interactions and their impacts on people. The Centers for Disease Control and Prevention conduct research and issue biennial reports on "Human Exposure to Environmental Chemicals." They monitor the residues of metabolites of two hundred and twelve chemicals found in human blood and urine, and compile statistical reports. Ecologist Sandra Steingraber explains why the persistent organic pollutants (POPs) remain in the bodies of living beings, and why they have spread globally:

> Because few organisms possess the enzymes sufficient to break POPs' molecules apart, they are resistant to degradation. This characteristic has several implications. One is that you accumulate POPs faster than you can excrete them. The older you get, the more POPs you amass. . . . And because their lifespans often exceed the length of a human generation – twenty to thirty years – POPs are passed from mother to child during pregnancy and breast-feeding. The older the mother, the more POPs her baby receives" (Faith 136 – 37).

Steingraber also elucidates an interesting characteristic of these chemicals; they tend to concentrate in the northern latitudes. Because these chemicals are semivolatile (their boiling point exceeds 150°C) “they evaporate slowly when the weather is warm and condense quickly when temperatures fall.” They evaporate from the tropical and temperate areas and then are carried by the wind, “precipitating back to earth in cooler zones. Semivolatility plus resistance to atmospheric breakdown means that northern regions are net accumulators of POPs. Indeed, the Arctic serves as their final repository” (Faith 137).

4. Surface Waters and Underground Seas

Of all our natural resources water has become the most precious (39).

Summary In this chapter Carson discusses the effects of pesticides on fish and the birds that eat them, and explains that pesticides travel in underground aquifers, and that they increase in concentration in animals higher on the food chain. Pesticides and herbicides sprayed on trees and vegetation enter the groundwater and travel to lakes and rivers. The central story of this chapter is the story of Clear Lake, California, a favorite fishing place north of San Francisco. In this case the chemical chosen was a relative of DDT. The chemical DDD, which was apparently less toxic to fish, was applied (at a rate of 1 part pesticide to 50 million parts of water, the equivalent of .02 parts per million) to the lake to kill small gnats that were harmless but annoying to people. When the western grebes that nested there were dying, it was discovered that the DDD was taken up by the plankton and concentrated as it ascended in the food chain so that the grebes were found to have as much as 1600 parts per million in their fatty tissues. Carson ends the chapter by reminding us that “in nature nothing exists alone” (51).

Discussion Fish are especially vulnerable to toxic chemicals. In an EPA summary of fish kills between 1961-1975 "pesticides ranked second, accounting for 18% of those kills whose causes were known" (Nimmo et. al 54). But in most cases (29% of the fish kills), the causes remain unknown.

Carsel and Smith discuss "Impact of Pesticides on Ground Water Contamination" in a book chapter published in 1987. They observe that before *Silent Spring* "the term 'ground water' was virtually unknown to the general public" (71). Ground water is important as it forms the aquifer that supplies drinking water for a large part of the population. Yet, as it remains hidden below the water table it is difficult to monitor.

Sandra Steingraber explains how the concentration of pollutants increases as they ascend the food chain, a process of bio-magnification.

> Fewer and fewer individuals can occupy each ascending link of the food chain because fewer and fewer calories (energy) are available to feed them. The total amount of a persistent pollutant (matter), however, doesn't change. Thus, as the rarer members of the higher links dine upon the commoners below them, poisons dispersed among the many are drawn up into the bodies of the few. . . . As a general rule, persistent toxic chemicals concentrate by a factor of 10 to 100 with every link ascended (Steingraber, Faith 250).

A tragic example of this process is the case of Minamata disease in Japan, when residents of a city near a chemical plant ate the fish that had concentrated levels of methylmercury (a neurotoxin) dumped as waste into the surrounding ocean. Those who ate the toxic fish suffered a range of neurological symptoms including slurred speech, loss of hearing and eventual paralysis among adults and severe congenital malformations in babies (Steingraber, Faith 43-50). And, as Steingraber points out, humans are at the top of the food chain, and breastfed infants are at the pinnacle.

5. Realms of the Soil

> Without soil, land plants as we know them could not grow, and without plants no animals could survive (53).

Summary Carson elucidates the development of soil and the intricate relationship of soil with living creatures. For, as she explains more fully in *The Sea Around Us,* planet Earth was a sterile, forbidding environment of rock and water, until living creatures began the process of turning rock into soil. Lichen, microscopic soil bacteria and fungi, and earthworms are all crucial for the continued health of the soil. Herbicides that interfere with the normal processes may upset the balance, resulting in unexpected and unwanted plant death.

6. Earth's Green Mantle

> I know well a stretch of road where nature's own landscaping has provided a border of alder, viburnum, sweet fern, and juniper with seasonally changing accents of bright flowers, or of fruits hanging in jeweled clusters in the fall(71).

Summary Carson laments the ugliness of roadsides when spraying destroyed the shrubs and wildflowers that grew there. She does not disapprove of all spraying, but suggests better ways to deal with unwanted vegetation. The purpose of roadside spraying is to eliminate plants that would interfere with drivers' views of the road. She points out that removing the trees and tall shrubs would achieve that goal more effectively, and removing the trees would need to be done only once in a generation, instead of annually.

Additionally, Carson offers several examples of successful biological controls to solve landscape problems. In Holland, marigolds were planted near roses to control the nematodes that had been killing the roses. Marigold roots excrete a substance that kills the nematodes so that the roses could flourish.

In addition to aesthetic considerations, destroying the natural vegetation removes the habitat and forage of wild bees and other pollinators. Moreover, the natural vegetation provides an indication of the soil's condition: certain plants prefer moist soil, others prefer acid soil, and so on.

Discussion Roadside spraying remains an issue. See Ingrid Lobet, "Herbicide along the Highway".

7. Needless Havoc

> As man proceeds toward his announced goal of the conquest of nature, he has written a depressing record of destruction (85).

Summary Carson reports on extensive spraying of dieldrin and aldrin in Michigan and Illinois to control (actually, the goal was to eradicate) Japanese beetles. These chemicals are non-selective poisons, and the spraying resulted in death of birds, cats, muskrats, rabbits, sheep, squirrels and other mammals. Carson points out that the funding for research to determine the effects of the spraying amounted to less than one percent of the cost of the spraying. Moreover, Japanese beetles have been brought under control in the Eastern states by a bacterial disease specific to these beetle grubs, "milky spore disease," but this was not used in the Western states.

Carson asks if "any civilization can wage relentless war on life without destroying itself, and without losing the right to be called civilized" (99).

8. And No Birds Sing

> Various scientific studies have established the critical role of birds in insect control in various situations (112).

Summary Spraying to destroy the bark beetles that carried the Dutch Elm disease fungus from tree to tree also killed the robins and other birds that ate the earthworms, insects, or leaves. The tragedy is that the insect-eating birds had been controlling the populations of many insect species, and once the birds were dead, the other insects could multiply. So there would be no elms and no birds!

Carson recounts the story of a group of farmers in Indiana who hired a spray plane to treat an area with parathion in order to kill the blackbirds that were eating their corn. Over 65,000 red-winged blackbirds died, and untold numbers of other birds and mammals. But there might have been a safer solution, such as planting a different variety of corn.

Examinations of dead robins and other birds found DDT in the testes of male robins, in ovaries of females and in bird embryos that did not hatch. Carson explains that eagles are not reproducing successfully, and suspects that pesticides are the cause (which indeed turned out to be the case).

Discussion The title of this chapter comes from a poem by John Keats that tells the story of a man bewitched by a fairy who seemed to promise happiness but abandoned him. The first stanza describes the knight, now bereft in a desolate and wintry landscape:

> O what can ail thee, knight-at-arms,
> Alone and palely loitering?
> The sedge has wither'd from the lake,
> And no birds sing.

Although Carson does not discuss the poem, it clearly offers an analogy to the pesticide situation. Like the bewitching fairy, pesticides seemed to promise a glorious future, but failed to live up to their promise, instead leading to wintry environmental desolation.

Subsequent research in the United States and Britain indeed revealed an inverse correlation between levels of DDT accumulated in the raptors' bodies and the thinness of their egg shells: "the higher the residue levels in the eggs the thinner the eggshell. And analyzed statistically the indications that this could have occurred by chance were one in a thousand times" (Hickey in Dunlap ed 84). Because they are higher in the food chain the raptors bio- accumulated the pesticide residues and were more affected than other birds. We have previously seen Steingraber's explanation of how this process works.

9. Rivers of Death

> The fisheries of fresh and salt water are a resource of great importance, involving the interests and the welfare of a very large number of people (152).

Summary This chapter relates the death of salmon that fed on poisoned insects. Runoff from fields carries chemicals such as herbicides and insecticides into lakes and streams, often resulting in fish kills. But even when fish are not immediately killed, questions remain about the quantities of chemicals in rivers, lakes and oceans.

Discussion Fish appear to be more sensitive than other animals to insecticides. Retired forester Tom Dupree told me that when he was growing up he lived near a river in Massachusetts. Each time the area was sprayed with DDT for insect control he would see dead fish floating in the river the next day.

10. Indiscriminately from the Skies

> The gypsy moth program shows what a vast amount of damage can be done when reckless large – scale treatment is substituted for local and moderate control (156).

Summary This chapter recounts the failure of the Department of Agriculture gypsy moth and fire ant eradication programs. Neither of these insects are native to the U.S., and did not have sufficient predators to keep them in balance. Yet there are far less drastic control measures than the massive, expensive spraying programs designed (albeit poorly designed) to wipe them completely out.

Gypsy moths were introduced by a scientist who hoped to mate them with silkworms and to develop a silk industry. However, several escaped from his laboratory in Medford, Massachusetts in 1869, and began to spread in the local deciduous forests, feeding on the leaves of oak and other hardwood trees. The moths were generally confined to the northeast where they found the forests to their taste. Carson notes that “thirteen parasites and predators were imported . . . and successfully established in New England” where they held the moths in check (157). But in 1957 the Department of Agriculture planned to spray an extensive area of three million acres including the New York City metropolitan area, where there are more skyscrapers than forests. Protests by concerned citizens led to court cases seeking injunctions to stop the spraying. Appeals eventually reached the U.S. Supreme Court, which declined to review the case. Justice William O. Douglas was the sole dissenter.

The spraying of DDT over a large area of Long Island including farms resulted in the death of bees and livestock. Troubling residues remained in food, especially in milk. Yet the spray program failed to eradicate the gypsy moths.

The fire ant eradication program in the southeast was another misguided project. Fire ants build mounds that are inconvenient for farmers, but individual mounds can be treated directly. The bite of fire ants is unpleasant, but does not cause severe injury to plants, people, or animals. In fact, fire ants feed on insects, some of which—such as the cotton boll weevil larva—are considered to be pests. Yet the Department of Agriculture undertook a massive spraying program in hopes of eradicating these ants. It sponsored "a propaganda movie (to gain support for its program) in which horror scenes were built around the fire ant's sting" (164). Although the sting is painful it is less dangerous than the stings of bees or wasps. Dieldrin and heptachlor, chemicals more toxic and stronger than DDT, were used. Conservation departments, ecologists and others asked in vain for delays until more research could be done on the toxicity of these chemicals. Chemical companies happily noted a "sales bonanza" as the program took place (162).

As a result of the spray programs in 1958 many birds died, especially ground feeders and worm eaters. Quail, woodcock, and wild turkey populations were decimated in the sprayed areas. When toxicity studies on rats which were published in 1952 became known, the FDA (Food and Drug Administration) banned "any residues of heptachlor" or its oxidation product in food (170). However, the FDA rulings apply only to interstate commerce, and therefore contaminated food was often marketed within states.

Discussion According to the Fish and Wildlife Service the gypsy moth population undergoes a fairly regular cyclical expansion and collapse, and has become susceptible to a virus and a fungus that keep the population in check.

> A variety of natural agents are known to kill gypsy moths in nature. These agents include over 20 insect parasitoids and predators that were introduced over the last 100 years from Asia and Europe. Small mammals are perhaps the most important gypsy moth predator, especially at low population densities. Birds are also known to prey on gypsy moths but at least in North America this does not substantially affect populations. A nucleopolyhedrosis virus usually causes the collapse of outbreak populations and recently an entomopathogenic fungus species has caused considerable mortality of populations in North America (U.S. Forest Service Web)

In the early 1980s spraying in New England was carried out with Sevin (carbaryl) to control gypsy moths. The story behind this is an interesting sidelight and indicates how interconnected the parts of the natural environment are, and how easily the delicate balance of nature that prevails at a particular time in a particular location may be upset. According to retired forester Tom Dupree, the gasoline crisis of 1979, when supplies of gasoline were short, raised fears of escalating fuel prices and shortages of fuel for home heating. This fear prompted many New Englanders to buy wood stoves. When they purchased cords of wood to fuel their stoves they inadvertently transported gypsy moth eggs on the wood to places where their natural predators had not been as prevalent. When the gypsy moths hatched out they multiplied rapidly and the population spike was much higher than it had previously been. Dupree remarked that the gypsy moths first defoliated the oaks, then the other trees, and then gobbled up even the briar and the poison ivy. In Rhode Island gypsy moth frass dropped onto people's porches, yards, and the clothes hanging on their lines. The sound was

like rain falling. People called the governor's office to complain, and the spraying program was put in place. After several years a balance was restored as natural predators were able to control the gypsy moth population and the cyclical expansion and collapse returned to its previous curve.

There is yet another piece of this story. When the gypsy moths defoliated and killed many trees the forest canopy opened up and understory plants flourished, making the forests more habitable for deer (personal communication by Tom Husband). Suburban development has contributed to this process, and consequently, the deer population has increased. Deer browse in the woods and in the lawns and gardens of suburban house lots. This has created a health problem for humans, the spread of Lyme disease, ehrlichiosis, Babesiosis, and other diseases which are transmitted by deer ticks.

Spraying for gypsy moths continues to take place in some areas of the country, and now the agent used is a bacterium, bacillus thuringiensis (BT), which, according to the Rachel Carson Council, is an acceptable pesticide for moths. There are variants of BT, each one specific for a few insect species and harmless to other animals (Dorothy Read, personal communication).

11. Beyond the Dreams of the Borgias

> In addition to making this change to less dangerous agricultural pesticides, we should diligently explore the possibilities of non-chemical means (184).

Summary The Borgias, a prominent family of Renaissance Italy, were known as ruthless, greedy people, who often poisoned their enemies. Carson uses the Borgias as a metaphor to indicate the spread of poisonous chemicals dispersed through the Department of Agriculture spraying programs and widely available for consumer use in the late 1950s. She notes that insecticides such as chlordane could be purchased in supermarkets even though known to be dangerous. Shelf paper for lining kitchen cabinets was often “impregnated with insecticides” (175). The U.S. Department of Agriculture Home and Garden Bulletin #24 published in1961 suggests that homeowners spray clothing with moth killers such as DDT, dieldrin, and chlordane. These chemicals were abundantly used on lawns and gardens. Carson remarks ruefully "the mores of suburbia now dictate that crabgrass must go at whatever cost" (177).

These chemicals were absorbed or ingested by those who used them or came in contact with them knowingly or unknowingly. In 1954 and 1955, an average sample of body fat in the general population of the U.S. contained between 5.3 to 7.4 parts per million of DDT. Carson notes that in Arctic Alaska at the time of her writing the native diet of fish, beaver, moose, caribou, seals and other animals contained no DDT, yet small residues of DDT were found in native Alaskans who went to hospitals in Anchorage and ate the food prepared there.

Carson points out that foods may contain pesticide residues because of an inadequate inspection force. Less than one per cent of food transported through interstate commerce, and therefore under the jurisdiction of the FDA, is inspected because of an insufficient number of federal inspectors. She comments wryly, “The luckless consumer pays his taxes but gets his poisons regardless” (183). The solutions she proposes include a zero tolerance of residues in foods, and use of less toxic chemicals such as pyrethrins, rotenones, and other plant-derived chemicals.

Discussion More recently, ecologist Sandra Steingraber reports that even though "there are no known sources of the persistent organic pesticide (POP) dioxin within Nunavut [in northern Canada] nor within 300 miles of its borders," Inuit mothers in Nunavut have a concentration of dioxin in their breast milk twice that of residents of southern Canada. Steingraber explains that POPs are carried north by the wind currents of the jet stream and "condense and fall" due to the cold temperatures (Faith 258).

Although Carson advocated zero tolerance of pesticide residues in food, that has proved impractical at the present time. (I will discuss this further in the chapter "After *Silent Spring*".)

12. The Human Price

> There is also an ecology of the world within our bodies. . . . When one is concerned with the mysterious and wonderful functioning of the human body, cause and effect are seldom simple and easily demonstrated relationships (189).

Summary Carson claims that ecology is not just the study of the external world; it must also be the study of the "world within our bodies" (189). After considering the pesticide residues that may remain in food and be stored in human bodies, Carson turns here to diseases affecting humans that may be attributed to these toxic chemicals commonly found in our bodies. She notes that poisons in the environment are "a problem of ecology, or interrelationships, of interdependencies" (189). She points out that fat-soluble insecticides persisting in our bodies "are in position to interfere with the most vital and necessary functions of oxidation and energy production" that take place in all the cells of our bodies (191). Some of the toxic pesticides are liver poisons; others affect the nervous system. Laboratory research analyzes the effects caused by single chemicals under controlled conditions. But we may have ingested and/ or absorbed a stew of different chemicals, and scientists still do not know how the various chemicals may interact in our bodies.

Discussion Cancer was the disease that Carson chiefly considered in the book. Cancer was a source of profound anxiety at the time Carson wrote *Silent Spring.* Moreover, perhaps she emphasized this dread disease because she herself had incurred cancer. Cancer remains a focus of toxicology studies of chemicals, although there is increasing concern for the possibility that some chemicals may produce endocrine disruption and interfere with embryonic development.

Carson remarks that diseases such as cholera, smallpox, and plague have been brought under control through "sanitation, better living conditions, and new drugs" (187). However, this is true of the developed, industrialized, Western world. People in many countries still suffer from a range of insect-borne infectious diseases, especially from the debilitating and life-threatening disease, malaria. Some critics have challenged Carson's work and blame her incorrectly for the reduced use of DDT in malaria prevention. (Please see the chapter "After *Silent Spring"* for further discussion of DDT and malaria.)

13. Through a Narrow Window

> This process by which the cell functions as a chemical factory is one of the wonders of the living world (201).

Summary This chapter deals with the ways that toxic chemicals may interfere with cellular functions, cause mutations, or perhaps initiate the development of cancer. The narrow window Carson refers to is the cell, and it is by looking through this window that we learn about cellular function and thereby "comprehend the most serious and far-reaching effects of the haphazard introduction of foreign chemicals into our internal environment" (200). The study of cell function has developed significantly since Carson researched and wrote this book.

Mustard gas was the first chemical found to be a mutagenic agent, in 1940. Carson observes that "our genetic heritage" is a prized possession, but this may be threatened by the unprecedented introduction of mutagenic chemicals into our environment.

Discussion When Carson published *Silent Spring* chemical manufacturers were not required to test their products for genetic effects. According to Joseph D. Rosen and Fred M. Gretch "mutagenicity and carcinogenicity data for new pesticide registrations are now required" (138). These studies have not been completed on all of the "grandfathered" pesticides which were developed before the Miller amendment of 1954 required proof of safety for pesticides when they are manufactured for sale. Toxicity studies now rely less on animals and more on computer modeling and on studies at the cellular level, using ongoing cell lines (Rainer Lohmann). Another issue of concern today is the effect of these chemicals on the endocrine system.

14. One in Every Four

> What happens in a cell to change its orderly multiplication into the wild and uncontrolled proliferation of cancer? When answers are found they will almost certainly be multiple (230).

Discussion The title for this chapter comes from Carson's assertion that cancer may strike one in every four people (242). Carson explains that the first awareness of carcinogens occurred in 1775 when Sir Percivall Pott, a London physician, believed that soot must cause the scrotal cancer so common in chimney sweeps of his time. She contends that new diseases, such as cancer, are replacing the diseases such as typhoid and tuberculosis that developed countries have reduced through better sanitation. As evidence for this contention, Carson notes that the Office of Vital Statistics of the U.S. recorded that 4% of deaths in 1900, and 15% of the deaths in 1958 were due to malignancy.

Carson discusses Otto Warburg's theory that interference with cell respiration would lead to cancer. She alludes to liver damage and hormonal imbalance as possible causes of cancer, and argues that repeated small doses of carcinogens may be more dangerous than single large doses (232). She advocates that prevention, not only cure, must be a goal, and suggests that ways to prevent cancer include a reduction in use of carcinogenic agents.

Discussion The causes of cancer are still unclear, although evidence is mounting that diet and environmental toxins are significant factors (see e.g. Steingraber, Servan-Schreiber). Carson discusses, but does not name, a pesticide used against mites and ticks that proved to be carcinogenic. Carson did not name specific pesticides that were produced by only a single manufacturer, so as to avoid lawsuits. Without institutional protection, defending herself against a lawsuit would have caused Carson significant financial harm. Her own lawyer as well as Houghton Mifflin and *New Yorker* lawyers advised her "to meticulously

avoid any mention of chemical trade names and only refer to the broad classes of pesticides. . . . Even frivolous harassment suits could wipe out her savings" (13 Lear CT Coll box 30 f 1).

According to the American Cancer Society, the risk of cancer is even greater today than when Carson wrote *Silent Spring*: "Half of all men and one-third of all women in the US will develop cancer during their lifetimes" (Web). According to the 2008 Vital Statistics preliminary Report, 18.62% of the deaths were attributed to cancer (vol 59 no.2). Servan-Schreiber refers to current cancer rates as "epidemic," and cites the increasing incidence of cancer in young people (54-65).

Do pesticides cause cancer? This question continues to be debated among researchers. As we have seen, Dr. Wayland Hayes, Jr. fed DDT to convicts and found no ill effects during the eighteen months of his study. Chemical biologist Dixy Lee Ray and freelance writer Louis Guzzo write about "The Blessings of Pesticides" and assert emphatically that the answer to the question DDT causes cancer "is an unequivocal no." I believe that the answer is less clear. Cancer is a complex disease that remains puzzling to physicians and researchers. There is much we do not know about the causes, prevention and treatment. Ray and Guzzo devote merely two paragraphs to discussion of the carcinogen question. Although their book was published in 1990 the only evidence they cite is a 1978 mouse experiment. On the basis of this they report that "The National Cancer Institute reviewed the mouse experiment results and, in 1978, declared DDT was not a carcinogen" (363 in Pojman). Strangely, counter to all the known scientific research on the subject, and without presenting any evidence at all, Ray and Guzzo also claim that DDT is not a persistent pesticide and "under normal environmental conditions" it would not remain in the environment (362). I believe that their book purveys the "little tranquilizing pills of half truth" and the "false assurances" fed to the public that Carson deplores (Silent Spring 13).

In fact, although much research has taken place since the 1978 study cited by Ray and Guzzo, questions remain. According to the National Toxicology Program 12th Report on Carcinogenesis many pesticides including DDT are on the list of substances that are "Reasonably anticipated to be human carcinogens" (www.cancer.org 6/29/2011). According to biologist Sandra Steingraber the pesticides "lindane, chlordane, dieldrin, aldrin, heptachlor . . . are now classified as known, probable, or possible carcinogens" (downstream 9). Atrazine is known to cause breast cancer and changes in the menstrual cycle (Downstream 159-72). Until 1983 these pesticides were still manufactured in the US and exported to other countries. Some of these countries may still spray pesticides on produce that may be imported into the U.S.

Steingraber cites a 1993 study by biochemist Mary Wolff and her colleagues. The researchers studied blood samples of 14,290 women in New York City. They found

> on average, the blood of breast cancer patients contained 35% more DDE [a metabolized form of DDT] than that of healthy women. . . . The women with the highest DDE levels in their blood were four times more likely to have breast cancer than women with the lowest levels. The authors concluded that residues of DDE "are strongly associated with breast cancer risk" (Downstream 11).

Some question whether the DDE found in blood serum is a cause or a result of disease, since when people lose weight from illness, some of the residues concentrate in their fatty remaining tissue (Wildavsky 61 cites Wayland J. Hayes). Also, note that "strongly

associated" is not the same as cause. Could DDT alone be the cause of these cancers or is there a synergistic process involving several chemicals? Does having DDT in one's body lead to disease, to endocrine disruption, or to other malfunctions? If so, what amount would be required to cause the disease or disruption of body processes? Apparently, at some levels DDT causes "changes in the liver enzymes that metabolize drugs and hormones," but, in studies conducted until 1984 "adverse health effects from exposure have not been unequivocally demonstrated" (Davies 116). More research is necessary on the causes of cancer and on the various chemical interactions that take place in our bodies.

According to historians of science Naomi Oreskes and Erik M. Conway early studies of DDT and cancer in women were flawed. "To really know whether or not DDT had an effect, you'd need to study women who'd been exposed to DDT early in life, at a time when environmental exposures were high" and you would need to follow these women as they aged, since cancers often take many years to develop. Dr. Barbara A. Cohn and her colleagues conducted such a study. They examined stored blood samples taken in the 1960s from 258 women who had been children or teenagers in the 1940s and 1950s when DDT was widely used. The researchers measured the amount of DDT in the blood samples in 2000 – 2001 and found "a fivefold increase in breast cancer risk among women with high levels of serum DDT or its metabolites" (Oreskes 229). More recent studies suggest a possible link between DDT and pancreatic cancer as well as endocrine disruption (Bouwman et al). However, our body burdens contain many organic chemicals which (as Carson noted) may interact synergistically with each other, and it is difficult to disentangle the actions of single chemicals. In fact, in 2005 the Centers for Disease Control found 148 toxic chemicals in the blood and urine of Americans of all ages (Servan-Schreiber 75). Steingraber explains the difficulty of assessing

> all possible cause-and-effect relationships. The reasons are many: there are no populations free from toxic exposures that can serve as control groups; conducting controlled experiments on human beings is unacceptable; and real-life variables are infinite because some individuals are more sensitive than others, because chemicals have multiple effects, and because chemicals interact in unpredictable ways with other chemicals (Faith 285).

In view of these issues, Steingraber urges that we use the Precautionary Principle in environmental decision-making: "Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation" (Faith 284). According to the Wingspread Statement, January 26, 1998 the

> Precautionary Principle shifts the burden of proof, insisting that those responsible for an activity must vouch for its harmlessness and be held responsible if damage occurs. The issues of scientific uncertainty, economics, environmental and public health protection which are embedded in the principle make this extremely complex (Science and Environmental Health Network Web).

I believe this is an admirable principle, but one that is clearly open to varying interpretations. What specifically constitutes a threat? How much damage is "serious"? How much scientific certainty is possible? It is impossible to establish the case for "no harm" because an isolated instance of harmful effects may show up in the future. Some argue that

we are losing substantial benefits because we do not want to incur small possible risks. For example Wildavsky asks if there may be other values that are reduced in an emphasis on health: "How much is a marginal gain in health worth compared with losses in other values such as freedom, justice, and excellence?" (429). Of course Wildavsky's question raises many others: whose freedom? If one's health is lost, what freedom remains? Similar questions arise about the other values he mentions, justice and excellence. This is another of the controversies linked with Carson's work.

15. Nature Fights Back

> Most of us walk unseeing through the world, unaware alike of its beauties, its wonders, and the strange and sometimes terrible intensity of the lives that are being lived about us (249).

Summary Carson discusses the balance of nature and how interference with this shifting balance may lead to unintended consequences. The balance of nature is not static, but fluid. Insect parasites and predators may hold a population of insects in check, whereas many human interventions have failed to achieve the goal of eradication of insect species perceived as pests because "insects are finding ways to circumvent our chemical attacks on them" (245). Carson explains that insect "populations are kept in check by something the ecologists call the resistance of the environment;" the environment can only support a certain amount of insects because of the availability of food, the weather, and competing insects (247).

Moreover, because of our limited knowledge of pesticides' effects sometimes an insecticide wreaks more havoc on beneficial insects than on the targeted undesirable ones. For example, in 1956 the U. S. Forest Service sprayed 885,000 acres with DDT to control the spruce budworm. But the spray killed the predators such as ladybugs and pirate bugs that kept the spider mite population in balance. After the spraying, there was "the most extensive and spectacular infestation of spider mites in history" with the resulting defoliation of Douglas firs (252-3). Another instance was the fire ant spray program. Spraying to eradicate fire ants did not eliminate them, but it killed the predators of the sugar cane borer, causing serious damage to sugar cane. Yet another example is the story of attempted control of the Japanese beetle through spraying with dieldrin in Illinois. After the spraying, corn borers multiplied. Whereas Japanese beetles caused about ten million dollars of damage, the corn borer damage was estimated at eighty-five million dollars of damage. Carson explains that a beetle called the vedalia imported in 1872 kept scale insects under control and saved the California citrus crop. But DDT spraying wiped out the vedalia and then the scale insect reappeared.

Carson concludes that attempts at chemical control are like getting on a treadmill: "once we have set foot on it we are unable to stop" (257). Chemical companies have invested a great deal of money into the research and development of pesticides. As a result, the majority of economic entomologists have turned from biological control to chemical control.

15. The Rumblings of an Avalanche

> No responsible person contends that insect-borne disease should be ignored (266).

Summary This chapter addresses the concept of insect populations developing resistance to pesticides. Carson begins the chapter with the observation that insect resistance proves Darwin's theories of evolution: the weaker ones succumb to the spray, while the stronger ones survive and multiply. It is the increasing occurrence of resistance that Carson terms an avalanche (borrowing the term from Dr. Charles Elton, one of the founders of animal ecology). Carson explains that development of resistance is especially problematic when trying to control insects that transmit disease. She documents resistance of insect vectors such as malarial mosquitoes and body lice to DDT and chlordane. To control these insects, stronger pesticides were added to the mix.

According to Carson

> in 1956, only five species of [Anopheles] mosquitoes displayed resistance [to DDT]; by early 1960 the number had risen from 5 to 28! The number includes very dangerous malaria vectors in West Africa, the Middle East, Central America, Indonesia, and the eastern European region (269).

Carson explains that resistance occurs through the phenomenon of natural selection: the resistant insects reproduce, generating "a population consisting entirely of tough, resistant strains" (273). Carson was not sure of the exact mechanism by which insects developed this resistance, speculating that it might be an enzyme that helps them detoxify the insecticide, or behavioral traits that help them reduce exposure. Carson ends the chapter with a quote from Dr. C. J. Briejèr, director of Holland's Plant Protection Service: "the resort to weapons such as insecticides to control [nature] is a proof of insufficient knowledge and of an incapacity so to guide the processes of nature that brute force becomes unnecessary. Humbleness is in order; there is no excuse for scientific conceit here" (275). (Please see the chapter "After *Silent Spring*" for a discussion of the use of DDT in malaria control.)

17. The Other Road

> A truly extraordinary variety of alternatives to the chemical control of insects is available (278).

Summary Carson advocates cultural and biological rather than chemical controls to solve problems caused by unwanted insects. She argues that we must choose this road because "the heedless and unrestrained use of chemicals is a greater menace to ourselves than to the targets" (279). This chapter reviews the pros and cons of different methods, including sterilization of male insects, or use of chemical lures that imitate female pheromones and prevent the males from fertilizing females.

Scientists have had some success developing a chemical lure for gypsy moths. For example, "in the laboratory, male gypsy moths have attempted copulation with chips of wood, vermiculite, and other small, inanimate objects" that had been sprayed with the lure (286). Carson reports on testing of ultrasonic sound, which has killed insects in laboratory tests (288).

She concludes with the admonition that

> the "control of nature" is a phrase conceived in arrogance . . . when it was supposed that nature exists for the convenience of man. . . . it is our alarming misfortune that so primitive a science has armed itself with the most modern and terrible weapons, and that in turning them against the insects it has also turned them against the earth (297).

Discussion Who is taking that "other road," the approach of non-chemical pest control? There are some indications that Carson's message is being taken seriously by some. Integrated Pest Management (IPM) is one strategy, combining different methods of dealing with pest problems. Organic farming is another solution. Last week National Public Radio's Living on Earth broadcast a story about one proposed example of IPM. To combat a recurrence of dengue fever in Key West, Florida the local government is seeking Federal approval to introduce genetically engineered sterile male mosquitoes into the environment. Ironically, they are facing a problem that Carson spoke about in a Congressional hearing: it remains unclear which Federal agency has the authority to grant the approval.

(Please see the chapter "After *Silent Spring*" for discussion of pesticide use in agriculture and malaria control.)

TIMELINESS OF THE BOOK

Whereas Carson's first book, *Under the Sea Wind*, had received little attention and sold only 1200 copies at first publication, *Silent Spring* was a great and immediate success. Carson's painstaking care and slowness in writing the book probably contributed in part to its public reception. The 1950s complacency and faith in government and science was being shaken by events that occurred just prior to the book's publication. For the book appeared soon after several widely publicized instances of chemical threats from dangerous drugs or contaminated foods. Radioactive strontium 90 was found in cows' milk, DDT accumulated in breast milk, and the pesticide aminotriazole was found in cranberries just before Thanksgiving. Thalidomide, a drug used to prevent morning sickness in pregnant women, turned out to have the previously unsuspected harmful effect of causing serious birth defects.

Silent Spring resonated with the recent public alarm over the element Strontium 90, one of the radioactive elements generated in atom bomb tests, now found to be present in milk. Cows ate the grass that had been contaminated, and Strontium 90 appeared in their milk. The public was wary: could it be that other chemicals posed similar unexpected dangers? Indeed, Carson addressed the issue of fallout and linked it symbolically to pesticides. Her first chapter speaks of a white granular powder that was connected to the death of songbirds. Ralph H. Lutts argues that Carson convincingly deployed the analogy of pesticides to radioactive fallout, thus linking pesticide contamination to another toxic long-acting and little-understood chemical substance. The analogy of pesticides to bombs had been developed originally by chemical companies to extol the power of DDT to "wage war" on insects. Carson could now utilize this analogy to tap in to the pervasive public apprehension about radioactive poisoning and thus strengthen the impact of her argument (see Lutts in Waddell).

Of even more concern, perhaps, was the discovery that DDT is found in breast milk. "DDT is highly soluble in fat, [does not break down easily in human bodies] and therefore

accumulates in body fat and in milk fat" (Dorothy Read, personal communication). Sandra Steingraber writes

> breast milk is the most contaminated of all human foods. It typically carries concentrations of organochlorine pollutants that are 10 to 20 times higher than those in cow's milk. . . Biomagnification means that breast-fed babies have a greater dietary exposure to toxic chemicals than their parents" (Faith 251).

Steingraber notes that nursing infants are at the top of the food chain. Steve Maguire observes "the symbolism of contaminating breast milk surely works against the chemical; DDT is associated with the defilement of an almost universal symbol of maternal love and comforting, of nourishment for innocent and helpless babies" (Maguire 206).

Also directly relevant was the 1959 "cranberry scare" when residues of the pesticide aminotriazole were found in cranberries just before Thanksgiving. The chemical had been approved for spraying in cranberry bogs after the harvest. However some growers had sprayed the bogs before harvesting the berries, and several lots of cranberries were contaminated. Secretary of Health, Education, and Welfare Arthur S. Flemming ordered that the contaminated fruit be destroyed and announced the decision at a news conference, urging people to refrain from eating cranberries for Thanksgiving that year. These health concerns made people question government reassurances about the safety of various chemicals (Dunlap 107).

Another alarm was the thalidomide tragedy that pointed to the possible dangers lurking in people's medicine cabinets. Thalidomide was a pill intended to prevent morning sickness in pregnancy. Unfortunatelly, it crossed the placental barrier and caused serious birth defects in babies born to European mothers. When women took the pill during the first trimester of pregnancy their babies were born with deformed or missing arms and legs. An estimated 10,000 to 20,000 infants were affected. Dr. Frances Oldham Kelsey, a medical officer newly hired by the FDA, "resisted pressure from the industry and her own bosses to speed approval for" pharmaceutical company Merrell's application to market thalidomide in the U.S. For this she received the President's Award for Distinguished Federal Civilian Service, the second woman to receive "the highest award for a federal employee" (Rossiter 289). Publicity about "her courage in denying the drug certification appeared in July 1962," just as Carson's *Silent Spring* began to be serialized in *The New Yorker*. The work of Carson and Kelsey "reinforced each other in making the public aware of the dangers of uncritical and indiscriminate use of the various postwar 'wonder drugs'" and helping to launch the modern environmental movement (Rossiter 289). As a result of Kelsey's work, stronger regulations for drug testing were approved. Kelsey was seven years younger than Carson, and had been able to avoid the job discrimination women in science usually faced because Dr. Eugene Geiling, a pharmacology Professor at the University of Chicago hired her sight unseen in 1936, misreading her name Frances as a man's Francis. As I am writing this chapter in September, 2010, Dr. Frances Kelsey, now 96 years old, will receive the first Dr. Frances O. Kelsey Award for Excellence and Courage in Protecting Public Health, to be given annually to an FDA staff member (Harris, New York Times 9/14/10 pp. D1, D6). Thalidomide is sometimes prescribed now to treat multiple myeloma, but it is carefully monitored to be sure that women taking this medication are not pregnant, and do not become pregnant while taking this medication (PubMed).

In 1962 Raymond Walters Jr., editor of *The New York Times Book Review,* asked Carson and other authors who were then currently popular to submit a brief statement to "tell what it is about their book or the climate of the times or both that has made their work so popular with Americans." Carson confessed that her book appeared two years later than originally planned. She believed that 1962 was a more propitious year for its appearance because recent events such as the thalidomide tragedy and the concern over nuclear fallout from weapons testing "have brought people to the uneasy realization that mankind is in many respects doing very badly in his self–imposed role of master of this planet." She noted that she received letters showing how the public understands these problems without panic, "but with a firm determination to bring under control the abuses I reported in *Silent Spring*" (YCAL MSS 46 box 74 F 1313).

REASONS FOR THE BOOK'S POPULARITY

How can we account for the immediate success and popularity of such a technical and controversial book? Of course there is no one single answer; multiple factors worked together. Part of the book's notoriety no doubt stemmed from its controversial nature. Ironically, the negative publicity generated by chemical companies to counteract the book had the unintended consequence of helping to publicize the book. Paul Brooks commented "the National Agricultural Chemical Association spent a quarter of a million dollars to discredit the book in the press and on television. The result was more publicity than Houghton Mifflin could possibly have afforded.… The showcase of the campaign was the television commercial featuring a sinister figure in a white lab coat, gloomily recounting the lethal effects of doing without these poisons" (Brooks in Waddell xvi).

What are some other reasons for *Silent Spring*'s meteoric rise to the top of the best-seller list? Carson's "impressive data" and her achievements and skills as a writer undoubtedly were major contributors to the book's success. Other books about environmental dangers (by Theron D. Randolph, Robert Rudd, and Murray Bookchin writing under the pseudonym Lewis Herber) published either slightly before or slightly after Carson's book, did not fare so well (Waddell 11). Another contributing factor was the recent popularity of *The Sea Around Us* and *The Edge of the Sea.* The success of these two books insured that her name would be widely recognized. The Book of the Month Club featured *Silent Spring* as its selection for the month of October. The noted Supreme Court judge, Justice William O Douglas wrote a brief summary in the Book-of-the-Month club's publicity brochure. The Consumer's Union issued a special edition of the book for its subscribers.

The book has remained in print since its first publication, and continues to sell an average of 20,000 copies a year (Doyle).

CHEMICAL COMPANY REACTIONS

Silent Spring generated a controversy. Some physicians, public health officials, and wildlife biologists heralded the book, urged halts on spraying programs and called for further research on DDT and similar pesticides. Thomas Dunlap points to three groups that opposed the book and spoke out against it: the chemical companies that manufactured pesticides; the

economic entomologists who had favored use of pesticides to solve insect problems; and people who believed that Carson's book was anti-science and technology. Chemical companies and economic entomologists saw *Silent Spring* as an attack on them, and responded to defend their products and their expertise, while the third group viewed *Silent Spring* as an attack on science, technology, and progress. One critic argued that Carson's book mandated a return to the dark ages, the giving up of all scientific and technological advances, "the end of all human progress . . . [the onset of] disease, starvation, misery" (Darby, qtd in Dunlap 112-13).

Another group, funded in part by chemical and tobacco companies, consists of members of libertarian think tanks and policy groups such as the Heartland Institute and the Competitive Enterprise Institute that view government regulations as unnecessary constraints on a free market and resist the idea that the free market is unable to prevent "negative externalities" such as environmental pollution (Oreskes and Conway 216-39). Carson's book was one of the impetuses to the switch in orientation of the American environmental movement "from aesthetic environmentalism toward legal regulation" and thus remains a target of those people and organizations that resist regulation. Some in this group continue to argue against Carson's work (for example see Dixy Lee Ray and Louis Guzzo, and the website http://rachelwaswrong.org). For if Carson was wrong then environmentalism is mistaken, and government regulation is excessive and unnecessary (Oreskes 230).

Even before *Silent Spring* appeared chemical companies sought to prevent its publication. They were concerned that publication of Carson's book would trigger more stringent government regulation of pesticides. They took various approaches to block publication.

Velsicol Chemical Corp, manufacturers of the pesticides chlordane and heptachlor, tried to prevent the book's publication, and to stop *The New Yorker* from printing the third installment. In a letter that Houghton Mifflin received on August 2, 1962 they implied that they might bring a lawsuit. (A photocopy of the letter's first page is on the website of Alma College, Alma Michigan (Alma College Web DDT)). The publishing firm released the letter to the *Audubon Magazine* which reprinted portions of it. Velsicol wrote: "The articles as they appear in *The New Yorker* present . . . questions of ethics and morality. . . . Inferences that the chemical industry . . . introduces untried pesticides for the sake of making a quick dollar, are untrue and unfair" (*Audubon Magazine* Nov-Dec 1962 p. 306 CT Coll box 18b f 174). The letter contended that there were

> sinister influences [involved] whose attacks on the chemical industry have a dual purpose (1) to create the false impression that all business is grasping and immoral, and (2) to reduce the use of agricultural chemicals in this country and in the countries of Western Europe so that our supply of food will be reduced to east–curtain parity. Many innocent groups are financed and led into attacks on the chemical industry by these sinister parties (Graham 49).

In other words Velsicol alleged Carson may have been the innocent dupe of a communist conspiracy to undermine the superiority of the Western world, and reduce us to starvation. Houghton Mifflin sent the manuscript out for verification and asked Carson to substantiate her claims. The reviewer's assessment and her proof convinced them that they could win any lawsuit. Milton Greenstein, legal counsel and vice president of *The New Yorker* told

Velsicol "everything in those articles has been checked and is true. Go ahead and sue" (Lear Witness 417 – 18).

When the attempt to prevent the book's publication failed, chemical companies worked to discredit the work and Carson herself by commissioning negative reviews and parodies of the book. But even if these strategies were to fail, the chemical companies consoled themselves that they were not likely to lose much revenue because the public would soon lose interest in the book:

> Industry can take heart from the fact that the main impact of the book will occur in the late fall and winter—seasons when consumers are not normally active buyers of insecticides. . . It is fairly safe to hope that by March or April *Silent Spring* will no longer be an interesting conversational subject (quoted by Carson in *The New Englander* April 1963 YCAL MSS 46 box 97 f 1778).

One of the agricultural chemical company reactions was to raise the threats of disease and famine if their products were not used. Biochemist Thomas H. Jukes, a researcher for Lederle Laboratories, a division of the American Cynamid Chemical Company, wrote a parody of Carson's first chapter "A Fable for Tomorrow." His version turns Carson's vision upside down, for "A Town in Harmony" is found to be at the mercy of bacteria, parasites, and the insects that carry them. According to Jukes, the balance of nature includes malarial mosquitoes and tuburcular cows. Lacking antibiotics, many of the children have died from "diphtheria, scarlet fever, whoping cough," and typhoid.Without pest control the grocery store is infested with weevils, cockroaches, flies and rats. Of course, this parody runs counter to the facts, for Carson asserted that we must use appropriate chemical agents to control disease.

When Carson's book appeared it generated great attention. At a meeting of The Federal Pest Control Review Board one board member remarked "I thought she was a spinster. What's she's so worried about genetics for?" (Graham 50). Graham cites a newspaper article describing the reactions of Farm Bureau personnel in two counties in Pennsylvania: "no one in either County farm office who was talked to today had read the book, but all disapproved of it heartily" (Graham 48). Newspapers throughout the U. S. published editorials and letters to the editor about the book. Many ran series of articles debating the issues raised by the book.

General critiques of *Silent Spring* by chemical companies complained that Carson's book is unbalanced because it reflects only one side of the pesticide story, the negative side, and does not portray all of the benefits of pesticides. They alleged that it contains inaccuracies of fact such as specifics of bird counts, or the amount of land affected by chemical spraying. They complained that Carson quotes from personal correspondence with scientists, rather than from published essays. (Scientific publications are usually reviewed by experts, a vetting process that takes time. Carson's information obtained from personal contacts and correspondence with experts in the field was often more current than published essays could be, since publication involves delays.) They argued that the balance of nature was already upset when the first field was plowed; they claimed that the government regulations already guarantee the safety of pesticides. They asserted that chemicals have done more good than harm; without pesticides we would have "natural contaminants" such as rat feces in our foods—and we would not have enough food to feed everyone. They made nasty ad hominem remarks, accusing Carson of being one of those "organic gardeners, bird lovers,

food faddists" and other "nature lovers" (as if those were bad or strange types of people). They argued that the book will scare ordinary people who don't have the scientific knowledge to understand the truth. They claimed that without chemicals and pesticides we will revert to a primitive state and will starve.

Monsanto Magazine published a take-off on Carson's book called "The Desolate Year" in October 1963. The essay contended that in a year without pesticide spraying, "the insect hosts descended in earnest." Cattle grubs, ticks, caterpillars, mosquitoes, gnats, and flies invaded; malaria returned; grasshoppers attacked alfalfa, soybeans, and other vegetables, with devastation of tomato, pepper, bean, corn, and cucumber crops (YCAL MSS 46 box 66 f 1190). Some chemical company magazines featured cartoons lampooning the book. *The Progressive Farmer* published a cartoon "Let's Not Upset the Balance of Nature" showing a woman sitting on a stool reading *Silent Spring*, oblivious to the bugs and mice that are cavorting happily in her kitchen, opening cupboards and boxes of food (December 1962 issue vol 21 #2, 10 CT Coll box 18b f 146).

Another chemical company strategy was to publicize the benefits of pesticides. Velsicol published *The Necessity, Value and Safety of Pesticides* by Louis A. McLean (secretary, Velsicol Chemical Corp, Chicago, Illinois), a 19 page pamphlet, undated. McLean alleges: "The superstitious and the impractical, giving voice to their desires to escape from reality, have repeatedly misasserted a benevolence in nature and have stated that man's efforts to bend nature to better serve man are themselves evil. These hermit-philosophies, because they are blind to reality, historically have led to abortive communal attempts to live with nature by those who had no real intent to reduce their standard of living to the animal-like level dictated by those philosophies—the level we see in the world's 'undeveloped' areas." The essay agrees that misuse of pesticides may cause problems; but points out that overdose of aspirin also kills people.

McLean is calling people who believe in the inherent goodness of nature ignorant and hypocritical. They seem to be hippies who claim they would like to live in accordance with nature, but they really would not like to do so, for it would "reduce their standard of living" to that of animals. It is interesting to contrast the language of this pamphlet to the clarity and readability of Carson's prose.

The *News and Pesticide Review* published by the National Agricultural Chemicals Association featured a guest editorial extolling the benefits that chemicals have bestowed on humans. Dr. Robert H. White-Stevens Assistant to the Director of Research of the American Cyanamid Company, wrote "What We Should Tell Others:"

> DDT alone has saved as many human lives over the past 15 years as all the wonder drugs combined. . . . This is the one moment in history when world hunger can be vanquished. . . . Our knowledge and control of the chemistry and function of the pesticides and additives we use vastly exceeds that of the natural compounds which invariably contaminate our food supply when it is unprotected. . . . The progress of a people is inversely proportional to the time and effort required to produce the necessaries of existence (October 1962 vol 21 #1 pages 2, 7 in CT Coll box 18B f 146).

White-Stevens appeared in person, wearing a white lab coat, in a CBS TV documentary *The Silent Spring of Rachel Carson.*

REVIEWS OF THE BOOK

Negative Reviews

Chemical companies and other advocates of pesticide use wished to discredit Carson and her findings. Frank Graham, Jr. excerpts some of these negative reviews. F. A. Soraci, director of the New Jersey Department of Agriculture, wrote in the *Conservation News*

> in any large-scale pest control program we are immediately confronted with the objection of a vociferous, misinformed group of nature–balancing, organic–gardening, bird–loving, unreasonable citizenry that has not been convinced of the important place of agricultural chemicals in our economy (Graham 55).

James Westman, Chairman of the Department of Wildlife Conservation at Rutgers University, quoted an old article explaining that DDT is strong medicine, "potentially dangerous to all forms of life. So is alcohol, "but we now know how to handle it." P. Rothberg, president of the Montrose Chemical Corporation of California, wrote that Carson was "a fanatic defender of the cult of the balance of nature." William J Darby, a nutritionist at the Vanderbilt University School of Medicine wrote in the *Chemical and Engineering News* that Carson's book mandated a return to the dark ages and "the end of all human progress. . . . She indicates that it is neither wise nor responsible to use pesticides in the control of insect–borne diseases." Carson, of course, had never suggested that pesticides not be used, only that they be used responsibly and only if less damaging alternatives had failed to solve the problem. Graham observes, "Dr. Darby, then, like most of the book's critics, made a great show of refuting statements that Rachel Carson had never made" (Graham 55 – 57).

One of the letters that *The New Yorker* received but did not publish until years later in its seventieth anniversary issue in February 1995, encompasses some of the anti-Communist, misogynistic, pro-business critiques being voiced:

> Miss Rachel Carson's reference to the selfishness of insecticide manufacturers probably reflects her communist sympathies, like a lot of our writers these days.We can live without birds and animals, but, as the current market slump shows, we cannot live without business.
>
> As for insects, isn't it just like a woman to be scared to death of a few little bugs! As long as we have the H-bomb everything will be okay PS. She's probably a peace-nut too.

However, given that *The New Yorker* readers tend to be a fairly liberal and sophisticated audience, I wonder if this letter was perhaps intended to satirize these viewpoints.

Shortly after Carson's death, *Time* magazine wrote

> Her critics, who included many eminent scientists, objected that the book's exaggerations and emotional tone played on the vague fears of city dwellers, the bulk of the U.S. population, who have little contact with uncontrolled nature and do not know how unpleasantly hostile it generally is (April 24, 1964).

This piece, like many critical of Carson, employs the rhetoric of human dominance over nature, asserting that “uncontrolled nature” is “unpleasantly hostile,” and, by implication, must be tamed and controlled. While accusing Carson of “playing on [people’s] fears,” *Time* is also stirring up fears, in this case fears of nature’s hostility. The article goes on to acknowledge the popularity of *Silent Spring,* but warns that “advanced modern societies” require pesticides.

> Nonetheless, *Silent Spring* was a runaway bestseller and an extremely effective polemic that stirred fierce arguments, from village councils to the halls of Congress. . . . In advanced modern societies, agriculture and public health can no longer manage without chemical pesticides" (April 24, 1964).

Edwin Diamond, the man whose proposed collaboration with Carson had ended with hard feelings, wrote a hostile review, "The Myth of the 'Pesticide Menace'" in the *Saturday Evening Post*. He charged that the book was emotional and claimed that although her facts were accurate, she had left out facts. Diamond cited several explanations for the popularity of Carson's book: her literary style, the timing of the book, and "the attention-getting quality inherent in any exaggeration." But he imagines that the most salient reason for the book's popularity is its appeal to "the latent demons of paranoia" from which many people suffer. One of the facts he took issue with was Carson's statement about a rain of pesticides. He quoted a review by Prof. I. L. Baldwin of the University of Wisconsin (the same Baldwin who had chaired the subcommittee of The National Association of Science that wrote a report favorable to pesticides) that supposedly refuted Carson’s claims of the extent of pesticide spraying (October 1963 pp 16-18). Baldwin had stated that "less than five percent of all the area of the United States is annually treated with insecticides" (*Science*). However, statements such as Baldwin's fail to take into account the fact that most of the pesticides sprayed on an area of ground do not land exactly where they were originally sprayed. Wind and water carry the spray to other places, and sprays may also be transported by plants or animals on which they have landed. In fact, when pesticides are sprayed, only a miniscule percentage actually reaches the target. And, as we now know, persistent organic pesticides have become part of the body burden of every human being.

Positive reviews

Craig Waddell read “approximately six hundred reviews of *Silent Spring* included in the Carson papers” at Yale University’s Beinecke Library. He found that many used superlatives and that they praised the book for a combination of factors including her “impressive data, . . . literary grace, . . . moderation and realism, . . . concern for human health” (Waddell 10).

Esquire Nov 1984 published an essay on Rachel Carson “Fifty Who Made the Difference” reprinted in Chatham Alumnae Magazine *the Recorder* spring 1985 as “The Flight of Rachel Carson” by Geoffrey Norman pp 4–8.

> What Carson did in *Silent Spring* was to introduce . . . the concept of ecology: the way the natural world fit together, the pieces so tightly and inextricably bound that you could not isolate cause and effect. The consequences of any action rippled through the whole system, affecting everything and sometimes even changing the system itself. So

> when we poisoned gypsy moths with massive sprayings of DDT, we were, ultimately, poisoning ourselves." (6) (CT Coll box 2 f 6)

The New Yorker assembled a scrapbook of reviews relating to its publication of the three articles and donated the scrapbook to the Beinecke Library in December, 1990. One review published in *The New York Times* July 2, 1962 read:

> Miss Carson will be accused of alarmism, or a lack of objectivity, showing only the bad side of pesticides while ignoring their benefits. But this, we suspect is her purpose as well as her method. We do not combat highway carelessness by reciting statistics only of the millions of motorists who return safely to their garages. . . . Miss Carson does not argue that chemical insecticides must never be used, but she warns of the dangers of misuse and overuse by a public that has become mesmerized by the notion that chemists are the possessors of divine wisdom and that nothing but benefit can emerge from their test tubes. If her series helps arouse enough public concern to immunize government agencies against the blandishments of the hucksters and [convince them to] enforce adequate controls, the author will be as deserving of the Nobel Prize as the inventor of DDT (New York Times 7/2/62 YCAL MSS 46 box 116).

In the *NY Times Book Review* Lorus and Margery Milne wrote "There's Poison all Around Us Now." They note that the dangers in the use of pesticides are vividly pictured by Rachel Carson. The review concludes: "It is high time for people to know about these rapid changes in their environment, and to take an effective part in the battle that may shape the future of all life on earth" (September 23, 1962 (section 7) pp. 1, 26).

CBS REPORTS

CBS Reports prepared an hour-long documentary about the pesticide issue. Because Carson was too ill to travel to the studio, host of the program Eric Sevareid visited her in her home to tape the interview. On the eve of its broadcast, three of the five sponsors of the program backed out, leaving the Kiwi Polish Company and the Brillo Manufacturing Company as the remaining sponsors. The three sponsors who withdrew were Lehn and Fink, manufacturers of cosmetics, and disinfectants; Standard Brands, manufacturers of liquor and foods; and Ralston Purina, manufacturers of livestock feed and food products.

Addressing the issue of sponsor withdrawal, one journalist wrote:

> Developments such as this point up the weakness of the sponsor system. The ideal program from the sponsor's point of view is a pleasant story about pleasant people with pleasant problems. . . Sponsors think controversy is unhealthy—for their business anyway. . . .Lehn and Fink's marketing people say that 'the show's audience wouldn't be the right one for our products.' I don't see why people who use these products wouldn't be interested in the program" (Warner Twyford, in the Norfolk VA *Virginia Pilot* April 3, 1963).

The program, "The Silent Spring of Rachel Carson" was broadcast on CBS on Wednesday, April 3, 1963 from 7:30 to 8:30 PM, Eastern Standard Time. Carson, in an advanced stage of breast cancer, wore a wig to cover up hair loss. She spoke simply and

calmly, whereas White-Stevens looming large in his white lab coat, was "overbearing" (McGillivray 71). Remembering the show forty-three years later, Carson's great-nephew Roger Christie commented "He would have made a great villain in a Bela Lugosi movie" (Johnson Legacy Web).

The show aired footage of aerial spraying, and of spray trucks driving through city streets followed by children running through the DDT mist. One scene displays students in a school cafeteria being sprayed with clouds of DDT.

Carson explained:

> These sprays, dusts and aerosols are now applied almost universally to farms, gardens, forests and homes — non-selective chemicals that have the power to kill every insect, the good and the bad, to still the song of birds and the leaping of fish in streams, to coat the leaves with a deadly film and to linger on in soil. . . . All this, though the intended target may be only a few weeds or insects.

The interviewers sought a balanced report, questioning Carson, government agents, and White- Stevens as the chemical company representative. "When CBS turned to government experts, the questions were many, but the answers few. Dr. Page Nicholson, water pollution expert, Public Health Service, wasn't able to answer how long pesticides persist in water once they enter or the extent to which pesticides contaminated groundwater supplies" (Johnson, Legacy). The government representatives were "ill informed, unconcerned, and evasive" ((McGillivray 71).

> They virtually made Carson's case for her. More befuddled than enlightened or enlightening, few of them expressed any sense of the potential hazards chemicals posed or of any plans to subject them to closer regulation or testing. Worse yet, they seemed unconcerned about the threat to human health or the environment. But if anyone truly frightened the public, it was Dr. White-Stevens. On television his assured tone came across as arrogant, his suave manner as oily. In ominous tones he described the plagues awaiting a world without chemical pesticides (Lytle 182).

Robert White-Stevens alleged

> The major claims in Miss Rachel Carson's book, *Silent Spring*, are gross distortions of the actual facts, completely unsupported by scientific experimental evidence and general practical experience in the field. . . . The real threat to the survival of man is not chemical but biological, in the shape of hordes of insects that can denude our forests, sweep over our crop lands, ravage our food supply and leave in their wake a train of destitution and hunger, conveying to an undernourished population the major diseases and scourges of mankind. If man were to faithfully follow the teachings of Miss Carson we would return to the Dark Ages and the insects and diseases and vermin would once again inherit the earth (transcript pp. 1–2). . . .Miss Carson maintains that the balance of nature is a major force in the survival of man; whereas the modern chemist, the modern biologist, the modern scientist, believes that man is steadily controlling nature, that he has already disrupted the balance of nature (transcript 30).

Rachel Carson countered:

> Now to these people apparently, the balance of nature was something that was repealed as soon as man came on the scene. . . .You might . . . as well assume that you could repeal the law of gravity. . . . . Now this doesn't mean . . . that we must not attempt to tilt that balance of nature in our favor; but when we do . . . we must know what we're doing, we must know the consequences" (transcript 30).

White-Stevens argued "If we had to investigate every possibility, we would never make any advances at all, because this would require an infinite time for experimental work" (transcript 31).

Carson had the last word in the program, noting:

> We still talk in terms of conquest. We still haven't become mature enough to think of ourselves as only a very tiny part of a vast and incredible universe. . . . Man is part of nature, and his war against nature is inevitably a war against himself. . . . We in this generation must come to terms with nature, and I think we're challenged . . . to prove our maturity and our mastery, not of nature, but of ourselves" (transcript 31).

An estimated 10 to 15 million people watched the program. The very next day Connecticut Senator Abraham Ribicoff began to set up a congressional committee to investigate pesticides, and invited Carson to testify at the hearings.

On May 15, 1963, the first day of the Ribicoff hearings, the Life Sciences Panel of the President's Science Advisory Committee (PSAC) issued a 43 typed page report "Use of Pesticides." They declared "the gypsy moth, fire ant, Japanese beetle, and white-fringed beetle programs, which have been continued for years, are examples of failures of the 'eradication' approach" (34). The committee advocated "a philosophy of control rather than eradication."

The committee explained that different government agencies were responsible for different aspects of pesticide regulation. The United States Department of Agriculture (USDA) is responsible for insuring that the marketed pesticides are properly labeled. The Food and Drug Administration (FDA) is responsible for insuring that tolerances are not exceeded. The panel found that not much research has been done on pesticide toxicity in humans and that decisions on safety are not as well based as those on efficacy. Until 1954 the evidence of safety was submitted in the form of testimony at public hearings. Amendments to the Federal Insecticide, Fungicide, Rodenticide Act (FIFRA) in 1954 improved these procedures. Another issue that the panel found was that federal regulations controlled food sold in interstate commerce, but food sold within the individual states was not well regulated, and higher pesticide residues were found in "three percent of the fresh fruits and vegetables offered for sale" within states. The panel made several recommendations, all of which were in line with Carson's ideas. It urged better cooperation among government agencies, better systems for monitoring spraying programs, more investigation of pesticide toxicity, and a research program to find nonchemical, as well as more selective and non-persistent chemicals to control unwanted insect pests. Even more gratifying to Carson and her friends was the comment that "until the publication of *Silent Spring*, people were generally unaware of the toxicity of pesticides" (YCAL MSS 46 box 74 f 1322).

When the report was released newspapers proclaimed that the panel proved that Carson was correct: the *Christian Science Monitor* headlined the May 15, 1963 issue in bold uppercase letters "Rachel Carson Stands Vindicated." CBS broadcast a program about the pesticide report called "The Verdict on The Silent Spring of Rachel Carson." It used some of the footage of its previous broadcast, and quoted Carson praising the report, especially "the fact that the public is entitled to the facts, which after all, was my reason for writing *Silent Spring"* (Lear Witness 452).

CARSON'S TESTIMONY IN CONGRESS

The immediate impact of *Silent Spring* was governmental attention to the pesticide issue, resulting in two separate Congressional inquiries in June. Despite her failing health Carson was able to testify twice before Congress. When she appeared before the Government Operations Subcommittee on June 4, 1963, Abraham Ribicoff, echoing Abraham Lincoln's comment about Harriet Beecher Stowe, introduced Carson as "the lady who started all this." During her forty minutes of testifying Carson described the recently accumulating evidence of pesticide pollution, and reiterated her point that freedom from poisons "is or should be one of the basic human rights."

Her second appearance before Congress on June 6 was to testify before the Senate Committee on Commerce. Senator Maurine Neuberger from Oregon had met Carson at a pre-publication lunch and learned about the problems Carson addressed. As a result, Neuberger proposed legislation that would require Federal agencies to consult with the Fish & Wildlife Service and with state wildlife agencies before beginning any Federal program "involving the use of pesticides or other chemicals designed for mass biological controls." Carson quickly established her professional credentials as "one who has had some 16 years' experience as a government biologist." She explained about the concentration of pesticides as they travel up the food chain, and told the stories of Clear Lake and Big Bear Lake in California. She recommended that there be a government pesticide commission made up of scientific experts outside of government and the chemical industries, in order that there be no conflict of interest. Another bill discussed at that hearing, S1251, mandates the Department of the Interior to conduct a program of evaluating biocides to determine their impact on wildlife and fish, and to publish the data it gathers. Carson explained that "this matter of conflicting interests, and of conflicting governmental mandates, lies at the heart of the problem this legislation is designed to solve." She expressed her insight about the interconnectedness of ecosystems:

> The application of pesticides is never a simple matter, involving only the chemical and the target organism.… Existing procedures… too often seem to assume such a simple relationship [whereas] a variety of interests are affected… such as the pollution of soil, water, air, and food products; protection of public health; and preservation of wildlife and fisheries (Pesticide Research and Controls p. 18).

She made six recommendations including requests for research and education in the field of pesticides, and legislation restricting the sale and use of pesticides to those capable of understanding the hazards. She felt that more agencies should be involved in the registration of chemicals because household chemicals also relate to health: therefore Health, Education and Welfare as well as the Department of Interior "should have a voice in the registration

and labeling of such chemicals." She asked for research "on new methods of pest control in which chemicals will be minimized or entirely eliminated." And she repeated again her belief in "the right of the citizen to be secure in his own home against the intrusion of poisons applied by other persons," and added "I speak not as a lawyer but as a biologist and human being" (YCAL MSS 46 box 73 f 1294).

RELATED SPEECHES AND ARTICLES: CONTEXTS FOR THE PESTICIDE ISSUE

Carson wrote an article for *The New Englander* documenting some of the detrimental effects of pesticides on wildlife such as the decline in eagle populations, and the fish in a Framingham MA reservoir which were found to have high levels of DDT. She places the problem in a larger context, expressing concern over the growing power and political influence of the chemical industry. She cites an income tax bill passed by the 87th Congress that permits "certain lobbying expenses to be considered a business expense deduction." She predicted that this provision would enable the chemical companies to "work at bargain rates to thwart future attempts at regulation," while non-profit organizations "stand to lose their tax-exempt status if they devote any 'substantial' part of their activities to attempts to influence legislation." She asked "what happens, then, when the public interest is pitted against large commercial interests?" Carson also cautioned about the increasing linkages "between professional organizations and industry, and between science and industry." For example, the American Medical Association (AMA) was referring physicians to a pesticide trade organization for information about the effects of pesticides. Carson would prefer a reference "to authoritative science or medical literature, not to a trade organization whose business it is to promote the sale of pesticides." She cited another cause for concern, "the increasing size and number of industry grants to universities. . . . Support of education seems desirable, but on reflection we see that this does not make for unbiased research." She concluded with a warning: "As you listen to the present controversy regarding pesticides, I recommend that you ask yourself, Who speaks? and why?" (38) ("Rachel Carson, author of the best-selling *Silent Spring* answers her critics" *The New Englander* April 1963 pp 13, 36-38 Reprinted in *Audubon Mag* Sep-Oct 1963; YCAL MSS 46 box 97 f 1778). Carson was wise to see these evolving trends, and we see the implications of them as they are currently being played out.

Although the pesticide industry took great issue with Carson's *Silent Spring*, many organizations recognized the importance and value of her work. For example, she received the Cullum medal of the American Geographic Society. (For excerpts from and discussion of Carson's awards please see chapter 4, "Other Writings.")

CARSON'S RHETORICAL STRATEGIES

Rhetorical analyses of *Silent Spring* lay bare the strategies that shaped the text. As previously noted, Carson was a careful researcher and writer. Because of the difficult subject matter, she particularly strove to make this book readable and accurate. She sent chapters out for review by experts in the fields, and she revised and rewrote repeatedly. What she achieved was a blend of scientific fact and striking anecdotes about real situations and ordinary people.

How could Carson capture and hold the attention of the non-scientist reader? As she wrestled with that question, Carson wrote and re-wrote the first chapter, the parable "A Fable for Tomorrow." Christine Oravec reviews Carson's working drafts of the first chapter, from its origin to the final version. First called "The Rain of Death," it was intended to be a summary of the book. In early versions Carson wrote the story as a narrative of a man returning to his home town and finding that it had been devastated. At some point she considered giving the town the name Green Meadows. However, leaving the town nameless, and eliminating the man turns the story into a more mythic, symbolic, one. The town now became Everywhere. Oravec points out that Carson continued her editing and revising even at the final stages of correcting the galley proofs (Oravec 42-59).

Another device Carson uses is her naming of the victims and the perpetrators. While she was careful not to name the names of particular companies or brands of pesticides, so as not to be liable for a lawsuit, she does make clear that chemical companies are the villains in her book, and ordinary citizens are the victims. Carson refers to the chemical companies, the scientists, and the government agents using the male pronoun "he." While this was the standard use for generic references to people at the time she wrote the book, nevertheless, it is clearly a predominantly male establishment that is responsible for the harm. In contrast, the people whose words she quotes approvingly are the "housewives" (her term), the women who care for the homes and the families that are being unwittingly, and unwillingly, targeted and harmed by the sprays.

Discussing Carson's distinctive style, Carol B. Gartner writes

> she begins each chapter with a strong thesis, which she gradually elaborates, moving with tight organization from general statements to supporting statements to specific details.… She uses the connotations of words, as well as their definitions, to build tone and ultimate meaning. . . . That Carson presented her material with scientific accuracy has been repeatedly validated, but she does use her own emotions – anger, bitterness – and occasional sensationalism to develop tone and thus appeal to the reader's emotions as well as intellect (Gartner 114 – 116).

As has been noted, Carson wrote the book at the height of the Cold War, when people feared a nuclear holocaust, and when the rhetoric of war was common, painting a conflict between two sharply demarcated ways of life, communism / totalitarianism and capitalism / democracy. The military metaphor was pervasive: entomologists, chemical companies, and public agencies conjured up images of a war on insects. Cheryll Gloltfelty notes that Carson did not deconstruct the war rhetoric, but used the metaphor of war, and took the role of "a smart general," suggesting better strategies of insect control (160). "*Silent Spring* likewise creates a bipolar, melodramatic picture, with the pesticide industry and its henchmen in the Department of Agriculture on one side; Carson and a few heroic biologists and concerned citizens on the other; and with fainting nature and the unsuspecting American public costarring as damsels in distress" (163). Whereas Gloltfelty wishes that Carson had changed the rhetorical framework, Christine Oravec sets the work into its temporal context: "Carson was writing in an era significantly different from our own: the environmental crisis was less perceptually apparent; there was a much greater faith in science as a benign enterprise; and almost no generic precedents existed, except perhaps that of the muckraking journalists at the turn of the century. Perhaps these situational factors account for the urgency, even the stridency, of her chosen format and style" (55).

Carson's most daring and dramatic strategy was to write from her perspective of the interconnectedness of the entire web of life, from the soil to the water, to the air, from the amoeba to the earthworm, to the sparrow, to the human. In her view humans are not objective observers, but are enmeshed in nature's totality. She "contextualized her alarming facts, figures, and rhetoric in a more fundamental reality [than previous nature writers and ecologists had invoked], one that insists that humans are not masters of the universe but integral elements of an ecosystem. It was a profound and humbling corrective, and if readers were willing to accept it, their worldview was bound to change radically" (McWilliams 197). Apparently many readers were willing to accept this viewpoint, and to recognize their integral connections to the ecosystem.

In any event "much of *Silent Spring*'s vocabulary has been incorporated into mainstream environmental-science education" (Peterson 72). Peterson and Peterson find that standard ecology textbooks have devoted more space to discussion of pesticides since Carson's book was published.

> Carson is able to offer a surprisingly optimistic vision in the face of such destruction because she portrays most human supporters of chemical warfare as ignorant rather than inherently evil. She writes that the unsuspecting public has been subjected to 'tranquilizing pills of half-truth.'… Ignorance can be remediated with knowledge, although such a process is rarely easy or pleasant'' (Peterson 80).

Al Gore writes that his mother read *Silent Spring* to her family at the dinner table, and this is what "first drew my attention to the environment" (McKibben xvii). The book "made an unmistakable impression" on him, and brought him to realize the links between government and the environment, and that "human actions could affect the ecosystems" (McKibben xvii).

> In 1969 the state of Michigan banned DDT from agricultural use. *The New York Times* magazine printed an obituary for DDT:

> Died, DDT, at age 95, a persistent pesticide and one time humanitarian. Considered to be one of World War II's greatest heroes, DDT saw its reputation fade after it was charged with murder by author Rachel Carson. Death came on June 27 in Michigan after a lingering illness. Survived by dieldrin, aldrin, endrin, chlordane, heptachlor, lindane and toxaphene. Please omit flowers (Hal Higdon, obituary for DDT, New York Times Magazine, July 6, 1969, 6 quoted in Maguire "Icons" 199).

The Environmental Protection Agency banned DDT from domestic use on June 14, 1972. Wildavsky details the steps in the process that led to the ban (77-78).

Silent Spring changed Americans' ideas about our relationship to the environment, and gave ecology a political basis (Hynes 14). The book exposed the underlying tensions of Cold War science, and brought a new voice into the public arena, changing the concept of who could speak about science and pointing out the links between science and public policy.

According to Edward O. Wilson, before Carson published *Silent Spring:*

> ecology was near the bottom of the scientific disciplines in prestige and support; few Americans even knew what the word meant. Conservation biology, later to become one of the most rapidly growing disciplines, did not exist... The environment was also excluded from the mainstream political agenda.... We rewarded science and

> technology with high esteem.... Environmental warnings were treated with irritable impatience... Arguments for limits and constraint seemed almost unpatriotic ("on *Silent Spring*" in Matthiessen, ed 28-29).

Wilson notes that Carson's book prompted the US "to turn away from wholesale toxic pollution," and to become more interested in "conservation of natural environments" (in Matthiessen 32- 33). The new interest in the environment prompted the passage of the Endangered Species Act by a near unanimous vote in Congress in 1973.

Carson gave new power to the "soft" science of ecology whereas the "hard" science of industrial chemistry had previously been dominant. Carson broadened the usage of the term ecology, and introduced ecology into the public discourse over environmental health. "Carson's ecological perspective provided a language to explain and articulate the sterile landscapes of Cold War America" (Kinkela 118, 121).

For many, the book posed a threat not only to the power and position of the chemical industry, but "to the nation as well," for the chemical companies had seemed to be at the forefront of progress and prosperity. Ecology now appeared to be "a subversive science" (Kinkela121).

> Ecology became the rallying cry of the modern American environmental movement, tying together the concepts of environmental protection and ecological science that were not necessarily, or historically, connected. . . . Therefore, ecology functioned within two realms, the scientific . . . and the popular. The popular movement reconnected humans to the natural world and encouraged them to question the authority of the expert (Kinkela 122).

In *Silent Spring* Carson "succeeded in making a book about death a celebration of life" (Brooks Foreword Silent Spring xiii). The book opened the eyes of Americans to the dangers of excessive pesticide use. It challenged government policies that responded too quickly to perceived dangers without adequate research into causes and effects. It challenged the belief that science is objective, benign, and free of errors. It engendered a shift in environmental thinking from preservation and conservation of individual species and local places to a broader view of an ecological whole. It politicized the environmental movement and changed its strategies to activism and to seeking regulation to control technology and remediation of ills. Chiefly, the book made Americans aware of their place as integral components of the natural world. The book is a classic of environmental thought that has retained its value and remained in print since its original publication.

NOTES

1. Much of the information in this section comes from the study by Thomas R. Dunlap. *DDT: Scientists, Citizens, and Public Policy*. Princeton, NJ: Princeton University Press, 1981. Print.

CHAPTER 6

AFTER *SILENT SPRING*: THE LEGACIES OF RACHEL CARSON

> On Friday October 26, 2012 the American Chemical Society, Chatham University and others honored *Silent Spring* as a National Historic Chemical Landmark

> The other fork of the road—the one "less traveled by"—offers, our last, our only chance to reach a destination that assures the preservation of our earth (Carson Silent Spring).

> Those who dwell . . . among the beauties and mysteries of the earth are never alone or weary of life. . . .Those who contemplate the beauty of the earth find reserves of strength that will endure as long as life lasts (Carson Wonder).

Rachel Carson's legacies are far-reaching. The controversies her work generated continue to simmer. Her work has numerous outcomes and manifestations that branch out like the spreading branches of a tree, influencing, challenging, and inspiring many people, agencies, and institutions. This chapter can barely begin to point to some of the branches. Chapter 8 contains a list of resources for those who wish to learn more about the topics broached here.

POSTHUMOUS AWARDS AND HONORS FOR CARSON

Awards and honors continued to accrue to Carson after her death. A National Wildlife Refuge along the Maine coast has been set aside in her name, thus both honoring her and preserving some of the Maine coast that she so dearly loved. At the dedication of the Rachel Carson National Wildlife Refuge on June 27, 1970, Secretary of the Interior Walter Hickel remarked, "She never bore children, but she was the mother of the Age of Ecology" (YCAL MSS 46 box 112 F 2188). According to the Fish and Wildlife Service:

> Rachel Carson National Wildlife Refuge was established in 1966 in cooperation with the State of Maine to protect valuable salt marshes and estuaries for migratory birds. Located along 50 miles of coastline in York and Cumberland counties, the refuge consists of eleven divisions between Kittery and Cape Elizabeth. It will contain approximately 14,600 acres when land acquisition is complete. The proximity of the refuge to the coast and its location between the eastern deciduous forest and the boreal forest creates a composition of plants and animals not found elsewhere in Maine. Major habitat types present on the refuge include forested upland, barrier beach/dune, coastal meadows, tidal salt marsh, and the distinctive rocky coast (U.S. Fish and Wildlife Service).

President Jimmy Carter awarded Carson posthumously the Presidential Medal of Freedom, the highest honor given to a civilian citizen of the U.S., in 1980 with these words, borrowing some of her language in awarding the honor:

> Never silent herself in the face of destructive trends, Rachel Carson fed a spring of awareness across America and beyond. She welcomed her audiences to her love of the sea, while with an equally clear voice she warned Americans of the danger human beings themselves pose for their own environment. Always concerned, always eloquent, she created a tide of environmental consciousness that has not ebbed (Pennsylvania Department of Environmental Protection Web).

In 1991 the US government issued a Rachel Carson postage stamp. Interestingly, other countries such as Palau, the Marshall Islands, and Zambia also have Rachel Carson postage stamps.

WOMEN MARINE BIOLOGISTS AND OCEANOGRAPHERS

Carson was one of the early women scientists to enter the male-dominated field of science, and of oceanography in particular. She was heroic in standing firm for her principles and challenging powerful opponents as she worked to defend the natural world, including humans, from one threat, the overuse of persistent organic pesticides. As an early woman marine biologist Carson demonstrated that women could be practicing professional scientists. Many women are following in her footsteps in the field of oceanography. Notable are Eugenie Clark, Kathleen Crane and Sylvia Earle. Clark and Earle are both accomplished divers, researchers, and writers of books about the sea for both adults and children.

Ichthyologist Eugenie Clark is known as the shark lady, because of her research with these voracious fish. Now at age 87 she is still scuba diving and traveling. She is the founding director of the Mote Marine Laboratory (established in 1955 as the Cape Haze Marine Laboratory), home of the Dr. Eugenie Clark Chair for Scientific Research established to provide the opportunity for young scientists to follow in her footsteps.

Sylvia Earle, winner of the 2009 TED Prize, made important studies of aquatic plant life. In 1969 she applied to participate in the Tektite project, sponsored by the U.S. Navy, the Department of the Interior, and the National Aeronautics and Space Administration (NASA). Scientists lived for weeks in an enclosed habitat on the ocean floor 50 feet below the surface, off the Virgin Islands. But the organizers were reluctant to admit a woman to live in a group of male scientists. Determined to participate in such a project, Earle led her own all-woman underwater expedition for two weeks. Since then Earle has pioneered in many important research projects, and worked with an engineer to design and build undersea exploration vehicles. She served for a time as Chief Scientist of the National Oceanographic and Atmospheric Administration. “There, among other duties, Sylvia Earle was responsible for monitoring the health of the nation's waters.” The TED Prize provided funding and publicity for her project to "establish a global network of marine protected areas" (Earle, and American Academy of Achievement). On July 17, 2012 Earle was featured on a National Public Radio program, *Morning Edition.* At age 76 she is participating in a “saturation dive” project at the U.S. Government undersea Aquarius Reef

Base off the Florida Keys. Earle remarked that robots and sophisticated technology have a place in underwater exploration, but "you can't surprise a machine" (NPR). The whole point of exploration is to find the unexpected. Aquarius Reef Base is the last remaining undersea research base, and government funding has been eliminated. Earle and other colleagues are seeking other funding sources to maintain the base.

Oceanographer Kathleen Crane developed maps that helped lead the expedition that discovered the remains of the Titanic (Sara MacSorley personal communication)

> Kathleen Crane is the U.S. mission coordinator for the RUSALCA Expedition [Russian-American Long-term Census of the Arctic]. She serves as program manager in the NOAA Arctic Research Office. In 1977, she received her PhD from Scripps Institution of Oceanography, where she took part in discovering the Galapagos hydrothermal vents. Dr. Crane later moved to Woods Hole Oceanographic Institution to study mid-ocean ridges. . . . Dr. Crane has been a visiting scientist at numerous institutions, including the University of California, Santa Barbara; the University of Hawaii; the University of Oslo, Norway; the University of Paris, France; and at the Environmental Defense Fund, where she helped to develop the Arctic At Risk Program. She has helped coordinate 18 international expeditions (with Russia, Japan, France, Italy, Norway, Sweden, Denmark, Germany, and Canada); has participated in 38 expeditions at sea; and has been the chief scientist of 18 (NOAA Ocean Explorer Web).

Science writers also follow in Carson's footsteps. Chapter 8 lists information about many of them, such as Annie Dillard, Elizabeth Kolbert, Mary Oliver, Sandra Steingraber, and Terry Tempest Williams.

CONTINUING CONTROVERSIES

This chapter will glance briefly at some of the outcomes of Carson's work, and point to some of the continuing controversies.

Carson's *Silent Spring* was a wake-up call reminding us that all of planet earth is an interconnected system, and damage to any one part may cause imbalance, perhaps even damage, to the entire delicate web of life. The book challenged the chemical industry, scientists and the U.S. government to consider the consequences of their assumptions that humans can control and manage nature for our convenience. It compelled us to ask questions about our impact on the ecosystem. The concerns that informed *Silent Spring* are still with us today, often in new forms.

The basis of the questions related to pesticide use is a risk / benefit analysis. What are the benefits and risks of using pesticides? Who benefits and who bears the burden of risk? In asking these questions we must take posterity into account: what benefits and risks will our generation bequeath to our successors? What impacts are the pesticides we use making on the land and the oceans, and even on the atmosphere? What impacts do these agents have on our bodies and the bodies of our children? What groups or entities make the risk/benefit determinations and what is the assessment process?

The two areas where insect control is most salient are food production and disease control. There are two main challenges in these areas. What quantity of synthetic chemical fertilizers, herbicides, and pesticides are required to achieve the goal of feeding the world's increasing population? Can we prevent or cure insect-borne diseases without toxic

chemicals? These questions are debated by proponents on both sides of the issues. The rest of this chapter will put these questions into context. I will consider briefly here two large issues—agrochemicals and malaria—glance at some of the central debates on these questions, and make a small detour to consider a perhaps surprising and seemingly trivial question: how green are our suburban lawns?

Before continuing the discussion, it is important to remember that Carson did not say that we must never use pesticides. Rather, she asserted that they must be used judiciously, cautiously, if there are no safer alternatives, and only if they are deemed necessary after research into the effects and due consideration of the possible consequences. Extensive spraying of pesticides over large areas with inadequate monitoring of the results was the immediate target of her book

UPDATES AND REVISITS: FROM THE DELANEY CLAUSE TO THE FOOD QUALITY PROTECTION ACT

Since Carson wrote *Silent Spring* many new agricultural and other widely-used chemicals have been brought to market, and regulatory agencies have been attempting to control them. But, as Michael Jacobs notes, “regulatory processes lag far behind scientific advance” (Jacobs Millenium introduction 8).

Government regulation of pesticides, along with the numerous other chemicals in common use, continues to be an issue. On July 23, 1992 the Committee on Government Operations of the House of Representatives published "Thirty Years after *Silent Spring*: Status of EPA's Review of Older Pesticides." This document reports a hearing before the Environment, Energy, and Natural Resources Subcommittee. The committee's chairman, Mike Synar commented

> the book made us aware not only of pesticide risks but in many ways it began the modern environmental movement by making us aware of some of the potential ecological impacts of our modern industrial age… If *Silent Spring* caused some to raise false fears of impending environmental doom, it also caused others to raise false fears of starvation in a world without DDT (1– 2).

The committee heard testimony and debated issues of risk assessment, tolerances, and reviews of pesticide regulations.

I will consider here one case of the changing regulations, the case of the Delaney Clause, a 1958 amendment to the Food, Drugs, and Cosmetic Act of 1938, named after Congressman James Delaney of New York. It said:

> The Secretary of the Food and Drug Administration shall not approve for use in food any chemical additive found to induce cancer in man, or, after tests, found to induce cancer in animals.

The clause proved difficult to enforce because pesticide residues persist in many food products. The U.S. Department of Health, Education and Welfare reports:

“If this clause were to be enforced for pesticide residues, it would outlaw most food of animal origin including all meat, all dairy products (milk, butter, ice cream, cheese, etc.), eggs, fowl, and fish. These foods presently contain and will continue to contain for years, traces of DDT despite any restrictions imposed on pesticides. Removal of these foods would

present a far worse hazard to health than uncertain carcinogenic risk of these trace amounts" U. S. Department of Health, Education and Welfare, Report of the Secretary's Commission on Pesticides and Their Relation to Environmental Health, pts. I and II, at i-xvii and 1-677 (Dec. 1969).

Because of this problem, EPA asked the National Academy of Sciences (NAS) to review the Delaney Clause in 1985.They issued a report, "Regulating Pesticides in Food: The Delaney Paradox," May 20, 1987. According to the Pesticide Management Education Program of Cornell University Cooperative Extension, "The NAS study report reached four principal conclusions:

1. All pesticides should be regulated on the basis of a consistent standard, so that there is no "double standard" for raw vs. processed foods or for old vs. new pesticides.
2. A uniform "negligible risk" rather than a "zero risk" standard for carcinogens in food, consistently applied, would best enable EPA to improve the overall safety of the food supply, and would result in only modest reductions in the benefits of pesticide use to farmers.
3. EPA should set its regulatory priorities by focusing first on the most worrisome pesticides used on the most-consumed crops.
4. The Agency should adopt a comprehensive analytical framework for forecasting the broad-scale impact of its pesticide-specific regulatory actions on the overall safety of the food supply (Cornell University Pesticide Management).

In 1996 Congress passed the Food Quality Protection Act based on the NAS report. This law replaced the Delaney Clause and replaced the "zero cancer risk standard" with a "safe standard" for pesticide residues. Additionally, the new law prohibits state and local governments "from setting pesticide tolerances more rigid than those established by the EPA" unless they petition for exceptions. The law also requires review of pesticide residue tolerances within ten years (Cornell University).

A subsequent report by the Rodale Institute finds that conventional agriculture has increased its use of herbicides as more weeds become herbicide-resistant. Consequently, the Environmental Protection Agency has increased the allowable residue in foods of the most commonly used herbicide, Roundup (glyphosate) (Rodale Institute). The National Public Radio Program *Living on Earth* reported on a recent French study linking genetically modified corn to an increased risk of cancer. This study used two hundred rats divided into treatment and control groups, and lasted for two years. The study found that "the rats that ate the genetically modified corn, whether or not it was spiked with Roundup, or drank the Roundup-laced water, got tumors earlier than rats in the control group. They also died much earlier: 50% of male rats died early, compared to 30% not fed the Roundup corn. And 70% of females died early compared to just 20% for the rats not fed the Roundup corn" ((Living on Earth September 28, 2012).

CANCER AND ENVIRONMENTAL TOXINS

Carson was especially concerned that toxic chemicals might cause cancer. While she was writing *Silent Spring* she herself was diagnosed with breast cancer that proved terminal. Based on research, especially the work of Dr. Wilhelm Hueper, Carson hypothesized that

some environmental toxins might interfere with cellular processes and lead to cancer. The causes of cancer are still in question. Ongoing research seeks to determine causes, prevention, and cures for cancer, but experts differ on the role of environmental factors. Questions remain: how much cancer is due to environmental toxins and how much is attributable to other causes such as greater longevity, eradication of other diseases, higher rates of detection, life-style factors such as smoking and obesity? And regarding environmental toxins, how best to untangle the twisted skeins, the possible synergetic interactions of toxic chemicals?

Introducing her discussion of environmental toxins Janisse Ray writes a fable about "a town without plastic" analogous to Carson's "Fable for Tomorrow" (Matthiessen 109-110). She explains that

> *Silent Spring* opened the flood gates of inquiry into environmental contaminants and their effects on wildlife and humans, an investigation that accelerated in the 1990s. . . . Study after study has shown what Rachel Carson predicted. Chemicals are disturbing normal hormone-controlled development, affecting gender, sex, and reproduction. . . . More than 200 animal species are known or suspected to have reproductive disorders that might be attributed to endocrine disrupting chemicals, synthetic compounds that, when absorbed in the body, disrupt its natural functions (Ray in Matthiessen 112- 113).

She cites the case of DES, diethylstilbestrol, a synthetic estrogen that was prescribed to pregnant women in the US from the 1940s to the 1970s to prevent miscarriages. DES was found to cause a rare form of vaginal cancer in the daughters of many of these women. Other suspect chemicals are certain PCBs, and phthalates. More research is needed to determine the safety and appropriate uses of these chemicals.

Rachel Carson speaks of the right of people to understand the implications of scientific advances and of technologies that are used in the public sphere. She argues that the present time is “an era dominated by industry, in which the right to make a dollar at whatever cost is seldom challenged.” When problems arise, the public is “fed little tranquilizing pills of half truth” while “unpalatable facts” are “sugar coat[ed]” (Silent Spring 13). She argues that the public must be given “full possession of the facts” so that it can make reasoned and reasonable decisions. Arguments like these probably caused as much consternation as Carson's warnings about pesticides.

Carson's insistence on the rights of the public to receive full information sets an important standard for government agencies and corporations. Unfortunately, these standards are sometimes ignored and contravened. Deregulation of industries and weak enforcement standards have resulted in much misinformation and disinformation about various dangers. Tobacco companies lied about the health impacts of cigarette smoking, and countless people died of lung cancer as a result. Oil companies have been involved in extensive advertising campaigns to deny that global warming is taking place.

Dr. Susan Love, a surgeon, started a research foundation dedicated to discovering the causes of cancer. One of her projects is the Army of Women, which aims to recruit one million women to serve as a pool of volunteers to participate in research studies when applicable and appropriate. Rather than focusing on cures, Dr. Love aims to find ways to prevent cancer.

The US government funds some cancer research, and one of its agencies reports on current findings annually. A President's Cancer Panel was created by an act of Congress in 1971; its charge is to "monitor the multi-billion-dollar National Cancer Program and [to issue] reports directly to the President every year." Whereas such reports typically advise non-smoking, self-examination, and doctor visits, the Panel issued a strongly-worded report in May 2010 that focused on toxic chemicals in the environment. It asked the President to take action, and "to use the power of your office to remove the carcinogens and other toxins from our food, water, and air that needlessly increase health care costs, cripple our nation's productivity, and devastate American lives." According to Marla Cone, the 240- page report "is the first to focus on environmental causes of cancer." Cone quotes from the report: "The true burden of environmentally induced cancers has been grossly underestimated" (Cone).

Nicholas D. Kristof summed up the report:

> It's an extraordinary document. It calls on America to rethink the way we confront cancer, including much more rigorous regulation of chemicals. . . . The Panel suggests . . . [filtering drinking water], giving preference to organic food, checking radon levels, and microwaving food in glass containers rather than plastic . . . The report blames weak laws, lax enforcement and fragmented authority, as well as the existing regulatory presumption that chemicals are safe unless strong evidence emerges to the contrary (*New York Times*).

Kristof quotes from the report: "Only a few hundred of the more than 80,000 chemicals in use in the United States have been tested for safety.… Many known or suspected carcinogens are completely unregulated'' (*New York Times* 5 May 2010).

In response the American Cancer Society argued that the Panel had overestimated the environmental causes of cancer and "estimates that about 6 percent of all cancers in the United States — 34,000 cases a year — are related to environmental causes (4 percent from occupational exposures, 2 percent from the community or other settings)."

One of Carson's legacies is a network of cancer activists who are exploring the links between cancer and environmental toxins. One example of this is *Rachel's Daughters,* a documentary film made by the parents of a young woman diagnosed with cancer (Klawiter 194-7). Allie Light and Irving Saraf worked with Nancy Evans, a medical writer, and women cancer activists to interview twenty-two prominent researchers about these toxin-cancer links.

> When she warned of the dangers that pesticides and other synthetic chemicals held for all life, especially future generations, her critics asked, "Why is she so concerned about the future; she has no children." We are all Rachel's daughters. . . . A primary goal in making *Rachel's Daughters* was to shift the focus of public attention from the detection and treatment of breast cancer to the known and suspected causes of the disease and the possible ways to prevent it or at least reduce the risk. The investigators wanted to know why so many women are getting this disease – why the lifetime risk of breast cancer has more than doubled in the last 50 years. Their search uncovered no single answer; instead it suggested many possibilities and raised many more questions (Women Make Movies).

Women Make Movies offers an on-line study guide with useful lists of resources. They describe the project:

> Seeing themselves as spiritual heirs of author Rachel Carson, . . . they focus on issues including chemical contamination, radiation, and electromagnetic exposure to find breast cancer's causes. Addressing environmental racism, inequalities in research funding, and disparities in cancer rates for women of color, they track the effects of social biases on cancer incidence and health care delivery (Women Make Movies).

Carson's name is invoked in other analyses of environmental toxins. Sandra Steingraber considers current implications. In *The Recurring Silent Spring* H. Patricia Hynes, who worked at the Environmental Protection Agency (EPA) and later was Chief of Environmental Management at the Massachusetts Port Authority, celebrates Carson's life and work, and takes the EPA to task for its failure to be more proactive in regulating the production of pesticides. According to Hynes "tens of thousands of synthetic chemicals are manufactured and in use everywhere, the majority of which are insufficiently tested for toxicity" (21). Hynes argues that the EPA's focus has shifted to "waste emissions" rather than "regulating industry at the point of generating chemicals" (21). Hynes addresses some issues that were current at the time she wrote the book, and remain current, such as bovine growth hormone, herbicide resistant crops and genetic engineering (187-97). Interestingly, Hynes speaks of the impact of bovine growth hormone on the animals' health, but does not ask what the effect on human health might be of drinking milk or eating meat from these cows (184).

ENVIRONMENTAL ETHICS

The entire field of environmental ethics did not exist when Carson wrote. In fact, her books, especially *Silent Spring* were instrumental in raising the ethical questions that led to the development of that field.

> When environmental ethics emerged as a new sub-discipline of philosophy in the early 1970s, it did so by posing a challenge to traditional anthropocentrism. In the first place, it questioned the assumed moral superiority of human beings to members of other species on earth. In the second place, it investigated the possibility of rational arguments for assigning intrinsic value to the natural environment and its nonhuman contents" (Stanford Encyclopedia of Philosophy Web)

> The field of environmental ethics is a new but now well-established sub-discipline of philosophy. Emerging in the mid-1970s, the field coalesced with the inaugural volume of the journal *Environmental Ethics* in 1979 and developed rapidly. By the turn of the century, most colleges and universities offered courses, if not major programs of study, in this important discipline (http://www.gale.cengage.com).

ENVIRONMENTAL JUSTICE

The environmental justice movement is not strictly speaking a direct outcome of Carson's work but it continues in the direction she marked when she argued for the rights of citizens to be secure in their homes and free from poisons applied by governments or other people. Environmental justice advocates argue that minority and low income areas frequently

become sites for toxic pollutants such as waste incinerators, sewage treatment plants and the like, thus leading to increases in rates of asthma and other respiratory ailments.

Many people are working for or writing about environmental justice in a variety of venues. Majora Carter is an outstanding leader in this area. She started a nonprofit organization, Sustainable South Bronx, in 2001 to train inner city unemployed and underemployed workers for green jobs. Some of the projects that the group has been involved with are a green roof project, a waterfront park at Hunts point in the Bronx, and an eleven mile Greenway project. Carter describes her projects in an eighteen minute TED talk. She received a MacArthur genius grant in 2005, and has since founded a for profit consulting firm, the Majora Carter group. She writes

> Money spent on public health and incarceration costs, imported energy resources, and conventional waste disposal, creates one-way movements of capital out of local economies. By contrast, robust distributed investments in Sustainable South Bronx-style solutions – adapted to local climactic conditions – pay for themselves through more effective economic multipliers as money works from the pockets of people who are presently regarded as problems(Majora Carter Group).

UPDATES AND REVISITS: *SILENT SPRING REVISITED* (1987)

Since the publication of *Silent Spring* environmentalists and scholars in a wide range of fields, as well as scientists and activists have published reconsiderations and assessments of Carson's book. In the following discussion I will review the insights provided by an essay collection that addresses questions Carson raised about pesticides in agriculture, and their consequences for health.

In 1984 a group of scientists from the American Chemical Society (who were members of the society's Pesticide Subcommittee of the Committee to Improve the Environment) held a symposium to consider the implications of Carson's book in light of current developments. One outcome of the symposium was a book, *Silent Spring Revisited*, published in 1987. It contains twelve chapters, each authored by one or more specialists in a relevant field. The chapters review the history of governmental pesticide regulations; address issues such as the impacts of pesticides on bird populations, ground water, and human health; and discuss the strategies for conducting a risk / benefit analysis of chemicals.

Shirley Briggs, a close associate of Carson, writes to counter the myths that have arisen about Carson as an "emotional polemicist." She offers a "comprehensive view" of Carson's book, noting that "few people use insecticides primarily to kill insects. How many people have much use for a great heap of dead bugs?" (Briggs Revisited 9). Rather, they wish to protect their homes, lawns, or plants from infestations of bugs. Fearing damage, they take action and spray, often before there is any damage. "Too short a focus on the immediate situation all too often leads to the quick fix instead of the lasting, and most effective, solution" (Briggs 9). Briggs points out that, in contrast, Carson took a long view. She would encourage people to ask if there might be better ways to solve their insect problems, such as planting a different variety of grass or flowers, using crop rotation, or companion planting, as when marigolds planted next to other plants control the nematodes in the soil. Briggs notes that the National Academy of Sciences estimated that only 10% of the "pesticides on the market . . . have been adequately tested to permit complete health hazard assessment" as

of 1984 (10). And, of course, since then many more such products have entered the market. Briggs comments that

> the Environmental Protection Agency seems to believe that the federal government is required to certify a toxic chemical solution for every conceivable pest problem. . . . But is this really necessary? Must a certain crop be grown under particular circumstances even if plants, soil, and water are to be seriously contaminated? Are there other ways to solve the problem? (10).

Food production

The overarching question here is: how can we grow enough food to supply the world's increasing population? Related issues arise. Can we promote the use of cultural, physical, and biological controls so as to minimize the use and the potential risks of chemicals? If we do continue to use chemicals, how much pesticide residue is permissible in foods and other products consumed by citizens? Are there safe limits? How can we measure the limits of toxicity? Could there ever be zero tolerance, that is, no toxic chemical residues in food? How accurate are our current assessments of the health risks of pesticides?

Gustave K. Kohn notes that until 1800 approximately 90% of the American population was engaged in agriculture and by the 1960s that percentage dropped to 5%. Use of farm machinery, improved crop varieties, and other techniques make it possible to produce sufficient food for most of the population (Kohn 159). But how safe are the pesticides and herbicides used in agriculture?

The authors who contributed to *Silent Spring Revisited* agree that substantially more pesticides were used at the time of their symposium (1984) than in Carson's time, and that the methods of analyzing pesticide residues had improved dramatically in the twenty years since *Silent Spring*. John A. Moore describes the review of chemicals for use on crops grown for human consumption required by the EPA: "product chemistry, residue chemistry, environmental fate, wildlife effects, and toxicology" (18). Yet despite greatly improved technology to detect residues at substantially lower levels, risk management is not a science, although based on scientific data. "Risk management . . . involves a series of value judgments by the regulator that can only be made after careful consideration of all scientific and other factors" (Wilkinson 30). The experiments required to conduct extensive animal testing require great expense and large numbers of animals. Thus, toxicology studies at that time typically used 100 animals and applied large doses of pesticides for a relatively short period of time to determine acute toxicity. Studies of oncogenicity typically involved 500 - 1000 animals. (Note: present methods use more computer modeling of chemical structures and experiments on cells rather than living animals.) Given the methods then available it was difficult to determine definitively the consequences of smaller amounts of pesticide residues over longer periods.

> The fundamental problem facing toxicologists is that the chronic adverse health effects of pesticides and other chemicals cannot be verified by direct experimentation. Consequently, their assessment invariably requires the extrapolation of data obtained under one set of laboratory conditions to those likely to be encountered under another different set of conditions (Wilkinson 39).

Wilkinson wonders if our concentrated focus on searching for cancer risk might even be preventing us from focusing on "other adverse effects [that] might prove . . . more serious threat[s] to human health" (Wilkinson 42).

In fact, current research is testing chemicals for endocrine disruption as well as cancer.

The authors in *Silent Spring Revisited* agree that current farming methods (such as monoculture) require pest control. Therefore pesticides, perhaps augmented by biological controls, are important tools. There is some disagreement about the effectiveness of pesticides. One author writes that "pesticides are synonymous with modern agriculture" and estimates that there would be 30% crop losses without pesticides (Wilkinson 26). Another author asserts that "an estimated 37% of all crops is lost annually to pests" comprised of insects, plant pathogens, and weeds (Pimental 1987, 183). According to David Pimentel losses due to insects have increased from 7% in the 1940s to about 13% in the early 1980s "in spite of a 10-fold increase in insecticide use" (Pimentel 1987, 183-84). Pimentel explains that this increased crop loss due to insects results from several factors including the increased practice of monoculture, reduction in crop rotation, "reduced tillage, leaving more crop remains on the land surface to harbor pests for subsequent crops," increased insect resistance, and "culturing crops in climatic regions where they are more susceptible to insect attack" (Pimentel 1987, 184). In 2008 Pimental writes that in the U.S. "approximately three pounds of pesticide are applied per acre per year to about four hundred million acres" in agriculture. However, homeowners apply "nearly three times the level that farmers apply per acre" to keep their lawns green and neat (Pimental 2008, 191). He notes that the newer classes of pesticides remain in the environment for about three months, whereas DDT and similar early pesticides persisted for "thirty to fifty years." However, the toxicity of the newer pesticides has increased "ten to twenty times." A "major problem" is that "it is estimated that less than 0.01 percent of the pesticides that are applied reach the target pests, which means that 99.99 percent . . . pollutes the environment" (Pimental 2008, 190-91). According to Pimental approximately "300,000 humans are poisoned with pesticides annually" in the U.S., and about "26 million [people] poisoned and about 220,000 deaths each year" worldwide.

> Estimates indicate over 10,000 cases of cancer resulting from pesticide exposure. Pesticides also disrupt the endocrine, immune, and neurological responses in humans and other animals. For example, endocrine disrupters tend to make male animals become female in structure. In addition, sperm production is greatly reduced or is entirely lost. (Pimental 2008, 191)

Pyrethroids were the most used insecticides in the 1980s, and rates of application were lower than they had previously been. Spraying was often used as part of a program of Integrated Pest Management (IPM) which includes biological controls as well as chemicals. However, IPM works more slowly than chemicals and requires more knowledge of environmental interactions and pest life cycles and processes (Kohn 165).

In the conclusion the editors note that there is still uncertainty about the carcinogenicity of many pesticides (195). Regarding agricultural productivity they explain how manufacturers and environmentalists view the same conditions differently. Manufacturers look for "better-growing high-yield plants that can lead to insect infestation requiring more control," hence more pesticides and more fertilizer, "leading to more weed problems," thus

requiring more herbicides but resulting in increased competitiveness and food production (196). The editors explain that looking at this situation, the environmentalist sees

> (1) more pest-susceptible plants in inherently unstable monocultures, leading to more crop losses; (2) newer classes of pesticides with unpredictable side effects introduced into the environment; (3) a degradation of the soil integrity by overfertilization and overproduction at a time of agricultural surplus; and (4) a high-density farming strategy with heavy agrochemical inputs, resulting in more hidden costs and delaying the application of biological and other nonchemical controls (197).

The manufacturer sees the situation as providing opportunities, while the environmentalist worries about "an increased assault on nature and human health" (197).

The editors note that developed nations are regulating the use and disposal of toxic chemicals, and producing new versions that may be less harmful. But underdeveloped nations face many of the problems about which Carson spoke. They must rely on the older, more persistent pesticides because they are cheaper than the new ones. "Poor storage conditions, overuse, misuse, uncontrolled disposal, and excessive worker exposure are routine in these societies" (Marco 197). When developed countries import food from the less developed ones, they may be re-introducing residues of pesticides banned in their own countries.

On balance they comment that Carson was wrong in a few respects, such as claiming that nature does not produce carcinogens, but correct in "many respects" (198).

They remark that "*Silent Spring* led society to evaluate the new technologies in terms of risk versus benefits rather than on the basis of benefits alone" (Marco et al 192). I shall return to this point at the conclusion of this chapter.

There have been substantial advances in knowledge, technology, and the amounts and kinds of new pesticides developed since the publication of Silent Spring Revisited. For example, toxicology studies now rely more on computer modeling, and cell lines are replacing animals in experiments. Questions about chemicals' effects also focus now on endocrine disruption, especially the impacts on embryos and reproduction. Given these developments I would argue that now at the fiftieth anniversary of Silent Spring's publication it is time for a similar volume with a range of experts addressing the continuing issues.

CARSON'S VIEW OF FACTORY FARMS

As we have seen, some of the practices of modern farming such as monoculture and continued growing of the same crops in the same locations deplete the soil of nutrients and increase the crops' susceptibility to insect infestation, thus requiring artificial fertilizers to maintain the soil, and pesticides or other practices to curtail insect damage. But what about the animals raised for food? What are the current practices in animal husbandry? Carson weighs in on this question in her foreword to Ruth Harrison's book, *Animal Machines*, which describes the modern system of industrial farming. Contemporary factory farms are another example of how, according to Carson, "the modern world worships the gods of speed and quantity, and of the quick and easy profit, and out of this idolatry monstrous evils have arisen" (Animal Machines vii).

Harrison's book describes English farms; however, many of their practices derive from American systems, and similar practices are rampant today. Carson worries about the effect of these systems on the animals and on the people who consume them. She points to the

same public faith in regard to animal farming that prevailed in regard to pesticide spraying, the citizens' innocent but mistaken belief that government agencies regulate the processes adequately to ensure their safety and protect the health of the animals, humans, and the environment.

Harrison describes the conditions under which poultry, veal calves, and pigs are raised. The goals are to shorten the time that animals are fattened and brought to market, and toreduce costs. The animals are crowded in sheds or, in the case of veal calves, in crates. When there is straw or litter it is often not replaced until one batch of animals is sent out for slaughter.

Harrison describes the poultry houses as “long, windowless houses punctuated only with [fans and vents]” (12). She quotes from a Ministry of Agriculture booklet, *The Broiler House*, "the atmosphere in the broiler house is dusty, humid, and charged with ammonia" from decomposing chicken manure (13). The animals are crowded together, with each bird allowed on average one-half to three quarters of a square foot of space. To keep the chickens from pecking each other they are de-beaked. In some cases chickens are even fitted with opaque eyeglasses or blinders to prevent them from seeing and pecking each other.

Special houses are set up for the laying hens that are kept for egg production. Their wattles and combs are removed, since this “resulted in less food being eaten and more eggs laid. . . . A yellow dye introduced into the feed would produce the golden yolk which the housewife associates with quality”). Laying hens were originally kept one bird in a 15 or 16 inch cage. But it was found that even three could be put together in one cage (43). Flies are a problem. When chickens are allowed free range or are kept on a solid floor, they eat the insect larvae, but in a henhouse they are kept in tiers of cages. Therefore, insecticides are necessary, although resistance to the insecticides is frequent.

Harrison writes that the lights in the henhouse

> go on and off for two hours round-the-clock. So the birds eat and sleep, eat and sleep, eat and sleep. At six weeks they are big enough to feel the intensity of crowding and too much light would mean too much fighting, so the lights are changed to 25 watt red and these go on and off round-the-clock every two hours (12 – 13).

Harrison explains that penicillin is added to the food to promote growth, and the amount is increased "as soon as any disease is suspected" (16). “This, of course leads to penicillin-resistant Salmonella in practically all factory-raised chickens” (Dorothy Read, personal communication).

Veal calves are males raised in special ways to keep their flesh white and tender. To achieve this result they must be kept anemic, gain weight quickly, have little exercise, and be slaughtered when they are three months old. Accordingly, they are kept in small narrow pens with their heads tethered to reduce the possibility of exercise. Solid foods would result in their synthesis of vitamin B12 in the rumen, one of the calves’ stomach compartments, and thus they would not be anemic (74). Accordingly, they are fed a liquid diet that does not contain iron. So that they can gain weight quickly they are fed a reconstituted milk substitute with a high fat concentration. Given that calves lack iron or B12 and are anemic, Harrison wonders “whether the finished product is of any value as a food” (73).

Harrison deplores the inhumane conditions to which the animals are subjected. She raises the question "how much do [those who raise the animals] really know of the ultimate effect on the human consumer of all the drugs they use?” (7).

Responding to Harrison's exposé of the realities of factory farming, Carson remarks

> as a biologist whose special interests lie in the field of ecology, or the relation between living things and their environment, I find it inconceivable that healthy animals can be produced under the artificial and damaging conditions that prevail in these modern factory-like installations, where animals are grown and turned out like so many inanimate objects (vii).

More recent works on factory farming include Michael Pollan's *The Omnivore's Dilemma.* Pollan describes the crowded, unsanitary conditions in which beef cattle are kept on the feedlots where they are fattened for slaughter. Previously, when cows were raised on grass they were four or five years old at slaughter; today they are slaughtered at age fourteen to sixteen months. To shorten the time for them to reach the desired weight their diet has been transformed. They now eat "tremendous quantities of corn, protein and fat supplements, and an arsenal of new drugs" (Pollan 71). The feedlot transforms the closed ecological recycling loop of the traditional farm into an environmental problem. On the traditional farm, animals eat "the waste products of your crops, and you can feed their waste products to your crops. In fact, when animals live on farms the very idea of waste ceases to exist" (67-68). On animal feedlots, "this elegant solution [becomes] . . . two new problems: a fertility problem on the farm (which must be remedied with chemical fertilizers) and a pollution problem on the feedlot (which seldom is remedied at all)" (67-68). Pollan explains how the switch in the steers' diet from grass, to which their four-part stomachs have adapted, to corn, which their stomachs have difficulty digesting, has caused problems both for the animals and for the humans that consume them. "There are higher concentrations of E. coli in the animals guts, which inevitably contaminates the meat" (Dorothy Read, personal communication). The diet of corn causes a range of ailments that must be treated with antibiotics and other drugs. Pollan explains that "animals exquisitely adapted by natural selection to live on grass must be adapted by us—at considerable cost to their health, to the health of the land, and ultimately to the health of their eaters—to live on corn" (68).

Meat from these animals contains residues of the antibiotics, and has less of the omega-3 three fatty acids than grass fed beef. The overuse of antibiotics is a concern because the bacteria are developing resistance to them, and thus some diseases or infections of cattle and of humans are becoming difficult to treat. The reduction of omega-3 three fatty acids in beef may be leading to an imbalance in the diet of many Americans with serious ramifications (see Servan-Schreiber 65-69). An additional problem of the feedlot system of corn-fed steers is the development of relatively new highly toxic bacteria. Because the rumen of modern factory farmed cattle has become acidified by their diets (whereas the rumen was neutral on grass diets), the new strain of bacteria has evolved to live in an acid environment, such as the acidic stomach of humans. Thus humans have lost some protection from diseases transmitted by cattle (Pollan 82).

It is interesting that the person who "revolutionized the meat-producing industry" with his discovery "that feeding small amounts of antibiotics to livestock increases growth" is Thomas H. Jukes, whom we have met previously as a researcher for American Cyanamid Company, and the author of "The Year in Harmony," his parody of Carson's "Fable for Tomorrow." Partly as a result of his innovation, factory farms, that "now use far more

antibiotics than does medicine," have replaced the traditional cattle farm (Obituary of Jukes from University of California).

A controversy over another innovation in the meat industry, "pink slime," erupted recently. I feel certain that Carson would have had something to say about this. The unsavory sounding substance is

> beef with an unappetizing history. It starts life as slaughterhouse trimmings, which once were relegated to pet food and cooking oil. . . . In the early 1990s, Eldon Roth, a savvy Midwest entrepreneur, came up with a way to turn meat trimmings into profit. He heated them, spun them in a centrifuge to separate the tiny particles of meat from fat, then treated the product with a puff of ammonium hydroxide gas to kill bacteria. It became known in the industry as 'lean, finely textured beef,' or LFTB, and Roth made a fortune selling frozen bricks of it to add to ground beef.

The term "pink slime" was coined by a USDA food inspector. Recently discussions of this substance appeared on the Internet and on YouTube, with the result that people have been asking that the substance be removed from hamburgers and hotdogs (*USA Today* April 1, 2012 online).

Other issues relating to food have to do with fish farming and genetically modified foods. How prevalent are they? How safe for human consumption and for the environment? My guess is that Carson would be researching and writing a book about these issues if she were alive now.

Carson ended *Silent Spring* with "The Other Road," her chapter about alternatives to the practices she deplored. It is appropriate here to sketch quickly some "other roads" or alternatives to the practices of chemical-intensive agriculture and factory farms.

What alternatives are there for the chemical intensive factory farms? At one extreme is the organic or natural, traditional farm. There is also a partial solution to the toxic bacteria problem in beef. Michael Pollan quotes USDA microbiologist Jim Russell who has found that "switching a cow's diet from corn to grass or hay for a few days prior to slaughter reduces the population of [the dangerous variety of *E. coli*] in the animal's gut by as much as 80 percent" (82). Although that is "considered wildly impractical" at the present time, perhaps that could become routine in the future. Pollan also describes a more natural, sustainable, grass-based farm where a variety of animals and vegetables are raised. Contrasting the two farms Pollan makes a list (130-31):

Industrial	Sustainable
Annual species (corn)	Perennial species (grasses)
Monoculture (corn)	Polyculture (grasses)
Fossil energy	Solar energy
Global market	Local market
Specialized	Diversified

Mechanical	Biological
Imported fertility	Local fertility
Myriad inputs	Chicken feed

ORGANIC FARMING

What is organic farming? The USDA sets the standards for organic farming. The Organic Foods Production Act (OFPA), enacted under Title 21 of the 1990 Farm Bill, served to establish uniform national standards for the production and handling of foods labeled as "organic."

> Organic agriculture is an ecological production management system that promotes and enhances biodiversity, biological cycles and soil biological activity. It is based on minimal use of off-farm inputs and on management practices that restore, maintain and enhance ecological harmony" (http://www.nal.usda.gov/afsic/pubs/ofp/ofp.shtml).

The Rodale Institute decided to conduct an experiment to compare organic farming to high-chemical-input farming. They have been conducting the Farming Systems Trial since 1981. They originally planted corn and soybeans, the most commonly grown crops in large-scale agriculture. Wheat was added in 2004. "After 30 years of a rigorous side-by-side comparison," the Rodale Institute has concluded that the organic methods produce equivalent yields of crops (and slightly better yields in drought years), improve the soil health, reduce soil erosion and water runoff, generate fewer greenhouse gas emissions, use fewer chemical inputs and less energy (Rodale Institute). It will become increasingly challenging to feed the world's inhabitants as the population increases.

McWilliams ends his discussion of "the losing war on insects" with the suggestion that local control may be more practical and successful than centralized control. Currently interest is growing in integrated pest management (IPM) which utilizes biological and cultural controls as well as chemical controls.

Another new development is the small but growing interest in local foods, offered in supermarkets and at farmers' markets. Transporting produce from distant places relies now on non-renewable energy sources and contributes to global warming. Clearly there are limits to the kinds of produce available in different regions. Would people in northern regions such as New England now be satisfied without oranges or bananas? Some restaurants feature varieties of more natural (organic, free-range, grass-fed, or antibiotic-free) and or locally-raised foods, so the demand for these may be growing.

Local fish have come to my town. Ironically, fish caught in area waters by the local fishing community are often sent away for processing and then brought back frozen to be sold to local markets and restaurants. But recently more of the local fishers have begun to sell their fresh catch to local restaurants and to community residents who purchase fish at the food co-op on Friday afternoons.

THE DDT AND MALARIA CONTROVERSY

A lingering controversy in connection with *Silent Spring* is how best to eradicate malaria, a debilitating and life-threatening disease affecting millions of people. Can we prevent or cure insect-borne diseases such as malaria without the use of powerful and possibly toxic chemicals? In 2007, when environmental organizations were celebrating the centennial of Carson's birth several websites posted articles critical of Carson. They accused her of mass murder because some critics claim that she was responsible for the cancellation of a program of Indoor Residual Spraying (IRS) for malaria control (see for example Rachelwaswrong.org). Here again, critics attribute to Carson a ban on DDT that she never advocated. Remember, of course, Carson wrote: "No responsible person contends that insect-borne disease should be ignored. . . . The question that has now urgently presented itself is whether it is either wise or responsible to attack the problem by methods that are rapidly making it worse" (Spring 266). In fact, the African spraying program was canceled even before the U.S. banned domestic use of DDT because insect resistance reduced its efficacy.

Why the sudden spurt of criticism of someone who had been dead for twenty-five years? According to Fairness and Accuracy in Reporting, right-wing conservative groups looked for a "politically correct" way to frame a position countering environmentalism, and settled on the DDT issue as their cause (Sept/Oct 2007 "Rachel Carson, Mass Murderer? The creation of an anti-environmental myth").

What is malaria and what are the ways to treat it? "Malaria is caused by a parasite that is passed from one human to another by the bite of infected *Anopheles* mosquitoes. After infection, the parasites (called sporozoites) travel through the bloodstream to the liver, where they mature and release another form, the merozoites. The parasites enter the bloodstream and infect red blood cells [causing] high fevers, shaking chills, flu-like symptoms, and anemia," and sometimes resulting in death (PubMed). DDT is the cheapest chemical killer of malarial mosquitoes. It was initially very effective in reducing the mosquito populations in Africa where it was used in a program of indoor residual spraying (IRS) in which DDT is applied to the interior walls of houses for the purpose of killing the malaria-carrying mosquitoes. This process works because the mosquitoes are active at night when residents are likely to be at home. After biting their human or animal victims, the mosquitoes rest on vertical surfaces and are killed by contact with the DDT on the walls.

But ironically, the widespread agricultural spraying of DDT led to the development of resistance in the mosquito populations, so that the pesticide was no longer working, and, in 1969 (before the U.S. had banned DDT for domestic use) the World Health Assembly canceled the program in Africa. In 2006, the World Health Organization again reinstated support for DDT in mosquito control through indoor residual spraying.

According to the Stockholm Convention (a treaty regulating persistent organic pesticides signed by 174 countries—but not the U.S—as of 2011), DDT may legally be used to prevent the spread of malaria, but it must be used with restraint. Overuse may lead to other problems such as insect resistance. In an unusual case in 1965 cats in the town of San Joaquin, Bolivia died from DDT poisoning. As a result, mouse-like rodents called lauchas proliferated, spreading a typhus epidemic. Interestingly the authority who confirmed that DDT was the cause of the cats' death was one of those who testified about the safety of DDT in trials and Congressional hearings, Dr. Wayland J. Hayes, Jr., chief of the toxicology section of the U. S. Public Health Service at Atlanta, Georgia (Graham 141-3).

Although DDT is a primary agent for malaria control, pesticides are only one facet of an effective program. The other aspects may include a variety of other approaches depending on the particular situation. Important parts of a system of malarial control and eradication include draining swamps and wetlands—indeed all standing water─when possible, using insect repellants, and installing netting over beds. Providing food security is also essential, as people who are properly nourished have greater resistance to disease. It is important to educate the people involved, who may not know the causes of malaria. For example, when bed netting was distributed without educating the recipients, many people had no ceiling hooks to hang the nets. Because many were unclear about the proper use, some used the nets for fishing rather than installing them over their beds (Tom Husband conversation).

Scientists are working to develop vaccines against malaria that may eventually be part or all of the solution. According to Fairness and Accuracy in Reporting (FAIR) "perhaps most importantly, the pro-DDT line is a vast distraction. There are numerous other techniques for dealing with malaria: alternative insecticides, bed nets and a combination of drugs called artemisinin-based combination therapy, or ACT. ACT actually kills the malaria parasite fast, allowing the patient a quick recovery, and has a success rate of 95 percent (World Health Organization, 2001). Rollouts of ACT in other countries have slashed malaria rates by 80 to 97 percent (Washington Monthly, 7/06) (FAIR Sept/Oct 2007). However, some strains of the malarial parasite are developing resistance to artemisinin, so a combination of drugs is necessary.

Many agencies are carrying out various projects to halt malaria. One of the more creative ones is the NightWatch initiative in Chad that includes a music video in several of the local languages reminding people to use the bednets and take other measures to prevent malaria (http://www.malarianomore.org).

CONTINUING RESEARCH ON DDT'S HEALTH EFFECTS

Studies of the effects of DDT on human health continue to accumulate. Alma College in the town of Alma, Michigan, (near a Velsicol Chemical Company hazardous waste site) at Pine River hosted the Eugene Kenaga International DDT Conference in 2008. A consensus emerged that was published as the Pine River Statement. The statement's authors reviewed 494 studies published between 2003 and 2008 and found that "DDT and its breakdown product DDE [dichlorodiphenyldichloroethylene] may be associated with adverse health outcomes such as breast cancer, diabetes, decreased semen quality, spontaneous abortion, and impaired neurodevelopment in children" (Eskenazi et al. 2009). This review, however, did not include information from regions where DDT is applied to indoor surfaces for mosquito control.

Subsequently, Bouwman and others reviewed an additional 22 epidemiological studies completed during 2009 and 2010, including some from developing countries using IRS. They found that twelve studies "showed significant associations of DDT and DDE with conditions or effects on type 2 diabetes, hormones in blood, infant birth weight, pancreatic ductal adenocarcinoma, and sperm parameters" (Bouwman). They point to the continued exposure of residents in houses treated with DDT. The chemical may remain in the air for up to eighty-four days; it may fall in dust to the floor and be swept outside depositing residues in the soil. Children who are likely to play on the floor and outside on the ground may be receiving a larger body burden of the toxic chemical. If those who apply the

chemical do not wear the required masks and protective clothing and take proper precautions, they are repeatedly exposed, and high concentrations of DDT and DDE have been measured in their blood.

The authors of the review note three positions with respect to DDT. People in favor of or opposed to continued use of the chemical are the "most vocal" in expressing their opinions and advocating for their positions. Situating themselves in the centrist position Bouwman and colleagues assert:

> The centrist-DDT point of view adopts an approach that pragmatically accepts the current need for DDT to combat malaria transmission using indoor residual spraying (IRS) but at the same time recognizes the risks inherent in using a toxic chemical in the immediate residential environment of millions of people (Bouwman).

They ask for further studies of the pesticide's health effects, and urge that research be conducted to reduce the exposure of people whose homes are treated with IRS, and to discover less hazardous methods of malaria prevention. They wonder: "one can only imagine the outcry that would follow if people in the developed world were forced to have 2 g/m^2 DDT applied to their inner residential walls once a year" (Bouwman).

SUSTAINABILITY

One of the primary issues Carson raises is what we would now term sustainability. This concern appears repeatedly in her work when she warns about contamination of the seas in *The Sea Around Us* or when she exhorts us to take "The Other Road" that "assures the preservation of our earth" (Silent Spring 277).

One facet of sustainability is sustainable design. In their book *Cradle to Cradle: Remaking the Way We Make Things* William McDonough and Michael Braungart explore the possibilities of creating products through technologies that do not pollute. The book itself is not made of paper, but from "plastic resins and inorganic fillers. . . .[It] is not only waterproof, extremely durable, and (in many localities) recyclable by conventional means; it is also a prototype for the book as a 'technical nutrient,' that is, as a product that can be broken down and circulated infinitely in industrial cycles" (5). By cradle to cradle they mean products made of components that can be reused repeatedly, rather than disposed of in landfills (as in cradle to grave). They suggest a new ethic of manufacture and building construction.

— buildings that, like trees, produce more energy than they consume and purify their own wastewater

— factories that produce effluents that are drinking water

— products that, when their useful life is over, do not become useless waste but can be tossed onto the ground to decompose and become food for plants and animals and nutrients for soil; or, alternately, that can return to industrial cycles to supply high – quality raw materials for new products . . .

— transportation that improves the quality of life while delivering goods and services

— a world of abundance, not one of limits, pollution, and waste (90-91).

They write about the designs they have produced that do not harm the environment. For a Swiss textile company they designed an upholstery fabric made of wool and ramie which could safely be composted when consumers finished using it. The factory had previously been told that the trimmings of the fabric it used to produce were hazardous waste and had to be shipped to Spain for disposal. After the factory switched to producing the compostable fabric its workers no longer had to wear gloves and masks to protect them from workplace toxins. Regulators tested the effluent (water leaving the factory) and found to their surprise that the water leaving was as clean as the water entering the factory (105-109).

Lawns

Moving to a seemingly more trivial issue, but one with which we in the U.S. have more familiarity and over which we have more control is the issue of lawns and landscaping. Why lawns?

Lawns are an important aspect of the pesticide issue because homeowners apply "nearly three times the level [of pesticides] that farmers apply per acre" to keep their lawns green and neat (Pimental 2008, 191). Residue from these pesticides runs off and is deposited in lakes, rivers, and the ocean. Questions to consider are:

What do we value in lawns? How much pesticide is necessary to maintain a beautiful lawn? Are there ways to reduce pesticide application and still have a healthy lawn? Are there ways to reduce the size or type of lawns and still have an attractively landscaped yard?

Carson sheds light on the subject of crabgrass, one undesirable plant in lawns. "Crabgrass exists only in an unhealthy lawn. It is a symptom, not a disease in itself" (80). When the soil is healthy and fertile it is an environment in which crabgrass cannot grow, because other grasses will prevent it from surviving (80). Grass specialist Dr. Rebecca Nelson Brown explains

> That is true, to a certain extent. Crabgrass is an annual. Our turfgrasses are perennials. When the turf is dense and healthy, the crabgrass seed cannot get enough light to germinate. Thus, crabgrass is a symptom of stressed turf. The stress is most often caused by poor soil, which provides too few nutrients for the perennial turf. However, crabgrass will grow in healthy soil if something else (insects, disease, drought, physical damage) has caused the turf to become thin, or to die completely.

Sometimes people routinely apply herbicides or pesticides as preventives that may in fact not be needed. Brown continues

> Application of herbicides to prevent crabgrass should not be the annual event that it has become. The problem is that the easiest herbicides to use – and the ones most available to homeowners – have to be applied before the crabgrass germinates, so people are reluctant to wait and see if crabgrass is going to appear, because once it does it is too late to treat for it. This leads both to unnecessary applications of herbicide – treating when crabgrass wouldn't appear anyway, because the turf is healthy – and a failure to solve the causes of stressed turf (personal communication).

Weeds such as crabgrass are undesirable plants, or plants in the wrong place. But perhaps a lawn with occasional small flowers such as buttercups and violets may be even more attractive than a monotonous stretch of green. Carson asks if the weed that "is taking

something from the soil [may also be] . . . contributing something to it" (silent spring 78). For one, weeds may be an indication of soil quality, as some prefer acid soil, some grow in wet places and so on. Nancy Gift, Assistant Professor of Environmental Studies at Carson's alma mater, Chatham University, wrote a book called *Good Weed, Bad Weed: Who's Who, What to Do, and Why Some Deserve a Second Chance.* Gift focuses on weeds that appear commonly in lawns. She notes "in our language and culture the idea of a 'good weed' is an oxymoron, since the weed is always a bad thing and thus can't possibly be good. However, the terms 'good weed' and 'bad weed' are actually common in other cultures" (ii). Gift points out that many weeds are edible, such as wood sorrel and purslane, which is rich in beta – carotene, magnesium and potassium and is actually one of the few plant sources of omega – 3 fatty acids (62, 73). Others such as daisies and buttercups produce attractive flowers that may provide pollen for insects and butterflies. Birdsfoot trefoil and clover help enrich soil by fixing nitrogen (63, 67).

Nancy Gift replaces the crabgrass patches in her yard with perennial grasses in the fall. She recommends corn gluten applied in early spring to prevent germination of crabgrass and other unwanted weeds.

Lawns are another instance of plants put into environments they do not naturally inhabit. People who seek neat lawns often utilize more water and chemicals than are needed in their attempts to keep lawns green and even. For example lawns in New England may turn brown in dry summer weather. The grass survives this dormancy undamaged, but people may find the brown grass an unappealing aesthetic issue, or they may fear that brown grass is dead. Watering the lawns will cause them to turn green, but much of this water is unnecessary and wasteful.

Many groups advocate reducing the size of lawns by replacing some of the lawn with wildflowers that are native to the local area. Native plants are acclimated to the local environment and therefore require less maintenance and chemical intervention. The internet provides plenty of information for people who wish to convert all or part of their lawns to meadows or vegetable and flower gardens. The Audubon Society has a website complete with a list of plants and pictures of a converted lawn in Philadelphia, Pennsylvania: http://web4.audubon.org/bird/at_home/pdf/aahpa-21-32-lawn.pdf Sally and Sadie Kneidel have a blog called "Veggie Revolution" that offers advice on lawn conversion with information from North Carolina: http://veggierevolution.blogspot.com/2006/06/how-to-convert-lawn-to-native-meadow.html The Pennsylvania State University College of Agricultural Sciences Agricultural Research and Cooperative Extension has a six page pamphlet on "Meadows and Prairies: Wildlife-Friendly Alternatives to Lawn" The pamphlet explains:

> In the United States, over 24 million acres of lawn surround our homes. As suburban development continues to spread into open and forested land alike, we lose more and more of our native vegetation and wildlife habitat. By replacing all or part of your lawn with native vegetation that provides food and cover, you can create a refuge to attract a variety of wildlife. This not only will provide much-needed habitat, but it also will create an opportunity for you to see and learn. (http://pubs.cas.psu.edu/freepubs/pdfs/uh117.pdf)

Lawns tend to be monocultures and to use resources wastefully.

> According to the EPA 30 to 60% of urban fresh water is used to water lawns each year. . . . A 2000 sq ft lawn produces 600-800 pounds of clippings per summer on average. . . . Lawns cover 20 million acres of residential land in the U.S., and lawnmowers account for 5% of the air pollution. A 3.5 horsepower lawnmower pollutes as much in one hour as an automobile driving 350 miles (eartheasy.com/article_lawn_reduce.htm).

The clippings may be composted either in a separate compost pile or in place in the lawn. Nevertheless, converting part of a lawn to a vegetable or flower bed, or to fruit trees, shrubs, native plants, or groundcover may reduce some of the negative impacts of a grass lawn. Native flowering plants may provide food for birds and wildlife. Eartheasy.com continues its article on lawn reduction with a section on “how to remove a lawn or lawn section,” followed by “what to replace the lawn with.”

Homeowners are sometimes too quick to assume that insects are destructive. Many insects are beneficial rather than harmful. For example, ladybugs munch on aphids and other insects, big eyed bugs (Geocoris species) attack aphids, cabbage loopers, caterpillars, and other lawn and garden pests. Bees, flies, butterflies, and hummingbirds are important pollinators. Currently bees nationwide are experiencing severe decimation of hives in a syndrome known as "colony collapse." As of this writing the cause of this disease is not known, but pesticides are among the suspected causes. Without pollinators, our food supply would dwindle dramatically. Keeping our yards free of pesticides, and planting flowering plants will help to sustain the pollinator population. Many flowering plants such as purple coneflower (Echinacea), bee balm (rudbeckia), and butterfly bush (Buddleia) beautify our gardens and attract butterflies. But it is important to plant host plants as well as nectar plants for butterflies. Butterflies sip nectar and pollinate a variety of flowers, but they lay their eggs only on plants specific to each species.

One idea for transforming a yard is permaculture, a principle of ecological gardens designed to maximize beauty and usefulness to humans and to the environment. Toby Hemenway explains that permaculture "is a set of techniques and principles for designing sustainable human settlements. The word, a contraction of both ‘permanent culture’ and ‘permanent agriculture’ was coined by Bill Mollison" (4). Hemenway informs us that

> the aim of permaculture is to create ecologically sound, economically prosperous human communities. It is guided by a set of ethical principles – care for the earth, care for people, and sharing the surplus. From these stem a set of design guidelines. Some of these guidelines are based on our understanding of nature. . . . For gardeners to be on the forefront of a better relationship between humans and nature seems only natural (4 – 5).

In conclusion let us return to the risk/ benefit questions related to pesticide use. What are the benefits and risks of using pesticides? Who benefits and who bears the burden of risk? In *Silent Spring* Carson analyzed extensively one case of a condition that has come to characterize contemporary society. Pesticide spraying programs (like the aminotriazole “cranberry scare,” the strontium 90 in milk, the thalidomide babies, and a host of subsequent environmental crises) are instances of a category of risks theorized by sociologist Ulrich

Beck. Extrapolating from examples of actual or probable ecological catastrophes Beck developed a theoretical model of our contemporary "risk society."

Beck argues that the risks of our new technologies are pervasive and difficult to contain or control. Because of the quality and the magnitude of its projects (such as nuclear energy and genetically engineered organisms) the technology of applied science must conduct its experiments in the world outside of laboratories. Many of these technological advances are embraced in the name of progress although the consequences of these technologies may be unknown. Thus

> nuclear reactors must first be built and used in order for their model assumptions, safety standards and so on to be tested. . . . Genetically altered plants must be . . . cultivated in order to check the theory behind them. . . . Society itself has become the laboratory" (global 24).

Although the consequences of errors may be far-reaching, unpredictable and hard to remediate, "regulatory processes lag far behind scientific advance" (Jacobs introduction 8). Moreover, "not only has society become a laboratory, there is also no longer anyone who could be made responsible for the results" (Beck global 24). These research projects are conducted by industries according to their own goals and ambitions, rather than according to the priorities of a socially determined agenda. Thus,

> the separation of powers cedes to industry the right to make decisions without a corresponding responsibility for the risks those decisions unleash for the public, while politics is assigned the task of democratically legitimating decisions it never really made and knows little about (Beck global 25).

And, should real catastrophes occur, the public bears the burdens and often the costs as well.

In her analysis of the pesticide spraying programs Carson was one of the first to point to an ongoing ecological crisis of modernity: industrial technology moves with its own inexorable progress, regulatory processes lag behind, and the consequences of errors are widespread and difficult to control or ameliorate. As a result we now live in what sociologist Ulrich Beck terms a "risk society." Carson's book prompted the realization that we are a part of the environment, not observers watching images of scenic beauty or damaged areas. The pesticide issue is one example, but its political, economic, scientific, and ethical implications resonate widely through contemporary society as new technology spawns vast new projects and produces vast quantities of new products that have not been fully tested before their release into the ecosystem.

ECOFEMINISM

Although Carson would not have called herself a feminist, and died before the second wave of feminism arose in the US, the term would certainly apply to her life and beliefs. In fact, I consider her an early ecofeminist in her concern for nature and her critique of governmental arrogance and disregard for the interconnectedness of all life. Ecofeminist theorist Glynis Carr explains "social eco- feminism . . . provides a powerful critique of capitalism's waste, violence, unsustainability (in its requirement for constant growth), and

faulty assumption that nature is an inexhaustible resource deriving its value solely from human needs" (17-18).

Carr points out that

> ecofeminist ideology is not confined to critique: ecofeminists also envision alternatives consisting of the transformation of everyday life in ways that properly value . . . the creation of political and social structures that liberate oppressed groups (not only women) as well as nature. Ecofeminists insist that their utopian visions are not mere pipe dreams, but they maintain a guarded optimism (17).

H. Patricia Hynes gives a feminist reading of Carson's *Silent Spring*, arguing that a male establishment seeks to dominate both women and nature.

Carson's story is one of courage and determination, of the individual speaking up for life against a bevy of powerful economic and political entities, and even the government. Although reluctantly, she became a whistle-blower, and a crusader speaking out for life. The many biographies and numerous children's books about Carson testify to the imaginative power of her story. Perhaps some of the young readers will find inspiration in her life and work.

Above all, I believe that Carson's gift to us is her deep passion for the natural world, and her lyrical evocations of its majesty and wonder. If people are moved by Carson's sea books and by *The Sense of Wonder*, they may be awakened to the mysteries and beauties of nature, and may be inspired to think about the impact they are making on our amazing planet.

Carson's Legacies in the Performing Arts

Carson's work, especially *Silent Spring,* has inspired musicians. In 1983 a group of musicians released a tribute album, *Songs for the Earth: A Tribute to Rachel Carson*. In 2012 Nellie McKay performed in a cabaret at the Regency Hotel in New York. Her performance is called *Silent Spring: It's Not Nice to Fool Mother Nature*. David Hajdu reviewed the show in *The New Republic* on March 23, 2012 and called it "a resolutely quirky performance piece by the singer, pianist, and songwriter Nellie McKay, who portrays Carson with . . .giddy fervor." Marian McPartland, jazz pianist and composer, wrote *A Portrait of Rachel Carson*.

> Inspired mainly by Carson's best-remembered book, *Silent Spring*, the composition is a partly improvised, largely programmatic work that movingly evokes and celebrates the sounds of the natural world. (McPartland produced it in collaboration with the arranger and composer Alan Broadbent) (Hajdu).

Carson's message led to major shifts in environmental thinking. The environmental movement now takes a more holistic approach that considers the risks as well as the benefits of technologies. Its strategies have become more political and shifted from conserving and protecting sensitive areas to seeking government regulations and controls that can serve to monitor and protect the entire environment with its plant, animal, and human inhabitants.

And, since teaching about Carson and her work is an excellent way to sustain her legacy, the next chapter will offer some suggestions about teaching the life, works, and impact of Rachel Carson.

CHAPTER 7

TEACHING RACHEL CARSON

> The public must decide whether it wishes to continue on the present road, and it can do so only when in full possession of the facts. In the words of Jean Rostand, "The obligation to endure gives us the right to know" (Rachel Carson Silent Spring 13).

This section will look at a range of courses where Carson's work might be relevant; it will consider some projects and topics for relating Carson's work to a larger context of environmental and social issues; and it will ask a question: what are the controversies, and how should we teach them? I will offer an explanation of one teaching strategy, Problem-based Learning, and suggest some appropriate problems.

Carson is an extraordinarily gifted writer whose work bridges the disciplines of science and literature. As a pioneering woman scientist, and one of the founders of the contemporary environmental movement, Carson merits inclusion in a range of courses:

— Courses on women and science could explore Carson's role as a woman scientist in a male-dominated field in the first half of the twentieth century
— Gender and women's studies could focus on her life and writings
— Her work should be included in courses about nature writing, natural history, and literature and the environment
— Her career should be noted when teaching the history of science
— Her ideas comprise foundational principles in the new field of sustainability studies
— Her work can be included in courses on biology, marine biology, ecology, and oceanography
— Philosophy courses may include her work in explorations of environmental ethics
— Urban studies programs could focus on Carson's work in conjunction with issues of environmental justice, when exploring the ways in which toxic pollutants may be concentrated in urban areas
— American history and political science courses could read *Silent Spring* in conjunction with examination and analysis of U.S. foreign policy during the Cold War period
— Courses in environmental studies, environmental economics, environmental law, agriculture, and forestry could include discussion of *Silent Spring* as it may be relevant to their themes and topics
— Special topic courses such as a course on "The 1960s in the U.S.," or on global health issues may include Carson's work

I believe it is essential to extrapolate from her work so as to address the issues her life and writings raise, such as the interconnectedness of all parts of the ecosystem. Her books encompass a breadth of topics about marine biology, the environment, environmental ethics,

environmental history, environmental justice, climate change, the ocean, pollution, agriculture, agrochemicals, factory farming, and so forth. A study of her life leads us to consider topics of female friendship, family and career choices, authorship, and the prevention, diagnosis and treatment of cancer.

At all academic levels it would be wise to follow the advice of Carson herself when she urges us to "help your child to wonder." Use whatever resources are available to bring students into direct contact with the natural world and to inspire them to wonder. If trips to organic farmers' markets, alternative energy sites, recycling centers (or even dumps), energy efficient buildings, planetariums, natural history museums, aquariums, vernal pools, marshes, rivers, lakes, oceans, or tide pools are possible, those are all excellent ways to involve students with questions of nature and sustainability. Is there a body of water or a vernal pool nearby so that students could monitor its temperature, cleanliness, pH, flora and fauna? Is there a plot of land that could be turned into an organic garden? Wherever your school is located, it may be possible to walk outside in different seasons to observe the local environment or to note the ways in which humans have impacted nature. There are changes of weather, clouds, plants, birds, insects, and small mammals to observe by day, and the moon and stars to watch by night. Are there teachers from other disciplines to collaborate with on projects? Perhaps your students could be involved with local civic groups dealing with environmental issues.

In introducing their students to Carson teachers may find information in a variety of sources. Chapter 8, "Resources," lists some organizations and books that may be useful to educators.

The internet is a valuable resource that offers many good sources of information and ideas, some of which are listed in chapter 8. Especially useful internet sites are the Smithsonian Institution, National Oceanic and Atmospheric Administration (NOAA), and the National Aeronautics and Space Administration (NASA). The Monterey Bay aquarium has short videos on topics such as underwater camouflage, jellyfish, and phosphorescence, as well as several webcams for watching otters and Monterey Bay itself.

When Carson spoke to a gathering of nature writers upon receiving the John Burroughs Medal for nature writing she urged them to share their expertise with the general public, and not just to write for each other. Nature writers tend to be accessible and to write for the public, but meteorologists, chemists, biologists, physicists, geneticists need to communicate effectively with the public as well. There are many good sources of accessible science writing. There is an annual collection of essays called *The Best American Science and Nature Writing* edited by scientists and science writers such as Edward O. Wilson (2001), Natalie Angier (2002), and Elizabeth Kolbert (2009). *The New York Times* publishes a science section on Tuesdays. *The Scientific American, The National Geographic Magazine*, and other monthly publications are fruitful sources of current information. *The New Yorker* magazine often has essays on science, and, of course, that was where a condensed version of *Silent Spring* first appeared. *Science Digest* appears weekly and contains summaries of the most current science news, culled from journals and conferences.

Teachers in the early grades could read one of the many biographies of Carson written for children of different ages. Older children could write short stories about her. I include Dr. Seuss's environmental children's story, *The Lorax*, in my college course on literature and the environment, and assign the students to write similar stories for children as a class project.

Carson's ideas could be the windows through which to view current environmental issues such as alternative energy, declining fish stocks, fish farming, genetically modified organisms, the increasing world population, meat and poultry production in factory farms, mining, ocean oil spills, and global climate change and the related increase of destructive weather events such as hurricanes, earthquakes, tornadoes, floods, droughts, and tsunamis. Newspapers and general magazines often report on current environmental issues. Some relevant organizations have blogs on related topics. For example, Weather Underground is a fascinating repository of weather information. Its founder, Jeff Masters, writes a blog that considers aspects of weather and climate change.

IDEAS FOR STUDENT PROJECTS

— Have students keep nature journals. They could visit a particular place every day for a week or two and write up their observations. Or they could take pictures every day and make a booklet, slide show, or brochure of a place.
— Collect a basket of random natural objects: shells, feathers, seed pods, twigs, leaves, and similar objects. Students each choose one object and write about it daily for two weeks, noting any changes. When I used this project one of the seed pods exploded, sending seeds flying in the student's dorm room. This incident provided a nice example of a phenomenon explained in an essay we read by Loren Eiseley, "How Flowers Changed the World".
— Read, discuss, and analyze case studies about environmental issues. Students could have debates or panel discussions or write position papers on issues related to current environmental issues. A series of books that might be useful for such case studies is the *Watersheds* series. Subtitled *Classic Cases in Environmental Ethics*, these books present detailed information followed by questions for discussion, notes, and suggestions for further reading. Some of the cases included are the pollution of Love Canal in Niagara, NY; the explosion at the Union Carbide factory in Bhopal, India; the radioactive emissions at Chernobyl; and Carson's *Silent Spring*. Environmental issues frequently appear in newspapers and in news programs on TV and the radio. Especially interesting is National Public Radio's weekly program "Living on Earth."
— Assign each of the different chapters of one of Carson's books to an individual or small group to analyze, update or annotate.
— Service or activism projects related to environmental issues.
— Students could make collections of natural objects such as leaves, shells, or rocks to identify and learn about.
— Learn about environmentalists, present the information to the class, make up quizzes, games, skits, puzzles or stories about them.

Here are 5 suggestions for topics for research, position papers, debates or panel discussions:

— Should we allow offshore oil exploration and drilling? If so, what controls and restrictions would we enforce?
— What sources of alternative energy would be most efficacious? What are the trade-offs in reducing our use of oil and coal energy sources?

— How best to move toward reducing our carbon footprint? Colin Beavan's *No Impact Man* was the motivating book for a two-week long class project in which students kept journals of their attempts to reduce their environmental impacts.
— Climate change: is it happening? What causes it? What is our role?
— How toxic are the agricultural chemicals in use in the U.S? What strategies are available for reducing or replacing them by substituting biological and other means of insect or weed control?

A Sample Course on Women and the Natural Sciences

Some years ago I taught excerpts from *Silent Spring* in a team-taught course on women and the natural sciences. The course had four main foci:
— How has science studied women? Women have often been viewed as aberrations, deviations from a male norm. For example, many important health and diet studies exclude women as subjects because women were assumed to have more complex hormonal changes.
— Who are the women scientists? Students study the biographies and achievements of significant women scientists.
— How is science socially constructed? Science claims to be objective, but is that really the case? For example, how does the language of science determine what problems scientists study and how scientific work is funded? What influences do political issues have? Would science be different if more women entered the scientific fields?
— What are the public policy debates about science and technology? How are they relevant to women? How can we prepare women for active participation as citizens and leaders in the public debates about science?

Participating faculty spoke about the history and current position of women in various science and technology fields: biology, chemistry, engineering, oceanography, and psychology. When we learned that women's reluctance to negotiate is one of the reasons why women's salaries in science fields are often lower than men's we set up a session to learn about and practice negotiating. A marine biologist spoke to the class about the bioaccumulation of pollutants through the food chain. Students presented brief biographies of women scientists to the class. They were required to carry out action projects. One group cleaned up a small pond in a local park. Two friends who tended bar together set up a bottle recycling system at the restaurant where they worked. Two biology and math majors organized science and math tutorial sessions in their sorority.

TEACH THE CONTROVERSIES

Carson's *Silent Spring* and the issues it raises are controversial because so much remains unknown about the effects on the soil, on the oceans, on wildlife and on humans of pesticides, and, indeed, of many other chemicals that we use in daily life. If one does extensive research about Rachel Carson or looks her up on the Internet one finds that many books, articles, and websites are either laudatory or condemnatory; while more moderate voices strive to achieve balance. Is *Silent Spring* a much–needed wake-up call for America? Did it help to protect wildlife and humans from toxic sprays? Is the book responsible for

saving robins, peregrine falcons, eagles, and ospreys from extinction? Or was the decline and return of their populations part of a natural cycle? Would the Precautionary Principle save lives and improve health? Or does it achieve minimal health benefits at the expense of values such as freedom? Is the environmental movement that her book helped to inspire guiding us toward a better future for ourselves and our posterity? Or is it making mountains out of mole hills and stifling technological innovation by urging greater governmental regulations? Did Carson overdo her opposition to chemical pesticides? How dangerous are they really? Are they benign or carcinogenic? How necessary are pesticides for agriculture and health? Did Carson's critique of the overuse of pesticides cause the deaths of millions from malaria in Africa? An excellent method of teaching about controversial subjects is Problem-Based Learning.

PROBLEM BASED LEARNING

A pedagogical approach well-suited to teaching about Carson is the practice of Problem Based Learning (PBL). This technique originated in medical and law schools when it was found that students did well on exams and tests of memorization, but were sometimes at a loss in applying what they had learned to actual situations. Growing knowledge about the learning process demonstrates that active learning which requires students to solve problems and actively engage in thinking and discussing issues is superior to the traditional approach in college of lectures and rote memorization.

PBL is a system of student centered learning that asks students to solve open-ended problems that have many possible answers. It encourages students to explore controversies and weigh the differing opinions as they use a variety of tools to formulate their answers to questions posed by the teacher. It is a student-centered pedagogy which encourages students to learn about a subject in the context of complex, multifaceted, and realistic problems. Working in groups, students identify what they already know, what they need to know, and how to access new information to solve the problem and answer the questions.

Characteristics of Problem Based Learning

—Learning is driven by challenging, open-ended, relatively unstructured problems
— Students work in collaborative groups
— Teachers are facilitators rather than lecturers

Students are encouraged to take responsibility for their groups and to organize and direct the learning process with support from a tutor or instructor. Advocates of Problem Based Learning claim it can be used to enhance content knowledge while simultaneously fostering the development of communication, problem-solving, and self-directed learning skills.

> PBL may position students in a simulated real world working and professional context which involves policy, process, and ethical problems that will need to be understood and resolved. . . . By working through a combination of learning strategies to discover the nature of a problem, understanding the constraints and options to its resolution, defining the input variables, and understanding the viewpoints involved, students learn to negotiate the complex sociological nature of

> the problem and how competing resolutions may inform decision – making (Canaday).

The teacher's role shifts from that of lecturer (the sage on the stage) to facilitator or problem designer (the guide on the side).

> No longer a 'content expert' who tells students what material they need to know to pass a test or write a paper (that has little relevance beyond the classroom), a teacher in a PBL class plays two roles: case designer and facilitator. As a facilitator the PBL teacher tries to help guide teams of students as they figure out how best to attack each new case. The students are encouraged to come up with effective, creative solutions to each case, to draw on their existing individual strengths as they cooperate in a group project (Canaday).

Maggi Savin-Baden asserts that Problem Based Learning is different from learning through more traditional problem-solving. In the traditional method, problems that have one correct answer are assigned and students use strategies from a particular discipline to solve them. In PBL the problems are more complex, less well-defined, open to different approaches, and have many possible solutions rather than a pre-determined single answer. Savin-Baden explains

> Problem-based learning can help students to *learn with complexity*, to see that there are no straightforward answers to problem scenarios, but that learning and life takes place in contexts, contexts which affect the kinds of solutions that are available and possible. Learning such as this is not just a straightforward method of solving problems, but it helps people to learn how to learn and to link learning with their own interests and motivations. It can help students to learn in the context of 'real life' and focus the explorations they undertake, when engaging with problem based learning (Problem Based Learning in higher education 2000, page 5, italics in original).

Some of the benefits attributed to Problem Based Learning are that it:

— Enhances retention of information through practice

— Develops critical thinking, writing and communication skills

— Provides a model for lifelong learning

— Prepares students for a cooperative work environment

— Promotes self-evaluation as well as evaluation of others

— Facilitates small group learning (Brandt, March 15, 2012 workshop)

In devising problems instructors begin with the content or concept that they wish students to learn, and then develop problems that will lead students to address the issue. They then

format the problem as a narrative that will focus the question. When possible, the class or the students are the main characters in the narrative. Small problems may be done during one or more class sessions using a variety of available technological resources, or they may require additional work done at home. Larger problems may become the basis of ongoing research projects done by individuals or by groups. The lower the class level the more information and the more structure instructors provide. (Brandt).

Problem Based Learning works from three starting questions:

— What do you know (about the topic in question)?
— What do you need to know?
— How can you find out?

Teachers may provide guidance, monitor group progress, and suggest new questions or sources of information. They may make information available, as for example on an internet website or class site.

Here is a project I have designed for my college freshmen "Literature and the Environment" course.

Rachel Carson Project In 2007 environmentalists celebrated the 100 year anniversary of Rachel Carson's birth, calling her a hero, a prophet, and an eco-saint. But some groups allege that Carson's book *Silent Spring* has led to the deaths of millions of people in Africa. Which side is the most persuasive? Why?

— What do you know about Rachel Carson and the issues she raised?

— What do you need to know?

— How can you find out what you need to know?

Students will discuss the question in small groups and report back to the class. Following the class discussion each student will write a one page paper with his or her analysis of the problem.

A SAMPLE COLLEGE COURSE ON LITERATURE AND THE ENVIRONMENT

This was a course of first year students from a range of majors. It was taught in conjunction with a Writing course on the topic of sustainability. Some of the assignments were coordinated between the two courses. We set up the course with several real-world situations as projects in an attempt to infuse Problem Based Learning techniques.

Our first project was to investigate groups of trees on our campus and to produce informative texts. First the class learned about the ways trees function in ecosystems. In addition to the biology of trees we read an essay that suggests the cultural values of trees ("Street Trees" by Melody Ermachild Chavis). Next we took a walk around campus guided by a faculty member from the College of Environment and Life Sciences who pointed out

some specimen trees and informed us of their natural history and their history on our campus. Then the students worked in groups to gather various kinds of information about other trees on our campus. Each group selected a particular area on campus and researched six of the trees found there. Then they presented their information to the class: they made scrapbooks or booklets, and one group even put a set of photos on a Facebook page. Then they led the class on guided tree walks. This project has become part of a larger interdisciplinary project involving students from art, computer science, botany, and English departments. Called "The Virtual Arboretum," this project will put information about the trees on our university campus into an application for smart phones and other devices. According to the "The Virtual Arboretum" project description, “The readily-accessible database of information detailing the botanical and historical information about our campus trees will be a dynamic repository of information that can be updated as required.”

Another project was based on our reading of Colin Beavan's book, *No Impact Man*. The book describes Manhattan, New York resident Beavan's attempt to live with his wife and infant daughter for a year with minimal environmental impact: no cars, no subway rides, no elevators, no TV, no take-out food in disposable containers, no disposable diapers. Beavan found that as a result of their project the family spent more time together, reading, cooking and eating, talking, playing games, visiting with friends, or being outdoors. Inspired by Beavan's efforts to reduce his carbon footprint, my class spent two weeks finding at least one way each student could reduce his or her own environmental impact. At the start of our “lower your impact” project the students visited the website http://www.informinc.org/global_footprint_network.php to calculate our approximate carbon footprints, and to compare our results with others in the class. We also compared the average U.S. carbon footprint with the average footprints of citizens in other less-developed countries as well as in developed countries. In their journals students wrote about buying re-usable mugs for coffee, turning their computers off when they were not in use, turning off lights in their dorm rooms, recycling as much as possible, combining errands when making trips off campus, walking to classes from their dorms, taking public transportation when possible, and enlisting their room-mates in these efforts. Some continued to be aware of their carbon footprints when they went home on weekends. One student wrote about convincing her family that they could watch the TV together in one room, rather than watching the same program each in his or her own room. Another wrote that his family drove to a sporting event in the same car, rather than each driving separately to the same event.

I used newspaper articles and current events to generate investigations and discussions. When I learned that a speaker was coming to our university to speak about the benefits of genetically modified salmon, we first read about fish farming and about genetically modified food products, and then debated the pros and cons of these practices. We read excerpts from Michael Pollan's book *The Omnivore's Dilemma* and discussed the way our food is produced. Another food issue that led to lively discussion was a newspaper article about in vitro meat (animal muscle tissue cultured in laboratories for human consumption). Would this be more humane and more environmentally friendly? Would vegetarians find in vitro meat acceptable, since no sentient beings are harmed when the food is produced?

The most successful project in that class was the “Design an Environment” project. I set this assignment up as a Problem Based Learning exercise. Four groups of students were each given a particular environment to design or re-design: a beach community in New England, a senior community in Arizona, an urban ghetto neighborhood, and a research park

on our campus. The project required them first to list the goals for their sites, the challenges they would face, and the sources of information that they would use to design the environments. The next phase was to collaboratively produce designs for sustainable environments. Each group researched the local flora and climate to develop landscaping plans.

The urban ghetto project had residents start roof gardens and a farmers' market to share produce. They started a residents' organization to clean up the litter, and persuaded the city to expand the bus routes serving the community; they found an empty lot to convert into a small park and playground. The Arizona senior community featured solar panels on the roofs, walking and bike paths shaded by trees native to the area, and a cactus garden. The beach community designed houses with low profiles, incorporated a wind power generator, and set up a small museum with native plants and animals.

The group working on the research park found the most environmentally friendly building materials (recycled steel beams; recycled wood or Forest Stewardship Council approved wood; ashcrete rather than concrete), and designed a building with solar panels, photovoltaic roof panels, large south-facing windows to maximize natural light, a rainwater utilization system, energy efficient ventilation, and low-flush toilets. They found an agency that rates laboratories for energy efficiency to evaluate the companies that would apply to rent space. They even found a company that offered the most effective waste disposal system.

Above all else, Carson was a dedicated researcher and a seeker after knowledge about the earth and all of its creatures. She persisted in her research and held to her convictions although she knew she would face opposition and controversy. As teachers we should lead our students to follow in her footsteps. We should help them to explore the many sides of the controversies that they will surely face because technology is often far ahead of our knowledge about its impacts. As we guide them, we would do well to remember that Carson's greatest legacy may be to inspire us to a "sense of wonder" and a love of the world of nature.

CHAPTER 8

RESOURCES

Here is a list with occasional brief commentaries of selected online and printed resources for teachers, students, and interested readers who would like to learn more about Rachel Carson or the topics she wrote about and related issues. Some of the web links may not be active; some of the web pages may cross-reference each other.

RACHEL CARSON

Biographies

Lear, L. (1997). *Rachel Carson: Witness for Nature*. New York: Henry Holt.
Lear's biography of Carson is the most complete and well-researched. It was awarded the prize for the best book on women in science by the History of Science Society in 1999. Print.

<http://www.rachelcarson.org/>. Web.
Lear's website, "The Life and Legacy of Rachel Carson," contains a time line of Carson's life, a bibliography of her work and works about her, guides for doing research on Carson, and information about her biographer, Linda Lear.

Gartner, C. B. (1983). *Rachel Carson*. New York: Ungar Pub. Co. Print.

Lytle, M. H. (2007). *The Gentle Subversive: Rachel Carson, Silent Spring, and the Rise of the Environmental Movement*. New York: Oxford University Press. Print.

McCay, M. A. (1993). *Rachel Carson*. New York: Twayne. Print.

Biographies for children and young adults

Ehrlich, A. (2003). *Rachel, The Story of Rachel Carson*. Orlando, FL: Harcourt. ill. Wendell Minor. Print.
A younger child's biography, beautifully illustrated.

Levine, E. (2007). *Rachel Carson: A Twentieth Century Life. Up Close Series*. New York: Viking Juvenile. Print.

Locker, T., & Bruchac, J. (2004). *Rachel Carson: Preserving a Sense of Wonder*. Golden, CO: Fulcrum. Print.

Quaratiello, A. R. (2004). *Rachel Carson: A Biography*. Westport, CT: Greenwood Press. Print.

Sterling, P. (1970). *Sea and Earth: The Life of Rachel Carson*. New York: Thomas Y. Crowell. Print

Rachel Carson's personal papers and related materials

Many university libraries contacted Carson requesting that she donate her papers and related materials to their collections. She donated her papers to the Beinecke Special collections Library at Yale University, New Haven, Connecticut. After her death, her agent Marie Rodell, donated other materials. The Yale collection consists of 118 boxes of materials, including but not limited to photos, notebooks, letters, and manuscripts.

The Linda Lear special collections at Shain Library, Connecticut College in New London, Connecticut includes materials collected during Lear's research for her biography of Carson. There is some overlap with the collection at the Beinecke Library at Yale, but also some other material.

The Dorothy Freeman Collection at the Edmund S. Muskie Archives and Special Collections Library at Bates College (Lewiston, ME) owns letters written by Rachel Carson and her summer neighbor, Dorothy Freeman. Highlights of this collection include early manuscript drafts and recordings of speeches.

Documents from Carson's graduate studies are held by the Ferdinand Hamburger Archives at the Milton Eisenhower Library located at Johns Hopkins University.

Scholarship on Carson

The following section includes a short listing of some of the scholarly resources on Carson that have been published over the past fifteen years. These journal articles, book chapters, and anthologies are suitable for college students and academic researchers.

Bekoff, Marc, and Jan Nystrom. "The Other Side of Silence: Rachel Carson's Views of Animals." *Zygon* 39.4 (2004): 861-883. Print.

Bell, Elizabeth S. "The Language of Discovery: The Seascapes of Rachel Carson and Jacques Cousteau." *CEA Critic: An Official Journal of the College English Association* 63.1 (2000): 5-13. Print.

Bellin, Joshua David. "Us or Them: *Silent Spring* and the 'Big Bug' Films of the 1950s." *Extrapolation: A Journal of Science Fiction and Fantasy* 50.1 (2009): 145-168. Print.

Corbett, Steven J. "Environmentalism (and Audience) Friendliness in Rachel Carson and Devra Davis: Where Ecocriticism and Rhetoric Meet." *ISLE: Interdisciplinary Studies in Literature and Environment* 16.3 (2009): 487-515. Print.

Foote, Bonnie. "The Narrative Interactions of *Silent Spring*: Bridging Literary Criticism and Ecocriticism." *New Literary History: A Journal of Theory and Interpretation* 38.4 (2007): 749-753. Print.

Kroll, Gary. "The 'Silent Springs' of Rachel Carson: Mass Media and the Origins of Modern Environmentalism." *Public Understanding of Science* 10.4 (2001): 403-420. Print.

Matthiessen, Peter, ed. *Courage for the Earth: Writers, Scientists, and Activists Celebrate the Life and Writing of Rachel Carson*. NY: Houghton Mifflin, 2007. Print.

Murphy, Priscilla. *What a Book Can Do: The Publication and Reception of Silent Spring*. U. of Massachusetts Press, 2005. Print.

Newell, Dianne. "Home Truths: Women Writing Science in the Nuclear Dawn." *European Journal of American Culture* 22.3 (2003): 193-203. Print.

Raglon, Rebecca. "Rachel Carson and Her Legacy." *Natural Eloquence: Women Reinscribe Science*. Ed. Barbara T. Gates and Ann B. Shteir. Madison: U of Wisconsin Press, 1997. 196-211. Print.

Ryden, Kent C. "Beneath the Surface: Natural Landscapes, Cultural Meanings, and Teaching about Place." *Teaching about Place: Learning from the Land*. Ed. Laird Christensen and Hal Crimmel. Reno: U of Nevada P, 2008. 125-136. Print.

Sideris, Lisa H. and Kathleen Dean Moore, eds. *Rachel Carson: Legacy and Challenge*. Albany: State U of NY Press, 2008. Print.

Other Resources About Carson

Doyle , J. (2012, Feburary 21). Power in the pen, silent spring: 1962. *PopHistoryDig.com*. Retrieved April 4, 2012.

http://www.nrdc.org/health/pesticides/hcarson.asp A brief summary of *Silent Spring* and related links by the Natural Resources Defense Council. Web.

The Fish and Wildlife Service celebration of the centennial of Carson's birth in 2007. These two posters were produced as part of the celebration. They contain links to the Rachel Carson National Wildlife Refuge, information about Carson's life and work, and short videos about Carson and the refuge.

http://www.fws.gov/rachelcarson

http://www.fws.gov/rachelcarson/RC_Conservation_Legacy.pdf

The Rachel Carson National Wildlife Refuge office Port Road, Wells, Maine

The refuge consists of 50 non-contiguous miles of coastline extending from the sandy beaches that characterize the mid-Atlantic region to the rocky shore of Maine.

http://www.fws.gov/northeast/rachelcarson/visit.html

Public Broadcasting System Bill Moyers interviewed an actor who role plays Carson. PBS *Bill Moyers' Journal* – This is a rich site with many links and it includes material designed for teachers.

Actor Kaiulani Lee has played the role of Rachel Carson. Here is a link to a transcript of Lee's interview with Bill Moyers on *Bill Moyers' Journal* on PBS September 21, 2007.

http://www.pbs.org/moyers/journal/09212007/transcript1.html

And: http://www.pbs.org/moyers/journal/educators/rachelcarson.html

Includes activities suitable for Science, English, and Civics classrooms, grades 6-12.

In 1993, the documentary "Rachel Carson's Silent Spring," was produced for the PBS *American Experience* series, with actress Meryl Streep narrating the voice of Carson

CHAPTER 8

ISSUES THAT CARSON ADDRESSED DIRECTLY, AND RELATED ISSUES

The Environment: General

The Encyclopedia of Earth. http://www.eoearth.org Web.
An electronic reference about the Earth, its natural environments, and their interaction with society. The EoE is a free, expert-reviewed collection of content contributed by scholars, professionals, educators, practitioners and other experts who collaborate and review each other's work. The content is presented in a style intended to be useful to students, educators, scholars, professionals, as well as to the general public.

Environment Health News, a daily on-line newspaper published by Environmental Health Sciences http://www.environmentalhealthnews.org/ehs Web

Brand, S. (2009). *Whole Earth Discipline: an Ecopragmatist Manifesto*. New York: Viking. Print.

Brower, D. (1996). *Let the Mountains Talk, Let the Rivers Run*. Harper. Print.

Chappell, T. D. J. (Ed.). (1997). *The Philosophy of the Environment*. Edinburgh: Edinburgh University Press. Print.

Dodds, W. K. (2008). *Humanity's Footprint: Momentum, Impact, and Our Global Environment*. New York: Columbia University Press. Print.

Dryzek, J. S. & Schlosberg, D. (Eds.), (2005). *Debating The Earth: The Environmental Politics Reader*. Oxford: New York: Oxford University Press. Print.

Garrard, G. (2004). *Ecocriticism*. London: Routledge. Print.

Keogh, M. (Ed.). (2010). *Hope Beneath Our Feet :Restoring Our Place in the Natural World*. North Atlantic Books. Print.

Margulis, L. (1992). *The Diversity of Life: The Five Kingdoms*. Hillside, NJ: Enslow. Print.

Steingraber, S. (1997). *Living Downstream: A Scientist's Personal Investigation of Cancer and the Environment*. New York: Vintage Books. Print.

Steingraber, S. (2001). *Having Faith: an Ecologist's Journey to Motherhood*. Cambridge, MA: Perseus. Print.

The Ecological Society of America. Web.
The Ecological Society of America (ESA)'s website hosts multiple online journals including *Ecology*, as well as special areas for educators and students. The educator resource area includes listservs, information on a "Planting Science" program, "Teaching Issues and Experiments in Ecology," workshops and ongoing activities, in addition to a digital library.

Climate Change

Crutzen, P. J. (2006) "Albedo enhancement by stratospheric sulfur injections: A contribution to resolve a policy dilemma?" (PDF). *Climatic Change* (Springer), *77*(3–4), 211–219. Web.

Gore, Al. (2008, March). "Al Gore's New Thinking on the climate Crisis" TED talks. Web.

Gore, Al. (2007). *An Inconvenient Truth: The Crisis of Global Warming*. New York: Viking. Print.Gore, Al. (2006). *An Inconvenient Truth*. Dir. David Guggenheim, Hollywood, CA: Paramount. Film.

Gore, Al. (2009). *Our Choice: A Plan to Solve the Climate Crisis*. Emmaus, PA: Rodale. Print.

Keith, D. (2007, September). "David Keith's unusual climate change idea" TED talks.. Geoengineering to solve climate change issue. Web

Kolbert, E. (2006, 2007). *Field Notes From a Catastrophe: Man, Nature, and Climate Change*. New York: Bloomsbury Pub. Print. There is also a sound recording of this book: New York : Simon & Schuster Audio, 2006.

The 11th Hour. Dir. Leila Conners Petersen and Nadia Conners. Warner Independent Pictures, 2007. Film..

Endangered Species

The Audubon Society has a watch list of endangered species with links. Web

The Mystic Aquarium Web

The Mystic Aquarium, like the Monterey Bay Aquarium, is also devoted to conserving ocean species. Learn more about current exhibits, the research of Dr. Robert Ballard and rescue efforts sponsored by the aquarium.

The World Wildlife Fund Web

The homepage is dedicated to an endangered species of the day.

Ecofeminism

Carr, Glynis, ed. *New Essays in Ecofeminist Literary Criticism*. Cranbury, NJ: Associated University Presses, 2000. Print.

Cudworth, Erika. *Developing Ecofeminist Theory: The Complexity of Difference*. New York: Palgrave, 2005. Print.

Gaard, Greta and Patrick D. Murphy, eds. *Ecofeminist Literary Criticism: Theory, Interpretation, Pedagogy*. Urbana, IL: U Illinois Press, 1998. Print.

Hynes, H. Patricia. *The Recurring Silent Spring*. NY: Pergamon Press, 1988. Print.

King, Ynestra. "The Ecology of Feminism and the Feminism of Ecology." *Feminist Theory: A Reader*. Eds.Wendy K. Kolmar and Frances Bartkowski. Boston: McGraw Hill, 2010. 407–13. Print.

Merchant, Carolyn. *The Death of Nature: Women, Ecology, and the Scientific Revolution*. San Francisco: Harper Collins, 1980. Print.

Plumwood, Val. *Environmental Culture : The Ecological Crisis of Reason*. London: Routledge, 2002. Print.

Primavesi, Anne. "Ecofeminism." *Encyclopedia of Science and Religion*. Web

Warren, Karen J., ed. *Ecological Feminism*. London: Routledge, 1994. Print.

Environmental Education

Wisconsin Department of Natural Resources Environmental Education for Kids Web.

Evolution

http://evolution.berkeley.edu/ Web
Includes a section "Understanding Evolution for Teachers" with lesson plans designed for different grades.

http://www.pbs.org/wgbh/evolution/ Web
This site offers information for teachers and students, and 7 short videos that answer questions such as "Who was Charles Darwin?" and "How do we know that evolution happens?"

Environmental Ethics

The majority of the texts in this section are anthologies geared toward more advanced readers.

Anglemyer, Mary. *A Search For Environmental Ethics : An Initial Bibliography.* Washington, D.C. : Smithsonian Institution Press, 1980. Print.

Bormann, F. Herbert and Stephen R. Kellert, eds. *Ecology, Economics, Ethics: the Broken Circle.* New Haven: Yale University Press, 1991. Print.

Callicott, J. Baird and Robert Frodeman, eds. *Encyclopedia of environmental ethics and philosophy.* Farmington Hills, MI : Macmillan, 2009. Print.

Clowney, David and Patricia Mosto, eds. *Earthcare: An Anthology in Environmental Ethics.* Lanham: MD: Rowman and Littlefield, 2009. Print.
15 chapters presenting a brief history of environmental ethics and a collection of essays by noted scholars on topics ranging from religions' views on the environment to environmental justice, ecofeminism, air pollution, and sustainability.

Curtin, Deane W. *Environmental Ethics For a Postcolonial World.* Lanham, Md: Rowman & Littlefield Publishers, 2005. Print.

Goldstein, Robert J. *Ecology and Environmental Ethics : Green Wood in the Bundle of Sticks* Burlington, VT: Ashgate, 2004. Print.

Goodstein, Eban. *Economics and the Environment.* 6th ed. Hoboken (NJ):Wiley, 2010. Print. A standard college textbook.

Gottlieb, Roger, ed. *The Oxford Handbook of Religion and Ecology.* Oxford: Oxford UP, 2006. Print.

Jamieson, Dale. *Ethics and the Environment: An Introduction.* Cambridge: Cambridge University P, 2008. Print.

Keller, David R. *Environmental Ethics: The Big Questions.* Malden (MA): Wiley-Blackwell, 2010. Print.

Light, Andrew and Holmes Rolston III., eds. *Environmental Ethics: An Anthology.* Malden (MA): Blackwell, 2003. Print.

Light, Andrew and Jonathan M. Smith, eds. *Space, place, and environmental ethics.* Lanham, MD: Roman & Littlefield Publishers, 1997. Print.

Naess, Arne. *Ecology, Community and Lifestyle.* Trans. and rev. by David Rothenberg. NY: Cambridge University Press, 1989. Print.

Nanda, Ved P, ed.. *Climate Change and Environmental Ethics.* New Brunswick : Transaction Publishers, 2011. Print.

Oelschlaeger, Max, ed. *Postmodern Environmental Ethics.* Albany, NY: SUNY Press, 1995. Print. A collection of fifteen essays first published in the journal *Environmental Ethics.*

Orr, David. *Ecological Literacy*. Albany, NY: SUNY Press, 1992. Print.

Peterson, Anna L. *Being Human: Ethics, Environment, and Our Place in the World.* Berkeley: U of California Press, 2001. Print.

Pojman, Paul and Louis P. Pojman, eds. *Environmental Ethics: Readings in Theory and Application*. 6th ed. Boston: Wadsworth/CENGAGE Learning, 2010. Print.
Designed for introductory law students. Selections range from John Stuart Mill's "Nature" to David Pimentel's "Is *Silent Spring* Behind Us?"; from Jonathan Rauch's "Can Frankenfood Save the Planet" to Stephen M. Gardiner's "Ethics and Global Climate Change." This reader also includes subcategories, such as "Race, Class, Gender: Environmental Justice, Ecofeminism, and Indigenous Rights." Basic introductions help to situate readers to each category.

Sandler, Ronald L. *Character and Environment: A Virtue-Oriented approach to Environmental Ethics.* NY: Columbia University P, 2007. Print.

Sessions, George and Bill Devall. *Deep Ecology: Living as if Nature Mattered.* Layton (UT): Gibbs Smith, 1985. Print.

Sideris, Lisa H. *Environmental Ethics, Ecological Theology, and Natural Selection.* New York: Columbia University Press, 2003. Print.

Smith, Pamela. *What are they saying about Environmental Ethics?* New York : Paulist Press, 1997. Print.
Concise overviews of major strands of environmental ethics: deep ecology, ecofeminism, animal rights, Leopoldian ethics, liberation ecotheology, and religious eco-ethics.

Traer, Robert Boulder, *Doing environmental ethics.* CO : Westview Press, 2009. Print.

Wenz, Peter S. *Environmental Ethics Today.* New York : Oxford University Press, 2001. Print.

Environmental History

Black, Brian. *Nature and the Environment in 19th-Century American Life.* Westport, CT: Greenwood P, 2006. Print.
A good general overview of American history, particularly in connection to the environment.

Conkin, Paul Keith. *The State of the Earth: Environmental Challenges on the Road to 2100.* Lexington (KY): The UP of Kentucky, 2007. Print.

Dunlap, Riley E. and Angela G. Mertig, eds. *American Environmentalism: The U.S. Environmental Movement, 1970-1990.* Philadelphia: Taylor and Francis, 1992. Print.

Glave, Dianne D. *Rooted in the Earth: Reclaiming the African American Environmental Heritage.* Chicago, IL: Lawrence Hill Books 2010. Print.

Guha, Ramachandra. *Environmentalism: a Global History.* NY: Longman, 2000. Print.

Hays, Samuel P. *Conservation and the Gospel of Efficiency: The Progressive Conservation Movement, 1890-1920.* Cambridge (MA): Harvard UP, 1959. Print.

Kline, Benjamin. *First Along The River: A Brief History of the U.S. Environmental Movement.* 3rd ed. Lanham (MD): Rowman & Littlefield, 2007. Print.

A standard college textbook.

Marx, Leo. *The Machine in the Garden: Technology and the Pastoral Ideal in America.* NY: Oxford UP, 1967. Print.
Marx's work is considered a classic study of tropes in 19th century American literature; appropriate for more advanced readers.

Merchant, Carolyn. *American Environmental History: An Introduction.* NY: Columbia University Press, 2007. Print.
Part I is a historical overview; Part II is "American Environmental History A to Z—Agencies, Concepts, Laws, and People"; Part III is a Chronology of environmental history; Part IV is an extensive resource guide of films and electronic resources, followed by a bibliographical essay and a bibliography.

Merchant, Carolyn and Thomas Paterson, eds. *Major Problems in American Environmental History.* 2nd ed. Boston: Wadsworth, 2006. Print.
A general reader; suitable for more advanced readers.

Nash, Roderick. *Wilderness and the American Mind.* 4th ed. New Haven, Yale UP, 2001. Print.
A history of the idea of wilderness and how it evolved in the American consciousness. For high school readers and above.

Smith, Kimberly K. *African American Environmental Thought: Foundations.* Lawrence, Kansas: University Press of Kansas, 2007. Print.

Tal, Alon. Ed. *Speaking of Earth: Environmental Speeches That Moved the World.* New Brunswick, NJ: Rutgers University Press, 2006. Print.

Warren, Louis S., ed. *American Environmental History.* Malden (MA): Blackwell, 2003. Print.
Blackwell Readers in American Social and Culture History Ser.
An anthology appropriate for high school readers and above

Environmental Justice

Adamson, Joni, Mei Mei Evans, and Rachel Stein, eds. *The Environmental Justice Reader: Politics, Poetics, and Pedagogy.* Tucson: U of Arizona Press, 2002. Print.

Carter, Majora. "Greening the Ghetto," TED talks. February 2006. Web.
http://www.ted.com/talks/lang/eng/majora_carter_s_tale_of_urban_renewal.html

Gibson, William E., ed. *Eco-Justice—The Unfinished Journey.* Albany, NY: State U of NY, 2004.

Stein, Rachel, ed. *New Perspectives on Environmental Justice: Gender, Sexuality, and Activism.* New Brunswick, NJ: Rutgers U Press, 2004. Print.

Environmental Law

The American Bar Association Web
The American Bar Association's website includes lists of books, discussions of recently enacted laws, newsletters (primarily for members of the legal community). and information regarding the "One Million Trees" project (goal:

plant one million trees by 2014). The ABA includes links to other environmental organizations, too, including the: Alliance for Community Trees, American Forests, Arbor Day Foundation, the Black Bear Conservation Coalition, TreeBank, and TreeLink.

Ferrey, Steven. *Environmental Law: Examples and Explanations*. 5th ed. NY: Aspen Publishers, 2010. Print.
A popular college textbook, Ferrey's work offers a good general overview of environmental law topics.

The Digest of Federal Resource Laws of Interest to the US Fish and Wildlife Service: <http://www.fws.gov/lawsdigest.indx.html>. Web

Gilpin, Alan. *Dictionary of Environmental Law*. Northampton, MA: Edward Elgar Publishing, 2000. Print.
Suitable for the general reader.

Kubasek, Nancy K. and Gary S. Silverman. *Environmental Law*. 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2000. Print.

Sandler, Ronald and Phaedra Pezzullo, eds. *Environmental Justice and Environmentalism: The Social Justice Challenge to the Environmental Movement*. Cambridge (MA): The MIT Press, 2007. Print.

Food

Bittman, M. TED talks. Filmed Dec 2007. Posted May 2008. Web.

Campbell, T. C., & Campbell, T. M., II. (2006). *The China Study: The Most comprehensive Study of Nutrition Ever Conducted and the Startling Implications for Diet, Weight Loss and Long-term Health*. Dallas, Texas: Benbella Books. Print.

Harrison, R. (1964). *Animal Machines: The New Factory Farming Industry*. London: Vincent Stuart Ltd. Print.

Pollan, M. (2006). *The Omnivore's Dilemma: A Natural History of Four Meals*. New York: Penguin Books. Print.

Pollan, M. (2008). *In Defense of Food: An Eater's Manifesto*. New York: Penguin. Print.

Pollan, M. TED talks Filmed March 2007. Posted Feb 2008.

Food, Inc. Dir. Robert Kenner. Magnolia Home Entertainment, 2008. Film.

Gardens And Gardening

Gardening is a way for people to relate intimately with their local environments. Good management of gardens with care for the environment can be a source of pleasure and of local foods. There are numerous books on all aspects of gardening. I have chosen a limited selection here.

Bartholomew, M. (1981). *Square Foot Gardening*. Emmaus, PA: Rodale Press. Print.
Raised bed gardening, much practical information.

Bartholomew, M. (2006). *All New Square Foot Gardening*. Franklin, TN: Cool Springs Press. Print.
A simpler method, less detail about the processes.

Bradley, F. M. & Ellis, B. W. (Eds.). (1992). *Rodale's All-New Encyclopedia of Organic Gardening*. Emmaus, PA: Rodale Press. Print.

Gift, N. (2011). *Good Weed, Bad Weed: Who's Who, What to Do, and Why Some Deserve a Second Chance*. Pittsburgh, PA: St. Lynn's Press. Illustrated. Print.

Hemenway, T. (2001). *Gaia's Garden: A Guide to Home-Scale Permaculture*. White River Junction, VT: Chelsea Green. Print.

Walliser, J. (2008). *Good Bug, Bad Bug: Who's Who, What They Do, and How to Manage Them Organically*. Pittsburgh, PA: St. Lynn's Press. Illustrated. Print.

Islands: the Galapagos Archipelago of Ecuador

In 1959 the Galapagos archipelago off the west coast of Ecuador was declared a national park, and in 1978 UNESCO declared the archipelago a world heritage site. Strange and unique birds live there, including the blue-footed booby, and flightless cormorants.

Kuffner, Stephan, "Postcard: Galapagos" *Time Magazine*. 31 May, 2010, 4
Tourism is threatening the fragile ecosystem that attracts tourists to the Galapagos.

Darwin Foundation Web
A website for the Charles Darwin Research Center explains current concerns and ongoing habitat restoration projects, including a new project to "link traditional biological research and conservation to actions by the human population; environmental education with local goals and values, and ecosystem restoration with the livelihoods of inhabitants."

<http://www.galapagosonline.com/Islands/Galapagos_Islands>.
Information about the Galapagos National Park, including sections on each of the islands and on the different types of animals and plants

NATURE WRITERS / ENVIRONMENTAL LITERATURE

Anthologies and Studies of Nature and Environmental Writing

Anderson, Lorraine et al, eds. *Literature and the Environment: A Reader on Nature and Culture*. NY: Longman, 1998. Print.

Brooks, P. (1980). *Speaking For Nature : How Literary Naturalists From Henry Thoreau To Rachel Carson Have Shaped America*. Boston: Houghton Mifflin. Print.

Elder, J. & Finch, R. (Eds.). (2002). *The Norton Book of Nature Writing*. New York: W.W. Norton & Co. Print.

Glotfelty, C. & Fromm, H. (Eds.). (1996). *The Ecocriticism Reader: Landmarks in Literary Ecology*. Athens, GA: University of Georgia Press. Print.

Hanley, W. (1977). *Natural History in America: From Mark Catesby to Rachel Carson*. New York: Quadrangle. Print.

Hart, G. & Slovic, S. (Eds.). (2004). *Literature and the Environment: Exploring Social Issues through Literature*. Westport, CT: Greenwood P. Print.

Lyon, T. J. "A Taxonomy of Nature Writing" in Glotfelty, Cheryll and Harold Fromm, 276–281.

Lyon, T. J. (2001). *This Incomparable Land: A Guide to American Nature Writing*. Milkweed Editions. Print.

Features a chronological listing of American nature writing and environmental milestones from 1492-2000; an extensive annotated bibliography of American nature writers from Edward Abbey through Ann Zwinger; and an annotated list of "Secondary Studies."

McKibben, B. (Ed.) (2008). *American Earth: Environmental Writing Since Thoreau*. New York: Library of America. Print.

Swearer, D. K. (Ed.). (2009). *Ecology and the Environment: Perspectives from the Humanities*. Cambridge, MA: Center for the Study of World Religions (Harvard UP). Print.

Waddell, C. (Ed.). (2000). *And No Birds Sing: Rhetorical Analyses of Rachel Carson's Silent Spring*. Carbondale, IL: Southern Illinois University Press. Print.

Waddell, C. (Ed.). (1998). *Landmark Essays on Rhetoric and the Environment*. Mahwah, NJ: Lawrence Erlbaum Associates. Print. Ser. Landmark Essays. Vol. 12. Print.

Wall, D. (Ed.). (1994). *Green History: A Reader in Environmental Literature, Philosophy and Politics*. London: Routledge. Print.

Critical Studies

Adamson, J. (2001). *American Indian Literature, Environmental Justice, and Ecocriticism: The Middle Place*. Tucson: University of Arizona Press. Print.

Bryson, M. A. (2002). *Visions of the Land: Science, Literature, and the American Environment from the Era of Exploration to the Age of Ecology*. Charlottesville, VA: University of Virginia P. Print. Under the Sign of Nature Ser. Print.

Bryson, M. A. (2003). Nature, narrative, and the scientist-writer: Rachel Carson's and Loren Eisley's critique of science. *Technical Communication Quarterly*, *12*(4), 369–387. Print.

Buell, L. (1995). *The Environmental Imagination: Thoreau, Nature Writing, and the Formation of American Culture*. Cambridge, MA: Harvard University Press. Print.

Buell, L. (2003). *Writing for an Endangered World: Literature, Culture, and Environment in the U.S. and Beyond*. Cambridge, MA: Belknap/Harvard University Press. Print.

Buell, L. (2005). *The Future of Environmental Criticism: Environmental Crisis and Literary Imagination*. Malden, MA: Blackwell. Print.

Cafaro, P. (2001). Thoreau, Leopold, and Carson: Toward an environmental virtue ethics. *Environmental Ethics*, *23*(1), 3–17. Print.

Christensen, L., et al. (Eds.). (2008). *Teaching North American Environmental Literature*. New York: Modern Language Association. Print.

Clark, T. (2011). *The Cambridge Introduction to Literature and the Environment*. New York: Cambridge University Press. Print.

Gatta, J. (2004). *Making Nature Sacred: Literature, Religion, and Environment in America from the Puritans to the Present*. Oxford: Oxford University Press. Print.

Love, G. A. (2003). *Practical Ecocriticism: Literature, Biology, and the Environment*. Charlottesville, VA: University of Virginia Press. Under the Sign of Nature Series. Print.

Ruffin, K. N. (2010). *Black on Earth: African American Ecoliterary Traditions*. Athens, GA: University of Georgia Press. Print.

Individual Authors:

Ackerman, D. (1990). *A Natural History of the Senses*. New York: Random House. Print.

Ackerman, D. (1995). *The Rarest of the Rare: Vanishing animals, Timeless Worlds*. New York: Random House. Print.

Audubon, J. J. (1831). *Ornithological Biography, or An Account of the Habits of the Birds of the United States of America* (Vol. 1). 5 vols. Philidelphia: J. Dobson; Boston: Hilliard, Gray and Co, 1835 (vol. 2); Edinburg: A. and C. Black, 1835-39 (vols. 3–5). Print.

Audubon, J. J. (1999). *John James Audubon: Writings and Drawings* (C. Irmscher, Ed.). New York: Library of America. Print.

Austin, M. H. (2003). *The Land of Little Rain*. (1903). Introd. Robert Hass. Modern Library. Print.

Austin, M. H. (2003). *The Land of Journeys' Ending*. (1924). Introd. Melody Graulich. Chicago: University of Illinois Press. Print.

Austin, M. H. (1996). *Beyond Borders: The Selected Essays of Mary Austin* (R. J. Ellis, Ed.). Carbondale, IL: Southern Illinois University Press. Print.

Austin, M. H. (2006). *Essential Mary Austin: A Selection of Mary Austin's Best Writing* (K. Hearle, Ed.). Berkeley: Heyday Press. Print.

Austin, M. H. (2005). *Mary Austin's Southwest. An Anthology of Her Literary Criticism* (C. Blackbird & B. Nelson, Eds.). Salt Lake City: University of Utah Press. Print.

Bailey, L. H. (1911). *The Country-Life Movement in the United States*. New York: Macmillan. Print.

Bailey, L. H. (1927). *The Harvest of the Year to the Tiller of the Soil*. New York: Macmillan. Print.

Bass, R. (1996). *The Book of Yaak*. Boston: Houghton Mifflin. Print.

Bass, R. (1995). *The Lost Grizzlies: A Search for Survivors in the Wilderness of Colorado*. Boston: Houghton Mifflin. Print.

Bates, M. (1960). *The Forest and the Sea: A Look at the Economy of Nature and the Ecology of Man*. New York: Random House. Print.

Beebe, W. (1928). *Beneath Tropic Seas: A Record of Diving Among the Coral Reefs of Haiti*. New York: G.P. Putnam's Sons. Print.

Beebe, W. (1921). *Edge of the Jungle*. New York: Henry Holt. Print.

Beebe, W. (1943). *Half Mile Down*. New York: Duell, Sloan and Pearce. Print.

Beebe, W. (1949). *High Jungle*. New York: Duell, Sloan and Pearce. Print.

Beston, H. (2003). *The Outermost House: A Year of Life on the Great Beach of Cape Cod*. (1928). New York: Holt. Print.

Beston, H. (1994). *The Northern Farm: A Glorious Year on a Small Maine Farm*. (1949). Holt. Print.

Burroughs, John.

Burroughs was a popular essayist, biographer, and poet. He continued publishing until 1922. The following are a small selection of his works, available online through Google Books or Project Gutenberg:

Burroughs, J. *Wake-Robin*. (1871) (An early work on ornithology.) Print.

Burroughs, J. (1877). *Birds and Poets*. Print.

Burroughs, J. (1896). *Birds and Bees and Other Studies in Nature*. Print.

Burroughs, J. (1902). *John James Audubon*. Print.

Burroughs, J. (1905). *Ways of Nature*. Print.

Burroughs, J. (2001). *The Art of Seeing Things and Other Essays* (C. Z. Walker, Ed.). Syracuse: Syracuse University Press. Print.
Includes essays "Speckled Trout," "Phases of Farm Life," "A Walk in the Fields," "Reading the Book of Nature," "Thoreau's Wildness," and "The Faith of a Naturalist," among others. The collection also includes illustrations and excerpts from several of Burroughs' books.

Walker, C. Z. (Ed.). (2000). *Sharp Eyes: John Burroughs and American Nature Writing*. Syracuse: Syracuse University Press. Print.

Websites devoted to Burroughs:

<http://www.catskillarchive.com/jb/youtube-john_burroughs.htm>.
A 1919 silent film entitled "A Day With John Burroughs" offers a rare archival view of Burroughs interacting with young visitors to his home.

<http://www.catskillarchive.com/jb/index.htm>.
A wonderful collection of materials relating to Burroughs, including links to various e-books available on Project Gutenberg, links to audio books available through the Librivox project, biographical articles by Burroughs scholars Edward Renehan Jr. and Burroughs contemporaries such as T. Morris Longstreth.

<http://research.amnh.org/burroughs/>.
The website of the John Burroughs Association; offers a slight teachers resource page and a much more ample bibliography; detailing of collections of Burroughs materials currently held in libraries across America.

<http://wakerobin.org/pages/Library.htm>.
A website assembled by Jack Eden which features electronic versions of several of Burroughs' writings.

Comstock, A. B. (1911). *Handbook of Nature- Study*. Ithaca, New York: Comstock Publishing; Ithaca: Cornell University Press, 1986. Print.

Dillard, A. (2007). *Pilgrim at Tinker Creek*. (1974). New York: Harper Perennial Modern Classics. Print.

Dillard, A. (1982). *Teaching a Stone to Talk*. New York: HarperCollins. Print.

Eiseley, L. (1957). *The Immense Journey: An Imaginative Naturalist Explores the Mysteries of Man and Nature*. (1957). New York: Random House. Print.

Eiseley, L. *The Star Thrower*. (1978). New York: Harvest. Print.

Eiseley, L. (1972). *The Unexpected Universe*. (1969). Orlando: Mariner/Harcourt Brace. Print.

Eiseley, L. (1960). *The Firmament of Time*. New York: Scribner. Print.

(1961 recipient of the John Burroughs Medal for the best published nature writing)
Eiseley, L. (1961). *Darwin's Century*. (1958). New York: Anchor. Print.
(1958 winner of the Phi Beta Kappa prize for best book on science.)
Ehrlich, G. (1991). *Islands, the Universe, Home*. New York: Penguin. Print.
Ehrlich, G. (1985). *The Solace of Open Spaces*. New York: Penguin. Print.
Ehrlich, G. (1994). *A Match to the Heart: One's Woman's Story of Being Struck by Lightning*. New York: Penguin. Print.
Ehrlich, G. (2005). *The Future of Ice: A Journey Into Cold*. New York: Vintage. Print.
Ehrlich, G. (2010). *In the Empire of Ice: Encounters in a Changing Landscape*. Washington: National Geographic Society. Print.
Ehrlich, G. (2003). *This Cold Heaven: Seven Seasons in Greenland*. New York: Vintage. Print.
Ehrlich, G. (2000). *John Muir: Nature's Visionary*. Washington: National Geographic. Print.
Klingel, G. (1951). *The Bay*. New York: Dodd, Mead & Co.
Naturalist's explorations of the Chesapeake Bay, Maryland, including dives into the Bay. Won the John Burroughs Medal.
Krakauer, J. (1999). *Into the Wild*. New York: Anchor/Random House.
Leopold, A. (2001). *Sand County Almanac*. (1949). Introd. Kenneth Brower. New York: Oxford University Press. Print. Won the John Burroughs Medal.
Leopold, A. (1999). *For the Health of the Land: Previously Unpublished Essays and Other Writings*. Illust. Abigail Rorer. Washington, DC: Island P. Print.

Surveys of Leopold's Work:
Scheese, D. (1990). 'Something More than Wood': Aldo Leopold and the language of landscape. *North Dakota Quarterly*, *58*(1), 72–89. Print.
Barillas, W. (1996). "Aldo Leopold and midwestern pastoralism." *American Studies*, *37*(2), 61–81. Print.
Stegner, W. (1987). "The legacy of Aldo Leopold." In J. Baird Callicott (Ed.), *Companion to A Sand County Almanac: Interpretive & Critical Essays* (pp. 233–245). Madison: University of Wisconsin Press. Print.

Websites Devoted to Leopold:
<http://www.dnr.state.wi.us/org/LAND/parks/trails/leopold/index.html/>.
Hike the Wisconsin State Park system's Leopold trail.
The website of the Aldo Leopold Foundation, features tools for educators and hosts many archival materials at its site.

Lewis, M., & Clark, W. (2002). *The Journals of Lewis and Clark* (7 Vols., G. E. Moulton, Ed.). Lincoln: Bison Books/U of Nebraska Press. Print.
Lopez, B. (1986). *Arctic Dreams: Imagination and Desire in a Northern Landscape*. New York: Scribner's. Print.
McCarthy, C. (2006). *The Road*. New York: Alfred A. Knopf. Print.
McCarthy, C. (1993). *All the Pretty Horses*. (1992). New York: Vintage. Print.
McKibben, B. (2003). *Enough: Staying Human in an Engineered Age*. New York: Henry Holt.
McKibben, B. (2006). *The End of Nature*. (1989). New York: Random House.

McKibben, B. (2007). *Fight Global Warming Now: The Handbook for Taking Action in Your Community*. New York: Henry Holt.
McKibben, B. (2008). *The Bill McKibben Reader: Pieces from an Active Life*. New York: Henry Holt.
McKibben, B. (2007). *Hope, Human and Wild: True Stories of Living Lightly on the Earth*. Minneapolis: Milkweed Editions.
McKibben, B. (2010). *Eaarth: Making a Life on a Tough New Planet*. New York: Times Books/Henry Holt.
Muir, J. (2001). *The Mountains of California*. (1894). Introd. Bill McKibben. New York: Modern Library.
Muir, J. (2003). *The Yosemite*. (1912). Introd. Gretel Ehrlich, New York: Modern Library.
Muir, J. (2003). *My First Summer in the Sierra*. (1911). Introd. Mike Davis. New York: Modern Library.
Muir, J. (2002). *Travels in Alaska*. (1915). Introd. Edward Hoaglund. New York: Modern Library.
The Sierra Club hosts a wide variety of materials relating to Muir at their website,
Oliver, M. (1983). *American Primitive*. Boston: Back Bay Books. Print.
Winner of the 1984 Pulitzer Prize for Poetry.
Oliver, M. (1992). *House of Light*. (1990). Boston: Beacon P. Print.
Oliver, M. (1994). *Dream Work*. (1986). New York: Atlantic Monthly P. Print.
Oliver, M. (2005). *Why I Wake Early: New Poems*. (2004). Boston: Beacon P. Print.
Oliver, M. (2006). *Blue Iris: Poems and Essays*. (2004). Boston: Beacon P. Print.
Oliver, M. (2006). *Owls and Other Fantasies: Poems and Essays*. (2003). Boston: Beacon P. Print.
Oliver, M. (2009). *Red Bird*. (2008). Boston: Beacon P. Print.
Oliver, M. (2009). *Evidence: Poems*. Boston: Beacon P. Print.
Oliver, M. (2010). *The Truro Bear and Other Adventures*. (2008). Boston: Beacon P. Print.
Smith, H. N. (2000). *Virgin Land: The American West as Symbol and Myth*. (1950, 1978). Cambridge, MA: Harvard University Press. Print.
Snyder, G. (1974). *Turtle Island*. New York: New Directions. Print.
Winner of the 1975 Pulitizer Prize for Poetry.
Thoreau, H. D. (2004). *Walden*. (1854). Introd. Bill McKibben. Boston: Beacon P. Print.
Thoreau, H. D. (1994). *Walking*. (1861). *Nature/Walking*. Introd. John Elder. (1991). Boston: Beacon P. Print.
Thoreau, H. D. (1985). "A Week on the Concord and Merrimack Rivers." (1849). *Thoreau*. New York: Library of America. Print.
Thoreau, H. D. (1985). "The Maine Woods." (1864). In *Thoreau*. New York: Library of America. Print.
Thoreau, H. D. (1985). "Cape Cod." (1865). In *Thoreau*. New York: Library of America. Print.
White, E. B. (1999). *The Essays of E.B. White*. (1977). New York: Harper Perennial. Print.
White, E. B. (2003). *Farewell to Model T: From Sea to Shining Sea*. (1936). New York: Little Bookroom. Print.
White, E. B. (1990). *Writings from The New Yorker 1927–1976* (R. M. Dale, Ed.). New York: Harper Collins. Print.

Whitman, W. (1975). *Leaves of Grass*. (1855). (M. Cowley, Ed.). (1959). New York: Viking/Penguin. Print.
Williams, T. T. (1984). *Pieces of White Shell: A Journey to Navajoland*. New York: Scribner's. Print.
Williams, T. T. (1991). *Refuge: An Unnatural History of Family and Place*. New York: Pantheon. Print.
Williams, T. T. (1994). *An Unspoken Hunger: Stories from the Field*. New York: Pantheon. Print.
Williams, T. T. (2001). *Red: Passion and Patience in the Desert*. New York: Pantheon. Print.
Williams, T. T. (2008). *Finding Beauty in a Broken World*. New York: Pantheon. Print.
Wilson, E. O. (1992). *The Diversity of Life*. Cambridge, MA: Belknap Press, Harvard. Print.

OCEANS AND OCEANOGRAPHY

Oceanography is a broad and interdisciplinary field encompassing biology, chemistry, geology, meteorology, ocean engineering, paleontology, and other disciplines. Any of the books by Tom S. Garrison provide excellent introductions to the field, with informative tables, charts and images. (college-level)

Fish and Fishing

Until the 1990s U.S. scientists and the general public believed that fish stocks were inexhaustible. In fact, "in the late nineteen-eighties, the total world [fish] catch topped out at around eighty-five million tons. . . .[and] for the past two decades, the global catch has been steadily declining" according to Elizabeth Kolbert in *The New Yorker*. Kolbert explains that improved technology such as purse seines (giant nets), echo-sounding sonar, and "factory fish trawlers . . . so gargantuan that they amounted to seafaring towns" have led to the reduction of fish stocks. She reviews several books about the declining fish stock.

Clover, C. (2008). *The End of the Line: How Overfishing is Changing the World and What We Eat*. Berkeley: The University of California Press. Print.
Kolbert, E. (2010, August 2). The scales fall: Is there any hope for our overfished oceans? *The New Yorker*, 70–73. Print.
Kurlansky, M. (1998). *Cod: A Biography of the Fish That Changed the World*. New York: Random House. Print.
Seaver, B. (2011). *For Cod and Country: Simple, Delicious, Sustainable Cooking*. New York: Sterling Epicure. Print.
A cookbook that focuses on sustainability, and advocates eating a wider variety of fish than Americans usually eat.

Marine life

When many of us visit the ocean we know very little about the teeming life in contains, and without special instrumentation it is hard for the ordinary person to learn more. Field trips to aquaria may be possible. But if not, a virtual visit may be effective.

The Monterey Bay Aquarium has a live webcam with views of the Pacific kelp forest, an aviary, otters, penguins, and the Bay. There are photos to guide observation of different creatures. Feeding times for the aquarium are posted. During feeding times there is an audio component as the interpreter at the site explains to viewers what is happening.

NOAA (National Oceanic and Atmospheric Administration) Web
Many resources for teachers and the public

Oceanography

Garrison, Tom. *Oceanography: An Invitation to Marine Science*, 6th ed. Canada: Thomson, Brooks/ Cole: 2007. Print.

Discovery of sound in the sea http://www.dosits.org/ Web
A gallery of photos and audio of marine mammals, teacher's resources, student resources, explanations of the technology used to hear and record sounds from the sea. Many useful links.

Maury, Matthew Fontaine. *Physical Geography of the Sea*. NY: Harper and Brothers, 1855. Print.

Piccard, Jacques and Robert S. Dietz. *Seven miles down; the story of the bathyscaph Trieste*. NY: Putnam, 1961. Print.

Oceans. Jacques Perrin and Jacques Cluzaud. Disney. 2010 DVD Release. Film.

Stewart, Robert. *Our Ocean Planet: Oceanography in the 21st Century, an On-line Textbook*. http://oceanworld.tamu.edu/resources/oceanography-book/contents2.htm

Inner Space Center, University of Rhode Island, Graduate School of Oceanography Web
This site partners with Dr. Robert Ballard's ship Nautilus; NOAA ship Okeanos Explorer; the Mystic Aquarium; National Geographic Society; The Ocean Exploration Trust; NOAA Office of National Marine Sanctuaries; NOAA Office of Ocean Exploration and Research. It has a variety of related educational projects, and live videos of cruises by the Nautilus and Okeanos when they are at sea.

http://oceanexplorer.noaa.gov/explorations/04etta/background/edu/edu.html
NOAA ocean explorations arranged by year, includes information for teachers

Virtual tide pool: http://www.ocparks.com/tidepools/
http://montereybay.noaa.gov/visitor/TidePool/VRTidepool/welcome.html
http://www.pbs.org/wnet/nature/episodes/life-at-the-edge-of-the-sea
http://www.montereyinstitute.org/noaa/lesson10/l10_ex.html
http://racphoto.com/Tidepool/Tidepool.html
Each of these sites has images of some of the creatures to be found in tide pools and descriptions of their adaptations.

The Deepwater Horizon Oil Spill

The Deepwater Horizon oil rig in the Gulf of Mexico exploded on April 20, 2010, killing 11 workers and injuring others, and causing as yet unknown damage on the fragile ecosystem of the coastal marshes and estuaries. Oil continued to pour from the well they were digging despite several attempts by British Petroleum to cap the well. Some of the oil was burned, some was contained in booms, and some was treated with chemical dispersants which broke the slick into smaller particles that sank to the ocean floor. For information about this and other oil spills see: NOAA, U.S. Fish and Wildlife Service, newspaper archives, and a U.S. whitehouse.gov blog.

ORNITHOLOGY

Books

Elphick, C., Dunning, J. B., Jr., & Sibley, D. A. (Eds.). (2001). Illus. David Allen Sibley. In *Sibley Guide to Bird Life & Behavior*. New York: Alfred A. Knopf. Print.

Halle, L. J. (1989). *The Appreciation of Birds*. Illus. Jens Gregersen. Baltimore: Johns Hopkins University Press. Print.

Sibley, D. A. (2000). Illus. David Allen Sibley. In *The Sibley Guide to Birds*. New York: Alfred A. Knopf. Print.

Sibley, D. A. (2002). Illus David Allen Sibley. In *Sibley's Birding Basics*. New York: Alfred A. Knopf. Print.

Field Guides

Dunn, J. L. & Alderfer, J. (Eds.). (2011). Maps by Paul Lehman. *National Geographic Field Guide to the Birds of North America*. Washington, DC: National Geographic Society. Print.

Kaufman, K. (2000). *Birds of North America*. New York: Houghton Mifflin. Print.

Peterson, R. T. (1947). Illus Roger Tory Peterson. In *A Field Guide to the Birds: Eastern Land and Water Birds*. Boston: Houghton Mifflin. Print.

Sibley, D. A. (2003). Illus David Allen Sibley. In *The Sibley Field Guide to Birds of Eastern North America*. New York: Alfred A. Knopf.

Websites

eBird Web

Founded in 2002 by the Cornell Lab of Ornithology and National Audubon Society, eBird is a real time on-line checklist for amateur birders and professional ornithologists. Members record bird sightings and counts, thus generating a large database of bird migration and bird populations throughout the western hemisphere.

http://www.whatbird.com/
Search by multiple terms, such as color, secondary color, size, bill size and shape.

Bird migration http://www.enature.com/birding/migration_home.asp
Maps of bird migrations, wildlife facts, quizzes and other information. Click on a region of the U.S. to follow the migration paths of the birds that may be found in that region.

http://www.closertonature.com/outdoors/bird-migration.htm
Many links to other sites such as the Smithsonian, and Field Guides (including podcasts about different birds)

http://www.npwrc.usgs.gov/resource/birds/migratio/index.htm
Brochure about bird migration by the Fish and Wildlife Service

OTHER ANIMALS AND PLANTS

Audubon Society Field Guide to North American Butterflies. NY: Alfred A. Knopf, 1981. Print.

Newcomb, Lawrence. Illus Gordon Morrison. *Newcomb's Wildflower Guide.* Boston, Little, Brown., 1977. Print.

Peterson, Roger Tory and Margaret McKenny. *A Field Guide to Wildflowers of Northeaster and North-central North America.* Boston: Houghton Mifflin, 1968. Print.

Peterson, Roger Tory. *Flash Guides.*
Laminated folding guides to birds of different regions in the U.S.

http://nationalzoo.si.edu/ Web
The Smithsonian National Zoo website offers curriculums, and webcams.

PESTICIDES

Beatty, R. G. (1973). *The DDT Myth: Triumph of the Amateurs.* New York: John Day.

Briggs, S. (1992). *Basic Guide to Pesticides: Their Characteristics and Hazards.* CRC Press. Print.

Competitive Enterprise Institute. "Rachelwaswrong.org" Web

Dunlap, T. R. (1981). *DDT: Scientists, Citizens, and Public Policy.* Princeton, NJ: Princeton University Press.

Dunlap, T. R. (Ed.). (2008). *DDT, Silent Spring, and the Rise of Environmentalism: Classic Texts.* Seattle: University of Washington Press.

Graham, F., Jr. (1970). *Since Silent Spring.* Boston: Houghton Mifflin.
Appendix I by Shirley a Briggs, "Safer Pesticides for Home and Garden," includes a descriptive list of safe and dangerous pesticides as well as some basic references. Appendix II by Harold G. Alford describes "Federal Registration Requirements for Pesticide Products."

Jukes, T., et al. (1973). *Effect of DDT on Man and other Mammals: I.* New York: MSS Information Corp.

Kinkela, D. (2011). *DDT and the American Century: Global Health, Environmental Politics, and the Pesticide that Changed the World.* Chapel Hill, N. Carolina: The University of North Carolina Press. Print.

CHAPTER 8

SUSTAINABILITY

McDonough, W., & Braungart, M. (2002). *Cradle to Cradle: Remaking the Way We Make Things*. New York: North Point Press. Print.

The book itself is made of a recyclable polymer, rather than paper. The authors advocate methods of production that use, and even produce, renewable resources.

McDounough, W. TED talks. Filmed Feb 2005. Posted April 2007.Web

Trainer, T. (Ed.). (1995). *The Conserver Society: Alternatives for Sustainability*. Zed. Print.

URBAN ENVIRONMENT

Carter, M. "Greening the Ghetto." TED talks Filmed Feb 2006. Posted June 2006. Web.

Lerner, J. (2007, March). "Jaime Lerner sings of the city," TED talks..

Miller, L. B. (2009). *Parks, Plants, and People: Beautifying the Urban Landscape*. New York: W. W. Norton

WEATHER EVENTS AND DISASTERS

The National Weather Service Web

The *National Geographic* Web

Dust Storms

http://www.livinghistoryfarm.org/farminginthe30s/water_02.html Web

http://www.ccccok.org/museum/dustbowl.html Web

Guthrie, Woody song lyrics Web

Hurricanes

As I read Carson's chapter about waves in *The Sea Around Us* I am reminded of the reports I have read about a dangerous hurricane that wrecked havoc on September 21, 1938, in Rhode Island, the state where I live. The Whale Rock lighthouse off the coast of Narragansett was swept off its base, carrying the unfortunate lighthouse keeper Walter Eberle to his death. You can read about hurricanes in the following sources:

Allen, E. S. (1976). *A Wind to Shake the World*. Boston: Little, Brown and Company. Print. (Reprinted by Commonwealth Editions, 2006.)

Brinkley, D. (2007). *The Great Deluge: Hurricane Katrina, New Orleans, and the Mississippi Gulf Coast*. New York: Harper Perennial. Print.

Burns, C. (2005). *The Great Hurricane: 1938*. New York: Grove/Atlantic. Print.

D'Entremont, J. (2001, April). Whale Rock's keeper Walter B. Eberle, 1898–1938. *Lighthouse Digest*. Print.

D'Entremont, J. (2006). *The Lighthouses of Rhode Island*. Beverly, MA: Commonwealth Editions. Print.

Federal Writers' Project. (1938). *New England Hurricane: A Factual, Pictorial Record*. Boston: Hale, Cushman, & Flint. Print.

R. Holmes has assembled a collection of information relating to Rhode Island lighthouses. Web.

Jacobson, P. & Long, R. dirs. *Violent Earth: New England's Killer Hurricane*. History Channel, 2006. DVD. (also *Nature's Fury: New England's Killer Hurricane*.)

Larson, E. (2000). *Isaac's Storm: A Man, A Time, and The Deadliest Hurricane in History*. New York: Vintage. Print.

Larry, A. Lovering has a collection of links relating to hurricanes, including the National Hurricane Center and the Saffir-Simpson Hurricane Scale. His general hurricane link collection can be found at http://www.southstation.org/hurricane.htm

Mandia, S. http://www2.sunysuffolk.edu/mandias/38hurricane/index.html

McCullough, D., narr. (2007). *The American Experience: The Hurricane of '38*. WGBH Educational Foundation. DVD. (The associated website, http://www.pbs.org/wgbh/americanexperience/films/hurricane/, includes a teacher's guide for multiple disciplines in addition to additional home videos of the aftermath of the hurricane, interviews with survivors, and a brief discussion of the National Weather Service.

The *NY Times* has a collection of resources dealing with the broad topic of hurricanes

Scotti, R. A. (2003). *Sudden Sea: The Great Hurricane of 1938*. New York: Little, Brown. Print.

Earthquakes

Google News (www.news.google.com) offers a search option that scans thousands of international and domestic newspapers. A quick recent search for "earthquakes" filtered to show results for only June 2011, for instance, returned more than 13,000 results. Students and teachers alike might be interested in using this resource to compare coverage of environmental disasters.

Incorporated Research Institutions for Seismology (IRIS) offers a very detailed website (http://www.iris.edu/hq/retm) which includes "teachable moment" PowerPoint presentations. The presentation on the March 2011 9.0 Japanese earthquake and resulting tsunami features graphs, newspaper photos, and a moment-by-moment recap of the earthquake.

National Earthquake Information Center at http://earthquake.usgs.gov/regional/neic/. A variety of educational resources available for both students and teachers, including information regarding specific earthquake events such as the devastating recent earthquakes in Haiti, Chile, New Zealand, and Japan.

Scholastic offers a teacher's resource website devoted to the topic of weather here: http://teacher.scholastic.com/activities/wwatch/earthquakes/famous.htm. The earthquake section includes a brief and accessible recap of several historically famous earthquakes and an experiment section that features a step-by-step process for building a personal quake detector.

Virtual Museum of the City of San Francisco offers a variety of information including eyewitness accounts of the 1906 earthquake and the resulting devastation. http://www.sfmuseum.org/1906/ew.html

Tornadoes

Federal Emergency Management Agency (FEMA)'s website hosts a variety of useful information, including a glossary of tornado terms. http://www.fema.gov/hazard/tornado/index.shtm

National Geographic's tornado website includes information on storm chasing, photo galleries, and simple explanations of how tornadoes are formed.

The *NY Times* "tornadoes" topic page, , offers many useful news articles and photos relating to tornadoes, including information on the deadly storm season of 2011.

NOAA's National Severe Storms Laboratory has two excellent websites filled with resources. The first site, http://www.nssl.noaa.gov/edu/safety/tornadoguide.html, includes information on tornadoes, their formation, and variations of tornadoes. The site also includes information that could be useful for developing a family disaster plan. The second site, http://www.nssl.noaa.gov/edu/, includes a comprehensive list of teacher resources, as well as children's coloring books featuring the adventures of Billy and Maria, two characters learning about staying safe during severe weather emergencies.

NOAA has another site, also devoted to severe weather. Learn more about tornadoes and lightning here: http://www.noaawatch.gov/themes/severe.php

The *Tornado Project* is a private company interested in all aspects of tornadoes, including the history of storms. Their website can be found at http://www.tornadoproject.com/.

Tsunamis

http://ptwc.weather.gov/

The home site of the Pacific Tsunami Weather Center, offers frequent warnings, charts of incidents, and a wide variety of information for site visitors.

http://www.ess.washington.edu/tsunami/index.html

a collection of basic materials collected by the Earth and Space Sciences department at the University of Washington. Particularly useful items for teachers include graphics on how to protect oneself in case of a tsunami.

http://walrus.wr.usgs.gov/tsunami/

Learn more about a variety of tsunami events, assembled by the US Geological Survey.

John McDaris and Monica Bruckner of Carleton College have assembled a fine collection of tsunami visualizations (animations and live video of actual events). http://serc.carleton.edu/NAGTWorkshops/ocean/visualizations/tsunami.html.

The American Red Cross website features many preparedness fact sheets, including one for tsunamis

To teach students how to search for news articles online could include going to www.news.google.com and searching either "March 2011 Japan tsunami" or "2004 Asian tsunami." Literally hundreds of search results will show up from a variety of both domestic and international news sources.

Wildfires

The Geospatial Multi-Agency Coordinator Group (GeoMAC) features maps of current forest fires at http://www.geomac.gov/index.shtml.

National Geographic's website features a section on their website devoted to the environment, organized by categories including one devoted to wildfires,

National Interagency Fire Center's website:

Scheese, Don. *Mountains of Memory: A Fire Lookout's Life In The River Of No Return Wilderness.* U of Iowa Press, 2001. Print.

The Ad Council, the National Association of State Foresters and the US Forest Service have teamed to offer resources for both educators and students at their Smokey Bear website, http://www.smokeybear.com/resources.asp and http://www.smokeybear.com/kids/?js=1

OTHER ISSUES

BREAST CANCER

Carson died of metastatic breast cancer. While all the causes of cancer, are not known, WHO (the World Health Organization) thinks that environmental pollution is a large causal factor. Prevention and treatment of breast cancer has been controversial, involving issues of gender, economics, medical ethics, and medical practice. The books and websites below address the topic of breast cancer, and one is by the doctor who supervised Carson's treatment after her mastectomy.

Websites

American Cancer Society http://www.cancer.org/

The Army of Women http://www.armyofwomen.org/

Dr. Susan Love is affiliated with a program to help eradicate cancer called the Army of Women. Women sign up to be notified of ongoing medical research programs that they may participate in if applicable and appropriate.

President's Cancer Panel. http://deainfo.nci.nih.gov/advisory/pcp/pcp.htm

PubMed http://www.ncbi.nlm.nih.gov/pubmed

The U.S. National Library of Medicine, National Institutes of Health. "PubMed comprises more than 21 million citations for biomedical literature from MEDLINE, life science journals, and online books.

Books

Casamayou, Maureen Hogan. *The Politics of Breast Cancer.* Washington, D. C.: Georgetown University Press, 2001. Print.

Crile, George, Jr. *What women should Know About the Breast Cancer Controversy.* NY: Macmillan, 1973. Print.
Crile was the doctor who supervised Carson's treatment after her mastectomy. His wife, Jane Halle Crile, died of breast cancer. He advocated the less invasive operations for breast cancer, the modified radical or partial mastectomy.

Light, Allie and Irving Saraf. *Rachel's Daughters: Searching for the Causes of Breast Cancer.* Light-Saraf Films, 1997. Film.

Klawiter, Maren. *The Biopolitics of Breast Cancer: Changing Cultures of Disease and Activism.* Minneapolis: University of Minnesota Press, 2008. Print.

Lerner, Barron H. *The Breast Cancer Wars: Fear, Hope, and the Pursuit of a Cure in Twentieth-Century America.* Oxford: Oxford University Press, 2001. Print.
History of breast cancer treatment in the U. S.

PEDAGOGY

Oracle ThinkQuest Projects by students for students information about ocean oil spills

Project Sea Grant
research initiative partnering government agencies and university researchers. resources for volunteers and teachers, available online http://seagrant.noaa.gov

TeacherServe by the National Humanities Center
resource for teachers. The section "Nature Transformed: the Environment in American History" contains material on "Native Americans and the Land," "Wilderness and American Identity" (has a section on Rachel Carson)," and "The Use of the Land."

Rosser, Sue V. *Female-Friendly Science: Applying Women's Studies Methods and Theories to Attract Students.* NY: Pergamon Press, 1990. Print.

Resources on Problem Based Learning

The Interdisciplinary Journal of Problem-Based Learning,

an open-access journal started publication in 2006. The journal "publishes relevant, interesting, and challenging articles of research, analysis, or promising practice related to all aspects of implementing problem-based learning (PBL) in K–12 and post-secondary classrooms."

Amador, Jose, Libby Miles, and C. B. Peters. *The Practice of Problem Based Learning: A Guide to Implementing PBL in the College Classroom.* Boston, MA: Anker Pub, 2006. Print.

Barell, John. *Problem Based Learning: An Inquiry Approach.* Thousand Oaks, CA: Corwin Press, 2007. Print.

Duch, Barbara J., Susan E. Groh, and Deborah E. Allen, eds. *The Power of Problem-based Learning: a Practical "How-To" for Teaching Undergraduate Courses in Any Discipline.* Sterling, VA: Stylus, 2001. Print.

Glasgow, Neal. *New Curriculum for New Times: A Guide to Student-Centered, Problem-Based Learning..,* Thousand Oaks, CA: Corwin Press, 1997. Print.

Knowlton, Dave S., David C. Sharp, eds, *Problem-Based Learning in the Information Age.* San Francisco, CA: Jossey -Bass, 2003. Print.

Savin-Baden, Maggi. *Problem Based Learning in Higher Education: Untold Stories.* Philadelpha, PA*:* Society for Research into Higher Education, 2000. Print.

WOMEN IN SCIENCE

Overviews

Gornick, V. (2009). *Women in Science: Then and Now*. New York: The Feminist Press. Print.

Interviews with women scientists, profiles with names changed.

Kass-Simon, G. & Farnes, P. (Eds.). (1990). *Women of Science: Righting the Record*. Bloomington, IN: Indiana University Press. Print.

Rosser, S. V. (1990). *Female-Friendly Science: Applying Women's Studies Methods and Theories to Attract Students*. Teachers College Press. Print.

Practical suggestions for teaching science more effectively so as to improve women's participation in science professions.

Rosser, S. V. (2004). *The Science Glass Ceiling: Academic Women Scientists and the Struggle to Succeed*. Routledge. Print.

Suggestions for changing the culture to promote equality.

Rossiter, M. (1982). *Women Scientists in America: Struggles and Strategies to 1940*. Baltimore: Johns Hopkins University Press. Print.

Rossiter, M. (1998). *Women Scientists in America: Before Affirmative Action, 1940–1972* (Vol. 2). Baltimore: Johns Hopkins University Press. Print.

Tang, J. (2006). *Scientific Pioneers: Women Succeeding in Science*. Lanham, MD: University Press of America. Print.

Valian, V. (1999). *Why So Slow? The Advancement of Women*. Cambridge, MA: The MIT Press. Print.

Based on theories and studies from psychology, economics, and sociology this book examines obstacles to women's advancement in the professions and offers strategies for improving the chilly climate women face.

Biographies and General Works by Women Scientists

Kass-Simon, G. & Farnes, P. (Eds.). (1993). *Women of Science: Righting the Record*. Indiana University Press. Print.

Chapters on the achievements of women in various branches of science including archaeology, astronomy, chemistry, crystallography, geology, medicine, physics.

McGrayne, S. B. (1998). *Nobel Prize Women in Science: Their Lives, Struggles, and Momentous Discoveries*. Washington, DC: Joseph Henry Press. Print.

Marie Curie

Curie, E. (2001). *Mme. Curie: A Biography* (V. Sheean, Trans.). Da Capo Press. Print.
The first biography of Curie, written by her daughter.
Quinn, S. (1996). *Marie Curie: A Life*. Da Capo Press. Print.
Mme. Curie DVD, 2007. Original film 1944 starring Greer Garson and Walter Pidgeon. Black and white.
Based on the Eve Curie biography. This film is unusual in its portrayal of a woman as a practicing scientist.

Other Women Scientists

Clark, E. (1953). *Lady With a Spear*. New York: Harper. Print.
Clark, E. (1969). *The Lady and the Sharks*. New York: Harper and Row. Print.
Mcgovern, A. (1978). *Shark Lady: True Adventures of Eugenie Clark*. Scholastic PaperbacksIll. Ruth Chew. Print.
Earle, S. (1996). *Sea Change: A Message of the Oceans*. Ballantine Books. Print.
Earle, S. (1999). *Dive: My Adventures In the Deep Frontier*. National Geographic Children's Books. Print.
Earle, S. (2001). *Hello, Fish!: Visiting The Coral Reef*. National Geographic Children's Books. Print.
Earle, S. (2009). *The World is Blue*. National Geographic Books. Print.
Earle, S. & Glover, L. K. (2008). *Ocean: An Illustrated Atlas (National Geographic Atlas)*. National Geographic. Print.
http://www.nationalgeographic.com/field/explorers/sylvia-earle/ Web
http://www.tedprize.org/sylvia-earle/
Earle won the TED Prize in 2009, which gave her resources to act on her "wish big enough to change the world." Her wish is "to ignite public support for a global network of marine protected areas, hope spots large enough to save and restore the ocean, the blue heart of the planet."

Primatologists Dian Fossey, Birute Galdikas, Jane Goodall

Gorillas in the Mist. Dir. Michael Apted, Perf. Sigourney Weaver. (1988). Universal Studios. Film.
Galdikas, B. (2008). *Great Ape Odyssey*. New York: Barnes and Noble. Print.
Galdikas, B. (1996). *Reflections of Eden: My Years with the Orangutans of Borneo*. Boston: Back Bay Books. Print.
Goodall, J. (2010). *In the Shadow of Man*. Mariner Books. Print.
Goodall, J. (2010). *Through a Window: My Thirty Years with the Chimpanzees of Gombe*. Mariner Books.
Montgomery, Sy. (1991). *Walking with the Great Apes: Jane Goodall, Dian Fossey, Birute Galdikas*. New York: Houghton Mifflin. Reprint Chelsea Green 2009. Print.
Mowat, F. (1988). *Woman in the Mists*. Grand Central Publishing. Print.

ENVIRONMENTAL AND RELATED ORGANIZATIONS

Organization: American Farmland Trust
Founded: 1980 by a group of farmers and citizens concerned about the rapid loss of farmland to development
Mission Statement: to help farmers and ranchers protect their land, produce a healthier environment and build successful communities.
Contact Information: 1200 18th Street NW, Suite 800, Washington, D.C. 20036 202-331-7300
Sources: http://www.farmland.org/default.asp

Organization: Association for the Study of Literature and the Environment (ASLE) www.asle.org
Founded: 1992
Mission Statement: to promote the understanding of nature and culture for a sustainable world by fostering a community of scholars, teachers, and writers who study the relationships among literature, culture, and the physical environment.
Contact Information: Fax: 603-357-7411info@asle.org
Publication: *ISLE: Interdisciplinary Studies in Literature and Environment* a quarterly academic journal

Organization: Center for Biological Diversity
Founded: In 1989 as a small group by the name of Greater Gila Biodiversity Project.
Mission Statement: Because diversity has intrinsic value, and because its loss impoverishes society, we work to secure a future for all species, great and small, hovering on the brink of extinction. We do so through science, law and creative media, with a focus on protecting the lands, waters and climate that species need to survive.
Contact Information: P.O. Box 710, Tucson, AZ 85702-0710, (520) 623.5252
Sources: http://www.biologicaldiversity.org/
http://en.wikipedia.org/wiki/Center_for_Biological_Diversity

Centers for Disease Control and Prevention (CDC)
Mission Statement: Collaborating to create the expertise, information, and tools that people and communities need to protect their health – through health promotion, prevention of disease, injury and disability, and preparedness for new health threats..
Contact Information: http://www.cdc.gov
Every 2 years the CDC issues a Report on Human Exposure to Environmental Chemicals.

http://www.cdc.gov/exposurereport/pdf/FourthReport_UpdatedTables_Feb2012.pdf

Organization: Center for International Environment Law (CIEL)
Founded: 1989

Mission Statement: Non-profit; works to strengthen and use international law and institutions to protect the environment, promote human health, and ensure a just and sustainable society. Provides legal counsel and advocacy, policy research and capacity building

Contact Information: 1350 Connecticut Avenue Northwest, Washington, DC 20036 (202) 785-8700

Organization: Environment America

Founded: November 2007, a federation formed of state environmental agencies

Mission Statement: "a 501(c)(3) organization. We are dedicated to protecting our air, water and open spaces. We investigate problems, craft solutions, educate the public and decision-makers, and help the public make their voices heard in local, state and national debates over the quality of our environment and our lives."Contact Information: http://www.environmentamerica.org/

Organization: Environmental Defense Fund

Founded: By a small group of scientists in 1967

Mission Statement: Environmental Defense Fund's mission is to preserve the natural systems on which all life depends. A commitment to sound science, efficient, market-based solutions.

Contact Information: Membership and Public Information

1875 Connecticut Ave, NW, Suite 600, Washington, DC 20009, 1-800-684-3322

Sources: http://www.edf.org/page.cfm?tagID=380

Organization: Fund For Wild Nature

Founded: In 1982 by Lance Christie, LaRue Christie, Abe Blank, Bill Bishop, Ken Sanders, and Bruce Hayse.

Mission Statement: The Fund for Wild Nature invests in cutting-edge grassroots organizations and innovative conservation efforts that meet emerging needs for protecting biodiversity and wilderness.

Fact: They are currently one of the leading endangered species protection organizations in the world.

Contact Information: Fund for Wild Nature, P.O. Box 42523, Portland, OR 97242 (360) 636-6030

Sources: http://www.fundwildnature.org/
http://fund-for-wild-nature.co.tv/

Organization: Green Zionist Alliance

Founded: In 2001 by Dr. Alon Tal, Dr. Eilon Schwartz and Rabbi Michael Cohen.

Mission Statement: The Green Zionist Alliance, a North America-based 501(c)3 nonprofit, offers a place for all people — regardless of political or religious affiliation — who care about humanity's responsibility to preserve the Earth and the special responsibility of the

Jewish people to preserve the ecology of Israel.
Sources: http://www.greenzionism.org/
http://en.wikipedia.org/wiki/Green_Zionist_Alliance#History

Organization: Greenpeace
Founded: By Dorothy Stowe, Dorothy Metcalfe, Zoe Hunter and others in 1971.
Mission Statement: Greenpeace is an independent ,campaigning organisation which uses nonviolent, creative confrontation to expose global environmental problems, and to force the solutions which are essential to a green and peaceful future. Greenpeace's goal is to ensure the ability of the earth to nurture life in all its diversity.
Contact Information: Greenpeace International
Sources: http://www.greenpeace.org/usa
http://en.wikipedia.org/wiki/Greenpeace#Criticism

Organization: John Burroughs Association
Founded: 1921
Mission Statement: fostering a love of nature as exemplified by Burroughs' life and work; preserving the places associated with his life, especially his house, Slabsides and the John Burroughs Sanctuary near West Park, NY. The John Burroughs Medal Award was created in 1926 to honor outstanding natural history writing, a genre perfected by Burroughs. Awards are also given for the John Burroughs List of Nature Books for Young Readers and the Outstanding Published Nature Essay.
Contact Information: The John Burroughs Association, Inc.at the American Museum of Natural History, 15 West 77th Street, New York, NY 10024
Web Address: http://john-burroughs.org/jb_assoc/assoc.htm

Organization: Keep America Beautiful
Founded: In 1953
Mission Statement: To engage individuals to take greater responsibility for improving their community's environment.
Contact Information: 1010 Washington Boulevard, Stamford, CT 06901
Sources: http://en.wikipedia.org/wiki/Keep_America_Beautiful
http://www.kab.org/site/PageServer?pagename=index

Organization: National Academy of Sciences
Founded: 1863 by an Act of Congress
Mission Statement: to "investigate, examine, experiment, and report upon any subject of science or art" whenever called upon to do so by any department of the government.
Fact: Provides "advice on the scientific and technological issues that frequently pervade policy decisions. The National Research Council (NRC), was created expressly for this purpose and provides a public service by working outside the framework of government to ensure independent advice on matters of science, technology, and medicine. The

NRC enlists committees of the nation's top scientists, engineers, and other experts, all of whom volunteer their time to study specific concerns."

Contact Information: National Academy of Sciences, 500 Fifth Street, NW Washington, DC 20001 http://www.nasonline.org

Organization: National Association of Environmental Professionals

Founded: In 1975.

Mission Statement: Our mission is to be the interdisciplinary organization dedicated to developing the highest standards of ethics and proficiency in the environmental professions.

Contact Information: NAEP Headquarters, PO Box 460, Collingswood, NJ 08108,

Sources: http://www.naep.org/mc/page.do?sitePageId=91299&orgId=naep

Organization: National Cancer Institute www.cancer.gov

Founded: 1937

Mission Statement: The National Cancer Institute coordinates the National Cancer Program, which conducts and supports research, training, health information dissemination, and other programs with respect to the cause, diagnosis, prevention, and treatment of cancer, rehabilitation from cancer, and the continuing care of cancer patients and the families of cancer patients. Publishes the SEER index annually: Surveillance Epidemiology and End Results.

Contact Information: 1-800-4-CANCER

Organization: National Endowment for the Oceans

Founded: March 8, 2012Mission Statement: the revenue that comes from the extraction and use of ocean resources is invested back into understanding and conserving our ocean. National competitive grants would be available through the Ocean Resources and Assistance Grant Program. State, local, regional and affected Indian tribal entities; non-profit organizations; and academic institutions would be eligible to apply for these grants.

Organization: National Institute of Environmental Health Sciences (National Institutes of Health · U.S. Department of Health and Human Services)

Mission Statement: "to reduce the burden of human illness and disability by understanding how the environment influences the development and progression of human disease."

Contact Information: http://www.niehs.nih.gov

National Institute of Environmental Health Sciences, P.O. Box 12233, MD K3-16 Research Triangle Park, North Carolina USA 27709-2233

Organization: Natural Resources Defense CouncilFounded: 1970Mission Statement: To safeguard the Earth: its people, its plants and animals and the natural systems on which

all life depends.Contact Information: Natural Resources Defense Council, 40 West 20th Street, NY

Organization: Nature Conservancy

Founded: In 1951

Mission Statement: works around the world to protect ecologically important lands and waters for nature and people.

Contact Information: The Nature Conservancy, 4245 North Fairfax Drive, Suite 100 Arlington, VA 22203-1606, +1 (703) 841-5300

Sources

EPILOGUE

The more I learned about Carson's life and works, the more questions I had about the issues she broached. How much have we learned since her book about the persistence of the organochlorine pesticides and their impact on the environment, on wildlife and on human health? What new pesticides and other persistent organic chemicals have been developed and how does their effectiveness and possible toxicity compare to the pesticides Carson discussed? Carson focused on cancer but are there other health problems, such as endocrine disruption, arising from persistent organic polluters?

To what extent have the trends Carson mentioned about rising ocean levels continued and what are their implications for coastal communities? How can we learn to manage the fish and other resources of the ocean? How polluted is the ocean, and how can it be cleaned?

What is the food value of crops raised with the new synthetic fertilizers, and of animals raised for human consumption in factory farms? How can we insure an adequate food supply for a growing human population?

What have been the successes and shortcomings of the environmental movement? What new trends and strategies are available for sustaining life on earth? How can we best deal with insect-borne diseases such as malaria?

How well have we learned the lessons of respect for the environment that Carson taught? How well are we teaching our young people to retain a sense of wonder and an appreciation for our planet? How sustainable are our current patterns of living and our use of environmental resources? How clean are our air, our land, and our water?

Twenty-five years after the book appeared, a group of chemists convened a symposium and published *Silent Spring Revisited,* a collection of twelve essays exploring questions about scientific studies of pesticides and of the effects of pesticides on agriculture, wildlife, and human health. It is a useful book. But there are more areas of inquiry that Carson's work raises: new environmental issues such as global warming, environmental ethics, environmental justice, and biological and chemical questions such as the current state of toxicology studies, carcinogenicity, factory farming and others.

It is now fifty years since the publication of *Silent Spring.* It is time for a new look at the meanings and implications of Carson's work for us in the second decade of a new century.

APPENDIX 1

RACHEL CARSON CHRONOLOGY

(Adapted from The Linda Lear Collection of Rachel Carson Books and Papers at the Shain Library, Connecticut College, New London, CT)

1907 Rachel Louise Carson born, 27 May, in Springdale, PA to Maria McLean and Robert Warden Carson

1913 Rachel Carson starts school

1918 "A Battle in the Clouds" in *St. Nicholas Magazine.* Carson receives Silver Badge, $10 prize

1919 Two more stories in *St. Nicholas Magazine*

1925 Starts Pennsylvania College for Women

1929 Graduates magna cum laude; summer at Woods Hole Biological Lab (on fellowship); starts MS in marine zoology at Johns Hopkins

1930-31 Teaches ½ time at University of Maryland College Park; teaches summer school at Johns Hopkins until 1936 and zoology at University of Maryland until 1933

1932 Receives M.A. degree in zoology from Johns Hopkins University

1935 Earns $19.25 a week writing radio scripts about fish

1936 Only woman to take junior aquatic biologist exam; scores 97%--the top score; sister Marian dies; Rachel and Maria Carson raise Marian's two daughters, Virginia and Marjorie

1936 Earns $10-20 per article for features in *Baltimore Sunday Sun*

1938 Spends summer at Woods Hole writing *Under the Sea Wind*

APPENDIX 1

1940 Bureau of fisheries merges with Bureau of Biological Survey to become Fish and Wildlife Service.

1942-3 moves to Chicago temporarily to work at Office of Information, Fish and Wildlife Service for wartime work

1942-52 Promotions to Assistant Aquatic Biologist (42-43); Associate Aquatic Biologist (43-45); Aquatic Biologist (45-46); Information specialist (46-49); Biologist and Chief Editor (49-52).

1942-44 Edits *Progressive Fish-Culturist*

1943-45 Writes and edits government booklets on eating fish (Conservation bulletins #33, 34, "Food from the Sea")

1946 Visits Chincoteague for "Conservation in Action" booklets; Rents cabin on Sheepscot River, Boothbay Maine

1948 Hires Marie Rodell as literary agent

1949 Diving in Florida; descends 15 feet; with Marie Rodell goes on voyage to Georges Bank 200 miles off coast of Boston on the Albatross III

1949 Receives Saxon Fellowship

1950 "Birth of an Island" in *Yale Review* wins $1000 George Westinghouse Award from American Association for the Advancement of Science for best science writing in a magazine in that year

1951 Guggenheim Fellowship for *Edge of the Sea* research

1951 *The Sea Around Us* voted by NY Times "outstanding book of the year" translated into more than 32 languages

1952 Wins National Book Award for best nonfiction book of 1951; The Henry Grier Bryant Gold Medal, Geographical Society New York; Zoological Society Gold Medal.

Awarded a Simon Guggenheim Fellowship for research on tidal life.

1952 Wins John Burroughs Medal for natural history book of outstanding literary quality.

1952 Receives honorary doctorate degrees from Pennsylvania College for Women, Oberlin College, and Drexel Institute for Technology

1952 Resigns from government job at Bureau of Fish and Wildlife

1954 Elected to Theta Sigma Phi the national fraternity of women in journalism and to the Royal Society of Literature in England

1955 *Edge of the Sea* serialized in *New Yorker*; condensed version in *Readers Digest;* best seller for 23 weeks

1956 Writes script for "Omnibus" television program on clouds; "Help Your Child to Wonder" published in *Woman's Home Companion*

1957 Niece Marjorie dies; Rachel Carson adopts her son Roger Christie, age 5

1958 *The Sea Around Us* edition for children; Publishes "Our Ever-Changing Shore" in *Holiday*

1958 Signs contract with Houghton Mifflin for pesticide book (tentative title "Control of Nature")

1958 (Dec) Maria Carson dies

1959 Lump in breast removed; Hires Jeanne Davis as assistant and secretary

1959 Cranberry pesticide scare indicates dangers that Carson points out

1960 Serves on Natural Resources Committee of the Democratic Advisory Council. Writes about pollution, radioactive waste in sea and chemical poisoning for the platform; Learns cancer is spreading— undergoes mastectomy and radiation treatment

1962 *New Yorker* serializes *Silent Spring;* Book of the Month Club selects *Silent Spring* for October; Chemical companies try to stop publication of *Silent Spring;* allocate $250,000 to discredit the book

1962 Jerome B. Wiesner sets up Pesticides Committee of the Office of Science and Technology, Special Panel of the President's Science Advisory Committee (PSAC) (report released May 15, 1963)

1963 Awarded Albert Schweitzer Medal of the Animal Welfare Institute; National Wildlife Foundation names Carson Conservationist of the Year; April 3 CBS broadcasts "Silent Spring of Rachel Carson"

1963 (June 4) Carson testifies before Senate Government Operations Committee on pesticides; June 6 testifies before Senate Commerce Committee

APPENDIX 1

1963 Monarch butterfly letter to Dorothy Freeman

1963 Audubon Soc Medal (1st to a woman); Cullum Medal , American Geographic Society; elected to American Academy of Arts and Letters (3 of 50 members are women)

1964 Houghton Mifflin publishes 600,000 copies of *Silent Spring* paperback printing

1964 (March 23) NY Herald Tribune photo of Rachel Carson and article on massive fish kills

1964 (April 14) Rachel Carson dies

1964 Rachel Carson Trust for the Living Environment set up by Marie Rodell, Ruth Scott, and Shirley Briggs. Name later changed to Rachel Carson Council

1965 *A Sense of Wonder* published with photos by Charles Pratt

1970 Rachel Carson National Wildlife Refuge established in Maine

1972 Environmental Protection Agency (EPA) established; DDT banned from sale in US

1980 President Jimmy Carter awards Presidential Medal of Freedom to Rachel Carson

1981 Ruth Scott establishes Rachel Carson Homestead and Trust; Rachel Carson stamp issued by Post Office, May 28

APPENDIX 2

HISTORY OF DDT (DICHLORO-DIPHENYL-TRICHLOROETHANE)

1874 PhD student Othmar Zeidler synthesizes DDT in a laboratory at the University of Strasbourg

1939 Paul Müller discovers insecticidal properties of DDT in laboratory of J. R. Geigy pharmaceutical company in Switzerland

1944 DDT used as part of the control of a typhus epidemic in Naples, Italy. Subsequently used widely on soldiers and civilians

1945 U.S. Fish and Wildlife Service undertakes experiments about DDT's effect on wildlife at Patuxent Research Refuge; Carson proposes article about this to *Reader's Digest;* the magazine does not answer

1945 On August 1, DDT released in U.S. for civilian use; September 9 Gimbels Department Store in New York advertises sale of DDT to public;

1945—1960s DDT widely used in sprays, powders, in homes and agriculture in U.S.

1948 Paul Müller receives Nobel Prize in Physiology and Medicine for work with DDT

1950s U.S. government agencies carry out insecticidal spraying programs against gypsy moths, beetles spreading Dutch Elm disease, and other insect pests

1950-51 U.S. House Select Committee to Investigate the Use of Chemicals in Food Products (Delaney Committee) holds hearings on pesticides; Beech-Nut Packing Company testifies that it can't find residue-free vegetables for the baby food it produces

1954 Miller Amendment requires manufacturers to provide evidence of the safety of agricultural chemicals

1955 World Health Organization (WHO) starts spraying interior walls of homes in anti-malarial campaign in Africa, Mexico, Sardinia and elsewhere. The campaign meets with initial success.

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1957 Marjorie Spock and others file suit in Long Island, New York court to prevent spraying for gypsy moths. Judge dismisses the suit.

1957 Olga Huckins writes letter to the *Boston Herald* about DDT spray killing birds at her bird feeder, sends Carson a copy. Carson begins work on *Silent Spring.*

1962 *Silent Spring* published; parts of it are serialized in *The New Yorker*; it is a Book of the Month Club selection; best-seller for 31 weeks

1963 CBS Broadcasts "The Silent Spring of Rachel Carson"; Carson testifies twice in Congress

1967 The Environmental Defense Fund formed; aims to stop DDT spraying

1969 WHO ends anti-malaria campaign; widespread agricultural use has contributed to development of DDT resistance in mosquito populations

1969 the state of Michigan bans DDT from agricultural use. *The New York Times* magazine prints an obituary for DDT: "died, DDT, at age 95"

1969-70 U.S. Department of Agriculture (USDA) cancels registration for most uses of DDT, with exceptions for disease prevention

1972 William Ruckelshaus, administrator of the Environmental Protection Agency (EPA) prohibits domestic use of DDT in the U.S., permits manufacture and export

1983 Last DDT manufacturing plant in U.S. dismantled and sold to Indonesia

2001 The Stockholm Convention, a global treaty to protect human health and the environment from DDT and eleven other persistent organic pollutants (POPs). The "Dirty Dozen": aldrin, chlordane, dieldrin, endrin, heptachlor, Hexachlorobenzene, mirex, toxaphene, Polychlorinated biphenyls (PCBs), DDT, dioxins. All but DDT are to be eliminated; DDT is allowed restricted use in vector control. The convention has met repeatedly and added other POPs to its list, after scientific investigations.

2004 Stockholm Convention enters into force on 17 May, when 50th country ratifies it.

2006 WHO declares support of indoor use of DDT (and 11 other pesticides) to control malaria in Africa, but restricts agricultural use

2009 Indonesia signs Stockholm agreement; no longer manufactures DDT

2008 Eugene Kenaga International DDT Conference issues the Pine River Statement on the Human Health Consequences of DDT

2011 174 countries and the European Union have ratified or acceded to the Stockholm Convention on POPs. As of this date, the U.S. still has not ratified the Convention.

APPENDIX 3

HISTORY OF U.S. ENVIRONMENTAL LEGISLATION

(adapted from H. Patricia Hynes, *The Recurring Silent Spring*, p. 48 and from Carolyn Merchant, *American Environmental History,* pp. 269–87*)*

1862 Morrill Land Grant Act: granted public land to states to sell so as to fund the establishment of agricultural colleges

1864 Yosemite State Park Created (it became a National Park in 1872)

1878 Free Timber Act: allowed residents of 9 western states the right to cut trees on public land

1887 Hatch Act set up agricultural experiment stations

1899 Refuse Act

1902 Reclamation Act: irrigation projects

1905 U.S. Forest Service Created

1906 Pure Food Act (set up monitoring of pesticide residues in foods)

1910 Insecticide Act

1938 Food, Drugs, and Cosmetics Act

1947 Federal Insecticide, Fungicide, Rodenticide Act (FIFRA)

1948 Water Pollution Control Act

1954 Miller Amendment (requires registration of pesticides and determinations of safety)

1958 Delaney Clause (amends the 1938 Food, Drugs, and Cosmetics Act: “the Secretary of the Food and Drug Administration shall not approve for use in food any chemical additive found to induce cancer in man, or, after tests, found to induce cancer in animals.")

1958 Magnuson-Metcalf Bill (Public Law 85-582) authorizes the U.S. Fish and Wildlife

Service to conduct studies of the effects of various pesticides on fish and other wildlife

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1963 Clean Air Act

1969 National Environmental Policy Act (NEPA) requires federal agencies to prepare Environmental Impact Statements for legislation or projects affecting the environment

1970 Environmental Protection Agency (EPA) founded

1970 Occupational Safety and Health Act

1971 Occupational Safety and Health Administration (OSHA) founded

1972 FIFRA Amended

1972 Federal Water Pollution Control Act

1972 Safe Drinking Water Act

1972 Coastal Zone Management Act

1972 Marine Protection, Research and Sanctuaries Act

1976 Toxic Substances Control Act (TOSCA) regulates public exposure to toxic materials

1976 Resource Conservation and Recovery Act (RCRA)

1980 Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Superfund: cleanup of hazardous waste dumps

1987 Montreal Protocol: phase-out of chlorofluorocarbons (CFCs) that break down the ozone layer

1989 North American Wetlands Conservation Act

1990 Clean Air Act amended

1996 Food Quality Protection Act replaced the Delaney Clause. Mandated a health-based standard for pesticides used in foods, provided special protections for babies and infants, streamlined the approval of safe pesticides, established incentives for the creation of safer pesticides, and required that pesticide registrations remain current

2012 National Endowment for the Oceans established

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